Ответы и официальные разборы

Том 7. Официальные оригиналы

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Shortlisted Problems

(with solutions)

Confidential until

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Contributing Countries

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Problems

Algebra

A1. Let n be an integer, and let A be a subset of $\{0, 1, 2, 3, \ldots, 5^n\}$ consisting of $4n + 2$ numbers. Prove that there exist $a, b, c \in A$ such that $a < b < c$ and $c + 2a > 3b$.

A2. For every integer $n \geqslant 1$ consider the $n \times n$ table with entry $\left\lfloor \frac{ij}{n+1} \right\rfloor$ at the intersection of row i and column j, for every $i = 1, \ldots, n$ and $j = 1, \ldots, n$. Determine all integers $n \geqslant 1$ for which the sum of the n^2 entries in the table is equal to $\frac{1}{4}n^2(n-1)$.

A3. Given a positive integer n, find the smallest value of $\left\lfloor \frac{a_1}{1} \right\rfloor + \left\lfloor \frac{a_2}{2} \right\rfloor + \cdots + \left\lfloor \frac{a_n}{n} \right\rfloor$ over all permutations $(a_1, a_2, \ldots, a_n)$ of $(1, 2, \ldots, n)$.

A4. Show that for all real numbers $x_1, \ldots, x_n$ the following inequality holds:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \sqrt{|x_i - x_j|} \leqslant \sum_{i=1}^{n} \sum_{j=1}^{n} \sqrt{|x_i + x_j|}.$$

A5. Let $n \geqslant 2$ be an integer, and let $a_1, a_2, \ldots, a_n$ be positive real numbers such that $a_1 + a_2 + \cdots + a_n = 1$. Prove that

$$\sum_{k=1}^{n} \frac{a_k}{1 - a_k} (a_1 + a_2 + \cdots + a_{k-1})^2 < \frac{1}{3}.$$

A6. Let A be a finite set of (not necessarily positive) integers, and let $m \geqslant 2$ be an integer. Assume that there exist non-empty subsets $B_1, B_2, B_3, \ldots, B_m$ of A whose elements add up to the sums $m^1, m^2, m^3, \ldots, m^m$, respectively. Prove that A contains at least $m/2$ elements.

A7. Let $n \geqslant 1$ be an integer, and let $x_0, x_1, \ldots, x_{n+1}$ be $n + 2$ non-negative real numbers that satisfy $x_i x_{i+1} - x_{i-1}^2 \geqslant 1$ for all $i = 1, 2, \ldots, n$. Show that

$$x_0 + x_1 + \cdots + x_n + x_{n+1} > \left(\frac{2n}{3} \right)^{3/2}.$$

A8. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy

$$\big(f(a) - f(b)\big)\big(f(b) - f(c)\big)\big(f(c) - f(a)\big) = f(ab^2 + bc^2 + ca^2) - f(a^2b + b^2c + c^2a)$$

for all real numbers a, b, c.

Combinatorics

C1. Let S be an infinite set of positive integers, such that there exist four pairwise distinct $a, b, c, d \in S$ with $\gcd(a, b) \neq \gcd(c, d)$. Prove that there exist three pairwise distinct $x, y, z \in S$ such that $\gcd(x, y) = \gcd(y, z) \neq \gcd(z, x)$.

C2. Let $n \geqslant 3$ be an integer. An integer $m \geqslant n + 1$ is called *n-colourful* if, given infinitely many marbles in each of n colours $C_1, C_2, \ldots, C_n$, it is possible to place m of them around a circle so that in any group of $n + 1$ consecutive marbles there is at least one marble of colour C_i for each $i = 1, \ldots, n$.

Prove that there are only finitely many positive integers which are not n-colourful. Find the largest among them.

C3. A thimblerigger has 2021 thimbles numbered from 1 through 2021. The thimbles are arranged in a circle in arbitrary order. The thimblerigger performs a sequence of 2021 moves; in the kth move, he swaps the positions of the two thimbles adjacent to thimble k.

Prove that there exists a value of k such that, in the kth move, the thimblerigger swaps some thimbles a and b such that $a < k < b$.

C4. The kingdom of Anisotropy consists of n cities. For every two cities there exists exactly one direct one-way road between them. We say that a *path from X to Y* is a sequence of roads such that one can move from X to Y along this sequence without returning to an already visited city. A collection of paths is called *diverse* if no road belongs to two or more paths in the collection.

Let A and B be two distinct cities in Anisotropy. Let N_{AB} denote the maximal number of paths in a diverse collection of paths from A to B. Similarly, let N_{BA} denote the maximal number of paths in a diverse collection of paths from B to A. Prove that the equality $N_{AB} = N_{BA}$ holds if and only if the number of roads going out from A is the same as the number of roads going out from B.

C5. Let n and k be two integers with $n > k \geqslant 1$. There are $2n + 1$ students standing in a circle. Each student S has $2k$ *neighbours*— namely, the k students closest to S on the right, and the k students closest to S on the left.

Suppose that $n + 1$ of the students are girls, and the other n are boys. Prove that there is a girl with at least k girls among her neighbours.

C6. A hunter and an invisible rabbit play a game on an infinite square grid. First the hunter fixes a colouring of the cells with finitely many colours. The rabbit then secretly chooses a cell to start in. Every minute, the rabbit reports the colour of its current cell to the hunter, and then secretly moves to an adjacent cell that it has not visited before (two cells are adjacent if they share a side). The hunter wins if after some finite time either

- the rabbit cannot move; or
- the hunter can determine the cell in which the rabbit started.

Decide whether there exists a winning strategy for the hunter.

C7. Consider a checkered $3m \times 3m$ square, where m is an integer greater than 1. A frog sits on the lower left corner cell S and wants to get to the upper right corner cell F. The frog can hop from any cell to either the next cell to the right or the next cell upwards.

Some cells can be sticky, and the frog gets trapped once it hops on such a cell. A set X of cells is called *blocking* if the frog cannot reach F from S when all the cells of X are sticky. A blocking set is *minimal* if it does not contain a smaller blocking set.

(a) Prove that there exists a minimal blocking set containing at least $3m^2 - 3m$ cells.

(b) Prove that every minimal blocking set contains at most $3m^2$ cells.

Note. An example of a minimal blocking set for $m = 2$ is shown below. Cells of the set X are marked by letters x.

					F
x	x				
		x			
			x		
				x	
S		x			

C8. Determine the largest N for which there exists a table T of integers with N rows and 100 columns that has the following properties:

(i) Every row contains the numbers $1, 2, \ldots, 100$ in some order.

(ii) For any two distinct rows r and s, there is a column c such that $|T(r, c) - T(s, c)| \geqslant 2$.

Here $T(r, c)$ means the number at the intersection of the row r and the column c.

Geometry

G1. Let $ABCD$ be a parallelogram such that $AC = BC$. A point P is chosen on the extension of the segment AB beyond B. The circumcircle of the triangle ACD meets the segment PD again at Q, and the circumcircle of the triangle APQ meets the segment PC again at R. Prove that the lines CD, AQ, and BR are concurrent.

G2. Let $ABCD$ be a convex quadrilateral circumscribed around a circle with centre I. Let ω be the circumcircle of the triangle ACI. The extensions of BA and BC beyond A and C meet ω at X and Z, respectively. The extensions of AD and CD beyond D meet ω at Y and T, respectively. Prove that the perimeters of the (possibly self-intersecting) quadrilaterals $ADTX$ and $CDYZ$ are equal.

G3.

Version 1. Let n be a fixed positive integer, and let S be the set of points (x, y) on the Cartesian plane such that both coordinates x and y are nonnegative integers smaller than $2n$ (thus $|\mathsf{S}| = 4n^2$). Assume that $\mathcal{F}$ is a set consisting of n^2 quadrilaterals such that all their vertices lie in S, and each point in S is a vertex of exactly one of the quadrilaterals in $\mathcal{F}$.

Determine the largest possible sum of areas of all n^2 quadrilaterals in $\mathcal{F}$.

Version 2. Let n be a fixed positive integer, and let S be the set of points (x, y) on the Cartesian plane such that both coordinates x and y are nonnegative integers smaller than $2n$ (thus $|\mathsf{S}| = 4n^2$). Assume that $\mathcal{F}$ is a set of polygons such that all vertices of polygons in $\mathcal{F}$ lie in S, and each point in S is a vertex of exactly one of the polygons in $\mathcal{F}$.

Determine the largest possible sum of areas of all polygons in $\mathcal{F}$.

G4. Let $ABCD$ be a quadrilateral inscribed in a circle Ω. Let the tangent to Ω at D intersect the rays BA and BC at points E and F, respectively. A point T is chosen inside the triangle ABC so that $TE \parallel CD$ and $TF \parallel AD$. Let $K \neq D$ be a point on the segment DF such that $TD = TK$. Prove that the lines AC, DT and BK intersect at one point.

G5. Let $ABCD$ be a cyclic quadrilateral whose sides have pairwise different lengths. Let O be the circumcentre of $ABCD$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at B_1 and D_1, respectively. Let O_B be the centre of the circle which passes through B and is tangent to AC at D_1. Similarly, let O_D be the centre of the circle which passes through D and is tangent to AC at B_1.

Assume that $BD_1 \parallel DB_1$. Prove that O lies on the line O_BO_D.

G6. Determine all integers $n \geqslant 3$ satisfying the following property: every convex n-gon whose sides all have length 1 contains an equilateral triangle of side length 1.

(Every polygon is assumed to contain its boundary.)

G7. A point D is chosen inside an acute-angled triangle ABC with $AB > AC$ so that $\angle BAD = \angle DAC$. A point E is constructed on the segment AC so that $\angle ADE = \angle DCB$. Similarly, a point F is constructed on the segment AB so that $\angle ADF = \angle DBC$. A point X is chosen on the line AC so that $CX = BX$. Let O_1 and O_2 be the circumcentres of the triangles ADC and DXE. Prove that the lines BC, EF, and O_1O_2 are concurrent.

G8. Let ω be the circumcircle of a triangle ABC, and let Ω_A be its excircle which is tangent to the segment BC. Let X and Y be the intersection points of ω and Ω_A. Let P and Q be the projections of A onto the tangent lines to Ω_A at X and Y, respectively. The tangent line at P to the circumcircle of the triangle APX intersects the tangent line at Q to the circumcircle of the triangle AQY at a point R. Prove that $AR \perp BC$.

Number Theory

N1. Determine all integers $n \geqslant 1$ for which there exists a pair of positive integers (a, b) such that no cube of a prime divides $a^2 + b + 3$ and

$$\frac{ab + 3b + 8}{a^2 + b + 3} = n.$$

N2. Let $n \geqslant 100$ be an integer. The numbers $n, n+1, \ldots, 2n$ are written on $n+1$ cards, one number per card. The cards are shuffled and divided into two piles. Prove that one of the piles contains two cards such that the sum of their numbers is a perfect square.

N3. Find all positive integers n with the following property: the k positive divisors of n have a permutation $(d_1, d_2, \ldots, d_k)$ such that for every $i = 1, 2, \ldots, k$, the number $d_1 + \cdots + d_i$ is a perfect square.

N4. Alice is given a rational number $r > 1$ and a line with two points $B \neq R$, where point R contains a red bead and point B contains a blue bead. Alice plays a solitaire game by performing a sequence of moves. In every move, she chooses a (not necessarily positive) integer k, and a bead to move. If that bead is placed at point X, and the other bead is placed at Y, then Alice moves the chosen bead to point X' with $\overrightarrow{YX'} = r^k \overrightarrow{YX}$.

Alice's goal is to move the red bead to the point B. Find all rational numbers $r > 1$ such that Alice can reach her goal in at most 2021 moves.

N5. Prove that there are only finitely many quadruples (a, b, c, n) of positive integers such that

$$n! = a^{n-1} + b^{n-1} + c^{n-1}.$$

N6. Determine all integers $n \geqslant 2$ with the following property: every n pairwise distinct integers whose sum is not divisible by n can be arranged in some order $a_1, a_2, \ldots, a_n$ so that n divides $1 \cdot a_1 + 2 \cdot a_2 + \cdots + n \cdot a_n$.

N7. Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers such that a_{n+2m} divides $a_n + a_{n+m}$ for all positive integers n and m. Prove that this sequence is eventually periodic, i.e. there exist positive integers N and d such that $a_n = a_{n+d}$ for all $n > N$.

N8. For a polynomial $P(x)$ with integer coefficients let $P^1(x) = P(x)$ and $P^{k+1}(x) = P(P^k(x))$ for $k \geqslant 1$. Find all positive integers n for which there exists a polynomial $P(x)$ with integer coefficients such that for every integer $m \geqslant 1$, the numbers $P^m(1), \ldots, P^m(n)$ leave exactly $\lceil n/2^m \rceil$ distinct remainders when divided by n.

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Solutions

Algebra

A1. Let n be an integer, and let A be a subset of $\{0, 1, 2, 3, \ldots, 5^n\}$ consisting of $4n + 2$ numbers. Prove that there exist $a, b, c \in A$ such that $a < b < c$ and $c + 2a > 3b$.

Solution 1. (By contradiction) Suppose that there exist $4n + 2$ non-negative integers $x_0 < x_1 < \cdots < x_{4n+1}$ that violate the problem statement. Then in particular $x_{4n+1} + 2x_i \leqslant 3x_{i+1}$ for all $i = 0, \ldots, 4n - 1$, which gives

$$x_{4n+1} - x_i \geqslant \frac{3}{2}(x_{4n+1} - x_{i+1}).$$

By a trivial induction we then get

$$x_{4n+1} - x_i \geqslant \left(\frac{3}{2}\right)^{4n-i}(x_{4n+1} - x_{4n}),$$

which for $i = 0$ yields the contradiction

$$x_{4n+1} - x_0 \geqslant \left(\frac{3}{2}\right)^{4n}(x_{4n+1} - x_{4n}) = \left(\frac{81}{16}\right)^{n}(x_{4n+1} - x_{4n}) > 5^n \cdot 1.$$

Solution 2. Denote the maximum element of A by c. For $k = 0, \ldots, 4n - 1$ let

$$A_k = \left\{x \in A : (1 - (2/3)^k)c \leqslant x < (1 - (2/3)^{k+1})c\right\}.$$

Note that

$$(1 - (2/3)^{4n})c = c - (16/81)^n c > c - (1/5)^n c \geqslant c - 1,$$

which shows that the sets $A_0, A_1, \ldots, A_{4n-1}$ form a partition of $A \setminus \{c\}$. Since $A \setminus \{c\}$ has $4n + 1$ elements, by the pigeonhole principle some set A_k does contain at least two elements of $A \setminus \{c\}$. Denote these two elements a and b and assume $a < b$, so that $a < b < c$. Then

$$c + 2a \geqslant c + 2(1 - (2/3)^k)c = (3 - 2(2/3)^k)c = 3(1 - (2/3)^{k+1})c > 3b,$$

as desired.

A2. For every integer $n \geqslant 1$ consider the $n \times n$ table with entry $\left\lfloor \frac{ij}{n+1} \right\rfloor$ at the intersection of row i and column j, for every $i = 1, \ldots, n$ and $j = 1, \ldots, n$. Determine all integers $n \geqslant 1$ for which the sum of the n^2 entries in the table is equal to $\frac{1}{4}n^2(n-1)$.

Answer: All integers n for which $n+1$ is a prime.

Solution 1. First, observe that every pair x, y of real numbers for which the sum $x+y$ is integer satisfies

$$\lfloor x \rfloor + \lfloor y \rfloor \geqslant x + y - 1. \tag{1}$$

The inequality is strict if x and y are integers, and it holds with equality otherwise.

We estimate the sum S as follows.

$$2S = \sum_{1 \leqslant i,j \leqslant n} \left(\left\lfloor \frac{ij}{n+1} \right\rfloor + \left\lfloor \frac{ij}{n+1} \right\rfloor \right) = \sum_{1 \leqslant i,j \leqslant n} \left(\left\lfloor \frac{ij}{n+1} \right\rfloor + \left\lfloor \frac{(n+1-i)j}{n+1} \right\rfloor \right)$$

$$\geqslant \sum_{1 \leqslant i,j \leqslant n} (j-1) = \frac{(n-1)n^2}{2}.$$

The inequality in the last line follows from (1) by setting $x = ij/(n+1)$ and $y = (n+1-i)j/(n+1)$, so that $x + y = j$ is integral.

Now $S = \frac{1}{4}n^2(n-1)$ if and only if the inequality in the last line holds with equality, which means that none of the values $ij/(n+1)$ with $1 \leqslant i, j \leqslant n$ may be integral.

Hence, if $n+1$ is composite with factorisation $n+1 = ab$ for $2 \leqslant a, b \leqslant n$, one gets a strict inequality for $i = a$ and $j = b$. If $n+1$ is a prime, then $ij/(n+1)$ is never integral and $S = \frac{1}{4}n^2(n-1)$.

Solution 2. To simplify the calculation with indices, extend the table by adding a phantom column of index 0 with zero entries (which will not change the sum of the table). Fix a row i with $1 \leqslant i \leqslant n$, and let $d := \gcd(i, n+1)$ and $k := (n+1)/d$. For columns $j = 0, \ldots, n$, define the remainder $r_j := ij \bmod (n+1)$. We first prove the following

Claim. For every integer g with $1 \leqslant g \leqslant d$, the remainders r_j with indices j in the range

$$(g-1)k \leqslant j \leqslant gk - 1 \tag{2}$$

form a permutation of the k numbers $0 \cdot d$, $1 \cdot d$, $2 \cdot d$, …, $(k-1) \cdot d$.

Proof. If $r_{j'} = r_j$ holds for two indices j' and j in (2), then $i(j'-j) \equiv 0 \bmod (n+1)$, so that $j'-j$ is a multiple of k; since $|j'-j| \leqslant k-1$, this implies $j' = j$. Hence, the k remainders are pairwise distinct. Moreover, each remainder $r_j = ij \bmod (n+1)$ is a multiple of $d = \gcd(i, n+1)$. This proves the claim.

We then have

$$\sum_{j=0}^{n} r_j = \sum_{g=1}^{d} \sum_{\ell=0}^{(n+1)/d-1} \ell d = d^2 \cdot \frac{1}{2} \left(\frac{n+1}{d} - 1 \right) \frac{n+1}{d} = \frac{(n+1-d)(n+1)}{2}. \tag{3}$$

By using (3), compute the sum S_i of row i as follows:

$$S_i = \sum_{j=0}^{n} \left\lfloor \frac{ij}{n+1} \right\rfloor = \sum_{j=0}^{n} \frac{ij - r_j}{n+1} = \frac{i}{n+1} \sum_{j=0}^{n} j - \frac{1}{n+1} \sum_{j=0}^{n} r_j$$

$$= \frac{i}{n+1} \cdot \frac{n(n+1)}{2} - \frac{1}{n+1} \cdot \frac{(n+1-d)(n+1)}{2} = \frac{(in - n - 1 + d)}{2}. \tag{4}$$

Equation (4) yields the following lower bound on the row sum S_i, which holds with equality if and only if $d = \gcd(i, n+1) = 1$:

$$S_i \geqslant \frac{(in - n - 1 + 1)}{2} = \frac{n(i-1)}{2}. \tag{5}$$

By summing up the bounds (5) for the rows $i = 1, \ldots, n$, we get the following lower bound for the sum of all entries in the table

$$\sum_{i=1}^{n} S_i \geqslant \sum_{i=1}^{n} \frac{n}{2}(i-1) = \frac{n^2(n-1)}{4}. \tag{6}$$

In (6) we have equality if and only if equality holds in (5) for each $i = 1, \ldots, n$, which happens if and only if $\gcd(i, n+1) = 1$ for each $i = 1, \ldots, n$, which is equivalent to the fact that $n+1$ is a prime. Thus the sum of the table entries is $\frac{1}{4}n^2(n-1)$ if and only if $n+1$ is a prime.

Comment. To simplify the answer, in the problem statement one can make a change of variables by introducing $m := n+1$ and writing everything in terms of m. The drawback is that the expression for the sum will then be $\frac{1}{4}(m-1)^2(m-2)$ which seems more artificial.

A3. Given a positive integer n, find the smallest value of $\left\lfloor \frac{a_1}{1} \right\rfloor + \left\lfloor \frac{a_2}{2} \right\rfloor + \cdots + \left\lfloor \frac{a_n}{n} \right\rfloor$ over all permutations $(a_1, a_2, \ldots, a_n)$ of $(1, 2, \ldots, n)$.

Answer: The minimum of such sums is $\lfloor \log_2 n \rfloor + 1$; so if $2^k \leqslant n < 2^{k+1}$, the minimum is $k+1$.

Solution 1. Suppose that $2^k \leqslant n < 2^{k+1}$ with some nonnegative integer k. First we show a permutation $(a_1, a_2, \ldots, a_n)$ such that $\left\lfloor \frac{a_1}{1} \right\rfloor + \left\lfloor \frac{a_2}{2} \right\rfloor + \cdots + \left\lfloor \frac{a_n}{n} \right\rfloor = k+1$; then we will prove that $\left\lfloor \frac{a_1}{1} \right\rfloor + \left\lfloor \frac{a_2}{2} \right\rfloor + \cdots + \left\lfloor \frac{a_n}{n} \right\rfloor \geqslant k+1$ for every permutation. Hence, the minimal possible value will be $k+1$.

I. Consider the permutation

$$(a_1) = (1), \quad (a_2, a_3) = (3, 2), \quad (a_4, a_5, a_6, a_7) = (7, 4, 5, 6), \quad \ldots$$
$$(a_{2^{k-1}}, \ldots, a_{2^k-1}) = (2^k - 1, 2^{k-1}, 2^{k-1} + 1, \ldots, 2^k - 2),$$
$$(a_{2^k}, \ldots, a_n) = (n, 2^k, 2^k + 1, \ldots, n-1).$$

This permutation consists of $k+1$ cycles. In every cycle $(a_p, \ldots, a_q) = (q, p, p+1, \ldots, q-1)$ we have $q < 2p$, so

$$\sum_{i=p}^{q} \left\lfloor \frac{a_i}{i} \right\rfloor = \left\lfloor \frac{q}{p} \right\rfloor + \sum_{i=p+1}^{q} \left\lfloor \frac{i-1}{i} \right\rfloor = 1;$$

The total sum over all cycles is precisely $k+1$.

II. In order to establish the lower bound, we prove a more general statement.

Claim. If $b_1, \ldots, b_{2^k}$ are distinct positive integers then

$$\sum_{i=1}^{2^k} \left\lfloor \frac{b_i}{i} \right\rfloor \geqslant k+1.$$

From the Claim it follows immediately that $\sum_{i=1}^{n} \left\lfloor \frac{a_i}{i} \right\rfloor \geqslant \sum_{i=1}^{2^k} \left\lfloor \frac{a_i}{i} \right\rfloor \geqslant k+1$.

Proof of the Claim. Apply induction on k. For $k = 1$ the claim is trivial, $\left\lfloor \frac{b_1}{1} \right\rfloor \geqslant 1$. Suppose the Claim holds true for some positive integer k, and consider $k+1$.

If there exists an index j such that $2^k < j \leqslant 2^{k+1}$ and $b_j \geqslant j$ then

$$\sum_{i=1}^{2^{k+1}} \left\lfloor \frac{b_i}{i} \right\rfloor \geqslant \sum_{i=1}^{2^k} \left\lfloor \frac{b_i}{i} \right\rfloor + \left\lfloor \frac{b_j}{j} \right\rfloor \geqslant (k+1) + 1$$

by the induction hypothesis, so the Claim is satisfied.

Otherwise we have $b_j < j \leqslant 2^{k+1}$ for every $2^k < j \leqslant 2^{k+1}$. Among the 2^{k+1} distinct numbers $b_1, \ldots, b_{2^{k+1}}$ there is some b_m which is at least 2^{k+1}; that number must be among $b_1 \ldots, b_{2^k}$. Hence, $1 \leqslant m \leqslant 2^k$ and $b_m \geqslant 2^{k+1}$.

We will apply the induction hypothesis to the numbers

$$c_1 = b_1, \ldots, c_{m-1} = b_{m-1}, \quad c_m = b_{2^k+1}, \quad c_{m+1} = b_{m+1}, \ldots, c_{2^k} = b_{2^k},$$

so take the first 2^k numbers but replace b_m with b_{2^k+1}. Notice that

$$\left\lfloor \frac{b_m}{m} \right\rfloor \geqslant \left\lfloor \frac{2^{k+1}}{m} \right\rfloor = \left\lfloor \frac{2^k + 2^k}{m} \right\rfloor \geqslant \left\lfloor \frac{b_{2^k+1} + m}{m} \right\rfloor = \left\lfloor \frac{c_m}{m} \right\rfloor + 1.$$

For the other indices i with $1 \leqslant i \leqslant 2^k$, $i \neq m$ we have $\left\lfloor \frac{b_i}{i} \right\rfloor = \left\lfloor \frac{c_i}{i} \right\rfloor$, so

$$\sum_{i=1}^{2^{k+1}} \left\lfloor \frac{b_i}{i} \right\rfloor = \sum_{i=1}^{2^k} \left\lfloor \frac{b_i}{i} \right\rfloor \geqslant \sum_{i=1}^{2^k} \left\lfloor \frac{c_i}{i} \right\rfloor + 1 \geqslant (k+1) + 1.$$

That proves the Claim and hence completes the solution. □

Solution 2. We present a different proof for the lower bound.

Assume again $2^k \leqslant n < 2^{k+1}$, and let $P = \{2^0, 2^1, \ldots, 2^k\}$ be the set of powers of 2 among $1, 2, \ldots, n$. Call an integer $i \in \{1, 2, \ldots, n\}$ and the interval $[i, a_i]$ *good* if $a_i \geqslant i$.

Lemma 1. The good intervals cover the integers $1, 2, \ldots, n$.

Proof. Consider an arbitrary $x \in \{1, 2 \ldots, n\}$; we want to find a good interval $[i, a_i]$ that covers x; i.e., $i \leqslant x \leqslant a_i$. Take the cycle of the permutation that contains x, that is $(x, a_x, a_{a_x}, \ldots)$. In this cycle, let i be the first element with $a_i \geqslant x$; then $i \leqslant x \leqslant a_i$. □

Lemma 2. If a good interval $[i, a_i]$ covers p distinct powers of 2 then $\left\lfloor \frac{a_i}{i} \right\rfloor \geqslant p$; more formally, $\left\lfloor \frac{a_i}{i} \right\rfloor \geqslant \left|[i, a_i] \cap P\right|$.

Proof. The ratio of the smallest and largest powers of 2 in the interval is at least 2^{p-1}. By Bernoulli's inequality, $\frac{a_i}{i} \geqslant 2^{p-1} \geqslant p$; that proves the lemma. □

Now, by Lemma 1, the good intervals cover P. By applying Lemma 2 as well, we obtain that

$$\sum_{i=1}^{n} \left\lfloor \frac{a_i}{i} \right\rfloor = \sum_{i \text{ is good}}^{n} \left\lfloor \frac{a_i}{i} \right\rfloor \geqslant \sum_{i \text{ is good}}^{n} \Big|[i, a_i] \cap P\Big| \geqslant \big|P\big| = k + 1.$$

Solution 3. We show yet another proof for the lower bound, based on the following inequality.

Lemma 3.

$$\left\lfloor \frac{a}{b} \right\rfloor \geqslant \log_2 \frac{a+1}{b}$$

for every pair a, b of positive integers.

Proof. Let $t = \left\lfloor \frac{a}{b} \right\rfloor$, so $t \leqslant \frac{a}{b}$ and $\frac{a+1}{b} \leqslant t + 1$. By applying the inequality $2^t \geqslant t + 1$, we obtain

$$\left\lfloor \frac{a}{b} \right\rfloor = t \geqslant \log_2(t+1) \geqslant \log_2 \frac{a+1}{b}.$$ □

By applying the lemma to each term, we get

$$\sum_{i=1}^{n} \left\lfloor \frac{a_i}{i} \right\rfloor \geqslant \sum_{i=1}^{n} \log_2 \frac{a_i + 1}{i} = \sum_{i=1}^{n} \log_2(a_i + 1) - \sum_{i=1}^{n} \log_2 i.$$

Notice that the numbers $a_1 + 1, a_2 + 1, \ldots, a_n + 1$ form a permutation of $2, 3, \ldots, n + 1$. Hence, in the last two sums all terms cancel out, except for $\log_2(n + 1)$ in the first sum and $\log_2 1 = 0$ in the second sum. Therefore,

$$\sum_{i=1}^{n} \left\lfloor \frac{a_i}{i} \right\rfloor \geqslant \log_2(n+1) > k.$$

As the left-hand side is an integer, it must be at least $k + 1$.

A4. Show that for all real numbers $x_1, \ldots, x_n$ the following inequality holds:

$$\sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i - x_j|} \leqslant \sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i + x_j|}.$$

Solution 1. If we add t to all the variables then the left-hand side remains constant and the right-hand side becomes

$$H(t) := \sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i + x_j + 2t|}.$$

Let T be large enough such that both $H(-T)$ and $H(T)$ are larger than the value L of the left-hand side of the inequality we want to prove. Not necessarily distinct points $p_{i,j} := -(x_i + x_j)/2$ together with T and $-T$ split the real line into segments and two rays such that on each of these segments and rays the function $H(t)$ is concave since $f(t) := \sqrt{|\ell + 2t|}$ is concave on both intervals $(-\infty, -\ell/2]$ and $[-\ell/2, +\infty)$. Let $[a, b]$ be the segment containing zero. Then concavity implies $H(0) \geqslant \min\{H(a), H(b)\}$ and, since $H(\pm T) > L$, it suffices to prove the inequalities $H(-(x_i + x_j)/2) \geqslant L$, that is to prove the original inequality in the case when all numbers are shifted in such a way that two variables x_i and x_j add up to zero. In the following we denote the shifted variables still by x_i.

If $i = j$, i.e. $x_i = 0$ for some index i, then we can remove x_i which will decrease both sides by $2\sum_k \sqrt{|x_k|}$. Similarly, if $x_i + x_j = 0$ for distinct i and j we can remove both x_i and x_j which decreases both sides by

$$2\sqrt{2|x_i|} + 2 \cdot \sum_{k \neq i,j} \left(\sqrt{|x_k + x_i|} + \sqrt{|x_k + x_j|} \right).$$

In either case we reduced our inequality to the case of smaller n. It remains to note that for $n = 0$ and $n = 1$ the inequality is trivial.

Solution 2. For real p consider the integral

$$I(p) = \int_0^\infty \frac{1 - \cos(px)}{x\sqrt{x}}\, dx,$$

which clearly converges to a strictly positive number. By changing the variable $y = |p|x$ one notices that $I(p) = \sqrt{|p|}I(1)$. Hence, by using the trigonometric formula $\cos(\alpha - \beta) - \cos(\alpha + \beta) = 2\sin\alpha\sin\beta$ we obtain

$$\sqrt{|a+b|} - \sqrt{|a-b|} = \frac{1}{I(1)} \int_0^\infty \frac{\cos((a-b)x) - \cos((a+b)x)}{x\sqrt{x}}\, dx = \frac{1}{I(1)} \int_0^\infty \frac{2\sin(ax)\sin(bx)}{x\sqrt{x}}\, dx,$$

from which our inequality immediately follows:

$$\sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i + x_j|} - \sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i - x_j|} = \frac{2}{I(1)} \int_0^\infty \frac{\left(\sum_{i=1}^{n} \sin(x_i x)\right)^2}{x\sqrt{x}}\, dx \geqslant 0.$$

Comment 1. A more general inequality

$$\sum_{i=1}^{n}\sum_{j=1}^{n}|x_i - x_j|^r \leqslant \sum_{i=1}^{n}\sum_{j=1}^{n}|x_i + x_j|^r$$

holds for any $r \in [0, 2]$. The first solution can be repeated verbatim for any $r \in [0, 1]$ but not for $r > 1$. In the second solution, by putting x^{r+1} in the denominator in place of $x\sqrt{x}$ we can prove the inequality for any $r \in (0, 2)$ and the cases $r = 0, 2$ are easy to check by hand.

Comment 2. In fact, the integral from Solution 2 can be computed explicitly, we have $I(1) = \sqrt{2\pi}$.

A5. Let $n \geqslant 2$ be an integer, and let $a_1, a_2, \ldots, a_n$ be positive real numbers such that $a_1 + a_2 + \cdots + a_n = 1$. Prove that

$$\sum_{k=1}^{n} \frac{a_k}{1-a_k}(a_1 + a_2 + \cdots + a_{k-1})^2 < \frac{1}{3}.$$

Solution 1. For all $k \leqslant n$, let

$$s_k = a_1 + a_2 + \cdots + a_k \qquad \text{and} \qquad b_k = \frac{a_k s_{k-1}^2}{1-a_k},$$

with the convention that $s_0 = 0$. Note that b_k is exactly a summand in the sum we need to estimate. We shall prove the inequality

$$b_k < \frac{s_k^3 - s_{k-1}^3}{3}. \tag{1}$$

Indeed, it suffices to check that

$$\begin{aligned}
(1) &\iff 0 < (1-a_k)\left((s_{k-1}+a_k)^3 - s_{k-1}^3\right) - 3a_k s_{k-1}^2 \\
&\iff 0 < (1-a_k)\left(3s_{k-1}^2 + 3s_{k-1}a_k + a_k^2\right) - 3s_{k-1}^2 \\
&\iff 0 < -3a_k s_{k-1}^2 + 3(1-a_k)s_{k-1}a_k + (1-a_k)a_k^2 \\
&\iff 0 < 3(1-a_k-s_{k-1})s_{k-1}a_k + (1-a_k)a_k^2
\end{aligned}$$

which holds since $a_k + s_{k-1} = s_k \leqslant 1$ and $a_k \in (0,1)$.

Thus, adding inequalities (1) for $k = 1, \ldots, n$, we conclude that

$$b_1 + b_2 + \cdots + b_n < \frac{s_n^3 - s_0^3}{3} = \frac{1}{3},$$

as desired.

Comment 1. There are many ways of proving (1) which can be written as

$$\frac{as^2}{1-a} - \frac{(a+s)^3 - s^3}{3} < 0, \tag{2}$$

for non-negative a and s satisfying $a + s \leqslant 1$ and $a > 0$.

E.g., note that for any fixed a the expression in (2) is quadratic in s with the leading coefficient $a/(1-a) - a > 0$. Hence, it is convex as a function in s, so it suffices to check the inequality at $s = 0$ and $s = 1 - a$. The former case is trivial and in the latter case the inequality can be rewritten as

$$as - \frac{3as(a+s) + a^3}{3} < 0,$$

which is trivial since $a + s = 1$.

Solution 2. First, let us define

$$S(a_1, \ldots, a_n) := \sum_{k=1}^{n} \frac{a_k}{1-a_k}(a_1 + a_2 + \cdots + a_{k-1})^2.$$

For some index i, denote $a_1 + \cdots + a_{i-1}$ by s. If we replace a_i with two numbers $a_i/2$ and $a_i/2$, i.e. replace the tuple $(a_1, \ldots, a_n)$ with $(a_1, \ldots, a_{i-1}, a_i/2, a_i/2, a_{i+1}, \ldots, a_n)$, the sum will increase by

$$\begin{aligned} S(a_1, \ldots, a_i/2, a_i/2, \ldots, a_n) - S(a_1, \ldots, a_n) &= \frac{a_i/2}{1 - a_i/2}\left(s^2 + (s + a_i/2)^2\right) - \frac{a_i}{1 - a_i}s^2 \\ &= a_i \frac{(1 - a_i)(2s^2 + sa_i + a_i^2/4) - (2 - a_i)s^2}{(2 - a_i)(1 - a_i)} \\ &= a_i \frac{(1 - a_i - s)sa_i + (1 - a_i)a_i^2/4}{(2 - a_i)(1 - a_i)}, \end{aligned}$$

which is strictly positive. So every such replacement strictly increases the sum. By repeating this process and making maximal number in the tuple tend to zero, we keep increasing the sum which will converge to

$$\int_0^1 x^2\, dx = \frac{1}{3}.$$

This completes the proof.

Solution 3. We sketch a probabilistic version of the first solution. Let x_1, x_2, x_3 be drawn uniformly and independently at random from the segment $[0, 1]$. Let $I_1 \cup I_2 \cup \cdots \cup I_n$ be a partition of $[0, 1]$ into segments of length $a_1, a_2, \ldots, a_n$ in this order. Let $J_k := I_1 \cup \cdots \cup I_{k-1}$ for $k \geqslant 2$ and $J_1 := \varnothing$. Then

$$\begin{aligned} \frac{1}{3} &= \sum_{k=1}^{n} \mathbb{P}\{x_1 \geqslant x_2, x_3;\ x_1 \in I_k\} \\ &= \sum_{k=1}^{n} \Big(\mathbb{P}\{x_1 \in I_k;\ x_2, x_3 \in J_k\} + 2 \cdot \mathbb{P}\{x_1 \geqslant x_2;\ x_1, x_2 \in I_k;\ x_3 \in J_k\} \\ &\qquad\qquad + \mathbb{P}\{x_1 \geqslant x_2, x_3;\ x_1, x_2, x_3 \in I_k\}\Big) \\ &= \sum_{k=1}^{n} \left(a_k(a_1 + \cdots + a_{k-1})^2 + 2 \cdot \frac{a_k^2}{2} \cdot (a_1 + \cdots + a_{k-1}) + \frac{a_k^3}{3}\right) \\ &> \sum_{k=1}^{n} \left(a_k(a_1 + \cdots + a_{k-1})^2 + a_k^2(a_1 + \cdots + a_{k-1}) \cdot \frac{a_1 + \cdots + a_{k-1}}{1 - a_k}\right), \end{aligned}$$

where for the last inequality we used that $1 - a_k \geqslant a_1 + \cdots + a_{k-1}$. This completes the proof since

$$a_k + \frac{a_k^2}{1 - a_k} = \frac{a_k}{1 - a_k}.$$

A6. Let A be a finite set of (not necessarily positive) integers, and let $m \geqslant 2$ be an integer. Assume that there exist non-empty subsets $B_1, B_2, B_3, \ldots, B_m$ of A whose elements add up to the sums $m^1, m^2, m^3, \ldots, m^m$, respectively. Prove that A contains at least $m/2$ elements.

Solution. Let $A = \{a_1, \ldots, a_k\}$. Assume that, on the contrary, $k = |A| < m/2$. Let

$$s_i := \sum_{j : a_j \in B_i} a_j$$

be the sum of elements of B_i. We are given that $s_i = m^i$ for $i = 1, \ldots, m$.

Now consider all m^m expressions of the form

$$f(c_1, \ldots, c_m) := c_1 s_1 + c_2 s_2 + \ldots + c_m s_m, \ c_i \in \{0, 1, \ldots, m-1\} \text{ for all } i = 1, 2, \ldots, m.$$

Note that every number $f(c_1, \ldots, c_m)$ has the form

$$\alpha_1 a_1 + \ldots + \alpha_k a_k, \ \alpha_i \in \{0, 1, \ldots, m(m-1)\}.$$

Hence, there are at most $(m(m-1)+1)^k < m^{2k} < m^m$ distinct values of our expressions; therefore, at least two of them coincide.

Since $s_i = m^i$, this contradicts the uniqueness of representation of positive integers in the base-m system.

Comment 1. For other rapidly increasing sequences of sums of B_i's the similar argument also provides lower estimates on $k = |A|$. For example, if the sums of B_i are equal to $1!$, $2!$, $3!$, …, $m!$, then for any fixed $\varepsilon > 0$ and large enough m we get $k \geqslant (1/2 - \varepsilon)m$. The proof uses the fact that the combinations $\sum c_i i!$ with $c_i \in \{0, 1, \ldots, i\}$ are all distinct.

Comment 2. The problem statement holds also if A is a set of real numbers (not necessarily integers), the above proofs work in the real case.

A7. Let $n \geqslant 1$ be an integer, and let $x_0, x_1, \ldots, x_{n+1}$ be $n+2$ non-negative real numbers that satisfy $x_i x_{i+1} - x_{i-1}^2 \geqslant 1$ for all $i = 1, 2, \ldots, n$. Show that

$$x_0 + x_1 + \cdots + x_n + x_{n+1} > \left(\frac{2n}{3}\right)^{3/2}.$$

Solution 1.

Lemma 1.1. If a, b, c are non-negative numbers such that $ab - c^2 \geqslant 1$, then

$$(a+2b)^2 \geqslant (b+2c)^2 + 6.$$

Proof. $(a+2b)^2 - (b+2c)^2 = (a-b)^2 + 2(b-c)^2 + 6(ab-c^2) \geqslant 6.$ □

Lemma 1.2. $\sqrt{1} + \cdots + \sqrt{n} > \frac{2}{3}n^{3/2}$.

Proof. Bernoulli's inequality $(1+t)^{3/2} > 1 + \frac{3}{2}t$ for $0 > t \geqslant -1$ (or, alternatively, a straightforward check) gives

$$(k-1)^{3/2} = k^{3/2}\left(1 - \frac{1}{k}\right)^{3/2} > k^{3/2}\left(1 - \frac{3}{2k}\right) = k^{3/2} - \frac{3}{2}\sqrt{k}. \qquad (*)$$

Summing up $(*)$ over $k = 1, 2, \ldots, n$ yields

$$0 > n^{3/2} - \frac{3}{2}\left(\sqrt{1} + \cdots + \sqrt{n}\right).$$ □

Now put $y_i := 2x_i + x_{i+1}$ for $i = 0, 1, \ldots, n$. We get $y_0 \geqslant 0$ and $y_i^2 \geqslant y_{i-1}^2 + 6$ for $i = 1, 2, \ldots, n$ by Lemma 1.1. Thus, an easy induction on i gives $y_i \geqslant \sqrt{6i}$. Using this estimate and Lemma 1.2 we get

$$3(x_0 + \ldots + x_{n+1}) \geqslant y_1 + \ldots + y_n \geqslant \sqrt{6}\left(\sqrt{1} + \sqrt{2} + \ldots + \sqrt{n}\right) > \sqrt{6} \cdot \frac{2}{3}n^{3/2} = 3\left(\frac{2n}{3}\right)^{3/2}.$$

Solution 2. Say that an index $i \in \{0, 1, \ldots, n+1\}$ is *good*, if $x_i \geqslant \sqrt{\frac{2}{3}i}$, otherwise call the index i *bad*.

Lemma 2.1. There are no two consecutive bad indices.

Proof. Assume the contrary and consider two bad indices j, $j+1$ with minimal possible j. Since 0 is good, we get $j > 0$, thus by minimality $j-1$ is a good index and we have

$$\frac{2}{3}\sqrt{j(j+1)} > x_j x_{j+1} \geqslant x_{j-1}^2 + 1 \geqslant \frac{2}{3}(j-1) + 1 = \frac{2}{3} \cdot \frac{j + (j+1)}{2}$$

that contradicts the AM–GM inequality for numbers j and $j+1$. □

Lemma 2.2. If an index $j \leqslant n-1$ is good, then

$$x_{j+1} + x_{j+2} \geqslant \sqrt{\frac{2}{3}}\left(\sqrt{j+1} + \sqrt{j+2}\right).$$

Proof. We have

$$x_{j+1} + x_{j+2} \geqslant 2\sqrt{x_{j+1}x_{j+2}} \geqslant 2\sqrt{x_j^2 + 1} \geqslant 2\sqrt{\frac{2}{3}j + 1} \geqslant \sqrt{\frac{2}{3}j + \frac{2}{3}} + \sqrt{\frac{2}{3}j + \frac{4}{3}},$$

the last inequality follows from concavity of the square root function, or, alternatively, from the AM–QM inequality for the numbers $\sqrt{\frac{2}{3}j + \frac{2}{3}}$ and $\sqrt{\frac{2}{3}j + \frac{4}{3}}$. □

Let $S_i = x_1 + \ldots + x_i$ and $T_i = \sqrt{\frac{2}{3}}(\sqrt{1} + \ldots + \sqrt{i})$.

Lemma 2.3. If an index i is good, then $S_i \geqslant T_i$.

Proof. Induction on i. The base case $i = 0$ is clear. Assume that the claim holds for good indices less than i and prove it for a good index $i > 0$.

If $i-1$ is good, then by the inductive hypothesis we get $S_i = S_{i-1} + x_i \geqslant T_{i-1} + \sqrt{\frac{2}{3}i} = T_i$.

If $i-1$ is bad, then $i > 1$, and $i-2$ is good by Lemma 2.1. Then using Lemma 2.2 and the inductive hypothesis we get

$$S_i = S_{i-2} + x_{i-1} + x_i \geqslant T_{i-2} + \sqrt{\frac{2}{3}}\left(\sqrt{i-1} + \sqrt{i}\right) = T_i. \qquad \square$$

Since either n or $n+1$ is good by Lemma 2.1, Lemma 2.3 yields in both cases $S_{n+1} \geqslant T_n$, and it remains to apply Lemma 1.2 from Solution 1.

Comment 1. Another way to get $(*)$ is the integral bound

$$k^{3/2} - (k-1)^{3/2} = \int_{k-1}^{k} \frac{3}{2}\sqrt{x}\,dx < \frac{3}{2}\sqrt{k}.$$

Comment 2. If $x_i = \sqrt{2/3} \cdot (\sqrt{i} + 1)$, the conditions of the problem hold. Indeed, the inequality to check is

$$(\sqrt{i}+1)(\sqrt{i+1}+1) - (\sqrt{i-1}+1)^2 \geqslant 3/2,$$

that rewrites as

$$\sqrt{i} + \sqrt{i+1} - 2\sqrt{i-1} \geqslant (i+1/2) - \sqrt{i(i+1)} = \frac{1/4}{i+1/2+\sqrt{i(i+1)}},$$

which follows from

$$\sqrt{i} - \sqrt{i-1} = \frac{1}{\sqrt{i}+\sqrt{i-1}} > \frac{1}{2i}.$$

For these numbers we have $x_0 + \ldots + x_{n+1} = \left(\frac{2n}{3}\right)^{3/2} + O(n)$, thus the multiplicative constant $(2/3)^{3/2}$ in the problem statement is sharp.

A8. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy

$$\big(f(a) - f(b)\big)\big(f(b) - f(c)\big)\big(f(c) - f(a)\big) = f(ab^2 + bc^2 + ca^2) - f(a^2b + b^2c + c^2a)$$

for all real numbers a, b, c.

Answer: $f(x) = \alpha x + \beta$ or $f(x) = \alpha x^3 + \beta$ where $\alpha \in \{-1, 0, 1\}$ and $\beta \in \mathbb{R}$.

Solution. It is straightforward to check that above functions satisfy the equation. Now let $f(x)$ satisfy the equation, which we denote $E(a, b, c)$. Then clearly $f(x) + C$ also does; therefore, we may suppose without loss of generality that $f(0) = 0$.We start with proving

Lemma. Either $f(x) \equiv 0$ or f is injective.

Proof. Denote by $\Theta \subseteq \mathbb{R}^2$ the set of points (a, b) for which $f(a) = f(b)$. Let $\Theta^* = \{(x, y) \in \Theta : x \neq y\}$. The idea is that if $(a, b) \in \Theta$, then by $E(a, b, x)$ we get

$$H_{a,b}(x) := (ab^2 + bx^2 + xa^2, a^2b + b^2x + x^2a) \in \Theta$$

for all real x. Reproducing this argument starting with $(a, b) \in \Theta^*$, we get more and more points in Θ. There are many ways to fill in the details, we give below only one of them.

Assume that $(a, b) \in \Theta^*$. Note that

$$g_-(x) := (ab^2 + bx^2 + xa^2) - (a^2b + b^2x + x^2a) = (a - b)(b - x)(x - a)$$

and

$$g_+(x) := (ab^2 + bx^2 + xa^2) + (a^2b + b^2x + x^2a) = (x^2 + ab)(a + b) + x(a^2 + b^2).$$

Hence, there exists x for which both $g_-(x) \neq 0$ and $g_+(x) \neq 0$. This gives a point $(\alpha, \beta) = H_{a,b}(x) \in \Theta^*$ for which $\alpha \neq -\beta$. Now compare $E(\alpha, 1, 0)$ and $E(\beta, 1, 0)$. The left-hand side expressions coincide, on right-hand side we get $f(\alpha) - f(\alpha^2) = f(\beta) - f(\beta^2)$, respectively. Hence, $f(\alpha^2) = f(\beta^2)$ and we get a point $(\alpha_1, \beta_1) := (\alpha^2, \beta^2) \in \Theta^*$ with both coordinates α_1, β_1 non-negative. Continuing squaring the coordinates, we get a point $(\gamma, \delta) \in \Theta^*$ for which $\delta > 5\gamma \geqslant 0$. Our nearest goal is to get a point $(0, r) \in \Theta^*$. If $\gamma = 0$, this is already done. If $\gamma > 0$, denote by x a real root of the quadratic equation $\delta\gamma^2 + \gamma x^2 + x\delta^2 = 0$, which exists since the discriminant $\delta^4 - 4\delta\gamma^3$ is positive. Also $x < 0$ since this equation cannot have non-negative root. For the point $H_{\delta,\gamma}(x) =: (0, r) \in \Theta$ the first coordinate is 0. The difference of coordinates equals $-r = (\delta - \gamma)(\gamma - x)(x - \delta) < 0$, so $r \neq 0$ as desired.

Now, let $(0, r) \in \Theta^*$. We get $H_{0,r}(x) = (rx^2, r^2x) \in \Theta$. Thus $f(rx^2) = f(r^2x)$ for all $x \in \mathbb{R}$. Replacing x to $-x$ we get $f(rx^2) = f(r^2x) = f(-r^2x)$, so f is even: $(a, -a) \in \Theta$ for all a. Then $H_{a,-a}(x) = (a^3 - ax^2 + xa^2, -a^3 + a^2x + x^2a) \in \Theta$ for all real a, x. Putting $x = \frac{1+\sqrt{5}}{2}a$ we obtain $(0, (1 + \sqrt{5})a^3) \in \Theta$ which means that $f(y) = f(0) = 0$ for every real y. □

Hereafter we assume that f is injective and $f(0) = 0$. By $E(a, b, 0)$ we get

$$f(a)f(b)(f(a) - f(b)) = f(a^2b) - f(ab^2). \qquad (\heartsuit)$$

Let $\kappa := f(1)$ and note that $\kappa = f(1) \neq f(0) = 0$ by injectivity. Putting $b = 1$ in $(\heartsuit)$ we get

$$\kappa f(a)(f(a) - \kappa) = f(a^2) - f(a). \qquad (\clubsuit)$$

Subtracting the same equality for $-a$ we get

$$\kappa(f(a) - f(-a))(f(a) + f(-a) - \kappa) = f(-a) - f(a).$$

Now, if $a \neq 0$, by injectivity we get $f(a) - f(-a) \neq 0$ and thus

$$f(a) + f(-a) = \kappa - \kappa^{-1} =: \lambda. \qquad (\spadesuit)$$

It follows that

$$f(a) - f(b) = f(-b) - f(-a)$$

for all non-zero a, b. Replace non-zero numbers a, b in ($\heartsuit$) with $-a, -b$, respectively, and add the two equalities. Due to ($\spadesuit$) we get

$$(f(a) - f(b))(f(a)f(b) - f(-a)f(-b)) = 0,$$

thus $f(a)f(b) = f(-a)f(-b) = (\lambda - f(a))(\lambda - f(b))$ for all non-zero $a \neq b$. If $\lambda \neq 0$, this implies $f(a) + f(b) = \lambda$ that contradicts injectivity when we vary b with fixed a. Therefore, $\lambda = 0$ and $\kappa = \pm 1$. Thus f is odd. Replacing f with $-f$ if necessary (this preserves the original equation) we may suppose that $f(1) = 1$.

Now, ($\clubsuit$) yields $f(a^2) = f^2(a)$. Summing relations ($\heartsuit$) for pairs (a, b) and $(a, -b)$, we get $-2f(a)f^2(b) = -2f(ab^2)$, i.e. $f(a)f(b^2) = f(ab^2)$. Putting $b = \sqrt{x}$ for each non-negative x we get $f(ax) = f(a)f(x)$ for all real a and non-negative x. Since f is odd, this multiplicativity relation is true for all a, x. Also, from $f(a^2) = f^2(a)$ we see that $f(x) \geqslant 0$ for $x \geqslant 0$. Next, $f(x) > 0$ for $x > 0$ by injectivity.

Assume that $f(x)$ for $x > 0$ does not have the form $f(x) = x^\tau$ for a constant τ. The known property of multiplicative functions yields that the graph of f is dense on $(0, \infty)^2$. In particular, we may find positive $b < 1/10$ for which $f(b) > 1$. Also, such b can be found if $f(x) = x^\tau$ for some $\tau < 0$. Then for all x we have $x^2 + xb^2 + b \geqslant 0$ and so $E(1, b, x)$ implies that

$$f(b^2 + bx^2 + x) = f(x^2 + xb^2 + b) + (f(b) - 1)(f(x) - f(b))(f(x) - 1) \geqslant 0 - ((f(b) - 1)^3/4$$

is bounded from below (the quadratic trinomial bound $(t - f(1))(t - f(b)) \geqslant -(f(b) - 1)^2/4$ for $t = f(x)$ is used). Hence, f is bounded from below on $(b^2 - \frac{1}{4b}, +\infty)$, and since f is odd it is bounded from above on $(0, \frac{1}{4b} - b^2)$. This is absurd if $f(x) = x^\tau$ for $\tau < 0$, and contradicts to the above dense graph condition otherwise.

Therefore, $f(x) = x^\tau$ for $x > 0$ and some constant $\tau > 0$. Dividing $E(a, b, c)$ by $(a - b)(b - c)(c - a) = (ab^2 + bc^2 + ca^2) - (a^2b + b^2c + c^2a)$ and taking a limit when a, b, c all go to 1 (the divided ratios tend to the corresponding derivatives, say, $\frac{a^\tau - b^\tau}{a - b} \to (x^\tau)'_{x=1} = \tau$), we get $\tau^3 = \tau \cdot 3^{\tau - 1}$, $\tau^2 = 3^{\tau - 1}$, $F(\tau) := 3^{\tau/2 - 1/2} - \tau = 0$. Since function F is strictly convex, it has at most two roots, and we get $\tau \in \{1, 3\}$.

Combinatorics

C1. Let S be an infinite set of positive integers, such that there exist four pairwise distinct $a, b, c, d \in S$ with $\gcd(a, b) \neq \gcd(c, d)$. Prove that there exist three pairwise distinct $x, y, z \in S$ such that $\gcd(x, y) = \gcd(y, z) \neq \gcd(z, x)$.

Solution. There exists $\alpha \in S$ so that $\{\gcd(\alpha, s) \mid s \in S,\ s \neq \alpha\}$ contains at least two elements. Since α has only finitely many divisors, there is a $d \mid \alpha$ such that the set $B = \{\beta \in S \mid \gcd(\alpha, \beta) = d\}$ is infinite. Pick $\gamma \in S$ so that $\gcd(\alpha, \gamma) \neq d$. Pick $\beta_1, \beta_2 \in B$ so that $\gcd(\beta_1, \gamma) = \gcd(\beta_2, \gamma) =: d'$. If $d = d'$, then $\gcd(\alpha, \beta_1) = \gcd(\gamma, \beta_1) \neq \gcd(\alpha, \gamma)$. If $d \neq d'$, then either $\gcd(\alpha, \beta_1) = \gcd(\alpha, \beta_2) = d$ and $\gcd(\beta_1, \beta_2) \neq d$ or $\gcd(\gamma, \beta_1) = \gcd(\gamma, \beta_2) = d'$ and $\gcd(\beta_1, \beta_2) \neq d'$.

Comment. The situation can be modelled as a complete graph on the infinite vertex set S, where every edge $\{s, t\}$ is colored by $c(s, t) := \gcd(s, t)$. For every vertex the incident edges carry only finitely many different colors, and by the problem statement at least two different colors show up on the edge set. The goal is to show that there exists a bi-colored triangle (a triangle, whose edges carry exactly two different colors).

For the proof, consider a vertex v whose incident edges carry at least two different colors. Let $X \subset S$ be an infinite subset so that $c(v, x) \equiv c_1$ for all $x \in X$. Let $y \in S$ be a vertex so that $c(v, y) \neq c_1$. Let $x_1, x_2 \in X$ be two vertices with $c(y, x_1) = c(y, x_2) = c_2$. If $c_1 = c_2$, then the triangle v, y, x_1 is bi-colored. If $c_1 \neq c_2$, then one of v, x_1, x_2 and y, x_1, x_2 is bi-colored.

C2. Let $n \geqslant 3$ be an integer. An integer $m \geqslant n+1$ is called *n-colourful* if, given infinitely many marbles in each of n colours $C_1, C_2, \ldots, C_n$, it is possible to place m of them around a circle so that in any group of $n+1$ consecutive marbles there is at least one marble of colour C_i for each $i = 1, \ldots, n$.

Prove that there are only finitely many positive integers which are not n-colourful. Find the largest among them.

Answer: $m_{max} = n^2 - n - 1$.

Solution. First suppose that there are $n(n-1)-1$ marbles. Then for one of the colours, say blue, there are at most $n-2$ marbles, which partition the non-blue marbles into at most $n-2$ groups with at least $(n-1)^2 > n(n-2)$ marbles in total. Thus one of these groups contains at least $n+1$ marbles and this group does not contain any blue marble.

Now suppose that the total number of marbles is at least $n(n-1)$. Then we may write this total number as $nk+j$ with some $k \geqslant n-1$ and with $0 \leqslant j \leqslant n-1$. We place around a circle $k-j$ copies of the colour sequence $[1, 2, 3, \ldots, n]$ followed by j copies of the colour sequence $[1, 1, 2, 3, \ldots, n]$.

C3. A thimblerigger has 2021 thimbles numbered from 1 through 2021. The thimbles are arranged in a circle in arbitrary order. The thimblerigger performs a sequence of 2021 moves; in the kth move, he swaps the positions of the two thimbles adjacent to thimble k.

Prove that there exists a value of k such that, in the kth move, the thimblerigger swaps some thimbles a and b such that $a < k < b$.

Solution. Assume the contrary. Say that the kth thimble is the *central thimble* of the kth move, and its position on that move is the *central position* of the move.

Step 1: Black and white colouring.

Before the moves start, let us paint all thimbles in white. Then, after each move, we repaint its central thimble in black. This way, at the end of the process all thimbles have become black.

By our assumption, in every move k, the two swapped thimbles have the same colour (as their numbers are either both larger or both smaller than k). At every moment, assign the colours of the thimbles to their current positions; then the only position which changes its colour in a move is its central position. In particular, each position is central for exactly one move (when it is being repainted to black).

Step 2: Red and green colouring.

Now we introduce a colouring of the *positions*. If in the kth move, the numbers of the two swapped thimbles are both less than k, then we paint the central position of the move in red; otherwise we paint that position in green. This way, each position has been painted in red or green exactly once. We claim that among any two adjacent positions, one becomes green and the other one becomes red; this will provide the desired contradiction since 2021 is odd.

Consider two adjacent positions A and B, which are central in the ath and in the bth moves, respectively, with $a < b$. Then in the ath move the thimble at position B is white, and therefore has a number greater than a. After the ath move, position A is green and the thimble at position A is black. By the arguments from Step 1, position A contains only black thimbles after the ath step. Therefore, on the bth move, position A contains a black thimble whose number is therefore less than b, while thimble b is at position B. So position B becomes red, and hence A and B have different colours.

Comment 1. Essentially, Step 1 provides the proof of the following two assertions (under the indirect assumption):

(1) Each position P becomes central in exactly one move (denote that move's number by k); and

(2) Before the kth move, position P always contains a thimble whose number is larger than the number of the current move, while after the kth move the position always contains a thimble whose number is smaller than the number of the current move.

Both (1) and (2) can be proved without introduction of colours, yet the colours help to visualise the argument.

After these two assertions have been proved, Step 2 can be performed in various ways, e.g., as follows.

At any moment in the process, the black positions are split into several groups consisting of one or more contiguous black positions each; different groups are separated by white positions. Now one can prove by induction on k that, after the kth move, all groups have odd sizes. Indeed, in every move, the new black position either forms a separate group, or merges two groups (say, of lengths a and b) into a single group of length $a + b + 1$.

However, after the 2020th move the black positions should form one group of length 2020. This is a contradiction.

This argument has several variations; e.g., one can check in a similar way that, after the process starts, at least one among the groups of *white* positions has an even size.

Comment 2. The solution above works equally well for any *odd* number of thimbles greater than 1, instead of 2021. On the other hand, a similar statement with an *even* number $n = 2k \geqslant 4$ of thimbles is wrong. To show that, the thimblerigger can enumerate positions from 1 through n clockwise, and then put thimbles $1, 2, \ldots, k$ at the odd positions, and thimbles $k+1, k+2, \ldots, 2k$ at the even positions.

C4. The kingdom of Anisotropy consists of n cities. For every two cities there exists exactly one direct one-way road between them. We say that a *path from X to Y* is a sequence of roads such that one can move from X to Y along this sequence without returning to an already visited city. A collection of paths is called *diverse* if no road belongs to two or more paths in the collection.

Let A and B be two distinct cities in Anisotropy. Let N_{AB} denote the maximal number of paths in a diverse collection of paths from A to B. Similarly, let N_{BA} denote the maximal number of paths in a diverse collection of paths from B to A. Prove that the equality $N_{AB} = N_{BA}$ holds if and only if the number of roads going out from A is the same as the number of roads going out from B.

Solution 1. We write $X \to Y$ (or $Y \leftarrow X$) if the road between X and Y goes from X to Y. Notice that, if there is any route moving from X to Y (possibly passing through some cities more than once), then there is a path from X to Y consisting of some roads in the route. Indeed, any cycle in the route may be removed harmlessly; after some removals one obtains a path.

Say that a path is *short* if it consists of one or two roads.

Partition all cities different from A and B into four groups, $\mathcal{I}$, $\mathcal{O}$, $\mathcal{A}$, and $\mathcal{B}$ according to the following rules: for each city C,

$$C \in \mathcal{I} \iff A \to C \leftarrow B; \qquad C \in \mathcal{O} \iff A \leftarrow C \to B;$$
$$C \in \mathcal{A} \iff A \to C \to B; \qquad C \in \mathcal{B} \iff A \leftarrow C \leftarrow B.$$

Lemma. Let $\mathcal{P}$ be a diverse collection consisting of p paths from A to B. Then there exists a diverse collection consisting of at least p paths from A to B and containing all short paths from A to B.

Proof. In order to obtain the desired collection, modify $\mathcal{P}$ as follows.

If there is a direct road $A \to B$ and the path consisting of this single road is not in $\mathcal{P}$, merely add it to $\mathcal{P}$.

Now consider any city $C \in \mathcal{A}$ such that the path $A \to C \to B$ is not in $\mathcal{P}$. If $\mathcal{P}$ contains at most one path containing a road $A \to C$ or $C \to B$, remove that path (if it exists), and add the path $A \to C \to B$ to $\mathcal{P}$ instead. Otherwise, $\mathcal{P}$ contains two paths of the forms $A \to C \dashrightarrow B$ and $A \dashrightarrow C \to B$, where $C \dashrightarrow B$ and $A \dashrightarrow C$ are some paths. In this case, we recombine the edges to form two new paths $A \to C \to B$ and $A \dashrightarrow C \dashrightarrow B$ (removing cycles from the latter if needed). Now we replace the old two paths in $\mathcal{P}$ with the two new ones.

After any operation described above, the number of paths in the collection does not decrease, and the collection remains diverse. Applying such operation to each $C \in \mathcal{A}$, we obtain the desired collection. □

Back to the problem, assume, without loss of generality, that there is a road $A \to B$, and let a and b denote the numbers of roads going out from A and B, respectively. Choose a diverse collection $\mathcal{P}$ consisting of N_{AB} paths from A to B. We will transform it into a diverse collection $\mathcal{Q}$ consisting of at least $N_{AB} + (b - a)$ paths from B to A. This construction yields

$$N_{BA} \geqslant N_{AB} + (b - a); \qquad \text{similarly, we get} \quad N_{AB} \geqslant N_{BA} + (a - b),$$

whence $N_{BA} - N_{AB} = b - a$. This yields the desired equivalence.

Apply the lemma to get a diverse collection $\mathcal{P}'$ of at least N_{AB} paths containing all $|\mathcal{A}| + 1$ short paths from A to B. Notice that the paths in $\mathcal{P}'$ contain no edge of a short path from B to A. Each non-short path in $\mathcal{P}'$ has the form $A \to C \dashrightarrow D \to B$, where $C \dashrightarrow D$ is a path from some city $C \in \mathcal{I}$ to some city $D \in \mathcal{O}$. For each such path, put into $\mathcal{Q}$ the

path $B \to C \dashrightarrow D \to A$; also put into $\mathcal{Q}$ all short paths from B to A. Clearly, the collection $\mathcal{Q}$ is diverse.

Now, all roads going out from A end in the cities from $\mathcal{I} \cup \mathcal{A} \cup \{B\}$, while all roads going out from B end in the cities from $\mathcal{I} \cup \mathcal{B}$. Therefore,

$$a = |\mathcal{I}| + |\mathcal{A}| + 1, \quad b = |\mathcal{I}| + |\mathcal{B}|, \qquad \text{and hence} \quad a - b = |\mathcal{A}| - |\mathcal{B}| + 1.$$

On the other hand, since there are $|\mathcal{A}| + 1$ short paths from A to B (including $A \to B$) and $|\mathcal{B}|$ short paths from B to A, we infer

$$|\mathcal{Q}| = |\mathcal{P}'| - (|\mathcal{A}| + 1) + |\mathcal{B}| \geqslant N_{AB} + (b - a),$$

as desired.

Solution 2. We recall some graph-theoretical notions. Let G be a finite graph, and let V be the set of its vertices; fix two distinct vertices $s, t \in V$. An (s,t)-*cut* is a partition of V into two parts $V = S \sqcup T$ such that $s \in S$ and $t \in T$. The *cut-edges* in the cut (S,T) are the edges going from S to T, and the *size* $e(S,T)$ of the cut is the number of cut-edges.

We will make use of the following theorem (which is a partial case of the Ford–Fulkerson "min-cut max-flow" theorem).

Theorem (Menger). Let G be a directed graph, and let s and t be its distinct vertices. Then the maximal number of edge-disjoint paths from s to t is equal to the minimal size of an (s,t)-cut.

Back to the problem. Consider a directed graph G whose vertices are the cities, and edges correspond to the roads. Then N_{AB} is the maximal number of edge-disjoint paths from A to B in this graph; the number N_{BA} is interpreted similarly.

As in the previous solution, denote by a and b the out-degrees of vertices A and B, respectively. To solve the problem, we show that for any (A,B)-cut (S_A, T_A) in our graph there exists a (B,A)-cut (S_B, T_B) satisfying

$$e(S_B, T_B) = e(S_A, T_A) + (b - a).$$

This yields

$$N_{BA} \leqslant N_{AB} + (b - a); \qquad \text{similarly, we get} \quad N_{AB} \leqslant N_{BA} + (a - b),$$

whence again $N_{BA} - N_{AB} = b - a$.

The construction is simple: we put $S_B = S_A \cup \{B\} \setminus \{A\}$ and hence $T_B = T_A \cup \{A\} \setminus \{B\}$. To show that it works, let A and B denote the sets of cut-edges in (S_A, T_A) and (S_B, T_B), respectively. Let a_s and $a_t = a - a_s$ denote the numbers of edges going from A to S_A and T_A, respectively. Similarly, denote by b_s and $b_t = b - b_s$ the numbers of edges going from B to S_B and T_B, respectively.

Notice that any edge incident to none of A and B either belongs to both A and B, or belongs to none of them. Denote the number of such edges belonging to A by c. The edges in A which are not yet accounted for split into two categories: those going out from A to T_A (including $A \to B$ if it exists), and those going from $S_A \setminus \{A\}$ to B — in other words, going from S_B to B. The numbers of edges in the two categories are a_t and $|S_B| - 1 - b_s$, respectively. Therefore,

$$|\mathsf{A}| = c + a_t + (|S_B| - b_s - 1). \qquad \text{Similarly, we get} \quad |\mathsf{B}| = c + b_t + (|S_A| - a_s - 1),$$

and hence

$$|\mathsf{B}| - |\mathsf{A}| = (b_t + b_s) - (a_t + a_s) = b - a,$$

since $|S_A| = |S_B|$. This finishes the solution.

C5. Let n and k be two integers with $n > k \geqslant 1$. There are $2n+1$ students standing in a circle. Each student S has $2k$ *neighbours*— namely, the k students closest to S on the right, and the k students closest to S on the left.

Suppose that $n+1$ of the students are girls, and the other n are boys. Prove that there is a girl with at least k girls among her neighbours.

Solution. We replace the girls by 1's, and the boys by 0's, getting the numbers $a_1, a_2, \ldots, a_{2n+1}$ arranged in a circle. We extend this sequence periodically by letting $a_{2n+1+k} = a_k$ for all $k \in \mathbb{Z}$. We get an infinite periodic sequence

$$\ldots, a_1, a_2, \ldots, a_{2n+1}, a_1, a_2, \ldots, a_{2n+1}, \ldots.$$

Consider the numbers $b_i = a_i + a_{i-k-1} - 1 \in \{-1, 0, 1\}$ for all $i \in \mathbb{Z}$. We know that

$$b_{m+1} + b_{m+2} + \cdots + b_{m+2n+1} = 1 \qquad (m \in \mathbb{Z}); \tag{1}$$

in particular, this yields that there exists some i_0 with $b_{i_0} = 1$. Now we want to find an index i such that

$$b_i = 1 \quad \text{and} \quad b_{i+1} + b_{i+2} + \cdots + b_{i+k} \geqslant 0. \tag{2}$$

This will imply that $a_i = 1$ and

$$(a_{i-k} + a_{i-k+1} + \cdots + a_{i-1}) + (a_{i+1} + a_{i+2} + \cdots + a_{i+k}) \geqslant k,$$

as desired.

Suppose, to the contrary, that for every index i with $b_i = 1$ the sum $b_{i+1} + b_{i+2} + \cdots + b_{i+k}$ is negative. We start from some index i_0 with $b_{i_0} = 1$ and construct a sequence i_0, i_1, i_2, …, where i_j $(j > 0)$ is the smallest possible index such that $i_j > i_{j-1} + k$ and $b_{i_j} = 1$. We can choose two numbers among i_0, i_1, …, i_{2n+1} which are congruent modulo $2n+1$ (without loss of generality, we may assume that these numbers are i_0 and i_T).

On the one hand, for every j with $0 \leqslant j \leqslant T-1$ we have

$$S_j := b_{i_j} + b_{i_j+1} + b_{i_j+2} + \cdots + b_{i_{j+1}-1} \leqslant b_{i_j} + b_{i_j+1} + b_{i_j+2} + \cdots + b_{i_j+k} \leqslant 0$$

since $b_{i_j+k+1}, \ldots, b_{i_{j+1}-1} \leqslant 0$. On the other hand, since $(i_T - i_0) \mid (2n+1)$, from (1) we deduce

$$S_0 + \cdots + S_{T-1} = \sum_{i=i_0}^{i_T-1} b_i = \frac{i_T - i_0}{2n+1} > 0.$$

This contradiction finishes the solution.

Comment 1. After the problem is reduced to finding an index i satisfying (2), one can finish the solution by applying the (existence part of) following statement.

Lemma (Raney). If $\langle x_1, x_2, \ldots, x_m \rangle$ is any sequence of integers whose sum is $+1$, exactly one of the cyclic shifts $\langle x_1, x_2, \ldots, x_m \rangle, \langle x_2, \ldots, x_m, x_1 \rangle, \ldots, \langle x_m, x_1, \ldots, x_{m-1} \rangle$ has all of its partial sums positive.

A (possibly wider known) version of this lemma, which also can be used in order to solve the problem, is the following

Claim (Gas stations problem). Assume that there are several fuel stations located on a circular route which together contain just enough gas to make one trip around. Then one can make it all the way around, starting at the right station with an empty tank.

Both Raney's theorem and the Gas stations problem admit many different (parallel) proofs. Their ideas can be disguised in direct solutions of the problem at hand (as it, in fact, happens in the above solution); such solutions may avoid the introduction of the b_i. Below, in Comment 2 we present a variant of such solution, while in Comment 3 we present an alternative proof of Raney's theorem.

Comment 2. Here is a version of the solution which avoids the use of the b_i.

Suppose the contrary. Introduce the numbers a_i as above. Starting from any index s_0 with $a_{s_0} = 1$, we construct a sequence $s_0, s_1, s_2, \ldots$ by letting s_i to be the smallest index larger than $s_{i-1}+k$ such that $a_{s_i} = 1$, for $i = 1, 2, \ldots$. Choose two indices among $s_1, \ldots, s_{2n+1}$ which are congruent modulo $2n+1$; we assume those two are s_0 and s_T, with $s_T - s_0 = t(2n+1)$. Notice here that $s_{T+1} - s_T = s_1 - s_0$.

For every $i = 0, 1, 2, \ldots, T$, put

$$L_i = s_{i+1} - s_i \quad \text{and} \quad S_i = a_{s_i} + a_{s_i+1} + \cdots + a_{s_{i+1}-1}.$$

Now, by the indirect assumption, for every $i = 1, 2, \ldots, T$, we have

$$a_{s_i-k} + a_{s_i-k+1} + \cdots + a_{s_i+k} \leqslant a_{s_i} + (k-1) = k.$$

Recall that $a_j = 0$ for all j with $s_i + k < j < a_{s_{i+1}}$. Therefore,

$$S_{i-1} + S_i = \sum_{j=s_{i-1}}^{s_i+k} a_j = \sum_{j=s_{i-1}}^{s_i-k-1} a_j + \sum_{j=s_i-k}^{s_i+k} a_j \leqslant (s_i - s_{i-1} - k) + k = L_{i-1}.$$

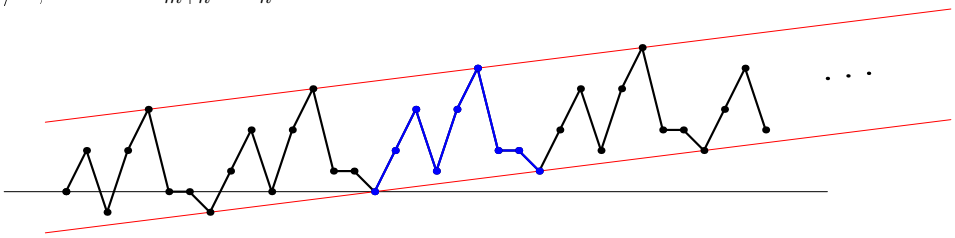
Summing up these equalities over $i = 1, 2, \ldots, T$ we get

$$2t(n+1) = \sum_{i=1}^{T} (S_{i-1} + S_i) \leqslant \sum_{i=1}^{T} L_{i-1} = (2n+1)t,$$

which is a contradiction.

Comment 3. Here we present a proof of Raney's lemma different from the one used above.

If we plot the partial sums $s_n = x_1 + \cdots + x_n$ as a function of n, the graph of s_n has an average slope of $1/m$, because $s_{m+n} = s_n + 1$.



The entire graph can be contained between two lines of slope $1/m$. In general these bounding lines touch the graph just once in each cycle of m points, since lines of slope $1/m$ hit points with integer coordinates only once per m units. The unique (in one cycle) lower point of intersection is the only place in the cycle from which all partial sums will be positive.

Comment 4. The following example shows that for different values of k the required girl may have different positions: 0110**0**110**1**.

C6. A hunter and an invisible rabbit play a game on an infinite square grid. First the hunter fixes a colouring of the cells with finitely many colours. The rabbit then secretly chooses a cell to start in. Every minute, the rabbit reports the colour of its current cell to the hunter, and then secretly moves to an adjacent cell that it has not visited before (two cells are adjacent if they share a side). The hunter wins if after some finite time either

- the rabbit cannot move; or
- the hunter can determine the cell in which the rabbit started.

Decide whether there exists a winning strategy for the hunter.

Answer: Yes, there exists a colouring that yields a winning strategy for the hunter.

Solution. A central idea is that several colourings $C_1, C_2, \ldots, C_k$ can be merged together into a single product colouring $C_1 \times C_2 \times \cdots \times C_k$ as follows: the colours in the product colouring are ordered tuples $(c_1, \ldots, c_n)$ of colours, where c_i is a colour used in C_i, so that each cell gets a tuple consisting of its colours in the individual colourings C_i. This way, any information which can be determined from one of the individual colourings can also be determined from the product colouring.

Now let the hunter merge the following colourings:

- The first two colourings C_1 and C_2 allow the tracking of the horizontal and vertical movements of the rabbit.

  The colouring C_1 colours the cells according to the residue of their x-coordinates modulo 3, which allows to determine whether the rabbit moves left, moves right, or moves vertically. Similarly, the colouring C_2 uses the residues of the y-coordinates modulo 3, which allows to determine whether the rabbit moves up, moves down, or moves horizontally.

- Under the condition that the rabbit's x-coordinate is unbounded, colouring C_3 allows to determine the exact value of the x-coordinate:

  In C_3, the columns are coloured white and black so that the gaps between neighboring black columns are pairwise distinct. As the rabbit's x-coordinate is unbounded, it will eventually visit two black cells in distinct columns. With the help of colouring C_1 the hunter can catch that moment, and determine the difference of x-coordinates of those two black cells, hence deducing the precise column.

  Symmetrically, under the condition that the rabbit's y-coordinate is unbounded, there is a colouring C_4 that allows the hunter to determine the exact value of the y-coordinate.

- Finally, under the condition that the sum $x + y$ of the rabbit's coordinates is unbounded, colouring C_5 allows to determine the exact value of this sum: The diagonal lines $x + y =$ const are coloured black and white, so that the gaps between neighboring black diagonals are pairwise distinct.

Unless the rabbit gets stuck, at least two of the three values x, y and $x+y$ must be unbounded as the rabbit keeps moving. Hence the hunter can eventually determine two of these three values; thus he does know all three. Finally the hunter works backwards with help of the colourings C_1 and C_2 and computes the starting cell of the rabbit.

Comment. There are some variations of the solution above: e.g., the colourings C_3, C_4 and C_5 can be replaced with different ones. However, such alternatives are more technically involved, and we do not present them here.

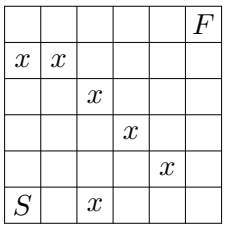
C7. Consider a checkered $3m \times 3m$ square, where m is an integer greater than 1. A frog sits on the lower left corner cell S and wants to get to the upper right corner cell F. The frog can hop from any cell to either the next cell to the right or the next cell upwards.

Some cells can be sticky, and the frog gets trapped once it hops on such a cell. A set X of cells is called *blocking* if the frog cannot reach F from S when all the cells of X are sticky. A blocking set is *minimal* if it does not contain a smaller blocking set.

(a) Prove that there exists a minimal blocking set containing at least $3m^2 - 3m$ cells.

(b) Prove that every minimal blocking set contains at most $3m^2$ cells.

Note. An example of a minimal blocking set for $m = 2$ is shown below. Cells of the set X are marked by letters x.



Solution for part (a). In the following example the square is divided into m stripes of size $3 \times 3m$. It is easy to see that X is a minimal blocking set. The first and the last stripe each contains $3m - 1$ cells from the set X; every other stripe contains $3m - 2$ cells, see Figure 1. The total number of cells in the set X is $3m^2 - 2m + 2$.

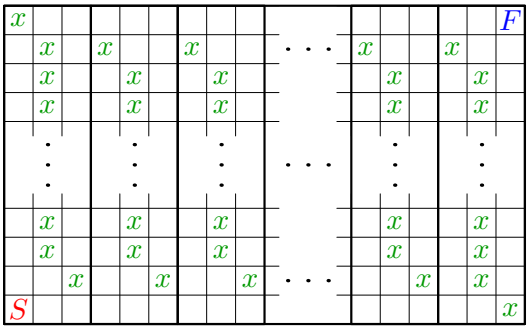


Figure 1

Solution 1 for part (b). For a given blocking set X, say that a non-sticky cell is *red* if the frog can reach it from S via some hops without entering set X. We call a non-sticky cell *blue* if the frog can reach F from that cell via hops without entering set X. One can regard the blue cells as those reachable from F by *anti-hops*, i.e. moves downwards and to the left. We also colour all cells in X *green*. It follows from the definition of the blocking set that no cell will be coloured twice. In Figure 2 we show a sample of a blocking set and the corresponding colouring.

Now assume that X is a minimal blocking set. We denote by R (resp., B and G) be the total number of red (resp., blue and green) cells.

We claim that $G \leqslant R + 1$ and $G \leqslant B + 1$. Indeed, there are at most $2R$ possible frog hops from red cells. Every green or red cell (except for S) is accessible by such hops. Hence $2R \geqslant G + (R - 1)$, or equivalently $G \leqslant R + 1$. In order to prove the inequality $G \leqslant B + 1$, we turn over the board and apply the similar arguments.

Therefore we get $9m^2 \geqslant B + R + G \geqslant 3G - 2$, so $G \leqslant 3m^2$.

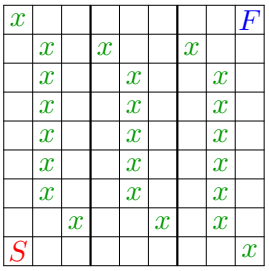


Figure 2 (a)

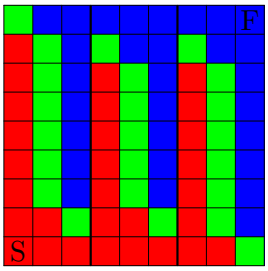
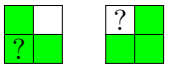


Figure 2 (b)

Solution 2 for part (b). We shall use the same colouring as in the above solution. Again, assume that X is a minimal blocking set.

Note that any 2×2 square cannot contain more than 2 green cells. Indeed, on Figure 3(a) the cell marked with "?" does not block any path, while on Figure 3(b) the cell marked with "?" should be coloured red and blue simultaneously. So we can split all green cells into chains consisting of three types of links shown on Figure 4 (diagonal link in the other direction is not allowed, corresponding green cells must belong to different chains). For example, there are 3 chains in Figure 2(b).



(a) (b)

Figure 3

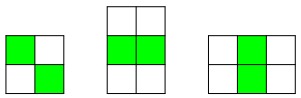


Figure 4

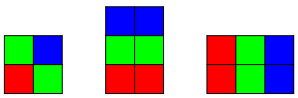
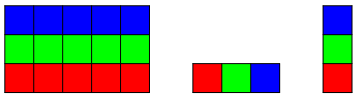


Figure 5

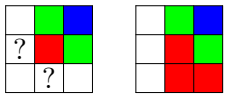
We will inscribe green chains in disjoint axis-aligned rectangles so that the number of green cells in each rectangle will not exceed 1/3 of the area of the rectangle. This will give us the bound $G \leqslant 3m^2$. Sometimes the rectangle will be the minimal bounding rectangle of the chain, sometimes minimal bounding rectangles will be expanded in one or two directions in order to have sufficiently large area.

Note that for any two consecutive cells in the chain the colouring of some neighbouring cells is uniquely defined (see Figure 5). In particular, this observation gives a corresponding rectangle for the chains of height (or width) 1 (see Figure 6(a)). A separate green cell can be inscribed in 1×3 or 3×1 rectangle with one red and one blue cell, see Figure 6(b)–(c), otherwise we get one of impossible configurations shown in Figure 3.



(a) (b) (c)

Figure 6



(a) (b)

Figure 7

Any diagonal chain of length 2 is always inscribed in a 2×3 or 3×2 rectangle without another green cells. Indeed, one of the squares marked with "?" in Figure 7(a) must be red. If it is the bottom question mark, then the remaining cell in the corresponding 2×3 rectangle must have the same colour, see Figure 7(b).

A longer chain of height (or width) 2 always has a horizontal (resp., vertical) link and can be inscribed into a $3 \times a$ rectangle. In this case we expand the minimal bounding rectangle across the long side which touches the mentioned link. On Figure 8(a) the corresponding expansion of the minimal bounding rectangle is coloured in light blue. The upper right corner cell must be also blue. Indeed it cannot be red or green. If it is not coloured in blue, see Figure 8(b), then all anti-hop paths from F to "?" are blocked with green cells. And these green cells are surrounded by blue ones, what is impossible. In this case the green chain contains a cells, which is exactly 1/3 of the area of the rectangle.

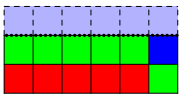


Figure 8 (a)

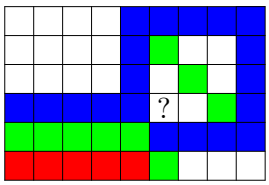


Figure 8 (b)

In the remaining case the minimal bounding rectangle of the chain is of size $a \times b$ where $a, b \geqslant 3$. Denote by ℓ the *length* of the chain (i.e. the number of cells in the chain).

If the chain has at least two diagonal links (see Figure 9), then $\ell \leqslant a + b - 3 \leqslant ab/3$.

If the chain has only one diagonal link then $\ell = a+b-2$. In this case the chain has horizontal and vertical end-links, and we expand the minimal bounding rectangle in two directions to get an $(a+1) \times (b+1)$ rectangle. On Figure 10 a corresponding expansion of the minimal bounding rectangle is coloured in light red. Again the length of the chain does not exceed 1/3 of the rectangle's area: $\ell \leqslant a + b - 2 \leqslant (a + 1)(b + 1)/3$.

On the next step we will use the following statement: all cells in constructed rectangles are coloured red, green or blue (the cells upwards and to the right of green cells are blue; the cells downwards and to the left of green cells are red). The proof repeats the same arguments as before (see Figure 8(b).)



Figure 9

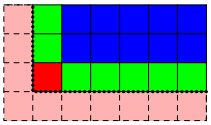
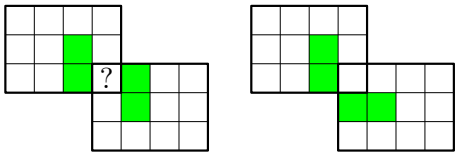


Figure 10



(a) (b)

Figure 11

Note that all constructed rectangles are disjoint. Indeed, assume that two rectangles have a common cell. Using the above statement, one can see that the only such cell can be a common corner cell, as shown in Figure 11. Moreover, in this case both rectangles should be expanded, otherwise they would share a green corner cell.

If they were expanded along the same axis (see Figure 11(a)), then again the common corner cannot be coloured correctly. If they were expanded along different axes (see Figure 11(b)) then the two chains have a common point and must be connected in one chain. (These arguments work for 2×3 and 1×3 rectangles in a similar manner.)

Comment 1. We do not *a priori* know whether all points are either red, or blue, or green. One might colour the remaining cells in black. The arguments from Solution 2 allow to prove that black cells do not exist. (One can start with a black cell which is nearest to S. Its left and downward neighbours must be coloured green or blue. In all cases one gets a configuration similar to Figure 8(b).)

Comment 2. The maximal possible size of a minimal blocking set in $3m \times 3m$ rectangle seems to be $3m^2 - 2m + 2$.

One can prove a more precise upper bound on the cardinality of the minimal blocking set: $G \leqslant 3m^2 - m + 2$. Denote by D_R the number of red branching cells (i.e. such cells which have 2 red subsequent neighbours). And let D_B be the number of similar blue cells. Then a double counting argument allows to prove that $G \leqslant R - D_R + 1$ and $G \leqslant B - D_B + 1$. Thus, we can bound G in terms of D_B and D_R as

$$9m^2 \geqslant R + B + G \geqslant 3G + D_R + D_B - 2.$$

Now one can estimate the number of branching cells in order to obtain that $G \leqslant 3m^2 - m + 2$.

Comment 3. An example with $3m^2 - 2m + 2$ green cells may look differently; see, e.g., Figure 12.

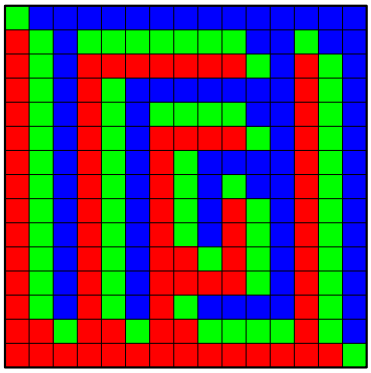


Figure 12

C8. Determine the largest N for which there exists a table T of integers with N rows and 100 columns that has the following properties:

(i) Every row contains the numbers $1, 2, \ldots, 100$ in some order.

(ii) For any two distinct rows r and s, there is a column c such that $|T(r,c) - T(s,c)| \geqslant 2$.

Here $T(r,c)$ means the number at the intersection of the row r and the column c.

Answer: The largest such integer is $N = 100!/2^{50}$.

Solution 1.

Non-existence of a larger table. Let us consider some fixed row in the table, and let us replace (for $k = 1, 2, \ldots, 50$) each of two numbers $2k-1$ and $2k$ respectively by the symbol x_k. The resulting *pattern* is an arrangement of 50 symbols $x_1, x_2, \ldots, x_{50}$, where every symbol occurs exactly twice. Note that there are $N = 100!/2^{50}$ distinct patterns $P_1, \ldots, P_N$.

If two rows $r \neq s$ in the table have the same pattern P_i, then $|T(r,c) - T(s,c)| \leqslant 1$ holds for all columns c. As this violates property (ii) in the problem statement, different rows have different patterns. Hence there are at most $N = 100!/2^{50}$ rows.

Existence of a table with N rows. We construct the table by translating every pattern P_i into a corresponding row with the numbers $1, 2, \ldots, 100$. We present a procedure that inductively replaces the symbols by numbers. The translation goes through steps $k = 1, 2, \ldots, 50$ in increasing order and at step k replaces the two occurrences of symbol x_k by $2k-1$ and $2k$.

- The left occurrence of x_1 is replaced by 1, and its right occurrence is replaced by 2.
- For $k \geqslant 2$, we already have the number $2k-2$ somewhere in the row, and now we are looking for the places for $2k-1$ and $2k$. We make the three numbers $2k-2, 2k-1, 2k$ show up (ordered from left to right) either in the order $2k-2, 2k-1, 2k$, or as $2k, 2k-2, 2k-1$, or as $2k-1, 2k, 2k-2$. This is possible, since the number $2k-2$ has been placed in the preceding step, and shows up before / between / after the two occurrences of the symbol x_k.

We claim that the N rows that result from the N patterns yield a table with the desired property (ii). Indeed, consider the r-th and the s-th row ($r \neq s$), which by construction result from patterns P_r and P_s. Call a symbol x_i *aligned*, if it occurs in the same two columns in P_r and in P_s. Let k be the largest index, for which symbol x_k is not aligned. Note that $k \geqslant 2$. Consider the column c' with $T(r,c') = 2k$ and the column c'' with $T(s,c'') = 2k$. Then $T(r,c'') \leqslant 2k$ and $T(s,c') \leqslant 2k$, as all symbols x_i with $i \geqslant k+1$ are aligned.

- If $T(r,c'') \leqslant 2k-2$, then $|T(r,c'') - T(s,c'')| \geqslant 2$ as desired.
- If $T(s,c') \leqslant 2k-2$, then $|T(r,c') - T(s,c')| \geqslant 2$ as desired.
- If $T(r,c'') = 2k-1$ and $T(s,c') = 2k-1$, then the symbol x_k is aligned; contradiction.

In the only remaining case we have $c' = c''$, so that $T(r,c') = T(s,c') = 2k$ holds. Now let us consider the columns d' and d'' with $T(r,d') = 2k-1$ and $T(s,d'') = 2k-1$. Then $d \neq d''$ (as the symbol x_k is not aligned), and $T(r,d'') \leqslant 2k-2$ and $T(s,d') \leqslant 2k-2$ (as all symbols x_i with $i \geqslant k+1$ are aligned).

- If $T(r,d'') \leqslant 2k-3$, then $|T(r,d'') - T(s,d'')| \geqslant 2$ as desired.
- If $T(s,c') \leqslant 2k-3$, then $|T(r,d') - T(s,d')| \geqslant 2$ as desired.

In the only remaining case we have $T(r,d'') = 2k-2$ and $T(s,d') = 2k-2$. Now the row r has the numbers $2k-2, 2k-1, 2k$ in the three columns d', d'', c'. As one of these triples violates the ordering property of $2k-2, 2k-1, 2k$, we have the final contradiction. □

Comment 1. We can identify rows of the table T with permutations of $\mathcal{M} := \{1, \ldots, 100\}$; also for every set $S \subset \mathcal{M}$ each row induces a subpermutation of S obtained by ignoring all entries not from S.

The example from Solution 1 consists of all permutations for which all subpermutations of the 50 sets $\{1,2\}, \{2,3,4\}, \{4,5,6\}, \ldots, \{98,99,100\}$ are even.

Solution 2. We provide a bit different proof why the example from Solution 1 (see also Comment 1) works.

Lemma. Let π_1 and π_2 be two permutations of the set $\{1, 2, \ldots, n\}$ such that $|\pi_1(i) - \pi_2(i)| \leqslant 1$ for every i. Then there exists a set of disjoint pairs $(i, i+1)$ such that π_2 is obtained from π_1 by swapping elements in each pair from the set.

Proof. We may assume that $\pi_1(i) = i$ for every i and proceed by induction on n. The case $n = 1$ is trivial. If $\pi_2(n) = n$, we simply apply the induction hypothesis. If $\pi_2(n) = n - 1$, then $\pi_2(i) = n$ for some $i < n$. It is clear that $i = n - 1$, and we can also use the induction hypothesis. □

Now let π_1 and π_2 be two rows (which we identify with permutations of $\{1, 2, \ldots, 100\}$) of the table constructed in Solution 1. Assume that $|\pi_1(i) - \pi_2(i)| \leqslant 1$ for any i. From the Lemma it follows that there exists a set $S \subset \{1, \ldots, 99\}$ such that any two numbers from S differ by at least 2 and π_2 is obtained from π_1 by applying the permutations $(j, j+1)$, $j \in S$. Let $r = \min(S)$. If $r = 2k - 1$ is odd, then π_1 and π_2 induce two subpermutations of $\{2k-2, 2k-1, 2k\}$ (or of $\{1,2\}$ for $k = 1$) of opposite parities. Thus $r = 2k$ is even. Since π_1 and π_2 induce subpermutations of the same (even) parity of $\{2k, 2k+1, 2k+2\}$, we must have $2k + 2 \in S$. Next, $2k + 4 \in S$ and so on, we get $98 \in S$, but then the parities of the subpermutations of $\{98, 99, 100\}$ in π_1, π_2 are opposite. A contradiction. □

Comment 2. In Solution 2 we only used that for each set from $\{1,2\}, \{2,3,4\}, \{4,5,6\}, \ldots, \{98,99,100\}$ any two rows of T induce a subpermutation of the same parity, not necessarily even.

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Geometry

G1. Let $ABCD$ be a parallelogram such that $AC = BC$. A point P is chosen on the extension of the segment AB beyond B. The circumcircle of the triangle ACD meets the segment PD again at Q, and the circumcircle of the triangle APQ meets the segment PC again at R. Prove that the lines CD, AQ, and BR are concurrent.

Common remarks. The introductory steps presented here are used in all solutions below.

Since $AC = BC = AD$, we have $\angle ABC = \angle BAC = \angle ACD = \angle ADC$. Since the quadrilaterals $APRQ$ and $AQCD$ are cyclic, we obtain

$$\angle CRA = 180^\circ - \angle ARP = 180^\circ - \angle AQP = \angle DQA = \angle DCA = \angle CBA,$$

so the points A, B, C, and R lie on some circle γ.

Solution 1. Introduce the point $X = AQ \cap CD$; we need to prove that B, R and X are collinear.

By means of the circle $(APRQ)$ we have

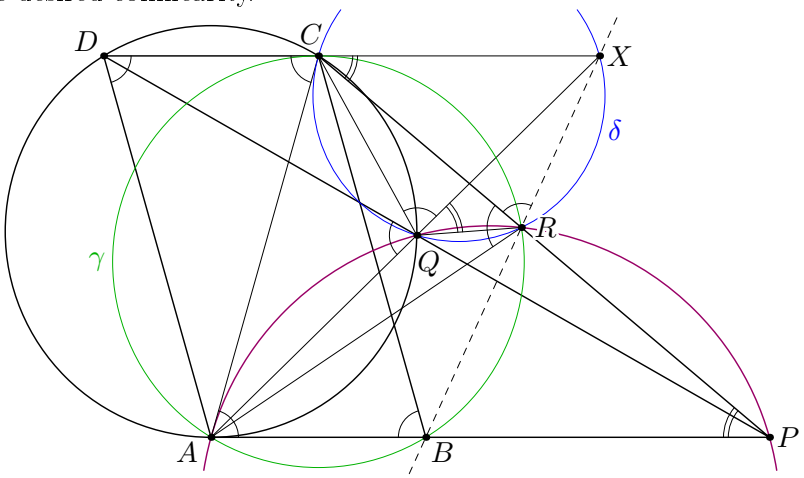
$$\angle RQX = 180^\circ - \angle AQR = \angle RPA = \angle RCX$$

(the last equality holds in view of $AB \parallel CD$), which means that the points C, Q, R, and X also lie on some circle δ.

Using the circles δ and γ we finally obtain

$$\angle XRC = \angle XQC = 180^\circ - \angle CQA = \angle ADC = \angle BAC = 180^\circ - \angle CRB,$$

that proves the desired collinearity.



Solution 2. Let α denote the circle $(APRQ)$. Since

$$\angle CAP = \angle ACD = \angle AQD = 180^\circ - \angle AQP,$$

the line AC is tangent to α.

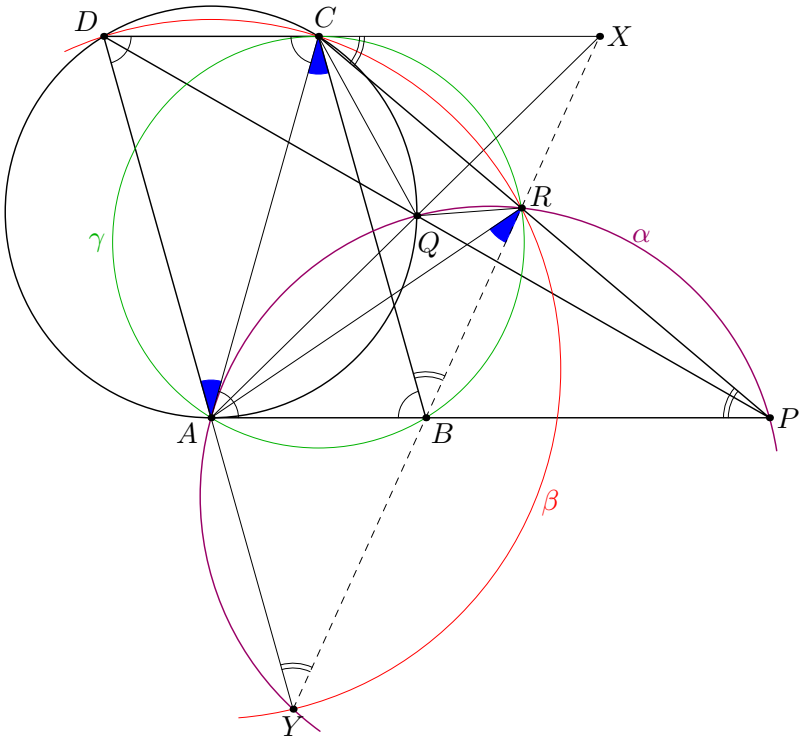
Now, let AD meet α again at a point Y (which necessarily lies on the extension of DA beyond A). Using the circle γ, along with the fact that AC is tangent to α, we have

$$\angle ARY = \angle CAD = \angle ACB = \angle ARB,$$

so the points Y, B, and R are collinear.

Applying Pascal's theorem to the hexagon $AAYRPQ$ (where AA is regarded as the tangent to α at A), we see that the points $AA \cap RP = C$, $AY \cap PQ = D$, and $YR \cap QA$ are collinear. Hence the lines CD, AQ, and BR are concurrent.

Comment 1. Solution 2 consists of two parts: (1) showing that BR and DA meet on α; and (2) showing that this yields the desired concurrency. Solution 3 also splits into those parts, but the proofs are different.



Solution 3. As in Solution 1, we introduce the point $X = AQ \cap CD$ and aim at proving that the points B, R, and X are collinear. As in Solution 2, we denote $\alpha = (APQR)$; but now we define Y to be the second meeting point of RB with α.

Using the circle α and noticing that CD is tangent to γ, we obtain

$$\angle RYA = \angle RPA = \angle RCX = \angle RBC. \tag{1}$$

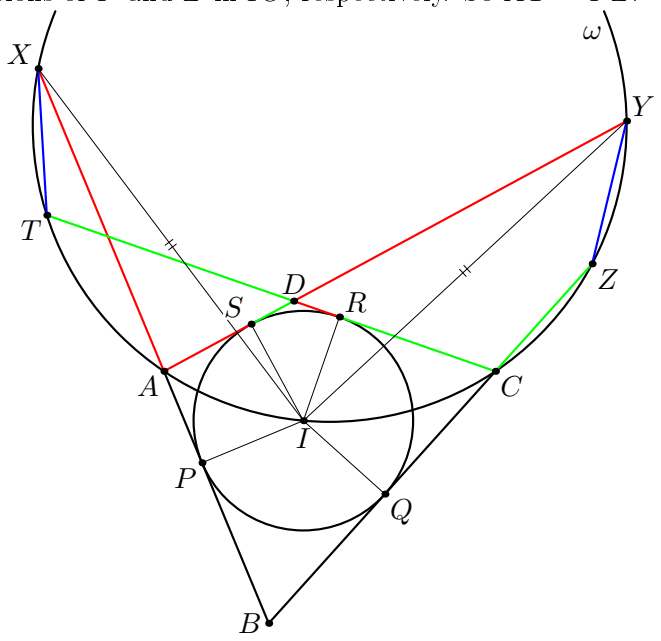
So $AY \parallel BC$, and hence Y lies on DA.

Now the chain of equalities (1) shows also that $\angle RYD = \angle RCX$, which implies that the points C, D, Y, and R lie on some circle β. Hence, the lines CD, AQ, and YBR are the pairwise radical axes of the circles $(AQCD)$, α, and β, so those lines are concurrent.

Comment 2. The original problem submission contained an additional assumption that $BP = AB$. The Problem Selection Committee removed this assumption as superfluous.

G2. Let $ABCD$ be a convex quadrilateral circumscribed around a circle with centre I. Let ω be the circumcircle of the triangle ACI. The extensions of BA and BC beyond A and C meet ω at X and Z, respectively. The extensions of AD and CD beyond D meet ω at Y and T, respectively. Prove that the perimeters of the (possibly self-intersecting) quadrilaterals $ADTX$ and $CDYZ$ are equal.

Solution. The point I is the intersection of the external bisector of the angle TCZ with the circumcircle ω of the triangle TCZ, so I is the midpoint of the arc TCZ and $IT = IZ$. Similarly, I is the midpoint of the arc YAX and $IX = IY$. Let O be the centre of ω. Then X and T are the reflections of Y and Z in IO, respectively. So $XT = YZ$.



Let the incircle of $ABCD$ touch AB, BC, CD, and DA at points P, Q, R, and S, respectively.

The right triangles IXP and IYS are congruent, since $IP = IS$ and $IX = IY$. Similarly, the right triangles IRT and IQZ are congruent. Therefore, $XP = YS$ and $RT = QZ$.

Denote the perimeters of $ADTX$ and $CDYZ$ by P_{ADTX} and P_{CDYZ} respectively. Since $AS = AP$, $CQ = RC$, and $SD = DR$, we obtain

$$\begin{aligned} P_{ADTX} &= XT + XA + AS + SD + DT = XT + XP + RT \\ &= YZ + YS + QZ = YZ + YD + DR + RC + CZ = P_{CDYZ}, \end{aligned}$$

as required.

Comment 1. After proving that X and T are the reflections of Y and Z in IO, respectively, one can finish the solution as follows. Since $XT = YZ$, the problem statement is equivalent to

$$XA + AD + DT = YD + DC + CZ. \tag{1}$$

Since $ABCD$ is circumscribed, $AB - AD = BC - CD$. Adding this to (1), we come to an equivalent equality $XA + AB + DT = YD + BC + CZ$, or

$$XB + DT = YD + BZ. \tag{2}$$

Let $\lambda = \frac{XZ}{AC} = \frac{TY}{AC}$. Since $XACZ$ is cyclic, the triangles ZBX and ABC are similar, hence

$$\frac{XB}{BC} = \frac{BZ}{AB} = \frac{XZ}{AC} = \lambda.$$

It follows that $XB = \lambda BC$ and $BZ = \lambda AB$. Likewise, the triangles TDY and ADC are similar, hence

$$\frac{DT}{AD} = \frac{DY}{CD} = \frac{TY}{AC} = \lambda.$$

Therefore, (2) rewrites as $\lambda BC + \lambda AD = \lambda CD + \lambda AB$.

This is equivalent to $BC + AD = CD + AB$ which is true as $ABCD$ is circumscribed.

Comment 2. Here is a more difficult modification of the original problem, found by the PSC.

Let $ABCD$ be a convex quadrilateral circumscribed around a circle with centre I. Let ω be the circumcircle of the triangle ACI. The extensions of BA and BC beyond A and C meet ω at X and Z, respectively. The extensions of AD and CD beyond D meet ω at Y and T, respectively. Let $U = BC \cap AD$ and $V = BA \cap CD$. Let I_U be the incentre of UYZ and let J_V be the V-excentre of VXT. Then $I_U J_V \perp BD$.

G3.

Version 1. Let n be a fixed positive integer, and let S be the set of points (x, y) on the Cartesian plane such that both coordinates x and y are nonnegative integers smaller than $2n$ (thus $|\mathsf{S}| = 4n^2$). Assume that $\mathcal{F}$ is a set consisting of n^2 quadrilaterals such that all their vertices lie in S, and each point in S is a vertex of exactly one of the quadrilaterals in $\mathcal{F}$.

Determine the largest possible sum of areas of all n^2 quadrilaterals in $\mathcal{F}$.

Version 2. Let n be a fixed positive integer, and let S be the set of points (x, y) on the Cartesian plane such that both coordinates x and y are nonnegative integers smaller than $2n$ (thus $|\mathsf{S}| = 4n^2$). Assume that $\mathcal{F}$ is a set of polygons such that all vertices of polygons in $\mathcal{F}$ lie in S, and each point in S is a vertex of exactly one of the polygons in $\mathcal{F}$.

Determine the largest possible sum of areas of all polygons in $\mathcal{F}$.

Answer for both Versions: The largest possible sum of areas is $\Sigma(n) := \frac{1}{3}n^2(2n+1)(2n-1)$.

Common remarks. Throughout all solutions, the area of a polygon P will be denoted by $[P]$.

We say that a polygon is *legal* if all its vertices belong to S. Let $O = \left(n - \frac{1}{2}, n - \frac{1}{2}\right)$ be the centre of S. We say that a legal square is *central* if its centre is situated at O. Finally, say that a set $\mathcal{F}$ of polygons is *acceptable* if it satisfies the problem requirements, i.e. if all polygons in $\mathcal{F}$ are legal, and each point in S is a vertex of exactly one polygon in $\mathcal{F}$. For an acceptable set $\mathcal{F}$, we denote by $\Sigma(\mathcal{F})$ the sum of areas of polygons in $\mathcal{F}$.

Solution 1, for both Versions. Each point in S is a vertex of a unique central square. Thus the set $\mathcal{G}$ of central squares is acceptable. We will show that

$$\Sigma(\mathcal{F}) \leqslant \Sigma(\mathcal{G}) = \Sigma(n), \tag{1}$$

thus establishing the answer.

We will use the following key lemma.

Lemma 1. Let $P = A_1A_2\ldots A_m$ be a polygon, and let O be an arbitrary point in the plane. Then

$$[P] \leqslant \frac{1}{2}\sum_{i=1}^{m} OA_i^2; \tag{2}$$

moreover, if P is a square centred at O, then the inequality (2) turns into an equality.

Proof. Put $A_{n+1} = A_1$. For each $i = 1, 2, \ldots, m$, we have

$$[OA_iA_{i+1}] \leqslant \frac{OA_i \cdot OA_{i+1}}{2} \leqslant \frac{OA_i^2 + OA_{i+1}^2}{4}.$$

Therefore,

$$[P] \leqslant \sum_{i=1}^{m}[OA_iA_{i+1}] \leqslant \frac{1}{4}\sum_{i=1}^{m}(OA_i^2 + OA_{i+1}^2) = \frac{1}{2}\sum_{i=1}^{m} OA_i^2,$$

which proves (2). Finally, all the above inequalities turn into equalities when P is a square centred at O. □

Back to the problem, consider an arbitrary acceptable set $\mathcal{F}$. Applying Lemma 1 to each element in $\mathcal{F}$ and to each element in $\mathcal{G}$ (achieving equality in the latter case), we obtain

$$\Sigma(\mathcal{F}) \leqslant \frac{1}{2}\sum_{A\in\mathsf{S}} OA^2 = \Sigma(\mathcal{G}),$$

which establishes the left inequality in (1).

It remains to compute $\Sigma(\mathcal{G})$. We have

$$\begin{aligned}\Sigma(\mathcal{G}) &= \frac{1}{2}\sum_{A\in\mathsf{S}} OA^2 = \frac{1}{2}\sum_{i=0}^{2n-1}\sum_{j=0}^{2n-1}\left(\left(n-\frac{1}{2}-i\right)^2+\left(n-\frac{1}{2}-j\right)^2\right)\\ &= \frac{1}{8}\cdot 4\cdot 2n\sum_{i=0}^{n-1}(2n-2i-1)^2 = n\sum_{j=0}^{n-1}(2j+1)^2 = n\left(\sum_{j=1}^{2n} j^2-\sum_{j=1}^{n}(2j)^2\right)\\ &= n\left(\frac{2n(2n+1)(4n+1)}{6}-4\cdot\frac{n(n+1)(2n+1)}{6}\right) = \frac{n^2(2n+1)(2n-1)}{3} = \Sigma(n).\end{aligned}$$

Comment. There are several variations of the above solution, also working for both versions of the problem. E.g., one may implement only the inequality $[OA_iA_{i+1}] \leqslant \frac{1}{2}OA_i\cdot OA_{i+1}$ to obtain

$$\Sigma(\mathcal{F}) \leqslant \frac{1}{2}\sum_{i=1}^{4n^2} OK_i\cdot OL_i,$$

where both (K_i) and (L_i) are permutations of all points in S. The right hand side can then be bounded from above by means of the rearrangement inequality; the bound is also achieved on the collection $\mathcal{G}$.

However, Version 2 seems to be more difficult than Version 1. First of all, the optimal model for this version is much less easy to guess, until one finds an idea for proving the upper bound. Moreover, Version 1 allows different solutions which do not seem to be generalized easily — such as Solution 2 below.

Solution 2, for Version 1. Let $\mathcal{F}$ be an accessible set of quadrilaterals. For every quadrilateral $ABCD$ in $\mathcal{F}$ write

$$[ABCD] = \frac{AC\cdot BD}{2}\sin\phi \leqslant \frac{AC^2+BD^2}{4}, \tag{3}$$

where ϕ is the angle between AC and BD. Applying this estimate to all members in $\mathcal{F}$ we obtain

$$\Sigma(\mathcal{F}) \leqslant \frac{1}{4}\sum_{i=1}^{2n^2} A_iB_i^2,$$

where $A_1, A_2, \ldots, A_{2n^2}, B_1, B_2, \ldots, B_{2n^2}$ is some permutation of S. For brevity, denote

$$f\big((A_i),(B_i)\big) := \sum_{i=1}^{2n^2} A_iB_i^2.$$

The rest of the solution is based on the following lemma.

Lemma 2. The maximal value of $f\big((A_i),(B_i)\big)$ over all permutations of S equals $\frac{4}{3}n^2(4n^2-1)$ and is achieved when A_i is symmetric to B_i with respect to O, for every $i = 1, 2, \ldots, 2n^2$.

Proof. Let $A_i = (p_i, q_i)$ and $B_i = (r_i, s_i)$, for $i = 1, 2, \ldots, 2n^2$. We have

$$f\big((A_i),(B_i)\big) = \sum_{i=1}^{2n^2}(p_i-r_i)^2 + \sum_{i=1}^{2n^2}(q_i-s_i)^2;$$

it suffices to bound the first sum, the second is bounded similarly. This can be done, e.g., by means of the QM–AM inequality as follows:

$$\sum_{i=1}^{2n^2}(p_i-r_i)^2=\sum_{i=1}^{2n^2}\bigl(2p_i^2+2r_i^2-(p_i+r_i)^2\bigr)=4n\sum_{j=0}^{2n-1}j^2-\sum_{i=1}^{2n^2}(p_i+r_i)^2$$
$$\leqslant 4n\sum_{j=0}^{2n-1}j^2-\frac{1}{2n^2}\left(\sum_{i=1}^{2n^2}(p_i+r_i)\right)^2=4n\sum_{j=0}^{2n-1}j^2-\frac{1}{2n^2}\left(2n\cdot\sum_{j=0}^{2n-1}j\right)^2$$
$$=4n\cdot\frac{2n(2n-1)(4n-1)}{6}-2n^2(2n-1)^2=\frac{2n^2(2n-1)(2n+1)}{3}.$$

All the estimates are sharp if $p_i+r_i=2n-1$ for all i. Thus,

$$f\bigl((A_i),(B_i)\bigr)\leqslant\frac{4n^2(4n^2-1)}{3},$$

and the estimate is sharp when $p_i+r_i=q_i+s_i=2n-1$ for all i, i.e. when A_i and B_i are symmetric with respect to O. □

Lemma 2 yields

$$\Sigma(\mathcal{F})\leqslant\frac{1}{4}\cdot\frac{4n^2(4n^2-1)}{3}=\frac{n^2(2n-1)(2n+1)}{3}.$$

Finally, all estimates are achieved simultaneously on the set $\mathcal{G}$ of central squares.

Comment 2. Lemma 2 also allows different proofs. E.g., one may optimize the sum $\sum_i p_ir_i$ step by step: if $p_i<p_j$ and $r_i<r_j$, then a swap $r_i\leftrightarrow r_j$ increases the sum. By applying a proper chain of such replacements (possibly swapping elements in some pairs (p_i,r_i)), one eventually comes to a permutation where $p_i+r_i=2n-1$ for all i.

Comment 3. Version 2 can also be considered for a square grid with odd number n of points on each side. If we allow a polygon consisting of one point, then Solution 1 is applied verbatim, providing an answer $\frac{1}{12}n^2(n^2-1)$. If such polygons are not allowed, then one needs to subtract $\frac{1}{2}$ from the answer.

G4. Let $ABCD$ be a quadrilateral inscribed in a circle Ω. Let the tangent to Ω at D intersect the rays BA and BC at points E and F, respectively. A point T is chosen inside the triangle ABC so that $TE \parallel CD$ and $TF \parallel AD$. Let $K \neq D$ be a point on the segment DF such that $TD = TK$. Prove that the lines AC, DT and BK intersect at one point.

Solution 1. Let the segments TE and TF cross AC at P and Q, respectively. Since $PE \parallel CD$ and ED is tangent to the circumcircle of $ABCD$, we have

$$\angle EPA = \angle DCA = \angle EDA,$$

and so the points A, P, D, and E lie on some circle α. Similarly, the points C, Q, D, and F lie on some circle γ.

We now want to prove that the line DT is tangent to both α and γ at D. Indeed, since $\angle FCD + \angle EAD = 180^\circ$, the circles α and γ are tangent to each other at D. To prove that T lies on their common tangent line at D (i.e., on their radical axis), it suffices to check that $TP \cdot TE = TQ \cdot TF$, or that the quadrilateral $PEFQ$ is cyclic. This fact follows from

$$\angle QFE = \angle ADE = \angle APE.$$

Since $TD = TK$, we have $\angle TKD = \angle TDK$. Next, as TD and DE are tangent to α and Ω, respectively, we obtain

$$\angle TKD = \angle TDK = \angle EAD = \angle BDE,$$

which implies $TK \parallel BD$.

Next we prove that the five points T, P, Q, D, and K lie on some circle τ. Indeed, since TD is tangent to the circle α we have

$$\angle EPD = \angle TDF = \angle TKD,$$

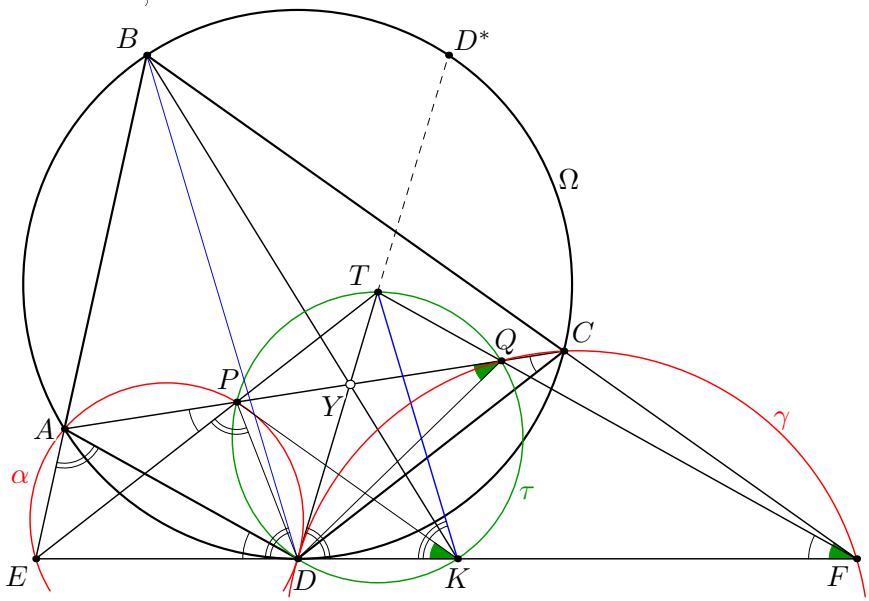
which means that the point P lies on the circle (TDK). Similarly, we have $Q \in (TDK)$.

Finally, we prove that $PK \parallel BC$. Indeed, using the circles τ and γ we conclude that

$$\angle PKD = \angle PQD = \angle DFC,$$

which means that $PK \parallel BC$.

Triangles TPK and DCB have pairwise parallel sides, which implies the fact that TD, PC and KB are concurrent, as desired.



Comment 1. There are several variations of the above solution.

E.g., after finding circles α and γ, one can notice that there exists a homothety h mapping the triangle TPQ to the triangle DCA; the centre of that homothety is $Y = AC \cap TD$. Since

$$\angle DPE = \angle DAE = \angle DCB = \angle DQT,$$

the quadrilateral $TPDQ$ is inscribed in some circle τ. We have $h(\tau) = \Omega$, so the point $D^* = h(D)$ lies on Ω.

Finally, by

$$\angle DCD^* = \angle TPD = \angle BAD,$$

the points B and D^* are symmetric with respect to the diameter of Ω passing through D. This yields $DB = DD^*$ and $BD^* \parallel EF$, so $h(K) = B$, and BK passes through Y.

Solution 2. Consider the spiral similarity ϕ centred at D which maps B to F. Recall that for any two points X and Y, the triangles $DX\phi(X)$ and $DY\phi(Y)$ are similar.

Define $T' = \phi(E)$. Then

$$\angle CDF = \angle FBD = \angle \phi(B)BD = \angle \phi(E)ED = \angle T'ED,$$

so $CD \parallel T'E$. Using the fact that DE is tangent to (ABD) and then applying ϕ we infer

$$\angle ADE = \angle ABD = \angle T'FD,$$

so $AD \parallel T'F$; hence T' coincides with T. Therefore,

$$\angle BDE = \angle FDT = \angle DKT,$$

whence $TK \parallel BD$.

Let $BK \cap TD = X$, $AC \cap TD = Y$, and $AC \cap TF = Q$. Notice that $TK \parallel BD$ implies

$$\frac{TX}{XD} = \frac{TK}{BD} = \frac{TD}{BD}.$$

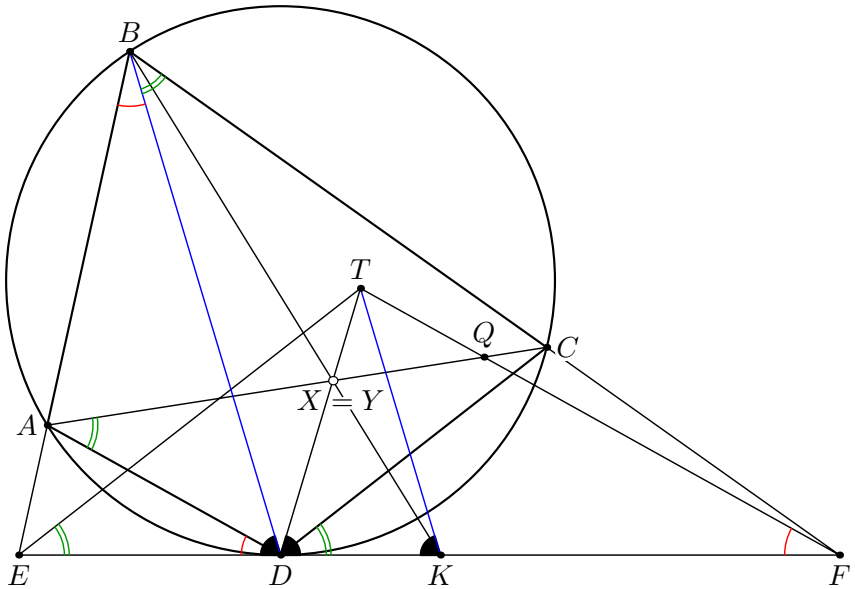
So we wish to prove that $\frac{TY}{YD}$ is equal to the same ratio.

We first show that $\phi(A) = Q$. Indeed,

$$\angle DA\phi(A) = \angle DBF = \angle DAC,$$

and so $\phi(A) \in AC$. Together with $\phi(A) \in \phi(EB) = TF$ this yields $\phi(A) = Q$. It follows that

$$\frac{TQ}{AE} = \frac{TD}{ED}.$$



Now, since $TF \parallel AD$ and $\triangle EAD \sim \triangle EDB$, we have

$$\frac{TY}{YD} = \frac{TQ}{AD} = \frac{TQ}{AE} \cdot \frac{AE}{AD} = \frac{TD}{ED} \cdot \frac{ED}{BD} = \frac{TD}{BD},$$

which completes the proof.

Comment 2. The point D is the Miquel point for any 4 of the 5 lines BA, BC, TE, TF and AC. Essentially, this is proved in both solutions by different methods.

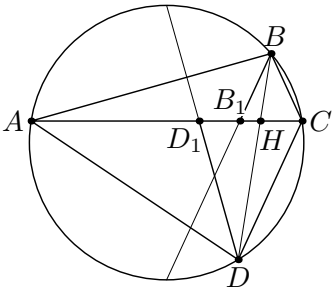
G5. Let $ABCD$ be a cyclic quadrilateral whose sides have pairwise different lengths. Let O be the circumcentre of $ABCD$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at B_1 and D_1, respectively. Let O_B be the centre of the circle which passes through B and is tangent to AC at D_1. Similarly, let O_D be the centre of the circle which passes through D and is tangent to AC at B_1.

Assume that $BD_1 \parallel DB_1$. Prove that O lies on the line O_BO_D.

Common remarks. We introduce some objects and establish some preliminary facts common for all solutions below.

Let Ω denote the circle $(ABCD)$, and let γ_B and γ_D denote the two circles from the problem statement (their centres are O_B and O_D, respectively). Clearly, all three centres O, O_B, and O_D are distinct.

Assume, without loss of generality, that $AB > BC$. Suppose that $AD > DC$, and let $H = AC \cap BD$. Then the rays BB_1 and DD_1 lie on one side of BD, as they contain the midpoints of the arcs ADC and ABC, respectively. However, if $BD_1 \parallel DB_1$, then B_1 and D_1 should be separated by H. This contradiction shows that $AD < CD$.



Let γ_B and γ_D meet Ω again at T_B and T_D, respectively. The common chord BT_B of Ω and γ_B is perpendicular to their line of centres O_BO; likewise, $DT_D \perp O_DO$. Therefore, $O \in O_BO_D \iff O_BO \parallel O_DO \iff BT_B \parallel DT_D$, and the problem reduces to showing that

$$BT_B \parallel DT_D. \tag{1}$$

Comment 1. It seems that the discussion of the positions of points is necessary for both Solutions below. However, this part automatically follows from the angle chasing in Comment 2.

Solution 1. Let the diagonals AC and BD cross at H. Consider the homothety h centred at H and mapping B to D. Since $BD_1 \parallel DB_1$, we have $h(D_1) = B_1$.

Let the tangents to Ω at B and D meet AC at L_B and L_D, respectively. We have

$$\angle L_BBB_1 = \angle L_BBC + \angle CBB_1 = \angle BAL_B + \angle B_1BA = \angle BB_1L_B,$$

which means that the triangle L_BBB_1 is isosceles, $L_BB = L_BB_1$. The powers of L_B with respect to Ω and γ_D are L_BB^2 and $L_BB_1^2$, respectively; so they are equal, whence L_B lies on the radical axis T_DD of those two circles. Similarly, L_D lies on the radical axis T_BB of Ω and γ_B.

By the sine rule in the triangle BHL_B, we obtain

$$\frac{HL_B}{\sin\angle HBL_B} = \frac{BL_B}{\sin\angle BHL_B} = \frac{B_1L_B}{\sin\angle BHL_B}; \tag{2}$$

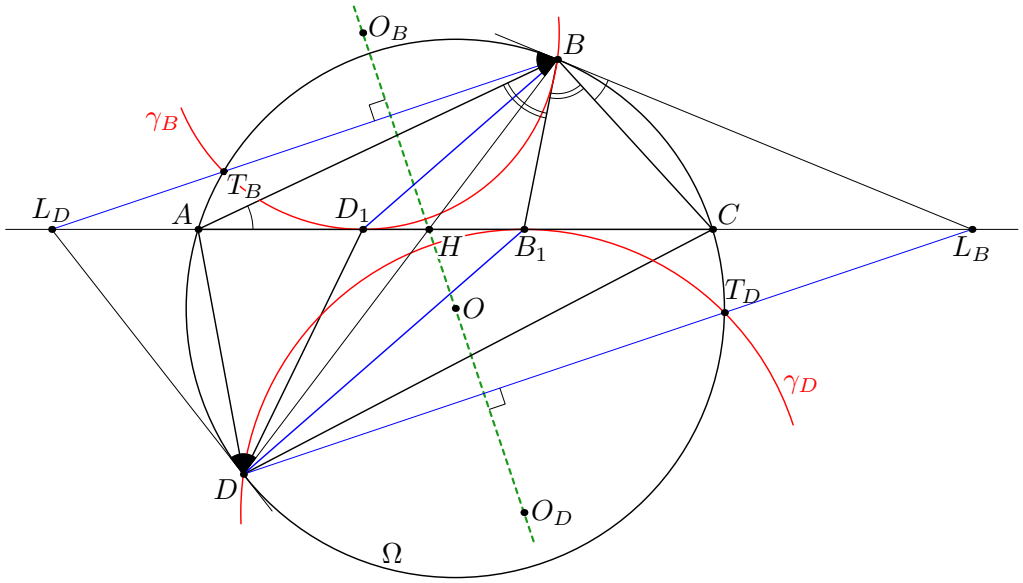
similarly,

$$\frac{HL_D}{\sin\angle HDL_D} = \frac{DL_D}{\sin\angle DHL_D} = \frac{D_1L_D}{\sin\angle DHL_D}. \tag{3}$$

Clearly, $\angle BHL_B = \angle DHL_D$. In the circle Ω, tangent lines BL_B and DL_D form equal angles with the chord BD, so $\sin\angle HBL_B = \sin\angle HDL_D$ (this equality does not depend on the picture). Thus, dividing (2) by (3) we get

$$\frac{HL_B}{HL_D} = \frac{B_1L_B}{D_1L_D}, \qquad \text{and hence} \quad \frac{HL_B}{HL_D} = \frac{HL_B - B_1L_B}{HL_D - D_1L_D} = \frac{HB_1}{HD_1}.$$

Since $h(D_1) = B_1$, the obtained relation yields $h(L_D) = L_B$, so h maps the line L_DB to L_BD, and these lines are parallel, as desired.



Comment 2. In the solution above, the key relation $h(L_D) = L_B$ was obtained via a short computation in sines. Here we present an alternative, pure synthetical way of establishing that.

Let the external bisectors of $\angle ABC$ and $\angle ADC$ cross AC at B_2 and D_2, respectively; assume that $\widehat{AB} > \widehat{CB}$. In the right-angled triangle BB_1B_2, the point L_B is a point on the hypothenuse such that $L_BB_1 = L_BB$, so L_B is the midpoint of B_1B_2.

Since DD_1 is the internal angle bisector of $\angle ADC$, we have

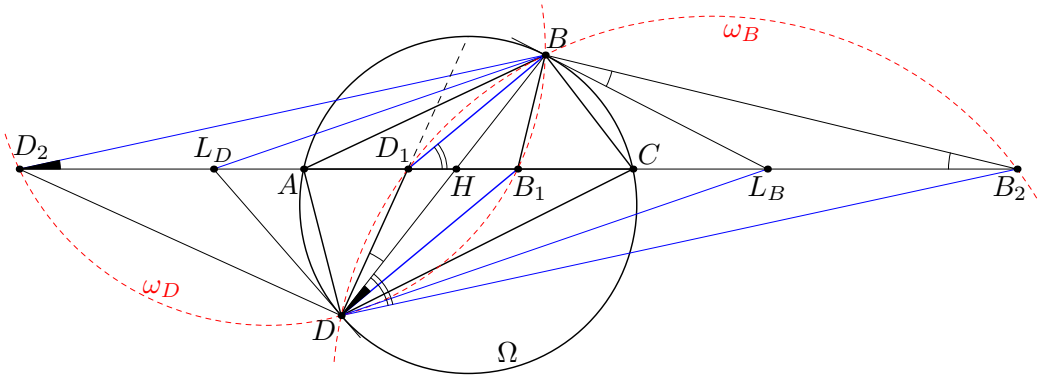
$$\angle BDD_1 = \frac{\angle BDA - \angle CDB}{2} = \frac{\angle BCA - \angle CAB}{2} = \angle BB_2D_1,$$

so the points B, B_2, D, and D_1 lie on some circle ω_B. Similarly, L_D is the midpoint of D_1D_2, and the points D, D_2, B, and B_1 lie on some circle ω_D.

Now we have

$$\angle B_2DB_1 = \angle B_2DB - \angle B_1DB = \angle B_2D_1B - \angle B_1D_2B = \angle D_2BD_1.$$

Therefore, the corresponding sides of the triangles DB_1B_2 and BD_1D_2 are parallel, and the triangles are homothetical (in H). So their corresponding medians DL_B and BL_D are also parallel.



Yet alternatively, after obtaining the circles ω_B and ω_D, one may notice that H lies on their radical axis BD, whence $HB_1 \cdot HD_2 = HD_1 \cdot HB_2$, or

$$\frac{HB_1}{HD_1} = \frac{HB_2}{HD_1}.$$

Since $h(D_1) = B_1$, this yields $h(D_2) = B_2$ and hence $h(L_D) = L_B$.

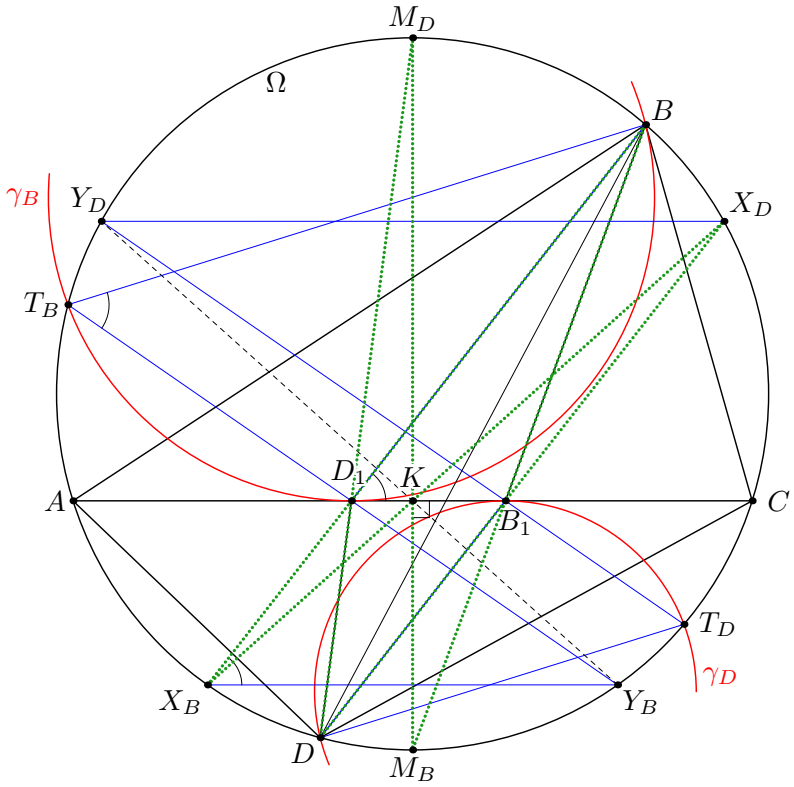
Comment 3. Since h preserves the line AC and maps $B \mapsto D$ and $D_1 \mapsto B_1$, we have $h(\gamma_B) = \gamma_D$. Therefore, $h(O_B) = O_D$; in particular, H also lies on O_BO_D.

Solution 2. Let BD_1 and T_BD_1 meet Ω again at X_B and Y_B, respectively. Then

$$\angle BD_1C = \angle BT_BD_1 = \angle BT_BY_B = \angle BX_BY_B,$$

which shows that $X_BY_B \parallel AC$. Similarly, let DB_1 and T_DB_1 meet Ω again at X_D and Y_D, respectively; then $X_DY_D \parallel AC$.

Let M_D and M_B be the midpoints of the arcs ABC and ADC, respectively; then the points D_1 and B_1 lie on DM_D and BM_B, respectively. Let K be the midpoint of AC (which lies on M_BM_D). Applying Pascal's theorem to $M_DDX_DX_BBM_B$, we obtain that the points $D_1 = M_DD \cap X_BB$, $B_1 = DX_D \cap BM_B$, and $X_DX_B \cap M_BM_D$ are collinear, which means that X_BX_D passes through K. Due to symmetry, the diagonals of an isosceles trapezoid $X_BY_BX_DY_D$ cross at K.



Let b and d denote the distances from the lines X_BY_B and X_DY_D, respectively, to AC. Then we get

$$\frac{X_BY_B}{X_DY_D} = \frac{b}{d} = \frac{D_1X_B}{B_1X_D},$$

where the second equation holds in view of $D_1X_B \parallel B_1X_D$. Therefore, the triangles $D_1X_BY_B$ and $B_1X_DY_D$ are similar. The triangles D_1T_BB and B_1T_DD are similar to them and hence to each other. Since $BD_1 \parallel DB_1$, these triangles are also homothetical. This yields $BT_B \parallel DT_D$, as desired.

Comment 4. The original problem proposal asked to prove that the relations $BD_1 \parallel DB_1$ and $O \in O_1O_2$ are equivalent. After obtaining $BD_1 \parallel DB_1 \Rightarrow O \in O_1O_2$, the converse proof is either repeated backwards *mutatis mutandis*, or can be obtained by the usual procedure of varying some points in the construction.

The Problem Selection Committee chose the current version, because it is less technical, yet keeps most of the ideas.

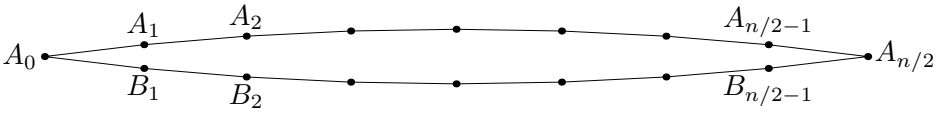
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G6. Determine all integers $n \geqslant 3$ satisfying the following property: every convex n-gon whose sides all have length 1 contains an equilateral triangle of side length 1.

(Every polygon is assumed to contain its boundary.)

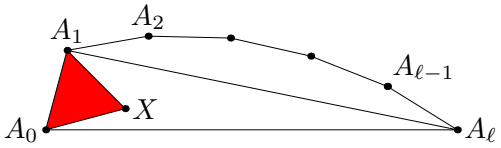
Answer: All odd $n \geqslant 3$.

Solution. First we show that for every even $n \geqslant 4$ there exists a polygon violating the required statement. Consider a regular k-gon $A_0A_1, \ldots A_{k-1}$ with side length 1. Let $B_1, B_2, \ldots, B_{n/2-1}$ be the points symmetric to $A_1, A_2, \ldots, A_{n/2-1}$ with respect to the line $A_0A_{n/2}$. Then $P = A_0A_1A_2 \ldots A_{n/2-1}A_{n/2}B_{n/2-1}B_{n/2-2} \ldots B_2B_1$ is a convex n-gon whose sides all have length 1. If k is big enough, P is contained in a strip of width $1/2$, which clearly does not contain any equilateral triangle of side length 1.



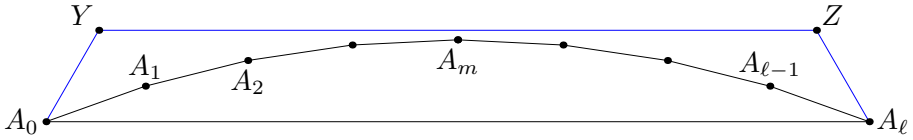
Assume now that $n = 2k+1$. As the case $k = 1$ is trivially true, we assume $k \geqslant 2$ henceforth. Consider a convex $(2k+1)$-gon P whose sides all have length 1. Let d be its longest diagonal. The endpoints of d split the perimeter of P into two polylines, one of which has length at least $k+1$. Hence we can label the vertices of P so that $P = A_0A_1 \ldots A_{2k}$ and $d = A_0A_\ell$ with $\ell \geqslant k+1$. We will show that, in fact, the polygon $A_0A_1 \ldots A_\ell$ contains an equilateral triangle of side length 1.

Suppose that $\angle A_\ell A_0 A_1 \geqslant 60^\circ$. Since d is the longest diagonal, we have $A_1A_\ell \leqslant A_0A_\ell$, so $\angle A_0A_1A_\ell \geqslant \angle A_\ell A_0 A_1 \geqslant 60^\circ$. It follows that there exists a point X inside the triangle $A_0A_1A_\ell$ such that the triangle A_0A_1X is equilateral, and this triangle is contained in P. Similar arguments apply if $\angle A_{\ell-1}A_\ell A_0 \geqslant 60^\circ$.



From now on, assume $\angle A_\ell A_0 A_1 < 60^\circ$ and $A_{\ell-1}A_\ell A_0 < 60^\circ$.

Consider an isosceles trapezoid A_0YZA_ℓ such that $A_0A_\ell \parallel YZ$, $A_0Y = ZA_\ell = 1$, and $\angle A_\ell A_0 Y = \angle ZA_\ell A_0 = 60^\circ$. Suppose that $A_0A_1 \ldots A_\ell$ is contained in A_0YZA_ℓ. Note that the perimeter of $A_0A_1 \ldots A_\ell$ equals $\ell + A_0A_\ell$ and the perimeter of A_0YZA_ℓ equals $2A_0A_\ell + 1$.



Recall a well-known fact stating that if a convex polygon P_1 is contained in a convex polygon P_2, then the perimeter of P_1 is at most the perimeter of P_2. Hence we obtain

$$\ell + A_0A_\ell \leqslant 2A_0A_\ell + 1, \quad \text{i.e.} \quad \ell - 1 \leqslant A_0A_\ell.$$

On the other hand, the triangle inequality yields

$$A_0A_\ell < A_\ell A_{\ell+1} + A_{\ell+1}A_{\ell+2} + \ldots + A_{2k}A_0 = 2k + 1 - \ell \leqslant \ell - 1,$$

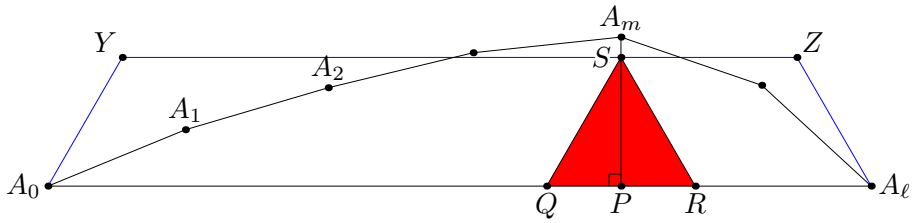
which gives a contradiction.

Therefore, there exists a vertex A_m of $A_0A_1 \ldots A_\ell$ which lies outside A_0YZA_ℓ. Since

$$\angle A_\ell A_0 A_1 < 60^\circ = \angle A_\ell A_0 Y \quad \text{and} \quad A_{\ell-1}A_\ell A_0 < 60^\circ = \angle ZA_\ell A_0, \tag{1}$$

the distance between A_m and A_0A_ℓ is at least $\sqrt{3}/2$.

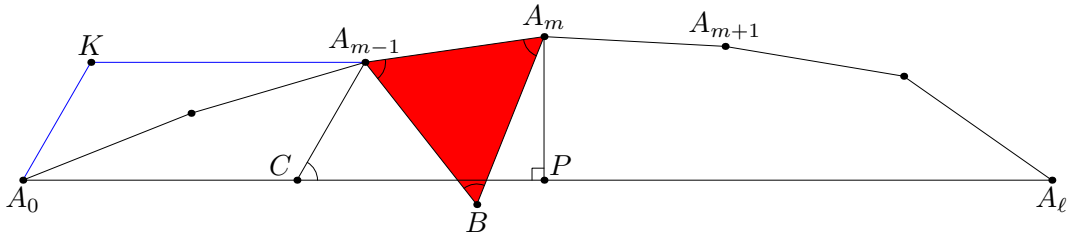
Let P be the projection of A_m to A_0A_ℓ. Then $PA_m \geqslant \sqrt{3}/2$, and by (1) we have $A_0P > 1/2$ and $PA_\ell > 1/2$. Choose points $Q \in A_0P$, $R \in PA_\ell$, and $S \in PA_m$ such that $PQ = PR = 1/2$ and $PS = \sqrt{3}/2$. Then QRS is an equilateral triangle of side length 1 contained in $A_0A_1 \ldots A_\ell$.



Comment. In fact, for every odd n a stronger statement holds, which is formulated in terms defined in the solution above: there exists an equilateral triangle $A_iA_{i+1}B$ contained in $A_0A_1 \ldots A_\ell$ for some $0 \leqslant i < \ell$. We sketch an indirect proof below.

As above, we get $\angle A_\ell A_0 A_1 < 60^\circ$ and $A_{\ell-1}A_\ell A_0 < 60^\circ$. Choose an index $m \in [1, \ell-1]$ maximising the distance between A_m and A_0A_ℓ. Arguments from the above solution yield $1 < m < \ell - 1$. Then $\angle A_0A_{m-1}A_m > 120^\circ$ and $\angle A_{m-1}A_mA_\ell > \angle A_0A_mA_\ell \geqslant 60^\circ$. We construct an equilateral triangle $A_{m-1}A_mB$ as in the figure below. If B lies in $A_0A_{m-1}A_mA_\ell$, then we are done. Otherwise B and A_m lie on different sides of A_0A_ℓ. As before, let P be the projection of A_m to A_0A_ℓ. We will show that

$$A_0A_1 + A_1A_2 + \ldots + A_{m-1}A_m < A_0P + 1/2. \tag{2}$$



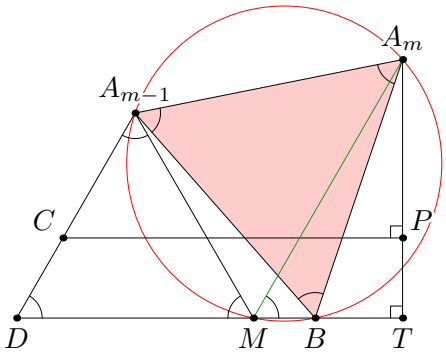
There exists a point C on the segment A_0P such that $\angle A_{m-1}CP = 60^\circ$. Construct a parallelogram $A_0CA_{m-1}K$. Then the polyline $A_0A_1 \ldots A_{m-1}$ is contained in the triangle $A_{m-1}KA_0$, so

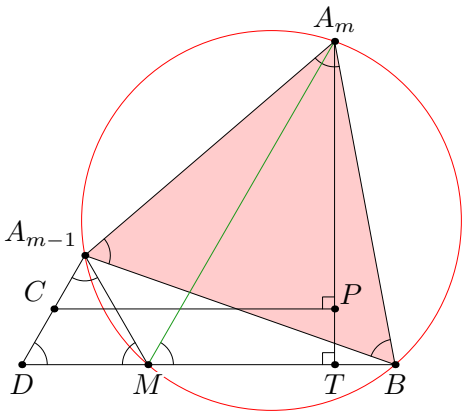
$$A_0A_1 + A_1A_2 + \ldots + A_{m-2}A_{m-1} + A_{m-1}A_m \leqslant A_0K + KA_{m-1} + A_{m-1}A_m = A_0C + CA_{m-1} + 1.$$

To prove (2), it suffices to show that $CA_{m-1} < CP - 1/2$. Let the line through B parallel to CP intersect the rays $A_{m-1}C$ and A_mP at D and T, respectively. It is easy to see that the desired inequality will follow from $DA_{m-1} \leqslant DT - 1/2$.

Two possible arrangements of points are shown in the figures below.

Observe that $\angle DA_{m-1}B \geqslant 60^\circ$, so there is a point M on the segment DB such that the triangle DMA_{m-1} is equilateral. Then $\angle A_{m-1}MD = 60^\circ = \angle A_{m-1}A_mB$, so $A_{m-1}MBA_m$ is a cyclic quadrilateral. Therefore, $\angle A_mMB = 60^\circ$. Thus, T lies on the ray MB and we have to show that $MT \geqslant 1/2$. Indeed, $MT = A_mM/2$ and $A_mM \geqslant A_mB = 1$. This completes the proof of the inequality (2).





Similarly, either there exists an equilateral triangle $A_mA_{m+1}B'$ contained in $A_0A_1\ldots A_\ell$, or

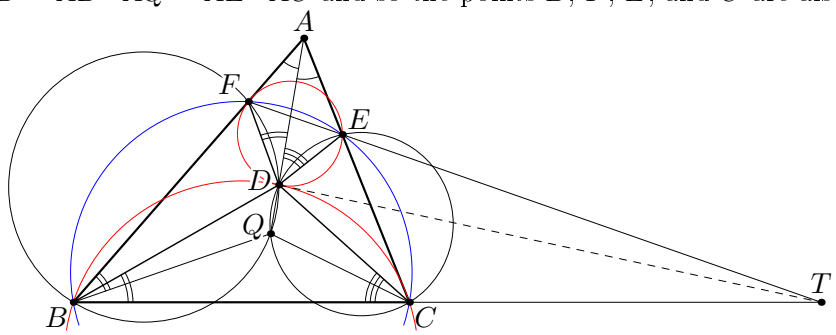
$$A_mA_{m+1} + A_{m+1}A_{m+2} + \ldots + A_{\ell-1}A_\ell < A_\ell P + 1/2. \tag{3}$$

Adding (2) and (3) yields $A_0A_1 + A_1A_2 + \ldots + A_{\ell-1}A_\ell < A_0A_\ell + 1$, which gives a contradiction.

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G7. A point D is chosen inside an acute-angled triangle ABC with $AB > AC$ so that $\angle BAD = \angle DAC$. A point E is constructed on the segment AC so that $\angle ADE = \angle DCB$. Similarly, a point F is constructed on the segment AB so that $\angle ADF = \angle DBC$. A point X is chosen on the line AC so that $CX = BX$. Let O_1 and O_2 be the circumcentres of the triangles ADC and DXE. Prove that the lines BC, EF, and O_1O_2 are concurrent.

Common remarks. Let Q be the isogonal conjugate of D with respect to the triangle ABC. Since $\angle BAD = \angle DAC$, the point Q lies on AD. Then $\angle QBA = \angle DBC = \angle FDA$, so the points Q, D, F, and B are concyclic. Analogously, the points Q, D, E, and C are concyclic. Thus $AF \cdot AB = AD \cdot AQ = AE \cdot AC$ and so the points B, F, E, and C are also concyclic.



Let T be the intersection of BC and FE.

Claim. $TD^2 = TB \cdot TC = TF \cdot TE$.

Proof. We will prove that the circles (DEF) and (BDC) are tangent to each other. Indeed, using the above arguments, we get

$$\angle BDF = \angle AFD - \angle ABD = (180^\circ - \angle FAD - \angle FDA) - (\angle ABC - \angle DBC)$$
$$= 180^\circ - \angle FAD - \angle ABC = 180^\circ - \angle DAE - \angle FEA = \angle FED + \angle ADE = \angle FED + \angle DCB,$$

which implies the desired tangency.

Since the points B, C, E, and F are concyclic, the powers of the point T with respect to the circles (BDC) and (EDF) are equal. So their radical axis, which coincides with the common tangent at D, passes through T, and hence $TD^2 = TE \cdot TF = TB \cdot TC$. □

Solution 1. Let TA intersect the circle (ABC) again at M. Due to the circles $(BCEF)$ and $(AMCB)$, and using the above Claim, we get $TM \cdot TA = TF \cdot TE = TB \cdot TC = TD^2$; in particular, the points A, M, E, and F are concyclic.

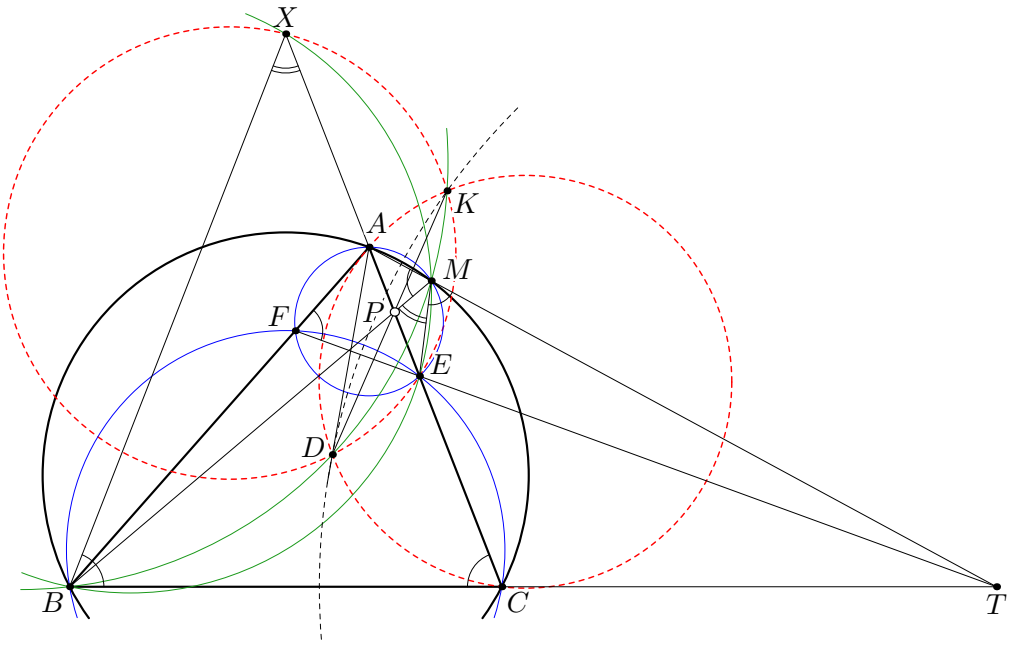
Under the inversion with centre T and radius TD, the point M maps to A, and B maps to C, which implies that the circle (MBD) maps to the circle (ADC). Their common point D lies on the circle of the inversion, so the second intersection point K also lies on that circle, which means $TK = TD$. It follows that the point T and the centres of the circles (KDE) and (ADC) lie on the perpendicular bisector of KD.

Since the center of (ADC) is O_1, it suffices to show now that the points D, K, E, and X are concyclic (the center of the corresponding circle will be O_2).

The lines BM, DK, and AC are the pairwise radical axes of the circles $(ABCM)$, $(ACDK)$ and $(BMDK)$, so they are concurrent at some point P. Also, M lies on the circle (AEF), thus

$$\measuredangle(EX, XB) = \measuredangle(CX, XB) = \measuredangle(XC, BC) + \measuredangle(BC, BX) = 2\measuredangle(AC, CB)$$
$$= \measuredangle(AC, CB) + \measuredangle(EF, FA) = \measuredangle(AM, BM) + \measuredangle(EM, MA) = \measuredangle(EM, BM),$$

so the points M, E, X, and B are concyclic. Therefore, $PE \cdot PX = PM \cdot PB = PK \cdot PD$, so the points E, K, D, and X are concyclic, as desired.



Comment 1. We present here a different solution which uses similar ideas.

Perform the inversion ι with centre T and radius TD. It swaps B with C and E with F; the point D maps to itself. Let $X' = \iota(X)$. Observe that the points E, F, X, and X' are concyclic, as well as the points B, C, X, and X'. Then

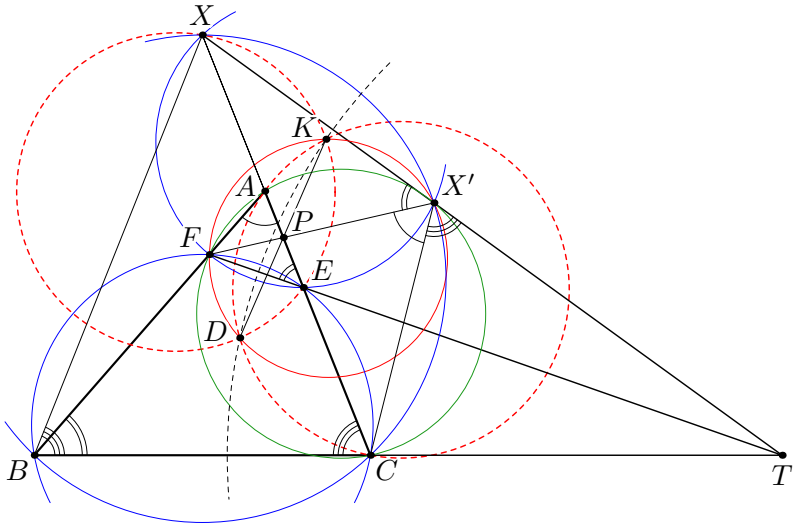
$$\begin{aligned}\measuredangle(CX', X'F) &= \measuredangle(CX', X'X) + \measuredangle(X'X, X'F) = \measuredangle(CB, BX) + \measuredangle(EX, EF)\\ &= \measuredangle(XC, CB) + \measuredangle(EC, EF) = \measuredangle(CA, CB) + \measuredangle(BC, BF) = \measuredangle(CA, AF),\end{aligned}$$

therefore the points C, X', A, and F are concyclic.

Let $X'F$ intersect AC at P, and let K be the second common point of DP and the circle (ACD). Then

$$PK \cdot PD = PA \cdot PC = PX' \cdot PF = PE \cdot PX;$$

hence, the points K, X, D, and E lie on some circle ω_1, while the points K, X', D, and F lie on some circle ω_2. (These circles are distinct since $\angle EXF + \angle EDF < \angle EAF + \angle DCB + \angle DBC < 180°$). The inversion ι swaps ω_1 with ω_2 and fixes their common point D, so it fixes their second common point K. Thus $TD = TK$ and the perpendicular bisector of DK passes through T, as well as through the centres of the circles $(CDKA)$ and $(DEKX)$.



Solution 2. We use only the first part of the Common remarks, namely, the facts that the tuples (C, D, Q, E) and (B, C, E, F) are both concyclic. We also introduce the point $T = BC \cap EF$. Let the circle (CDE) meet BC again at E_1. Since $\angle E_1CQ = \angle DCE$, the arcs DE and QE_1 of the circle (CDQ) are equal, so $DQ \parallel EE_1$.

Since $BFEC$ is cyclic, the line AD forms equal angles with BC and EF, hence so does EE_1. Therefore, the triangle EE_1T is isosceles, $TE = TE_1$, and T lies on the common perpendicular bisector of EE_1 and DQ.

Let U and V be the centres of circles (ADE) and $(CDQE)$, respectively. Then UO_1 is the perpendicular bisector of AD. Moreover, the points U, V, and O_2 belong to the perpendicular bisector of DE. Since $UO_1 \parallel VT$, in order to show that O_1O_2 passes through T, it suffices to show that

$$\frac{O_2U}{O_2V} = \frac{O_1U}{TV}. \tag{1}$$

Denote angles A, B, and C of the triangle ABC by α, β, and γ, respectively. Projecting onto AC we obtain

$$\frac{O_2U}{O_2V} = \frac{(XE - AE)/2}{(XE + EC)/2} = \frac{AX}{CX} = \frac{AX}{BX} = \frac{\sin(\gamma - \beta)}{\sin\alpha} \tag{2}$$

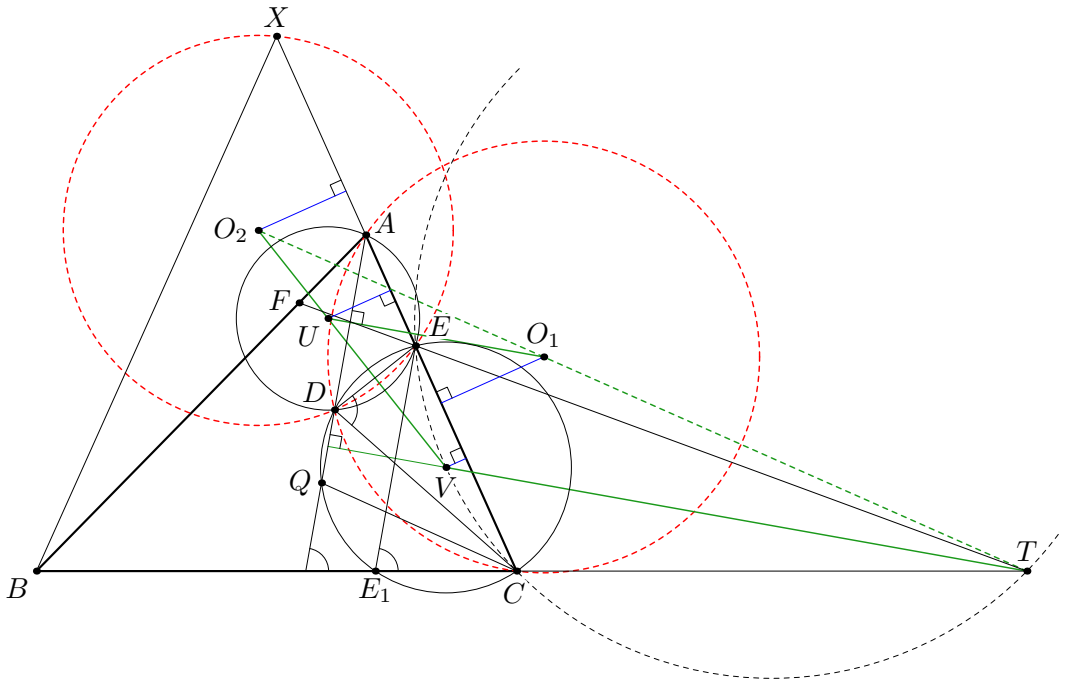
The projection of O_1U onto AC is $(AC - AE)/2 = CE/2$; the angle between O_1U and AC is $90^\circ - \alpha/2$, so

$$\frac{O_1U}{EC} = \frac{1}{2\sin(\alpha/2)} \tag{3}$$

Next, we claim that E, V, C, and T are concyclic. Indeed, the point V lies on the perpendicular bisector of CE, as well as on the internal angle bisector of $\angle CTF$. Therefore, V coincides with the midpoint of the arc CE of the circle (TCE).

Now we have $\angle EVC = 2\angle EE_1C = 180^\circ - (\gamma - \beta)$ and $\angle VET = \angle VE_1T = 90^\circ - \angle E_1EC = 90^\circ - \alpha/2$. Therefore,

$$\frac{EC}{TV} = \frac{\sin\angle ETC}{\sin\angle VET} = \frac{\sin(\gamma - \beta)}{\cos(\alpha/2)}. \tag{4}$$



Recalling (2) and multiplying (3) and (4) we establish (1):

$$\frac{O_2U}{O_2V} = \frac{\sin(\gamma-\beta)}{\sin\alpha} = \frac{1}{2\sin(\alpha/2)} \cdot \frac{\sin(\gamma-\beta)}{\cos(\alpha/2)} = \frac{O_1U}{EC} \cdot \frac{EC}{TV} = \frac{O_1U}{TV}.$$

Solution 3. Notice that $\angle AQE = \angle QCB$ and $\angle AQF = \angle QBC$; so, if we replace the point D with Q in the problem set up, the points E, F, and T remain the same. So, by the Claim, we have $TQ^2 = TB \cdot TC = TD^2$.

Thus, there exists a circle Γ centred at T and passing through D and Q. We denote the second meeting point of the circles Γ and (ADC) by K. Let the line AC meet the circle (DEK) again at Y; we intend to prove that $Y = X$. As in Solution 1, this will yield that the point T, as well as the centres O_1 and O_2, all lie on the perpendicular bisector of DK.

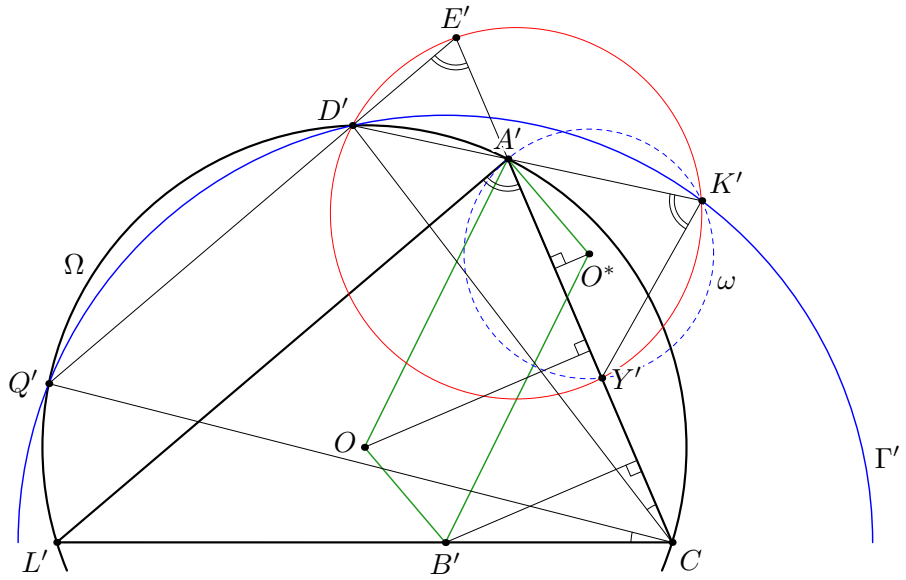
Let $L = AD \cap BC$. We perform an inversion centred at C; the images of the points will be denoted by primes, e.g., A' is the image of A. We obtain the following configuration, constructed in a triangle $A'CL'$.

The points D' and Q' are chosen on the circumcircle Ω of $A'L'C$ such that $\measuredangle(L'C, D'C) = \measuredangle(Q'C, A'C)$, which means that $A'L' \parallel D'Q'$. The lines $D'Q'$ and $A'C$ meet at E'.

A circle Γ' centred on CL' passes through D' and Q'. Notice here that B' lies on the segment CL', and that $\angle A'B'C = \angle BAC = 2\angle LAC = 2\angle A'L'C$, so that $B'L' = B'A'$, and B' lies on the perpendicular bisector of $A'L'$ (which coincides with that of $D'Q'$). All this means that B' is the centre of Γ'.

Finally, K' is the second meeting point of $A'D'$ and Γ', and Y' is the second meeting point of the circle $(D'K'E')$ and the line $A'E'$, We have $\measuredangle(Y'K', K'A') = \measuredangle(Y'E', E'D') = \measuredangle(Y'A', A'L')$, so $A'L'$ is tangent to the circumcircle ω of the triangle $Y'A'K'$.

Let O and O^* be the centres of Ω and ω, respectively. Then $O^*A' \perp A'L' \perp B'O$. The projections of vectors $\overrightarrow{O^*A'}$ and $\overrightarrow{B'O}$ onto $K'D'$ are equal to $\overrightarrow{K'A'}/2 = \overrightarrow{K'D'}/2 - \overrightarrow{A'D'}/2$. So $\overrightarrow{O^*A'} = \overrightarrow{B'O}$, or equivalently $\overrightarrow{A'O} = \overrightarrow{O^*B'}$. Projecting this equality onto $A'C$, we see that the projection of $\overrightarrow{O^*B'}$ equals $\overrightarrow{A'C}/2$. Since O^* is projected to the midpoint of $A'Y'$, this yields that B' is projected to the midpoint of CY', i.e., $B'Y' = B'C$ and $\angle B'Y'C = \angle B'CY'$. In the original figure, this rewrites as $\angle CBY = \angle BCY$, so Y lies on the perpendicular bisector of BC, as desired.



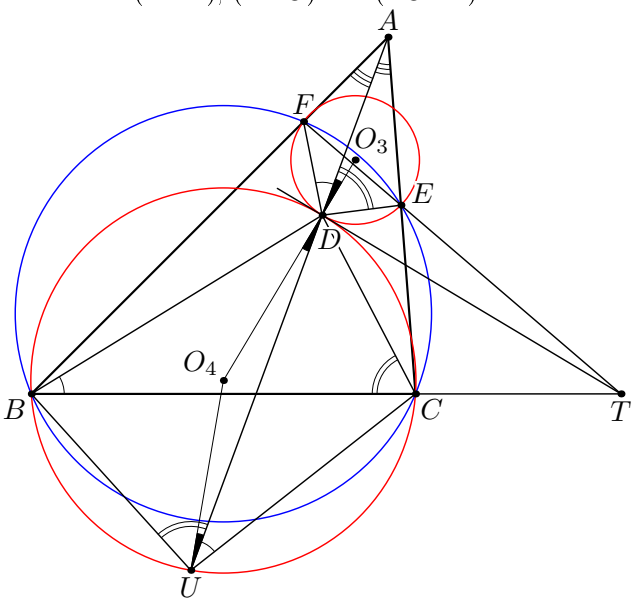
Comment 2. The point K appears to be the same in Solutions 1 and 3 (and Comment 1 as well). One can also show that K lies on the circle passing through A, X, and the midpoint of the arc BAC.

Comment 3. There are different proofs of the facts from the Common remarks, namely, the cyclicity of B, C, E, and F, and the Claim. We present one such alternative proof here.

We perform the composition ϕ of a homothety with centre A and the reflection in AD, which maps E to B. Let $U = \phi(D)$. Then $\measuredangle(BC, CD) = \measuredangle(AD, DE) = \measuredangle(BU, UD)$, so the points B, U, C, and D are concyclic. Therefore, $\measuredangle(CU, UD) = \measuredangle(CB, BD) = \measuredangle(AD, DF)$, so $\phi(F) = C$. Then the coefficient of the homothety is $AC/AF = AB/AE$, and thus points C, E, F, and B are concyclic.

Denote the centres of the circles (EDF) and $(BUCD)$ by O_3 and O_4, respectively. Then $\phi(O_3) = O_4$, hence $\measuredangle(O_3D, DA) = -\measuredangle(O_4U, UA) = \measuredangle(O_4D, DA)$, whence the circle (BDC) is tangent to the circle (EDF).

Now, the radical axes of circles (DEF), (BDC) and $(BCEF)$ intersect at T, and the claim follows.



This suffices for Solution 1 to work. However, Solutions 2 and 3 need properties of point Q, established in Common remarks before Solution 1.

Comment 4. In the original problem proposal, the point X was hidden. Instead, a circle γ was constructed such that D and E lie on γ, and its center is collinear with O_1 and T. The problem requested to prove that, in a fixed triangle ABC, independently from the choice of D on the bisector of $\angle BAC$, all circles γ pass through a fixed point.

G8. Let ω be the circumcircle of a triangle ABC, and let Ω_A be its excircle which is tangent to the segment BC. Let X and Y be the intersection points of ω and Ω_A. Let P and Q be the projections of A onto the tangent lines to Ω_A at X and Y, respectively. The tangent line at P to the circumcircle of the triangle APX intersects the tangent line at Q to the circumcircle of the triangle AQY at a point R. Prove that $AR \perp BC$.

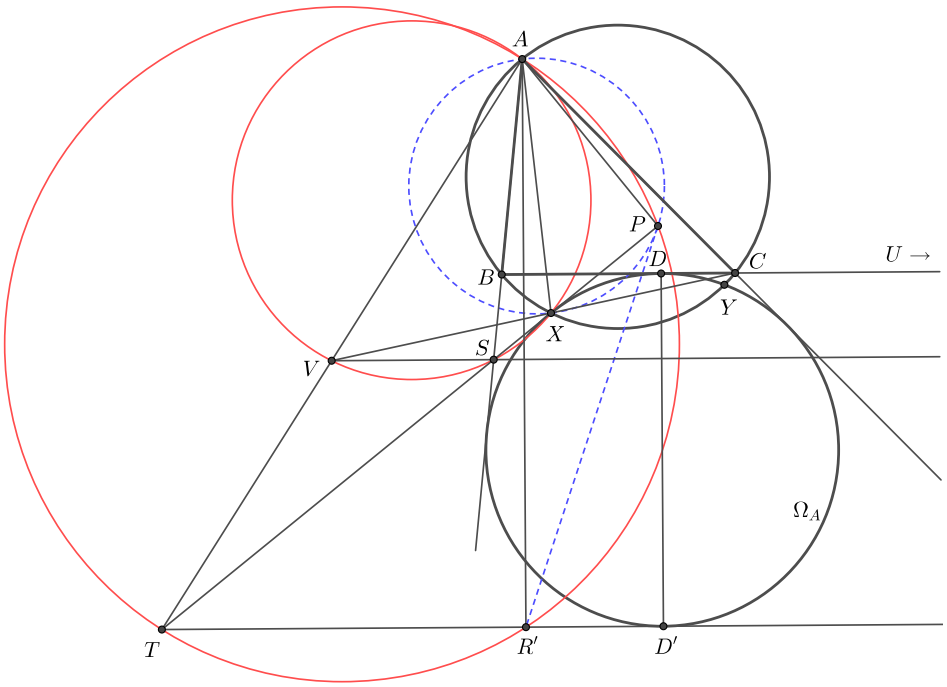
Solution 1. Let D be the point of tangency of BC and Ω_A. Let D' be the point such that DD' is a diameter of Ω_A. Let R' be (the unique) point such that $AR' \perp BC$ and $R'D' \parallel BC$. We shall prove that R' coincides with R.

Let PX intersect AB and $D'R'$ at S and T, respectively. Let U be the ideal common point of the parallel lines BC and $D'R'$. Note that the (degenerate) hexagon $ASXTUC$ is circumscribed around Ω_A, hence by the Brianchon theorem AT, SU, and XC concur at a point which we denote by V. Then $VS \parallel BC$. It follows that $\measuredangle(SV, VX) = \measuredangle(BC, CX) = \measuredangle(BA, AX)$, hence $AXSV$ is cyclic. Therefore, $\measuredangle(PX, XA) = \measuredangle(SV, VA) = \measuredangle(R'T, TA)$. Since $\angle APT = \angle AR'T = 90^\circ$, the quadrilateral $APR'T$ is cyclic. Hence,

$$\measuredangle(XA, AP) = 90^\circ - \measuredangle(PX, XA) = 90^\circ - \measuredangle(R'T, TA) = \measuredangle(TA, AR') = \measuredangle(TP, PR').$$

It follows that PR' is tangent to the circle (APX).

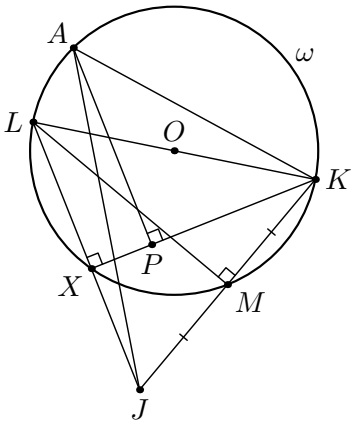
Analogous argument shows that QR' is tangent to the circle (AQY). Therefore, $R = R'$ and $AR \perp BC$.

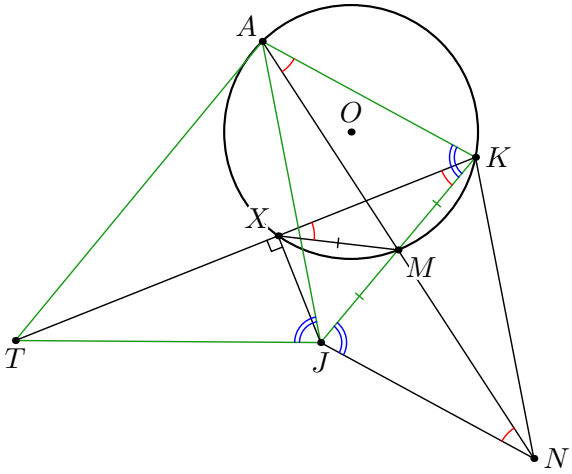


Comment 1. After showing $\measuredangle(PX, XA) = \measuredangle(R'T, TA)$ one can finish the solution as follows. There exists a spiral similarity mapping the triangle ATR' to the triangle AXP. So the triangles ATX and $AR'P$ are similar and equioriented. Thus, $\measuredangle(TX, XA) = \measuredangle(R'P, PA)$, which implies that PR' is tangent to the circle (APX).

Solution 2. Let J and r be the center and the radius of Ω_A. Denote the diameter of ω by d and its center by O. By Euler's formula, $OJ^2 = (d/2)^2 + dr$, so the power of J with respect to ω equals dr.

Let JX intersect ω again at L. Then $JL = d$. Let LK be a diameter of ω and let M be the midpoint of JK. Since $JL = LK$, we have $\angle LMK = 90°$, so M lies on ω. Let R' be the point such that $R'P$ is tangent to the circle (APX) and $AR' \perp BC$. Note that the line AR' is symmetric to the line AO with respect to AJ.



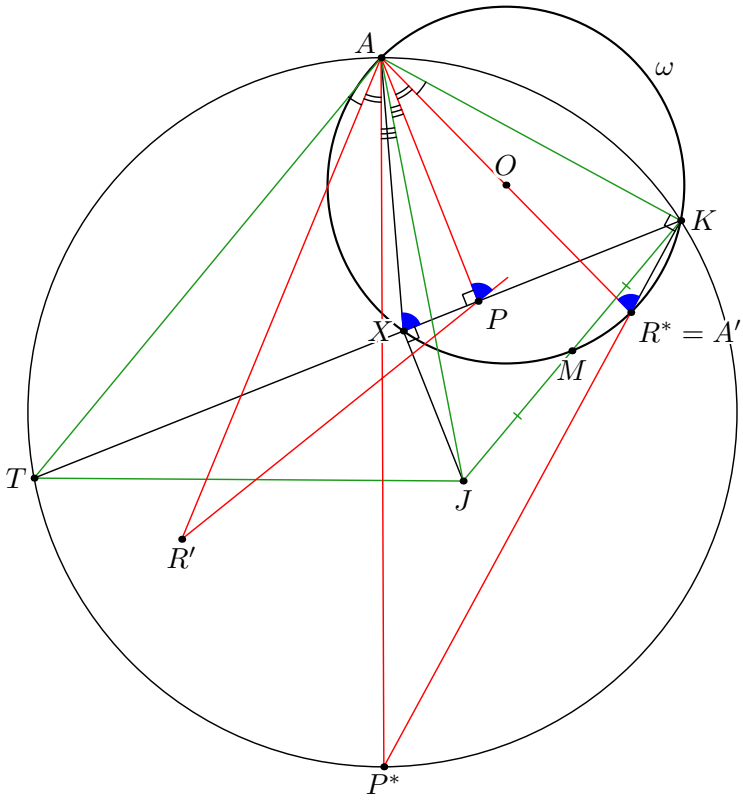


Lemma. Let M be the midpoint of the side JK in a triangle AJK. Let X be a point on the circle (AMK) such that $\angle JXK = 90°$. Then there exists a point T on the line KX such that the triangles AKJ and AJT are similar and equioriented.

Proof. Note that $MX = MK$. We construct a parallelogram $AJNK$. Let T be a point on KX such that $\measuredangle(NJ, JA) = \measuredangle(KJ, JT)$. Then

$$\measuredangle(JN, NA) = \measuredangle(KA, AM) = \measuredangle(KX, XM) = \measuredangle(MK, KX) = \measuredangle(JK, KT).$$

So there exists a spiral similarity with center J mapping the triangle AJN to the triangle TJK. Therefore, the triangles NJK and AJT are similar and equioriented. It follows that the triangles AKJ and AJT are similar and equioriented. □



Back to the problem, we construct a point T as in the lemma. We perform the composition ϕ of inversion with centre A and radius AJ and reflection in AJ. It is known that every triangle AEF is similar and equioriented to $A\phi(F)\phi(E)$.

So $\phi(K) = T$ and $\phi(T) = K$. Let $P^* = \phi(P)$ and $R^* = \phi(R')$. Observe that $\phi(TK)$ is a circle with diameter AP^*. Let AA' be a diameter of ω. Then $P^*K \perp AK \perp A'K$, so A' lies on P^*K. The triangles $AR'P$ and AP^*R^* are similar and equioriented, hence

$$\sphericalangle(AA', A'P^*) = \sphericalangle(AA', A'K) = \sphericalangle(AX, XP) = \sphericalangle(AX, XP) = \sphericalangle(AP, PR') = \sphericalangle(AR^*, R^*P^*),$$

so A, A', R^*, and P^* are concyclic. Since A' and R^* lie on AO, we obtain $R^* = A'$. So $R' = \phi(A')$, and $\phi(A')P$ is tangent to the circle (APX).

An identical argument shows that $\phi(A')Q$ is tangent to the circle (AQY). Therefore, $R = \phi(A')$ and $AR \perp BC$.

Comment 2. One of the main ideas of Solution 2 is to get rid of the excircle, along with points B and C. After doing so we obtain the following fact, which is, essentially, proved in Solution 2.

Let ω be the circumcircle of a triangle AK_1K_2. Let J be a point such that the midpoints of JK_1 and JK_2 lie on ω. Points X and Y are chosen on ω so that $\angle JXK_1 = \angle JYK_2 = 90^\circ$. Let P and Q be the projections of A onto XK_1 and YK_2, respectively. The tangent line at P to the circumcircle of the triangle APX intersects the tangent line at Q to the circumcircle of the triangle AQY at a point R. Then the reflection of the line AR in AJ passes through the centre O of ω.

Number Theory

N1. Determine all integers $n \geqslant 1$ for which there exists a pair of positive integers (a, b) such that no cube of a prime divides $a^2 + b + 3$ and

$$\frac{ab + 3b + 8}{a^2 + b + 3} = n.$$

Answer: The only integer with that property is $n = 2$.

Solution. As $b \equiv -a^2 - 3 \pmod{a^2 + b + 3}$, the numerator of the given fraction satisfies

$$ab + 3b + 8 \equiv a(-a^2 - 3) + 3(-a^2 - 3) + 8 \equiv -(a + 1)^3 \pmod{a^2 + b + 3}.$$

As $a^2 + b + 3$ is not divisible by p^3 for any prime p, if $a^2 + b + 3$ divides $(a + 1)^3$ then it does also divide $(a + 1)^2$. Since

$$0 < (a + 1)^2 < 2(a^2 + b + 3),$$

we conclude $(a+1)^2 = a^2 + b + 3$. This yields $b = 2(a-1)$ and $n = 2$. The choice $(a, b) = (2, 2)$ with $a^2 + b + 3 = 9$ shows that $n = 2$ indeed is a solution.

N2. Let $n \geqslant 100$ be an integer. The numbers $n, n+1, \ldots, 2n$ are written on $n+1$ cards, one number per card. The cards are shuffled and divided into two piles. Prove that one of the piles contains two cards such that the sum of their numbers is a perfect square.

Solution. To solve the problem it suffices to find three squares and three cards with numbers a, b, c on them such that pairwise sums $a+b, b+c, a+c$ are equal to the chosen squares. By choosing the three consecutive squares $(2k-1)^2$, $(2k)^2$, $(2k+1)^2$ we arrive at the triple

$$(a, b, c) = \left(2k^2 - 4k, \quad 2k^2 + 1, \quad 2k^2 + 4k\right).$$

We need a value for k such that

$$n \leqslant 2k^2 - 4k, \quad \text{and} \quad 2k^2 + 4k \leqslant 2n.$$

A concrete k is suitable for all n with

$$n \in \left[k^2 + 2k, 2k^2 - 4k + 1\right] =: I_k.$$

For $k \geqslant 9$ the intervals I_k and I_{k+1} overlap because

$$(k+1)^2 + 2(k+1) \leqslant 2k^2 - 4k + 1.$$

Hence $I_9 \cup I_{10} \cup \ldots = [99, \infty)$, which proves the statement for $n \geqslant 99$.

Comment 1. There exist approaches which only work for sufficiently large n.

One possible approach is to consider three cards with numbers $70k^2, 99k^2, 126k^2$ on them. Then their pairwise sums are perfect squares and so it suffices to find k such that $70k^2 \geqslant n$ and $126k^2 \leqslant 2n$ which exists for sufficiently large n.

Another approach is to prove, arguing by contradiction, that a and $a-2$ are in the same pile provided that n is large enough and a is sufficiently close to n. For that purpose, note that every pair of neighbouring numbers in the sequence $a, x^2 - a, a + (2x+1), x^2 + 2x + 3 - a, a - 2$ adds up to a perfect square for any x; so by choosing $x = \lfloor \sqrt{2a} \rfloor + 1$ and assuming that n is large enough we conclude that a and $a-2$ are in the same pile for any $a \in [n+2, 3n/2]$. This gives a contradiction since it is easy to find two numbers from $[n+2, 3n/2]$ of the same parity which sum to a square.

It then remains to separately cover the cases of small n which appears to be quite technical.

Comment 2. An alternative formulation for this problem could ask for a proof of the statement for all $n > 10^6$. An advantage of this formulation is that some solutions, e.g. those mentioned in Comment 1 need not contain a technical part which deals with the cases of small n. However, the original formulation seems to be better because the bound it gives for n is almost sharp, see the next comment for details.

Comment 3. The statement of the problem is false for $n = 98$. As a counterexample, the first pile may contain the even numbers from 98 to 126, the odd numbers from 129 to 161, and the even numbers from 162 to 196.

N3. Find all positive integers n with the following property: the k positive divisors of n have a permutation $(d_1, d_2, \ldots, d_k)$ such that for every $i = 1, 2, \ldots, k$, the number $d_1 + \cdots + d_i$ is a perfect square.

Answer: $n = 1$ and $n = 3$.

Solution. For $i = 1, 2, \ldots, k$ let $d_1 + \ldots + d_i = s_i^2$, and define $s_0 = 0$ as well. Obviously $0 = s_0 < s_1 < s_2 < \ldots < s_k$, so

$$s_i \geqslant i \quad \text{and} \quad d_i = s_i^2 - s_{i-1}^2 = (s_i + s_{i-1})(s_i - s_{i-1}) \geqslant s_i + s_{i-1} \geqslant 2i - 1. \tag{1}$$

The number 1 is one of the divisors $d_1, \ldots, d_k$ but, due to $d_i \geqslant 2i - 1$, the only possibility is $d_1 = 1$.

Now consider d_2 and $s_2 \geqslant 2$. By definition, $d_2 = s_2^2 - 1 = (s_2 - 1)(s_2 + 1)$, so the numbers $s_2 - 1$ and $s_2 + 1$ are divisors of n. In particular, there is some index j such that $d_j = s_2 + 1$.

Notice that

$$s_2 + s_1 = s_2 + 1 = d_j \geqslant s_j + s_{j-1}; \tag{2}$$

since the sequence $s_0 < s_1 < \ldots < s_k$ increases, the index j cannot be greater than 2. Hence, the divisors $s_2 - 1$ and $s_2 + 1$ are listed among d_1 and d_2. That means $s_2 - 1 = d_1 = 1$ and $s_2 + 1 = d_2$; therefore $s_2 = 2$ and $d_2 = 3$.

We can repeat the above process in general.

Claim. $d_i = 2i - 1$ and $s_i = i$ for $i = 1, 2, \ldots, k$.

Proof. Apply induction on i. The Claim has been proved for $i = 1, 2$. Suppose that we have already proved $d = 1$, $d_2 = 3$, …, $d_i = 2i - 1$, and consider the next divisor d_{i+1}:

$$d_{i+1} = s_{i+1}^2 - s_i^2 = s_{i+1}^2 - i^2 = (s_{i+1} - i)(s_{i+1} + i).$$

The number $s_{i+1} + i$ is a divisor of n, so there is some index j such that $d_j = s_{i+1} + i$.

Similarly to (2), by (1) we have

$$s_{i+1} + s_i = s_{i+1} + i = d_j \geqslant s_j + s_{j-1}; \tag{3}$$

since the sequence $s_0 < s_1 < \ldots < s_k$ increases, (3) forces $j \leqslant i + 1$. On the other hand, $d_j = s_{i+1} + i > 2i > d_i > d_{i-1} > \ldots > d_1$, so $j \leqslant i$ is not possible. The only possibility is $j = i + 1$.

Hence,

$$s_{i+1} + i = d_{i+1} = s_{i+1}^2 - s_i^2 = s_{i+1}^2 - i^2;$$

$$s_{i+1}^2 - s_{i+1} = i(i + 1).$$

By solving this equation we get $s_{i+1} = i + 1$ and $d_{i+1} = 2i + 1$, that finishes the proof. □

Now we know that the positive divisors of the number n are $1, 3, 5, \ldots, n-2, n$. The greatest divisor is $d_k = 2k - 1 = n$ itself, so n must be odd. The second greatest divisor is $d_{k-1} = n - 2$; then $n - 2$ divides $n = (n - 2) + 2$, so $n - 2$ divides 2. Therefore, n must be 1 or 3.

The numbers $n = 1$ and $n = 3$ obviously satisfy the requirements: for $n = 1$ we have $k = 1$ and $d_1 = 1^2$; for $n = 3$ we have $k = 2$, $d_1 = 1^2$ and $d_1 + d_2 = 1 + 3 = 2^2$.

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N4. Alice is given a rational number $r > 1$ and a line with two points $B \neq R$, where point R contains a red bead and point B contains a blue bead. Alice plays a solitaire game by performing a sequence of moves. In every move, she chooses a (not necessarily positive) integer k, and a bead to move. If that bead is placed at point X, and the other bead is placed at Y, then Alice moves the chosen bead to point X' with $\overrightarrow{YX'} = r^k \overrightarrow{YX}$.

Alice's goal is to move the red bead to the point B. Find all rational numbers $r > 1$ such that Alice can reach her goal in at most 2021 moves.

Answer: All $r = (b+1)/b$ with $b = 1, \ldots, 1010$.

Solution. Denote the red and blue beads by $\mathcal{R}$ and $\mathcal{B}$, respectively. Introduce coordinates on the line and identify the points with their coordinates so that $R = 0$ and $B = 1$. Then, during the game, the coordinate of $\mathcal{R}$ is always smaller than the coordinate of $\mathcal{B}$. Moreover, the distance between the beads always has the form r^ℓ with $\ell \in \mathbb{Z}$, since it only multiplies by numbers of this form. Denote the value of the distance after the mth move by $d_m = r^{\alpha_m}$, $m = 0, 1, 2, \ldots$ (after the 0th move we have just the initial position, so $\alpha_0 = 0$).

If some bead is moved in two consecutive moves, then Alice could instead perform a single move (and change the distance from d_i directly to d_{i+2}) which has the same effect as these two moves. So, if Alice can achieve her goal, then she may as well achieve it in fewer (or the same) number of moves by alternating the moves of $\mathcal{B}$ and $\mathcal{R}$. In the sequel, we assume that Alice alternates the moves, and that $\mathcal{R}$ is shifted altogether t times.

If $\mathcal{R}$ is shifted in the mth move, then its coordinate increases by $d_m - d_{m+1}$. Therefore, the total increment of $\mathcal{R}$'s coordinate, which should be 1, equals

$$\text{either} \qquad (d_0 - d_1) + (d_2 - d_3) + \cdots + (d_{2t-2} - d_{2t-1}) = 1 + \sum_{i=1}^{t-1} r^{\alpha_{2i}} - \sum_{i=1}^{t} r^{\alpha_{2i-1}},$$

$$\text{or} \qquad (d_1 - d_2) + (d_3 - d_4) + \cdots + (d_{2t-1} - d_{2t}) = \sum_{i=1}^{t} r^{\alpha_{2i-1}} - \sum_{i=1}^{t} r^{\alpha_{2i}},$$

depending on whether $\mathcal{R}$ or $\mathcal{B}$ is shifted in the first move. Moreover, in the former case we should have $t \leqslant 1011$, while in the latter one we need $t \leqslant 1010$. So both cases reduce to an equation

$$\sum_{i=1}^{n} r^{\beta_i} = \sum_{i=1}^{n-1} r^{\gamma_i}, \qquad \beta_i, \gamma_i \in \mathbb{Z}, \tag{1}$$

for some $n \leqslant 1011$. Thus, if Alice can reach her goal, then this equation has a solution for $n = 1011$ (we can add equal terms to both sums in order to increase n).

Conversely, if (1) has a solution for $n = 1011$, then Alice can compose a corresponding sequence of distances $d_0, d_1, d_2, \ldots, d_{2021}$ and then realise it by a sequence of moves. So the problem reduces to the solvability of (1) for $n = 1011$.

Assume that, for some rational r, there is a solution of (1). Write r in lowest terms as $r = a/b$. Substitute this into (1), multiply by the common denominator, and collect all terms on the left hand side to get

$$\sum_{i=1}^{2n-1} (-1)^i a^{\mu_i} b^{N-\mu_i} = 0, \qquad \mu_i \in \{0, 1, \ldots, N\}, \tag{2}$$

for some $N \geqslant 0$. We assume that there exist indices j_- and j_+ such that $\mu_{j_-} = 0$ and $\mu_{j_+} = N$.

Reducing (2) modulo $a-b$ (so that $a \equiv b$), we get

$$0 = \sum_{i=1}^{2n-1} (-1)^i a^{\mu_i} b^{N-\mu_i} \equiv \sum_{i=1}^{2n-1} (-1)^i b^{\mu_i} b^{N-\mu_i} = -b^N \mod (a-b).$$

Since $\gcd(a-b, b) = 1$, this is possible only if $a - b = 1$.

Reducing (2) modulo $a+b$ (so that $a \equiv -b$), we get

$$0 = \sum_{i=1}^{2n-1} (-1)^i a^{\mu_i} b^{N-\mu_i} \equiv \sum_{i=1}^{2n-1} (-1)^i (-1)^{\mu_i} b^{\mu_i} b^{N-\mu_i} = S b^N \mod (a+b)$$

for some odd (thus nonzero) S with $|S| \leqslant 2n-1$. Since $\gcd(a+b, b) = 1$, this is possible only if $a + b \mid S$. So $a + b \leqslant 2n - 1$, and hence $b = a - 1 \leqslant n - 1 = 1010$.

Thus we have shown that any sought r has the form indicated in the answer. It remains to show that for any $b = 1, 2, \ldots, 1010$ and $a = b + 1$, Alice can reach the goal. For this purpose, in (1) we put $n = a$, $\beta_1 = \beta_2 = \cdots = \beta_a = 0$, and $\gamma_1 = \gamma_2 = \cdots = \gamma_b = 1$.

Comment 1. Instead of reducing modulo $a + b$, one can reduce modulo a and modulo b. The first reduction shows that the number of terms in (2) with $\mu_i = 0$ is divisible by a, while the second shows that the number of terms with $\mu_i = N$ is divisible by b.

Notice that, in fact, $N > 0$, as otherwise (2) contains an alternating sum of an odd number of equal terms, which is nonzero. Therefore, all terms listed above have different indices, and there are at least $a + b$ of them.

Comment 2. Another way to investigate the solutions of equation (1) is to consider the Laurent polynomial

$$L(x) = \sum_{i=1}^{n} x^{\beta_i} - \sum_{i=1}^{n-1} x^{\gamma_i}.$$

We can pick a sufficiently large integer d so that $P(x) = x^d L(x)$ is a polynomial in $\mathbb{Z}[x]$. Then

$$P(1) = 1, \tag{3}$$

and

$$1 \leqslant |P(-1)| \leqslant 2021. \tag{4}$$

If $r = p/q$ with integers $p > q \geqslant 1$ is a rational number with the properties listed in the problem statement, then $P(p/q) = L(p/q) = 0$. As $P(x)$ has integer coefficients,

$$(p - qx) \mid P(x). \tag{5}$$

Plugging $x = 1$ into (5) gives $(p - q) \mid P(1) = 1$, which implies $p = q + 1$. Moreover, plugging $x = -1$ into (5) gives $(p + q) \mid P(-1)$, which, along with (4), implies $p + q \leqslant 2021$ and $q \leqslant 1010$. Hence $x = (q + 1)/q$ for some integer q with $1 \leqslant q \leqslant 1010$.

N5. Prove that there are only finitely many quadruples (a, b, c, n) of positive integers such that

$$n! = a^{n-1} + b^{n-1} + c^{n-1}.$$

Solution. For fixed n there are clearly finitely many solutions; we will show that there is no solution with $n > 100$. So, assume $n > 100$. By the AM–GM inequality,

$$\begin{aligned} n! &= 2n(n-1)(n-2)(n-3) \cdot (3 \cdot 4 \cdots (n-4)) \\ &\leqslant 2(n-1)^4 \left(\frac{3 + \cdots + (n-4)}{n-6}\right)^{n-6} = 2(n-1)^4 \left(\frac{n-1}{2}\right)^{n-6} < \left(\frac{n-1}{2}\right)^{n-1}, \end{aligned}$$

thus $a, b, c < (n-1)/2$.

For every prime p and integer $m \neq 0$, let $\nu_p(m)$ denote the p-adic valuation of m; that is, the greatest non-negative integer k for which p^k divides m. Legendre's formula states that

$$\nu_p(n!) = \sum_{s=1}^{\infty} \left\lfloor \frac{n}{p^s} \right\rfloor,$$

and a well-know corollary of this formula is that

$$\nu_p(n!) < \sum_{s=1}^{\infty} \frac{n}{p^s} = \frac{n}{p-1}. \qquad (\heartsuit)$$

If n is odd then $a^{n-1}, b^{n-1}, c^{n-1}$ are squares, and by considering them modulo 4 we conclude that a, b and c must be even. Hence, $2^{n-1} \mid n!$ but that is impossible for odd n because $\nu_2(n!) = \nu_2((n-1)!) < n-1$ by $(\heartsuit)$.

From now on we assume that n is even. If all three numbers $a+b, b+c, c+a$ are powers of 2 then a, b, c have the same parity. If they all are odd, then $n! = a^{n-1} + b^{n-1} + c^{n-1}$ is also odd which is absurd. If all a, b, c are divisible by 4, this contradicts $\nu_2(n!) \leqslant n-1$. If, say, a is not divisible by 4, then $2a = (a+b) + (a+c) - (b+c)$ is not divisible by 8, and since all $a+b$, $b+c$, $c+a$ are powers of 2, we get that one of these sums equals 4, so two of the numbers of a, b, c are equal to 2. Say, $a = b = 2$, then $c = 2^r - 2$ and, since $c \mid n!$, we must have $c \mid a^{n-1} + b^{n-1} = 2^n$ implying $r = 2$, and so $c = 2$, which is impossible because $n! \equiv 0 \not\equiv 3 \cdot 2^{n-1} \pmod 5$.

So now we assume that the sum of two numbers among a, b, c, say $a+b$, is not a power of 2, so it is divisible by some odd prime p. Then $p \leqslant a + b < n$ and so $c^{n-1} = n! - (a^{n-1} + b^{n-1})$ is divisible by p. If p divides a and b, we get $p^{n-1} \mid n!$, contradicting $(\heartsuit)$. Next, using $(\heartsuit)$ and the Lifting the Exponent Lemma we get

$$\nu_p(1) + \nu_p(2) + \cdots + \nu_p(n) = \nu_p(n!) = \nu_p(n! - c^{n-1}) = \nu_p\left(a^{n-1} + b^{n-1}\right) = \nu_p(a+b) + \nu_p(n-1). \quad (\diamondsuit)$$

In view of $(\diamondsuit)$, no number of $1, 2, \ldots, n$ can be divisible by p, except $a + b$ and $n - 1 > a + b$. On the other hand, $p | c$ implies that $p < n/2$ and so there must be at least two such numbers. Hence, there are two multiples of p among $1, 2, \ldots, n$, namely $a + b = p$ and $n - 1 = 2p$. But this is another contradiction because $n - 1$ is odd. This final contradiction shows that there is no solution of the equation for $n > 100$.

Comment 1. The original version of the problem asked to find all solutions to the equation. The solution to that version is not much different but is more technical.

Comment 2. To find all solutions we can replace the bound $a, b, c < (n-1)/2$ for all n with a weaker bound $a, b, c \leqslant n/2$ only for even n, which is a trivial application of AM–GM to the tuple $(2, 3, \ldots, n)$. Then we may use the same argument for odd n (it works for $n \geqslant 5$ and does not require any bound on a, b, c), and for even n the same solution works for $n \geqslant 6$ unless we have $a + b = n - 1$ and $2\nu_p(n-1) = \nu_p(n!)$. This is only possible for $p = 3$ and $n = 10$ in which case we can consider the original equation modulo 7 to deduce that $7 \mid abc$ which contradicts the fact that $7^9 > 10!$. Looking at $n \leqslant 4$ we find four solutions, namely,

$$(a, b, c, n) = (1, 1, 2, 3), (1, 2, 1, 3), (2, 1, 1, 3), (2, 2, 2, 4).$$

Comment 3. For sufficiently large n, the inequality $a, b, c < (n-1)/2$ also follows from Stirling's formula.

N6. Determine all integers $n \geqslant 2$ with the following property: every n pairwise distinct integers whose sum is not divisible by n can be arranged in some order $a_1, a_2, \ldots, a_n$ so that n divides $1 \cdot a_1 + 2 \cdot a_2 + \cdots + n \cdot a_n$.

Answer: All odd integers and all powers of 2.

Solution. If $n = 2^k a$, where $a \geqslant 3$ is odd and k is a positive integer, we can consider a set containing the number $2^k + 1$ and $n - 1$ numbers congruent to 1 modulo n. The sum of these numbers is congruent to 2^k modulo n and therefore is not divisible by n; for any permutation $(a_1, a_2, \ldots, a_n)$ of these numbers

$$1 \cdot a_1 + 2 \cdot a_2 + \cdots + n \cdot a_n \equiv 1 + \cdots + n \equiv 2^{k-1} a(2^k a + 1) \not\equiv 0 \pmod{2^k}$$

and *a fortiori* $1 \cdot a_1 + 2 \cdot a_2 + \cdots + n \cdot a_n$ is not divisible by n.

From now on, we suppose that n is either odd or a power of 2. Let S be the given set of integers, and s be the sum of elements of S.

Lemma 1. If there is a permutation (a_i) of S such that (n, s) divides $\sum_{i=1}^{n} i a_i$, then there is a permutation (b_i) of S such that n divides $\sum_{i=1}^{n} i b_i$.

Proof. Let $r = \sum_{i=1}^{n} i a_i$. Consider the permutation (b_i) defined by $b_i = a_{i+x}$, where $a_{j+n} = a_j$. For this permutation, we have

$$\sum_{i=1}^{n} i b_i = \sum_{i=1}^{n} i a_{i+x} \equiv \sum_{i=1}^{n} (i - x) a_i \equiv r - sx \pmod{n}.$$

Since (n, s) divides r, the congruence $r - sx \equiv 0 \pmod{n}$ admits a solution.

Lemma 2. Every set T of km integers, $m > 1$, can be partitioned into m sets of k integers so that in every set either the sum of elements is not divisible by k or all the elements leave the same remainder upon division by k.

Proof. The base case, $m = 2$. If T contains k elements leaving the same remainder upon division by k, we form one subset A of these elements; the remaining elements form a subset B. If k does not divide the sum of all elements of B, we are done. Otherwise it is enough to exchange any element of A with any element of B not congruent to it modulo k, thus making sums of both A and B not divisible by k. This cannot be done only when all the elements of T are congruent modulo k; in this case any partition will do.

If no k elements of T have the same residue modulo k, there are three elements $a, b, c \in T$ leaving pairwise distinct remainders upon division by k. Let t be the sum of elements of T. It suffices to find $A \subset T$ such that $|A| = k$ and $\sum_{x \in A} x \not\equiv 0, t \pmod{k}$: then neither the sum of elements of A nor the sum of elements of $B = T \setminus A$ is divisible by k. Consider $U' \subset T \setminus \{a, b, c\}$ with $|U'| = k - 1$. The sums of elements of three sets $U' \cup \{a\}$, $U' \cup \{b\}$, $U' \cup \{c\}$ leave three different remainders upon division by k, and at least one of them is not congruent either to 0 or to t.

Now let $m > 2$. If T contains k elements leaving the same remainder upon division by k, we form one subset A of these elements and apply the inductive hypothesis to the remaining $k(m - 1)$ elements. Otherwise, we choose any $U \subset T$, $|U| = k - 1$. Since all the remaining elements cannot be congruent modulo k, there is $a \in T \setminus U$ such that $a \not\equiv -\sum_{x \in U} x \pmod{k}$. Now we can take $A = U \cup \{a\}$ and apply the inductive hypothesis to $T \setminus A$.

Now we are ready to prove the statement of the problem for all odd n and $n = 2^k$. The proof is by induction.

If n is prime, the statement follows immediately from Lemma 1, since in this case $(n, s) = 1$. Turning to the general case, we can find prime p and an integer t such that $p^t \mid n$ and $p^t \nmid s$. By Lemma 2, we can partition S into p sets of $\frac{n}{p} = k$ elements so that in every set either the sum of numbers is not divisible by k or all numbers have the same residue modulo k.

For sets in the first category, by the inductive hypothesis there is a permutation (a_i) such that $k \mid \sum_{i=1}^{k} ia_i$.

If n (and therefore k) is odd, then for each permutation (b_i) of a set in the second category we have

$$\sum_{i=1}^{k} ib_i \equiv b_1 \frac{k(k+1)}{2} \equiv 0 \pmod{k}.$$

By combining such permutation for all sets of the partition, we get a permutation (c_i) of S such that $k \mid \sum_{i=1}^{n} ic_i$. Since this sum is divisible by k, and k is divisible by (n, s), we are done by Lemma 1.

If $n = 2^s$, we have $p = 2$ and $k = 2^{s-1}$. Then for each of the subsets there is a permutation $(a_1, \ldots, a_k)$ such that $\sum_{i=1}^{k} ia_i$ is divisible by $2^{s-2} = \frac{k}{2}$: if the subset belongs to the first category, the expression is divisible even by k, and if it belongs to the second one,

$$\sum_{i=1}^{k} ia_i \equiv a_1 \frac{k(k+1)}{2} \equiv 0 \pmod{\frac{k}{2}}.$$

Now the numbers of each permutation should be multiplied by all the odd or all the even numbers not exceeding n in increasing order so that the resulting sums are divisible by k:

$$\sum_{i=1}^{k} (2i-1)a_i \equiv \sum_{i=1}^{k} 2ia_i \equiv 2\sum_{i=1}^{k} ia_i \equiv 0 \pmod{k}.$$

Combining these two sums, we again get a permutation (c_i) of S such that $k \mid \sum_{i=1}^{n} ic_i$, and finish the case by applying Lemma 1.

Comment. We cannot dispense with the condition that n does not divide the sum of all elements. Indeed, for each $n > 1$ and the set consisting of 1, -1, and $n - 2$ elements divisible by n the required permutation does not exist.

N7. Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers such that a_{n+2m} divides $a_n + a_{n+m}$ for all positive integers n and m. Prove that this sequence is eventually periodic, i.e. there exist positive integers N and d such that $a_n = a_{n+d}$ for all $n > N$.

Solution. We will make repeated use of the following simple observation:

Lemma 1. If a positive integer d divides a_n and a_{n-m} for some m and $n > 2m$, it also divides a_{n-2m}. If d divides a_n and a_{n-2m}, it also divides a_{n-m}.

Proof. Both parts are obvious since a_n divides $a_{n-2m} + a_{n-m}$. □

Claim. The sequence (a_n) is bounded.

Proof. Suppose the contrary. Then there exist infinitely many indices n such that a_n is greater than each of the previous terms a_1, a_2, …, a_{n-1}. Let $a_n = k$ be such a term, $n > 10$. For each $s < \frac{n}{2}$ the number $a_n = k$ divides $a_{n-s} + a_{n-2s} < 2k$, therefore

$$a_{n-s} + a_{n-2s} = k.$$

In particular,

$$a_n = a_{n-1} + a_{n-2} = a_{n-2} + a_{n-4} = a_{n-4} + a_{n-8},$$

that is, $a_{n-1} = a_{n-4}$ and $a_{n-2} = a_{n-8}$. It follows from Lemma 1 that a_{n-1} divides a_{n-1-3s} for $3s < n-1$ and a_{n-2} divides a_{n-2-6s} for $6s < n-2$. Since at least one of the numbers a_{n-1} and a_{n-2} is at least $a_n/2$, so is some a_i with $i \leqslant 6$. However, a_n can be arbitrarily large, a contradiction. □

Since (a_n) is bounded, there exist only finitely many i for which a_i appears in the sequence finitely many times. In other words, there exists N such that if $a_i = t$ and $i > N$, then $a_j = t$ for infinitely many j.

Clearly the sequence $(a_{n+N})_{n>0}$ satisfies the divisibility condition, and it is enough to prove that this sequence is eventually periodic. Thus truncating the sequence if necessary, we can assume that each number appears infinitely many times in the sequence. Let k be the maximum number appearing in the sequence.

Lemma 2. If a positive integer d divides a_n for some n, then the numbers i such that d divides a_i form an arithmetical progression with an odd difference.

Proof. Let $i_1 < i_2 < i_3 < \ldots$ be all the indices i such that d divides a_i. If $i_s + i_{s+1}$ is even, it follows from Lemma 1 that d also divides $a_{\frac{i_s+i_{s+1}}{2}}$, impossible since $i_s < \frac{i_s+i_{s+1}}{2} < i_{s+1}$. Thus i_s and i_{s+1} are always of different parity, and therefore $i_s + i_{s+2}$ is even. Applying Lemma 1 again, we see that d divides $a_{\frac{i_s+i_{s+2}}{2}}$, hence $\frac{i_s+i_{s+2}}{2} = i_{s+1}$, □

We are ready now to solve the problem.

The number of positive divisors of all terms of the progression is finite. Let d_s be the difference of the progression corresponding to s, that is, s divides a_n if and only if it divides a_{n+td_s} for any positive integer t. Let D be the product of all d_s. Then each s dividing a term of the progression divides a_n if and only if it divides a_{n+D}. This means that the sets of divisors of a_n and a_{n+D} coincide, and $a_{n+D} = a_n$. Thus D is a period of the sequence.

Comment. In the above solution we did not try to find the exact structure of the periodic part of (a_n). A little addition to the argument above shows that the period of the sequence has one of the following three forms:

(i) t (in this case the sequence is eventually constant);

(ii) t, $2t$, $3t$ or $2t$, t, $3t$ (so the period is 3);

(iii) t, t, …, $2t$ (the period can be any odd number).

In these three cases t can be any positive integer. It is easy to see that all three cases satisfy the original condition.

We again denote by k be the maximum number appearing in the sequence. All the indices i such that $a_i = k$ form an arithmetical progression. If the difference of this progression is 1, the sequence (a_n) is constant, and we get the case (i). Assume that the difference T is at least 3.

Take an index n such that $a_n = k$ and let $a = a_{n-2}$, $b = a_{n-1}$. We have $a, b < k$ and therefore $k = a_n = a_{n-1} + a_{n-2} = a + b$. If $a = b = \frac{k}{2}$, then all the terms a_1, a_2, ..., a_n are divisible by $k/2$, that is, are equal to k or $k/2$. Since the indices i such that $a_i = k$ form an arithmetical progression with odd diference, we get the case (iii).

Suppose now that $a \neq b$.

Claim. For $\frac{n}{2} < m < n$ we have $a_m = a$ if $m \equiv n - 2 \pmod 3$ and $a_m = b$ if $m \equiv n - 1 \pmod 3$.

Proof. The number $k = a_n$ divides $a_{n-2} + a_{n-1} = a + b$ and $a_{n-4} + a_{n-2} = a_{n-4} + a$ and is therefore equal to these sums (since $a, b < k$ and $a_i \leqslant k$ for all i). Therefore $a_{n-1} = a_{n-4} = b$, that is, $a_{n-4} < k$, $a_{n-4} + a_{n-8} = k$ and $a_{n-8} = a_{n-2} = a$. One of the numbers a and b is greater than $k/2$.

If $b = a_{n-1} = a_{n-4} > \frac{k}{2}$, it follows from Lemma 1 that a_{n-1} divides a_{n-1-3s} when $3s < n - 1$, and therefore $a_{n-1-3s} = b$ when $3s < n-1$. When $6s < n-4$, k also divides $a_{n-4-6s}+a_{n-2-3s} = b+a_{n-2-3s}$, thus, $a_{n-2-3s} = k - b = a$.

If $a = a_{n-2} = a_{n-8} > \frac{k}{2}$, all the terms a_{n-2-6s} with $6s < n - 2$ are divisible by a, that is, the indices i for which a divides a_i form a progression with difference dividing 6. Since this difference is odd and greater than 1, it must be 3, that is, $a_{n-2-3s} = a$ when $3s < n - 2$. Similarly to the previous case, we have $a_{n-1-3s} = a_n - a_{n-2-6s} = k - a = b$ when $6s < n - 2$. □

Let a_n and a_{n+T} be two consecutive terms of the sequence equal to k. If n is large enough, $\frac{n+T}{2} < n - 2$, and applying the claim to $n + T$ instead of n we see that the three consecutive terms $a_{n-2} = a$, $a_{n-1} = b$, $a_n = k$ must be equal to a_{n+T-2}, a_{n+T-1} and a_{n+T} respectively. Thus, for some i we have $a_{i+3s} = a$ and $a_{i+1+3s} = b$ for all s. Truncating the sequence again if necessary, we may assume that $a_{3s+1} = a$ and $a_{3s+2} = b$ for all s. We know also that $a_n = k$ if and only if n is divisible by T (incidentally, this proves that T is divisible by 3).

If $a_{3s} = c$ for some integer s, each of the numbers a, b, c divides the sum of the other two. It is easy to see that these numbers are proportional to one of the triplets (1, 1, 1), (1, 1, 2) and (1, 2, 3) in some order. It follows that the greater of the two numbers a and b is the smaller multiplied by 2, 3 or 3/2. The last two cases are impossible because then c cannot be the maximum element in the triplet (a, b, c), while $c = k = a + b$ for infinitely many s. Thus the only possible case is 2, the numbers a and b are $k/3$ and $2k/3$ in some order, and the only possible values of c are k and $k/3$. Suppose that $a_{3s} = k/3$ for some $s > 1$. We can choose s so that $a_{3s+3} = k$. Therefore T, which we already know to be odd and divisible by 3, is greater than 3, that is, at least 9. Then $a_{3s-3} \neq k$, and the only other possibility is $a_{3s-3} = k/3$. However, $a_{3s+3} = k$ must divide $a_{3s} + a_{3s-3} = 2k/3$, which is impossible. We have proved then that $a_{3s} = k$ for all $s > 1$, which is the case (ii).

N8. For a polynomial $P(x)$ with integer coefficients let $P^1(x) = P(x)$ and $P^{k+1}(x) = P(P^k(x))$ for $k \geqslant 1$. Find all positive integers n for which there exists a polynomial $P(x)$ with integer coefficients such that for every integer $m \geqslant 1$, the numbers $P^m(1), \ldots, P^m(n)$ leave exactly $\lceil n/2^m \rceil$ distinct remainders when divided by n.

Answer: All powers of 2 and all primes.

Solution. Denote the set of residues modulo ℓ by $\mathbb{Z}_\ell$. Observe that P can be regarded as a function $\mathbb{Z}_\ell \to \mathbb{Z}_\ell$ for any positive integer ℓ. Denote the cardinality of the set $P^m(\mathbb{Z}_\ell)$ by $f_{m,\ell}$. Note that $f_{m,n} = \lceil n/2^m \rceil$ for all $m \geqslant 1$ if and only if $f_{m+1,n} = \lceil f_{m,n}/2 \rceil$ for all $m \geqslant 0$.

Part 1. The required polynomial exists when n is a power of 2 or a prime.

If n is a power of 2, set $P(x) = 2x$.

If $n = p$ is an odd prime, every function $f\colon \mathbb{Z}_p \to \mathbb{Z}_p$ coincides with some polynomial with integer coefficients. So we can pick the function that sends $x \in \{0, 1, \ldots, p-1\}$ to $\lfloor x/2 \rfloor$.

Part 2. The required polynomial does not exist when n is not a prime power.

Let $n = ab$ where $a, b > 1$ and $\gcd(a, b) = 1$. Note that, since $\gcd(a, b) = 1$,

$$f_{m,ab} = f_{m,a} f_{m,b}$$

by the Chinese remainder theorem. Also, note that, if $f_{m,\ell} = f_{m+1,\ell}$, then P permutes the image of P^m on $\mathbb{Z}_\ell$, and therefore $f_{s,\ell} = f_{m,\ell}$ for all $s > m$. So, as $f_{m,ab} = 1$ for sufficiently large m, we have for each m

$$f_{m,a} > f_{m+1,a} \quad \text{or} \quad f_{m,a} = 1, \qquad f_{m,b} > f_{m+1,b} \quad \text{or} \quad f_{m,b} = 1.$$

Choose the smallest m such that $f_{m+1,a} = 1$ or $f_{m+1,b} = 1$. Without loss of generality assume that $f_{m+1,a} = 1$. Then $f_{m+1,ab} = f_{m+1,b} < f_{m,b} \leqslant f_{m,ab}/2 \leqslant f_{m+1,ab}$, a contradiction.

Part 3. The required polynomial does not exist when n is an odd prime power that is not a prime.

Let $n = p^k$, where $p \geqslant 3$ is prime and $k \geqslant 2$. For $r \in \mathbb{Z}_p$ let S_r denote the subset of $\mathbb{Z}_{p^k}$ consisting of numbers congruent to r modulo p. We denote the cardinality of a set S by $|S|$.

Claim. For any residue r modulo p, either $|P(S_r)| = p^{k-1}$ or $|P(S_r)| \leqslant p^{k-2}$.

Proof. Recall that $P(r+h) = P(r) + hP'(r) + h^2 Q(r, h)$, where Q is an integer polynomial.

If $p \mid P'(r)$, then $P(r + ps) \equiv P(r) \pmod{p^2}$, hence all elements of $P(S_r)$ are congruent modulo p^2. So in this case $|P(S_r)| \leqslant p^{k-2}$.

Now we show that $p \nmid P'(r)$ implies $|P(S_r)| = p^{k-1}$ for all k.

Suppose the contrary: $|P(S_r)| < p^{k-1}$ for some $k > 1$. Let us choose the smallest k for which this is so. To each residue in $P(S_r)$ we assign its residue modulo p^{k-1}; denote the resulting set by $\overline{P}(S, r)$. We have $|\overline{P}(S, r)| = p^{k-2}$ by virtue of minimality of k. Then $|P(S_r)| < p^{k-1} = p \cdot |\overline{P}(S, r)|$, that is, there is $u = P(x) \in P(S_r)$ ($x \equiv r \pmod{p}$) and $t \not\equiv 0 \pmod{p}$ such that $u + p^{k-1}t \notin P(S_r)$.

Note that $P(x + p^{k-1}s) \equiv u + p^{k-1} sP'(x) \pmod{p^k}$. Since $P(x + p^{k-1}s) \not\equiv u + p^{k-1}t \pmod{p^k}$, the congruence $p^{k-1}sP'(x) \equiv p^{k-1}t \pmod{p^k}$ has no solutions. So the congruence $sP'(x) \equiv t \pmod{p}$ has no solutions, which contradicts $p \nmid P'(r)$. □

Since the image of P^m consists of one element for sufficiently large m, we can take the smallest m such that $|P^{m-1}(S_r)| = p^{k-1}$ for some $r \in \mathbb{Z}_p$, but $|P^m(S_q)| \leqslant p^{k-2}$ for all $q \in \mathbb{Z}_p$.

From now on, we fix m and r.

Since the image of $P^{m-1}(\mathbb{Z}_{p^k})\backslash P^{m-1}(S_r)$ under P contains $P^m(\mathbb{Z}_{p^k})\backslash P^m(S_r)$, we have

$$a := |P^m(\mathbb{Z}_{p^k})\backslash P^m(S_r)| \leqslant |P^{m-1}(\mathbb{Z}_{p^k})\backslash P^{m-1}(S_r)|,$$

thus

$$a + p^{k-1} \leqslant f_{m-1,p^k} \leqslant 2f_{m,p^k} \leqslant 2p^{k-2} + 2a,$$

so

$$(p-2)p^{k-2} \leqslant a.$$

Since $f_{i,p} = 1$ for sufficiently large i, there is exactly one $t \in \mathbb{Z}_p$ such that $P(t) \equiv t \pmod p$. Moreover, as i increases, the cardinality of the set $\{s \in \mathbb{Z}_p \mid P^i(s) \equiv t \pmod p\}$ increases (strictly), until it reaches the value p. So either

$$|\{s \in \mathbb{Z}_p \mid P^{m-1}(s) \equiv t \pmod p\}| = p \quad \text{or} \quad |\{s \in \mathbb{Z}_p \mid P^{m-1}(s) \equiv t \pmod p\}| \geqslant m.$$

Therefore, either $f_{m-1,p} = 1$ or there exists a subset $X \subset \mathbb{Z}_p$ of cardinality at least m such that $P^{m-1}(x) \equiv t \pmod p$ for all $x \in X$.

In the first case $|P^{m-1}(\mathbb{Z}_{p^k})| \leqslant p^{k-1} = |P^{m-1}(S_r)|$, so $a = 0$, a contradiction.

In the second case let Y be the set of all elements of $\mathbb{Z}_{p^k}$ congruent to some element of X modulo p. Let $Z = \mathbb{Z}_{p^k}\backslash Y$. Then $P^{m-1}(Y) \subset S_t$, $P(S_t) \subsetneq S_t$, and $Z = \bigcup_{i \in \mathbb{Z}_p \backslash X} S_i$, so

$$|P^m(Y)| \leqslant |P(S_t)| \leqslant p^{k-2} \quad \text{and} \quad |P^m(Z)| \leqslant |\mathbb{Z}_p\backslash X| \cdot p^{k-2} \leqslant (p-m)p^{k-2}.$$

Hence,

$$(p-2)p^{k-2} \leqslant a < |P^m(\mathbb{Z}_{p^k})| \leqslant |P^m(Y)| + |P^m(Z)| \leqslant (p-m+1)p^{k-2}$$

and $m < 3$. Then $|P^2(S_q)| \leqslant p^{k-2}$ for all $q \in \mathbb{Z}_p$, so

$$p^k/4 \leqslant |P^2(\mathbb{Z}_{p^k})| \leqslant p^{k-1},$$

which is impossible for $p \geqslant 5$. It remains to consider the case $p = 3$.

As before, let t be the only residue modulo 3 such that $P(t) \equiv t \pmod 3$.

If $3 \nmid P'(t)$, then $P(S_t) = S_t$ by the proof of the Claim above, which is impossible.

So $3 \mid P'(t)$. By substituting $h = 3^i s$ into the formula $P(t+h) = P(t) + hP'(t) + h^2Q(t,h)$, we obtain $P(t+3^i s) \equiv P(t) \pmod{3^{i+1}}$. Using induction on i we see that all elements of $P^i(S_t)$ are congruent modulo 3^{i+1}. Thus, $|P^{k-1}(S_t)| = 1$.

Note that $f_{1,3} \leqslant 2$ and $f_{2,3} \leqslant 1$, so $P^2(\mathbb{Z}_{3^k}) \subset S_t$. Therefore, $|P^{k+1}(\mathbb{Z}_{3^k})| \leqslant |P^{k-1}(S_t)| = 1$. It follows that $3^k \leqslant 2^{k+1}$, which is impossible for $k \geqslant 2$.

Comment. Here is an alternative version of the problem.

A function $f\colon \mathbb{Z} \to \mathbb{Z}$ is chosen so that $a - b \mid f(a) - f(b)$ for all $a, b \in \mathbb{Z}$ with $a \neq b$. Let $S_0 = \mathbb{Z}$, and for each positive integer m, let S_m denote the image of f on the set S_{m-1}. It is given that, for each nonnegative integer m, there are exactly $\lceil n/2^m \rceil$ distinct residues modulo n in the set S_m. Find all possible values of n.

Answer: All powers of primes.

Solution. Observe that f can be regarded as a function $\mathbb{Z}_\ell \to \mathbb{Z}_\ell$ for any positive integer ℓ. We use notations f^m and $f_{m,\ell}$ as in the above solution.

Part 1. There exists a function $f\colon \mathbb{Z}_{p^k} \to \mathbb{Z}_{p^k}$ satisfying the desired properties.

For $x \in \mathbb{Z}_{p^k}$, let $\mathrm{rev}(x)$ denote the reversal of the base-p digits of x (we write every $x \in \mathbb{Z}_{p^k}$ with exactly k digits, adding zeroes at the beginning if necessary). Choose

$$f(x) = \mathrm{rev}\left(\left\lfloor \frac{\mathrm{rev}(x)}{2} \right\rfloor\right)$$

where, for dividing by 2, $\mathrm{rev}(x)$ is interpreted as an integer in the range $[0, p^k)$. It is easy to see that $f_{m+1,k} = \lceil f_{m,k}/2 \rceil$.

We claim that if $a, b \in \mathbb{Z}_{p^k}$ so that $p^m | a - b$, then $p^m | f(a) - f(b)$. Let $x = \mathrm{rev}(a)$, $y = \mathrm{rev}(b)$. The first m digits of x and y are the same, i.e $\lfloor x/p^{m-k} \rfloor = \lfloor y/p^{m-k} \rfloor$. For every positive integers c, d and z we have $\lfloor \lfloor z/c \rfloor / d \rfloor = \lfloor z/(cd) \rfloor = \lfloor \lfloor z/d \rfloor / c \rfloor$, so

$$\left\lfloor \lfloor x/2 \rfloor / p^{m-k} \right\rfloor = \left\lfloor \lfloor x/p^{m-k} \rfloor / 2 \right\rfloor = \left\lfloor \lfloor y/p^{m-k} \rfloor / 2 \right\rfloor = \left\lfloor \lfloor y/2 \rfloor / p^{m-k} \right\rfloor.$$

Thus, the first m digits of $\lfloor x/2 \rfloor$ and $\lfloor y/2 \rfloor$ are the same. So the last m digits of $f(a)$ and $f(b)$ are the same, i.e. $p^m | f(a) - f(b)$.

Part 2. Lifting the function $f\colon \mathbb{Z}_{p^k} \to \mathbb{Z}_{p^k}$ to a function on all of $\mathbb{Z}$.

We show that, for any function $f\colon \mathbb{Z}_{p^k} \to \mathbb{Z}_{p^k}$ for which $\gcd(p^k, a - b) \mid f(a) - f(b)$, there is a corresponding function $g\colon \mathbb{Z} \to \mathbb{Z}$ for which $a - b \mid g(a) - g(b)$ for all distinct integers a, b and $g(x) \equiv f(x) \pmod{p^k}$ for all $x \in \mathbb{Z}$, whence the proof will be completed. We will construct the values of such a function inductively; assume that we have constructed it for some interval $[a, b)$ and wish to define $g(b)$. (We will define $g(a-1)$ similarly.)

For every prime $q \leqslant |a - b|$, we choose the maximal α_q for which there exists $c_q \in [a, b)$, such that $b - c_q \vdots q^{\alpha_q}$, and choose one such c_q.

We apply Chinese remainder theorem to find $g(b)$ satisfying the following conditions:

$$g(b) \equiv g(c_q) \pmod{q^{\alpha_q}} \quad \text{for} \quad q \neq p, \quad \text{and}$$
$$g(b) \equiv g(c_p) \pmod{q^{\alpha_p}} \quad \text{if} \quad \alpha_p \geqslant k, \qquad g(b) \equiv f(b) \pmod{p^k} \quad \text{if} \quad \alpha_p < k.$$

It is not hard to verify that $b - c \mid g(b) - g(c)$ for every $c \in [a, b)$ and $g(b) \equiv f(b) \pmod{p^k}$.

Part 3. The required function does not exist if n has at least two different prime divisors.

The proof is identical to the polynomial version.

IMO 2020

61st International
Mathematical
Olympiad
Saint Petersburg
Russia

SHORTLISTED PROBLEMS

(with solutions)

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Shortlisted Problems

(with solutions)

61st International Mathematical Olympiad

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Note of Confidentiality

The Shortlist has to be kept strictly confidential until the conclusion of the following International Mathematical Olympiad.

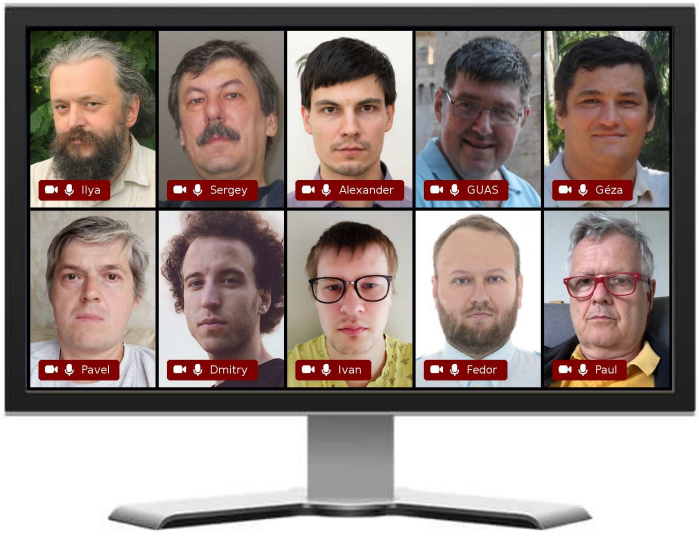
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Contributing Countries

The Organising Committee and the Problem Selection Committee of IMO 2020 thank the following 39 countries for contributing 149 problem proposals:

Armenia, Australia, Austria, Belgium, Brazil, Canada, Croatia, Cuba, Cyprus, Czech Republic, Denmark, Estonia, France, Georgia, Germany, Hong Kong, Hungary, India, Iran, Ireland, Israel, Kosovo, Latvia, Luxembourg, Mongolia, Netherlands, North Macedonia, Philippines, Poland, Slovakia, Slovenia, South Africa, Taiwan, Thailand, Trinidad and Tobago, Ukraine, United Kingdom, USA, Venezuela

Problem Selection Committee



Ilya I. Bogdanov, Sergey Berlov, Alexander Gaifullin, Alexander S. Golovanov, Géza Kós, Pavel Kozhevnikov, Dmitry Krachun, Ivan Mitrofanov, Fedor Petrov, Paul Vaderlind

Problems

Algebra

A1. *Version 1.* Let n be a positive integer, and set $N = 2^n$. Determine the smallest real number a_n such that, for all real x,

$$\sqrt[N]{\frac{x^{2N}+1}{2}} \leqslant a_n(x-1)^2 + x.$$

Version 2. For every positive integer N, determine the smallest real number b_N such that, for all real x,

$$\sqrt[N]{\frac{x^{2N}+1}{2}} \leqslant b_N(x-1)^2 + x.$$

(Ireland)

A2. Let $\mathcal{A}$ denote the set of all polynomials in three variables x, y, z with integer coefficients. Let $\mathcal{B}$ denote the subset of $\mathcal{A}$ formed by all polynomials which can be expressed as

$$(x+y+z)P(x,y,z) + (xy+yz+zx)Q(x,y,z) + xyzR(x,y,z)$$

with $P, Q, R \in \mathcal{A}$. Find the smallest non-negative integer n such that $x^i y^j z^k \in \mathcal{B}$ for all non-negative integers i, j, k satisfying $i + j + k \geqslant n$.

(Venezuela)

A3. Suppose that a, b, c, d are positive real numbers satisfying $(a+c)(b+d) = ac + bd$. Find the smallest possible value of

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

(Israel)

A4. Let a, b, c, d be four real numbers such that $a \geqslant b \geqslant c \geqslant d > 0$ and $a + b + c + d = 1$. Prove that

$$(a + 2b + 3c + 4d)\, a^a\, b^b\, c^c\, d^d < 1.$$

(Belgium)

A5. A magician intends to perform the following trick. She announces a positive integer n, along with $2n$ real numbers $x_1 < \ldots < x_{2n}$, to the audience. A member of the audience then secretly chooses a polynomial $P(x)$ of degree n with real coefficients, computes the $2n$ values $P(x_1), \ldots, P(x_{2n})$, and writes down these $2n$ values on the blackboard in non-decreasing order. After that the magician announces the secret polynomial to the audience.

Can the magician find a strategy to perform such a trick?

(Luxembourg)

A6. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$f^{a^2+b^2}(a+b) = af(a) + bf(b) \qquad \text{for every } a, b \in \mathbb{Z}.$$

Here, f^n denotes the nth iteration of f, i.e., $f^0(x) = x$ and $f^{n+1}(x) = f(f^n(x))$ for all $n \geqslant 0$.

(Slovakia)

A7. Let n and k be positive integers. Prove that for $a_1, \ldots, a_n \in [1, 2^k]$ one has

$$\sum_{i=1}^{n} \frac{a_i}{\sqrt{a_1^2 + \ldots + a_i^2}} \leqslant 4\sqrt{kn}.$$

(Iran)

A8. Let $\mathbb{R}^+$ be the set of positive real numbers. Determine all functions $f\colon \mathbb{R}^+ \to \mathbb{R}^+$ such that, for all positive real numbers x and y,

$$f\big(x + f(xy)\big) + y = f(x)f(y) + 1.$$

(Ukraine)

Combinatorics

C1. Let n be a positive integer. Find the number of permutations $a_1, a_2, \ldots, a_n$ of the sequence $1, 2, \ldots, n$ satisfying

$$a_1 \leqslant 2a_2 \leqslant 3a_3 \leqslant \ldots \leqslant na_n.$$

(United Kingdom)

C2. In a regular 100-gon, 41 vertices are colored black and the remaining 59 vertices are colored white. Prove that there exist 24 convex quadrilaterals Q_1, ..., Q_{24} whose corners are vertices of the 100-gon, so that

- the quadrilaterals Q_1, ..., Q_{24} are pairwise disjoint, and
- every quadrilateral Q_i has three corners of one color and one corner of the other color.

(Austria)

C3. Let n be an integer with $n \geqslant 2$. On a slope of a mountain, n^2 checkpoints are marked, numbered from 1 to n^2 from the bottom to the top. Each of two cable car companies, A and B, operates k cable cars numbered from 1 to k; each cable car provides a transfer from some checkpoint to a higher one. For each company, and for any i and j with $1 \leqslant i < j \leqslant k$, the starting point of car j is higher than the starting point of car i; similarly, the finishing point of car j is higher than the finishing point of car i. Say that two checkpoints are *linked* by some company if one can start from the lower checkpoint and reach the higher one by using one or more cars of that company (no movement on foot is allowed).

Determine the smallest k for which one can guarantee that there are two checkpoints that are linked by each of the two companies.

(India)

C4. The Fibonacci numbers $F_0, F_1, F_2, \ldots$ are defined inductively by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \geqslant 1$. Given an integer $n \geqslant 2$, determine the smallest size of a set S of integers such that for every $k = 2, 3, \ldots, n$ there exist some $x, y \in S$ such that $x - y = F_k$.

(Croatia)

C5. Let p be an odd prime, and put $N = \frac{1}{4}(p^3 - p) - 1$. The numbers $1, 2, \ldots, N$ are painted arbitrarily in two colors, red and blue. For any positive integer $n \leqslant N$, denote by $r(n)$ the fraction of integers in $\{1, 2, \ldots, n\}$ that are red.

Prove that there exists a positive integer $a \in \{1, 2, \ldots, p-1\}$ such that $r(n) \neq a/p$ for all $n = 1, 2, \ldots, N$.

(Netherlands)

C6. $4n$ coins of weights $1, 2, 3, \ldots, 4n$ are given. Each coin is colored in one of n colors and there are four coins of each color. Show that all these coins can be partitioned into two sets with the same total weight, such that each set contains two coins of each color.

(Hungary)

C7. Consider any rectangular table having finitely many rows and columns, with a real number $a(r, c)$ in the cell in row r and column c. A pair (R, C), where R is a set of rows and C a set of columns, is called a *saddle pair* if the following two conditions are satisfied:

(i) For each row r', there is $r \in R$ such that $a(r, c) \geqslant a(r', c)$ for all $c \in C$;

(ii) For each column c', there is $c \in C$ such that $a(r, c) \leqslant a(r, c')$ for all $r \in R$.

A saddle pair (R, C) is called a *minimal pair* if for each saddle pair (R', C') with $R' \subseteq R$ and $C' \subseteq C$, we have $R' = R$ and $C' = C$.

Prove that any two minimal pairs contain the same number of rows.

(Thailand)

C8. Players A and B play a game on a blackboard that initially contains 2020 copies of the number 1. In every round, player A erases two numbers x and y from the blackboard, and then player B writes one of the numbers $x + y$ and $|x - y|$ on the blackboard. The game terminates as soon as, at the end of some round, one of the following holds:

(1) one of the numbers on the blackboard is larger than the sum of all other numbers;

(2) there are only zeros on the blackboard.

Player B must then give as many cookies to player A as there are numbers on the blackboard. Player A wants to get as many cookies as possible, whereas player B wants to give as few as possible. Determine the number of cookies that A receives if both players play optimally.

(Austria)

Geometry

G1. Let ABC be an isosceles triangle with $BC = CA$, and let D be a point inside side AB such that $AD < DB$. Let P and Q be two points inside sides BC and CA, respectively, such that $\angle DPB = \angle DQA = 90^\circ$. Let the perpendicular bisector of PQ meet line segment CQ at E, and let the circumcircles of triangles ABC and CPQ meet again at point F, different from C.

Suppose that P, E, F are collinear. Prove that $\angle ACB = 90^\circ$.

(Luxembourg)

G2. Let $ABCD$ be a convex quadrilateral. Suppose that P is a point in the interior of $ABCD$ such that

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC.$$

The internal bisectors of angles ADP and PCB meet at a point Q inside the triangle ABP. Prove that $AQ = BQ$.

(Poland)

G3. Let $ABCD$ be a convex quadrilateral with $\angle ABC > 90^\circ$, $\angle CDA > 90^\circ$, and $\angle DAB = \angle BCD$. Denote by E and F the reflections of A in lines BC and CD, respectively. Suppose that the segments AE and AF meet the line BD at K and L, respectively. Prove that the circumcircles of triangles BEK and DFL are tangent to each other.

(Slovakia)

G4. In the plane, there are $n \geqslant 6$ pairwise disjoint disks $D_1, D_2, \ldots, D_n$ with radii $R_1 \geqslant R_2 \geqslant \ldots \geqslant R_n$. For every $i = 1, 2, \ldots, n$, a point P_i is chosen in disk D_i. Let O be an arbitrary point in the plane. Prove that

$$OP_1 + OP_2 + \ldots + OP_n \geqslant R_6 + R_7 + \ldots + R_n.$$

(A disk is assumed to contain its boundary.)

(Iran)

G5. Let $ABCD$ be a cyclic quadrilateral with no two sides parallel. Let K, L, M, and N be points lying on sides AB, BC, CD, and DA, respectively, such that $KLMN$ is a rhombus with $KL \parallel AC$ and $LM \parallel BD$. Let ω_1, ω_2, ω_3, and ω_4 be the incircles of triangles ANK, BKL, CLM, and DMN, respectively. Prove that the internal common tangents to ω_1 and ω_3 and the internal common tangents to ω_2 and ω_4 are concurrent.

(Poland)

G6. Let I and I_A be the incenter and the A-excenter of an acute-angled triangle ABC with $AB < AC$. Let the incircle meet BC at D. The line AD meets BI_A and CI_A at E and F, respectively. Prove that the circumcircles of triangles AID and I_AEF are tangent to each other.

(Slovakia)

G7. Let P be a point on the circumcircle of an acute-angled triangle ABC. Let D, E, and F be the reflections of P in the midlines of triangle ABC parallel to BC, CA, and AB, respectively. Denote by ω_A, ω_B, and ω_C the circumcircles of triangles ADP, BEP, and CFP, respectively. Denote by ω the circumcircle of the triangle formed by the perpendicular bisectors of segments AD, BE and CF.

Show that ω_A, ω_B, ω_C, and ω have a common point.

(Denmark)

G8. Let Γ and I be the circumcircle and the incenter of an acute-angled triangle ABC. Two circles ω_B and ω_C passing through B and C, respectively, are tangent at I. Let ω_B meet the shorter arc AB of Γ and segment AB again at P and M, respectively. Similarly, let ω_C meet the shorter arc AC of Γ and segment AC again at Q and N, respectively. The rays PM and QN meet at X, and the tangents to ω_B and ω_C at B and C, respectively, meet at Y.

Prove that the points A, X, and Y are collinear.

(Netherlands)

G9. Prove that there exists a positive constant c such that the following statement is true:

Assume that n is an integer with $n \geqslant 2$, and let $\mathcal{S}$ be a set of n points in the plane such that the distance between any two distinct points in $\mathcal{S}$ is at least 1. Then there is a line ℓ separating $\mathcal{S}$ such that the distance from any point of $\mathcal{S}$ to ℓ is at least $cn^{-1/3}$.

(A line ℓ *separates* a point set $\mathcal{S}$ if some segment joining two points in $\mathcal{S}$ crosses ℓ.)

(Taiwan)

Number Theory

N1. Given a positive integer k, show that there exists a prime p such that one can choose distinct integers $a_1, a_2, \ldots, a_{k+3} \in \{1, 2, \ldots, p-1\}$ such that p divides $a_i a_{i+1} a_{i+2} a_{i+3} - i$ for all $i = 1, 2, \ldots, k$.

(South Africa)

N2. For each prime p, there is a kingdom of p-Landia consisting of p islands numbered 1, 2, ..., p. Two distinct islands numbered n and m are connected by a bridge if and only if p divides $(n^2 - m + 1)(m^2 - n + 1)$. The bridges may pass over each other, but cannot cross. Prove that for infinitely many p there are two islands in p-Landia not connected by a chain of bridges.

(Denmark)

N3. Let n be an integer with $n \geqslant 2$. Does there exist a sequence $(a_1, \ldots, a_n)$ of positive integers with not all terms being equal such that the arithmetic mean of every two terms is equal to the geometric mean of some (one or more) terms in this sequence?

(Estonia)

N4. For any odd prime p and any integer n, let $d_p(n) \in \{0, 1, \ldots, p-1\}$ denote the remainder when n is divided by p. We say that $(a_0, a_1, a_2, \ldots)$ is a *p-sequence*, if a_0 is a positive integer coprime to p, and $a_{n+1} = a_n + d_p(a_n)$ for $n \geqslant 0$.

(a) Do there exist infinitely many primes p for which there exist p-sequences $(a_0, a_1, a_2, \ldots)$ and $(b_0, b_1, b_2, \ldots)$ such that $a_n > b_n$ for infinitely many n, and $b_n > a_n$ for infinitely many n?

(b) Do there exist infinitely many primes p for which there exist p-sequences $(a_0, a_1, a_2, \ldots)$ and $(b_0, b_1, b_2, \ldots)$ such that $a_0 < b_0$, but $a_n > b_n$ for all $n \geqslant 1$?

(United Kingdom)

N5. Determine all functions f defined on the set of all positive integers and taking non-negative integer values, satisfying the three conditions:

(i) $f(n) \neq 0$ for at least one n;

(ii) $f(xy) = f(x) + f(y)$ for every positive integers x and y;

(iii) there are infinitely many positive integers n such that $f(k) = f(n-k)$ for all $k < n$.

(Croatia)

N6. For a positive integer n, let $d(n)$ be the number of positive divisors of n, and let $\varphi(n)$ be the number of positive integers not exceeding n which are coprime to n. Does there exist a constant C such that

$$\frac{\varphi(d(n))}{d(\varphi(n))} \leqslant C$$

for all $n \geqslant 1$?

(Cyprus)

N7. Let $\mathcal{S}$ be a set consisting of $n \geqslant 3$ positive integers, none of which is a sum of two other distinct members of $\mathcal{S}$. Prove that the elements of $\mathcal{S}$ may be ordered as $a_1, a_2, \ldots, a_n$ so that a_i does not divide $a_{i-1} + a_{i+1}$ for all $i = 2, 3, \ldots, n-1$.

(Ukraine)

Solutions

Algebra

A1. *Version 1.* Let n be a positive integer, and set $N = 2^n$. Determine the smallest real number a_n such that, for all real x,

$$\sqrt[N]{\frac{x^{2N}+1}{2}} \leqslant a_n(x-1)^2 + x.$$

Version 2. For every positive integer N, determine the smallest real number b_N such that, for all real x,

$$\sqrt[N]{\frac{x^{2N}+1}{2}} \leqslant b_N(x-1)^2 + x.$$

(Ireland)

Answer for both versions : $a_n = b_N = N/2$.

Solution 1 (for Version 1). First of all, assume that $a_n < N/2$ satisfies the condition. Take $x = 1 + t$ for $t > 0$, we should have

$$\frac{(1+t)^{2N}+1}{2} \leqslant (1 + t + a_n t^2)^N.$$

Expanding the brackets we get

$$(1 + t + a_n t^2)^N - \frac{(1+t)^{2N}+1}{2} = \left(Na_n - \frac{N^2}{2}\right) t^2 + c_3 t^3 + \ldots + c_{2N} t^{2N} \tag{1}$$

with some coefficients $c_3, \ldots, c_{2N}$. Since $a_n < N/2$, the right hand side of (1) is negative for sufficiently small t. A contradiction.

It remains to prove the following inequality

$$\sqrt[N]{\frac{1+x^{2N}}{2}} \leqslant x + \frac{N}{2}(x-1)^2, \tag{$\mathcal{I}(N,x)$}$$

where $N = 2^n$.

Use induction in n. The base case $n = 0$ is trivial: $N = 1$ and both sides of $\mathcal{I}(N,x)$ are equal to $(1 + x^2)/2$. For completing the induction we prove $\mathcal{I}(2N,x)$ assuming that $\mathcal{I}(N,y)$ is established for all real y. We have

$$\begin{aligned}\left(x + N(x-1)^2\right)^2 &= x^2 + N^2(x-1)^4 + N(x-1)^2\frac{(x+1)^2-(x-1)^2}{2} \\ &= x^2 + \frac{N}{2}(x^2-1)^2 + \left(N^2 - \frac{N}{2}\right)(x-1)^4 \geqslant x^2 + \frac{N}{2}(x^2-1)^2 \geqslant \sqrt[N]{\frac{1+x^{4N}}{2}},\end{aligned}$$

where the last inequality is $\mathcal{I}(N, x^2)$. Since

$$x + N(x-1)^2 \geqslant x + \frac{(x-1)^2}{2} = \frac{x^2+1}{2} \geqslant 0,$$

taking square root we get $\mathcal{I}(2N,x)$. The inductive step is complete.

Solution 2.1 (for Version 2). Like in Solution 1 of Version 1, we conclude that $b_N \geqslant N/2$. It remains to prove the inequality $\mathcal{I}(N, x)$ for an arbitrary positive integer N.

First of all, $\mathcal{I}(N, 0)$ is obvious. Further, if $x > 0$, then the left hand sides of $\mathcal{I}(N, -x)$ and $\mathcal{I}(N, x)$ coincide, while the right hand side of $\mathcal{I}(N, -x)$ is larger than that of $\mathcal{I}(N, -x)$ (their difference equals $2(N-1)x \geqslant 0$). Therefore, $\mathcal{I}(N, -x)$ follows from $\mathcal{I}(N, x)$. So, hereafter we suppose that $x > 0$.

Divide $\mathcal{I}(N, x)$ by x and let $t = (x-1)^2/x = x - 2 + 1/x$; then $\mathcal{I}(n, x)$ reads as

$$f_N := \frac{x^N + x^{-N}}{2} \leqslant \left(1 + \frac{N}{2}t\right)^N. \tag{2}$$

The key identity is the expansion of f_N as a polynomial in t:

Lemma.

$$f_N = N \sum_{k=0}^{N} \frac{1}{N+k} \binom{N+k}{2k} t^k. \tag{3}$$

Proof. Apply induction on N. We will make use of the straightforward recurrence relation

$$f_{N+1} + f_{N-1} = (x + 1/x) f_N = (2 + t) f_N. \tag{4}$$

The base cases $N = 1, 2$ are straightforward:

$$f_1 = 1 + \frac{t}{2}, \quad f_2 = \frac{1}{2}t^2 + 2t + 1.$$

For the induction step from $N-1$ and N to $N+1$, we compute the coefficient of t^k in f_{N+1} using the formula $f_{N+1} = (2+t)f_N - f_{N-1}$. For $k = 0$ that coefficient equals 1, for $k > 0$ it equals

$$\begin{aligned}
&2\frac{N}{N+k}\binom{N+k}{2k} + \frac{N}{N+k-1}\binom{N+k-1}{2k-2} - \frac{N-1}{N+k-1}\binom{N+k-1}{2k} \\
&= \frac{(N+k-1)!}{(2k)!(N-k)!}\left(2N + \frac{2k(2k-1)N}{(N+k-1)(N-k+1)} - \frac{(N-1)(N-k)}{N+k-1}\right) \\
&= \frac{(N+k-1)!}{(2k)!(N-k+1)!}\left(2N(N-k+1) + 3kN + k - N^2 - N\right) = \frac{\binom{N+k+1}{2k}}{(N+k+1)}(N+1),
\end{aligned}$$

that completes the induction. □

Turning back to the problem, in order to prove (2) we write

$$\left(1 + \frac{N}{2}t\right)^N - f_N = \left(1 + \frac{N}{2}t\right)^N - N \sum_{k=0}^{N} \frac{1}{N+k}\binom{N+k}{2k} t^k = \sum_{k=0}^{N} \alpha_k t^k,$$

where

$$\begin{aligned}
\alpha_k &= \left(\frac{N}{2}\right)^k \binom{N}{k} - \frac{N}{N+k}\binom{N+k}{2k} \\
&= \left(\frac{N}{2}\right)^k \binom{N}{k}\left(1 - 2^k \frac{(1+1/N)(1+2/N)\cdot\ldots\cdot(1+(k-1)/N)}{(k+1)\cdot\ldots\cdot(2k)}\right) \\
&\geqslant \left(\frac{N}{2}\right)^k \binom{N}{k}\left(1 - 2^k \frac{2\cdot 3\cdot\ldots\cdot k}{(k+1)\cdot\ldots\cdot(2k)}\right) = \left(\frac{N}{2}\right)^k \binom{N}{k}\left(1 - \prod_{j=1}^{k}\frac{2j}{k+j}\right) \geqslant 0,
\end{aligned}$$

and (2) follows.

Solution 2.2 (for Version 2). Here we present another proof of the inequality (2) for $x > 0$, or, equivalently, for $t = (x-1)^2/x \geqslant 0$. Instead of finding the coefficients of the polynomial $f_N = f_N(t)$ we may find its roots, which is in a sense more straightforward. Note that the recurrence (4) and the initial conditions $f_0 = 1$, $f_1 = 1 + t/2$ imply that f_N is a polynomial in t of degree N. It also follows by induction that $f_N(0) = 1$, $f_N'(0) = N^2/2$: the recurrence relations read as $f_{N+1}(0) + f_{N-1}(0) = 2f_N(0)$ and $f_{N+1}'(0) + f_{N-1}'(0) = 2f_N'(0) + f_N(0)$, respectively.

Next, if $x_k = \exp(\frac{i\pi(2k-1)}{2N})$ for $k \in \{1, 2, \ldots, N\}$, then

$$-t_k := 2 - x_k - \frac{1}{x_k} = 2 - 2\cos\frac{\pi(2k-1)}{2N} = 4\sin^2\frac{\pi(2k-1)}{4N} > 0$$

and

$$f_N(t_k) = \frac{x_k^N + x_k^{-N}}{2} = \frac{\exp\left(\frac{i\pi(2k-1)}{2}\right) + \exp\left(-\frac{i\pi(2k-1)}{2}\right)}{2} = 0.$$

So the roots of f_N are $t_1, \ldots, t_N$ and by the AM–GM inequality we have

$$f_N(t) = \left(1 - \frac{t}{t_1}\right)\left(1 - \frac{t}{t_2}\right)\ldots\left(1 - \frac{t}{t_N}\right) \leqslant \left(1 - \frac{t}{N}\left(\frac{1}{t_1} + \ldots + \frac{1}{t_n}\right)\right)^N = \left(1 + \frac{tf_N'(0)}{N}\right)^N = \left(1 + \frac{N}{2}t\right)^N.$$

Comment. The polynomial $f_N(t)$ equals to $\frac{1}{2}T_N(t+2)$, where T_n is the nth Chebyshev polynomial of the first kind: $T_n(2\cos s) = 2\cos ns$, $T_n(x + 1/x) = x^n + 1/x^n$.

Solution 2.3 (for Version 2). Here we solve the problem when $N \geqslant 1$ is an arbitrary real number. For a real number a let

$$f(x) = \left(\frac{x^{2N}+1}{2}\right)^{\frac{1}{N}} - a(x-1)^2 - x.$$

Then $f(1) = 0$,

$$f'(x) = \left(\frac{x^{2N}+1}{2}\right)^{\frac{1}{N}-1} x^{2N-1} - 2a(x-1) - 1 \quad \text{and} \quad f'(1) = 0;$$

$$f''(x) = (1-N)\left(\frac{x^{2N}+1}{2}\right)^{\frac{1}{N}-2} x^{4N-2} + (2N-1)\left(\frac{x^{2N}+1}{2}\right)^{\frac{1}{N}-1} x^{2N-2} - 2a \quad \text{and} \quad f''(1) = N - 2a.$$

So if $a < \frac{N}{2}$, the function f has a strict local minimum at point 1, and the inequality $f(x) \leqslant 0 = f(1)$ does not hold. This proves $b_N \geqslant N/2$.

For $a = \frac{N}{2}$ we have $f''(1) = 0$ and

$$f'''(x) = \frac{1}{2}(1-N)(1-2N)\left(\tfrac{x^{2N}+1}{2}\right)^{\frac{1}{N}-3} x^{2N-3}(1-x^{2N}) \quad \begin{cases} > 0 & \text{if } 0 < x < 1 \text{ and} \\ < 0 & \text{if } x > 1. \end{cases}$$

Hence, $f''(x) < 0$ for $x \neq 1$; $f'(x) > 0$ for $x < 1$ and $f'(x) < 0$ for $x > 1$, finally $f(x) < 0$ for $x \neq 1$.

Comment. Version 2 is much more difficult, of rather A5 or A6 difficulty. The induction in Version 1 is rather straightforward, while all three above solutions of Version 2 require some creativity.

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A2. Let $\mathcal{A}$ denote the set of all polynomials in three variables x, y, z with integer coefficients. Let $\mathcal{B}$ denote the subset of $\mathcal{A}$ formed by all polynomials which can be expressed as

$$(x+y+z)P(x,y,z)+(xy+yz+zx)Q(x,y,z)+xyzR(x,y,z)$$

with $P, Q, R \in \mathcal{A}$. Find the smallest non-negative integer n such that $x^i y^j z^k \in \mathcal{B}$ for all non-negative integers i, j, k satisfying $i+j+k \geqslant n$.

(Venezuela)

Answer: $n = 4$.

Solution. We start by showing that $n \leqslant 4$, i.e., any monomial $f = x^i y^j z^k$ with $i+j+k \geqslant 4$ belongs to $\mathcal{B}$. Assume that $i \geqslant j \geqslant k$, the other cases are analogous.

Let $x+y+z = p$, $xy+yz+zx = q$ and $xyz = r$. Then

$$0 = (x-x)(x-y)(x-z) = x^3 - px^2 + qx - r,$$

therefore $x^3 \in \mathcal{B}$. Next, $x^2y^2 = xyq - (x+y)r \in \mathcal{B}$.

If $k \geqslant 1$, then r divides f, thus $f \in \mathcal{B}$. If $k = 0$ and $j \geqslant 2$, then x^2y^2 divides f, thus we have $f \in \mathcal{B}$ again. Finally, if $k = 0, j \leqslant 1$, then x^3 divides f and $f \in \mathcal{B}$ in this case also.

In order to prove that $n \geqslant 4$, we show that the monomial x^2y does not belong to $\mathcal{B}$. Assume the contrary:

$$x^2y = pP + qQ + rR \tag{1}$$

for some polynomials P, Q, R. If polynomial P contains the monomial x^2 (with nonzero coefficient), then $pP+qQ+rR$ contains the monomial x^3 with the same nonzero coefficient. So P does not contain x^2, y^2, z^2 and we may write

$$x^2y = (x+y+z)(axy+byz+czx)+(xy+yz+zx)(dx+ey+fz)+gxyz,$$

where a, b, c; d, e, f; g are the coefficients of xy, yz, zx; x, y, z; xyz in the polynomials P; Q; R, respectively (the remaining coefficients do not affect the monomials of degree 3 in $pP+qQ+rR$). By considering the coefficients of xy^2 we get $e = -a$, analogously $e = -b$, $f = -b$, $f = -c$, $d = -c$, thus $a = b = c$ and $f = e = d = -a$, but then the coefficient of x^2y in the right hand side equals $a + d = 0 \neq 1$.

Comment 1. The general question is the following. Call a polynomial $f(x_1, \ldots, x_n)$ with integer coefficients *nice*, if $f(0, 0, \ldots, 0) = 0$ and $f(x_{\pi_1}, \ldots, x_{\pi_n}) = f(x_1, \ldots, x_n)$ for any permutation π of $1, \ldots, n$ (in other words, f is symmetric and its constant term is zero.) Denote by $\mathcal{I}$ the set of polynomials of the form

$$p_1q_1 + p_2q_2 + \ldots + p_mq_m, \tag{2}$$

where m is an integer, $q_1, \ldots, q_m$ are polynomials with integer coefficients, and $p_1, \ldots, p_m$ are nice polynomials. Find the least N for which any monomial of degree at least N belongs to $\mathcal{I}$.

The **answer** is $n(n-1)/2+1$. The lower bound follows from the following claim: the polynomial

$$F(x_1, \ldots, x_n) = x_2x_3^2x_4^3 \cdot \ldots \cdot x_n^{n-1}$$

does not belong to $\mathcal{I}$.

Assume that $F = \sum p_iq_i$, according to (2). By taking only the monomials of degree $n(n-1)/2$, we can additionally assume that every p_i and every q_i is homogeneous, $\deg p_i > 0$, and $\deg p_i + \deg q_i = \deg F = n(n-1)/2$ for all i.

Consider the alternating sum

$$\sum_{\pi} \operatorname{sign}(\pi) F(x_{\pi_1}, \ldots, x_{\pi_n}) = \sum_{i=1}^{m} p_i \sum_{\pi} \operatorname{sign}(\pi) q_i(x_{\pi_1}, \ldots, x_{\pi_n}) := S, \tag{3}$$

where the summation is done over all permutations π of $1, \dots n$, and $\operatorname{sign}(\pi)$ denotes the sign of the permutation π. Since $\deg q_i = n(n-1)/2 - \deg p_i < n(n-1)/2$, in any monomial Q of q_i, there are at least two variables, say x_α and x_β, with equal exponents. Therefore $\sum_\pi \operatorname{sign}(\pi) Q(x_{\pi_1}, \dots, x_{\pi_n}) = 0$, because each pair of terms that corresponds to permutations which differ by the transposition of α and β, cancels out. This holds for any $i = 1, \dots, m$ and any monomial of q_i, so $S = 0$. But the left hand side of (3) is a non-zero polynomial. This is a contradiction.

Let us now prove, using induction on n, that any monomial $h = x_1^{c_1} \dots x_n^{c_n}$ of degree $n(n-1)/2+1$ belongs to $\mathcal{I}$, and additionally all p_i, q_i in the representation (2) can be chosen homogeneous with sum of degrees equal to $n(n-1)/2+1$. (Obviously, any monomial of degree at least $n(n-1)/2+1$ is divisible by a monomial of degree exactly $n(n-1)/2+1$, thus this suffices.) The proposition is true for $n = 1$, so assume that $n > 1$ and that the proposition is proved for smaller values of n.

We proceed by an internal induction on $S := |\{i \colon c_i = 0\}|$. In the base case $S = 0$ the monomial h is divisible by the nice polynomial $x_1 \cdot \ldots \cdot x_n$, therefore $h \in \mathcal{I}$. Now assume that $S > 0$ and that the claim holds for smaller values of S. Let $T = n - S$. We may assume that $c_{T+1} = \ldots = c_n = 0$ and $h = x_1 \cdot \ldots \cdot x_T g(x_1, \dots, x_{n-1})$, where $\deg g = n(n-1)/2 - T + 1 \geqslant (n-1)(n-2)/2 + 1$. Using the outer induction hypothesis we represent g as $p_1 q_1 + \ldots + p_m q_m$, where $p_i(x_1, \dots, x_{n-1})$ are nice polynomials in $n-1$ variables. There exist nice homogeneous polynomials $P_i(x_1, \dots, x_n)$ such that $P_i(x_1, \dots, x_{n-1}, 0) = p_i(x_1, \dots, x_{n-1})$. In other words, $\Delta_i := p_i(x_1, \dots, x_{n-1}) - P_i(x_1, \dots, x_{n-1}, x_n)$ is divisible by x_n, let $\Delta_i = x_n g_i$. We get

$$h = x_1 \cdot \ldots \cdot x_T \sum p_i q_i = x_1 \cdot \ldots \cdot x_T \sum (P_i + x_n g_i) q_i = (x_1 \cdot \ldots \cdot x_T x_n) \sum g_i q_i + \sum P_i q_i \in \mathcal{I}.$$

The first term belongs to $\mathcal{I}$ by the inner induction hypothesis. This completes both inductions.

Comment 2. The solutions above work smoothly for the versions of the original problem and its extensions to the case of n variables, where all polynomials are assumed to have real coefficients. In the version with integer coefficients, the argument showing that $x^2 y \notin \mathcal{B}$ can be simplified: it is not hard to show that in every polynomial $f \in \mathcal{B}$, the sum of the coefficients of $x^2 y$, $x^2 z$, $y^2 x$, $y^2 z$, $z^2 x$ and $z^2 y$ is even. A similar fact holds for any number of variables and also implies that $N \geqslant n(n-1)/2+1$ in terms of the previous comment.

A3. Suppose that a, b, c, d are positive real numbers satisfying $(a+c)(b+d) = ac+bd$. Find the smallest possible value of

$$S = \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

(Israel)

Answer: The smallest possible value is 8.

Solution 1. To show that $S \geqslant 8$, apply the AM–GM inequality twice as follows:

$$\left(\frac{a}{b}+\frac{c}{d}\right)+\left(\frac{b}{c}+\frac{d}{a}\right) \geqslant 2\sqrt{\frac{ac}{bd}}+2\sqrt{\frac{bd}{ac}} = \frac{2(ac+bd)}{\sqrt{abcd}} = \frac{2(a+c)(b+d)}{\sqrt{abcd}} \geqslant 2\cdot\frac{2\sqrt{ac}\cdot 2\sqrt{bd}}{\sqrt{abcd}} = 8\,.$$

The above inequalities turn into equalities when $a = c$ and $b = d$. Then the condition $(a+c)(b+d) = ac+bd$ can be rewritten as $4ab = a^2+b^2$. So it is satisfied when $a/b = 2\pm\sqrt{3}$. Hence, S attains value 8, e.g., when $a = c = 1$ and $b = d = 2+\sqrt{3}$.

Solution 2. By homogeneity we may suppose that $abcd = 1$. Let $ab = C$, $bc = A$ and $ca = B$. Then a, b, c can be reconstructed from A, B and C as $a = \sqrt{BC/A}$, $b = \sqrt{AC/B}$ and $c = \sqrt{AB/C}$. Moreover, the condition $(a+c)(b+d) = ac+bd$ can be written in terms of A, B, C as

$$A + \frac{1}{A} + C + \frac{1}{C} = bc + ad + ab + cd = (a+c)(b+d) = ac+bd = B + \frac{1}{B}.$$

We then need to minimize the expression

$$\begin{aligned} S := \frac{ad+bc}{bd} + \frac{ab+cd}{ac} &= \left(A+\frac{1}{A}\right)B + \left(C+\frac{1}{C}\right)\frac{1}{B} \\ &= \left(A+\frac{1}{A}\right)\left(B-\frac{1}{B}\right) + \left(A+\frac{1}{A}+C+\frac{1}{C}\right)\frac{1}{B} \\ &= \left(A+\frac{1}{A}\right)\left(B-\frac{1}{B}\right) + \left(B+\frac{1}{B}\right)\frac{1}{B}. \end{aligned}$$

Without loss of generality assume that $B \geqslant 1$ (otherwise, we may replace B by $1/B$ and swap A and C, this changes neither the relation nor the function to be maximized). Therefore, we can write

$$S \geqslant 2\left(B-\frac{1}{B}\right) + \left(B+\frac{1}{B}\right)\frac{1}{B} = 2B + \left(1-\frac{1}{B}\right)^2 =: f(B).$$

Clearly, f increases on $[1,\infty)$. Since

$$B + \frac{1}{B} = A + \frac{1}{A} + C + \frac{1}{C} \geqslant 4,$$

we have $B \geqslant B'$, where $B' = 2+\sqrt{3}$ is the unique root greater than 1 of the equation $B' + 1/B' = 4$. Hence,

$$S \geqslant f(B) \geqslant f(B') = 2\left(B'-\frac{1}{B'}\right) + \left(B'+\frac{1}{B'}\right)\frac{1}{B'} = 2B' - \frac{2}{B'} + \frac{4}{B'} = 8.$$

It remains to note that when $A = C = 1$ and $B = B'$ we have the equality $S = 8$.

Solution 3. We present another proof of the inequality $S \geqslant 8$. We start with the estimate

$$\left(\frac{a}{b}+\frac{c}{d}\right)+\left(\frac{b}{c}+\frac{d}{a}\right) \geqslant 2\sqrt{\frac{ac}{bd}}+2\sqrt{\frac{bd}{ac}}.$$

Let $y=\sqrt{ac}$ and $z=\sqrt{bd}$, and assume, without loss of generality, that $ac \geqslant bd$. By the AM–GM inequality, we have

$$y^2+z^2=ac+bd=(a+c)(b+d) \geqslant 2\sqrt{ac}\cdot 2\sqrt{bd}=4yz.$$

Substituting $x=y/z$, we get $4x \leqslant x^2+1$. For $x \geqslant 1$, this holds if and only if $x \geqslant 2+\sqrt{3}$.

Now we have

$$2\sqrt{\frac{ac}{bd}}+2\sqrt{\frac{bd}{ac}}=2\left(x+\frac{1}{x}\right).$$

Clearly, this is minimized by setting $x(\geqslant 1)$ as close to 1 as possible, i.e., by taking $x=2+\sqrt{3}$. Then $2(x+1/x)=2((2+\sqrt{3})+(2-\sqrt{3}))=8$, as required.

A4. Let a, b, c, d be four real numbers such that $a \geqslant b \geqslant c \geqslant d > 0$ and $a + b + c + d = 1$. Prove that

$$(a + 2b + 3c + 4d)\, a^a\, b^b\, c^c\, d^d < 1.$$

(Belgium)

Solution 1. The weighted AM–GM inequality with weights a, b, c, d gives

$$a^a b^b c^c d^d \leqslant a \cdot a + b \cdot b + c \cdot c + d \cdot d = a^2 + b^2 + c^2 + d^2,$$

so it suffices to prove that $(a + 2b + 3c + 4d)(a^2 + b^2 + c^2 + d^2) < 1 = (a + b + c + d)^3$. This can be done in various ways, for example:

$$\begin{aligned}(a + b + c + d)^3 &> a^2(a + 3b + 3c + 3d) + b^2(3a + b + 3c + 3d) \\ &\quad + c^2(3a + 3b + c + 3d) + d^2(3a + 3b + 3c + d) \\ &\geqslant (a^2 + b^2 + c^2 + d^2) \cdot (a + 2b + 3c + 4d).\end{aligned}$$

Solution 2. From $b \geqslant d$ we get

$$a + 2b + 3c + 4d \leqslant a + 3b + 3c + 3d = 3 - 2a.$$

If $a < \frac{1}{2}$, then the statement can be proved by

$$(a + 2b + 3c + 4d)\, a^a b^b c^c d^d \leqslant (3 - 2a) a^a a^b a^c a^d = (3 - 2a)a = 1 - (1 - a)(1 - 2a) < 1.$$

From now on we assume $\frac{1}{2} \leqslant a < 1$.

By $b, c, d < 1 - a$ we have

$$b^b c^c d^d < (1 - a)^b \cdot (1 - a)^c \cdot (1 - a)^d = (1 - a)^{1-a}.$$

Therefore,

$$(a + 2b + 3c + 4d) a^a b^b c^c d^d < (3 - 2a)\, a^a\, (1 - a)^{1-a}.$$

For $0 < x < 1$, consider the functions

$$f(x) = (3 - 2x) x^x (1 - x)^{1-x} \quad \text{and} \quad g(x) = \log f(x) = \log(3 - 2x) + x \log x + (1 - x) \log(1 - x);$$

hereafter, log denotes the natural logarithm. It is easy to verify that

$$g''(x) = -\frac{4}{(3 - 2x)^2} + \frac{1}{x} + \frac{1}{1 - x} = \frac{1 + 8(1 - x)^2}{x(1 - x)(3 - 2x)^2} > 0,$$

so g is strictly convex on $(0, 1)$.

By $g\left(\frac{1}{2}\right) = \log 2 + 2 \cdot \frac{1}{2} \log \frac{1}{2} = 0$ and $\lim\limits_{x \to 1-} g(x) = 0$, we have $g(x) \leqslant 0$ (and hence $f(x) \leqslant 1$) for all $x \in \left[\frac{1}{2}, 1\right)$, and therefore

$$(a + 2b + 3c + 4d) a^a b^b c^c d^d < f(a) \leqslant 1.$$

Comment. For a large number of variables $a_1 \geqslant a_2 \geqslant \ldots \geqslant a_n > 0$ with $\sum_i a_i = 1$, the inequality

$$\left(\sum_i i a_i\right) \prod_i a_i^{a_i} \leqslant 1$$

does not necessarily hold. Indeed, let $a_2 = a_3 = \ldots = a_n = \varepsilon$ and $a_1 = 1 - (n - 1)\varepsilon$, where n and $\varepsilon \in (0, 1/n)$ will be chosen later. Then

$$\left(\sum_i i a_i\right) \prod_i a_i^{a_i} = \left(1 + \frac{n(n - 1)}{2}\varepsilon\right) \varepsilon^{(n-1)\varepsilon} (1 - (n - 1)\varepsilon)^{1-(n-1)\varepsilon}. \tag{1}$$

If $\varepsilon = C/n^2$ with an arbitrary fixed $C > 0$ and $n \to \infty$, then the factors $\varepsilon^{(n-1)\varepsilon} = \exp((n - 1)\varepsilon \log \varepsilon)$ and $(1 - (n - 1)\varepsilon)^{1-(n-1)\varepsilon}$ tend to 1, so the limit of (1) in this set-up equals $1 + C/2$. This is not simply greater than 1, but it can be arbitrarily large.

A5. A magician intends to perform the following trick. She announces a positive integer n, along with $2n$ real numbers $x_1 < \ldots < x_{2n}$, to the audience. A member of the audience then secretly chooses a polynomial $P(x)$ of degree n with real coefficients, computes the $2n$ values $P(x_1), \ldots, P(x_{2n})$, and writes down these $2n$ values on the blackboard in non-decreasing order. After that the magician announces the secret polynomial to the audience.

Can the magician find a strategy to perform such a trick?

(Luxembourg)

Answer: No, she cannot.

Solution. Let $x_1 < x_2 < \ldots < x_{2n}$ be real numbers chosen by the magician. We will construct two distinct polynomials $P(x)$ and $Q(x)$, each of degree n, such that the member of audience will write down the same sequence for both polynomials. This will mean that the magician cannot distinguish P from Q.

Claim. There exists a polynomial $P(x)$ of degree n such that $P(x_{2i-1}) + P(x_{2i}) = 0$ for $i = 1, 2, \ldots, n$.

Proof. We want to find a polynomial $a_n x^n + \ldots + a_1 x + a_0$ satisfying the following system of equations:

$$\begin{cases} (x_1^n + x_2^n)a_n + (x_1^{n-1} + x_2^{n-1})a_{n-1} + \ldots + 2a_0 = 0 \\ (x_3^n + x_4^n)a_n + (x_3^{n-1} + x_4^{n-1})a_{n-1} + \ldots + 2a_0 = 0 \\ \ldots \\ (x_{2n-1}^n + x_{2n}^n)a_n + (x_{2n-1}^{n-1} + x_{2n}^{n-1})a_{n-1} + \ldots + 2a_0 = 0 \end{cases}$$

We use the well known fact that a homogeneous system of n linear equations in $n+1$ variables has a nonzero solution. (This fact can be proved using induction on n, via elimination of variables.) Applying this fact to the above system, we find a nonzero polynomial $P(x)$ of degree not exceeding n such that its coefficients $a_0, \ldots, a_n$ satisfy this system. Therefore $P(x_{2i-1}) + P(x_{2i}) = 0$ for all $i = 1, 2, \ldots, n$. Notice that P has a root on each segment $[x_{2i-1}, x_{2i}]$ by the Intermediate Value theorem, so n roots in total. Since P is nonzero, we get $\deg P = n$. □

Now consider a polynomial $P(x)$ provided by the Claim, and take $Q(x) = -P(x)$. The properties of $P(x)$ yield that $P(x_{2i-1}) = Q(x_{2i})$ and $Q(x_{2i-1}) = P(x_{2i})$ for all $i = 1, 2, \ldots, n$. It is also clear that $P \neq -P = Q$ and $\deg Q = \deg P = n$.

Comment. It can be shown that for any positive integer n the magician can choose $2n+1$ distinct real numbers so as to perform such a trick. Moreover, she can perform such a trick with almost all (in a proper sense) $(2n+1)$-tuples of numbers.

A6. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$f^{a^2+b^2}(a+b) = af(a) + bf(b) \qquad \text{for every } a, b \in \mathbb{Z}.$$

Here, f^n denotes the nth iteration of f, i.e., $f^0(x) = x$ and $f^{n+1}(x) = f(f^n(x))$ for all $n \geqslant 0$.

(Slovakia)

Answer: Either $f(x) = 0$ for all $x \in \mathbb{Z}$, or $f(x) = x + 1$ for all $x \in \mathbb{Z}$.

Solution. Refer to the main equation as $E(a, b)$.

$E(0, b)$ reads as $f^{b^2}(b) = bf(b)$. For $b = -1$ this gives $f(-1) = 0$.

Now $E(a, -1)$ reads as

$$f^{a^2+1}(a-1) = af(a) = f^{a^2}(a). \tag{1}$$

For $x \in \mathbb{Z}$ define the *orbit of* x by $\mathcal{O}(x) = \{x, f(x), f(f(x)), \ldots\} \subseteq \mathbb{Z}$. We see that the orbits $\mathcal{O}(a-1)$ and $\mathcal{O}(a)$ differ by finitely many terms. Hence, any two orbits differ by finitely many terms. In particular, this implies that either all orbits are finite or all orbits are infinite.

Case 1: All orbits are finite.

Then $\mathcal{O}(0)$ is finite. Using $E(a, -a)$ we get

$$a\big(f(a) - f(-a)\big) = af(a) - af(-a) = f^{2a^2}(0) \in \mathcal{O}(0).$$

For $|a| > \max\limits_{z \in \mathcal{O}(0)} |z|$, this yields $f(a) = f(-a)$ and $f^{2a^2}(0) = 0$. Therefore, the sequence $\big(f^k(0)\colon k = 0, 1, \ldots\big)$ is purely periodic with a minimal period T which divides $2a^2$. Analogously, T divides $2(a+1)^2$, therefore, $T \mid \gcd\big(2a^2, 2(a+1)^2\big) = 2$, i.e., $f(f(0)) = 0$ and $a\big(f(a) - f(-a)\big) = f^{2a^2}(0) = 0$ for all a. Thus,

$$f(a) = f(-a) \qquad \text{for all } a \neq 0; \tag{♣}$$

$$\text{in particular,} \quad f(1) = f(-1) = 0. \tag{♠}$$

Next, for each $n \in \mathbb{Z}$, by $E(n, 1-n)$ we get

$$nf(n) + (1-n)f(1-n) = f^{n^2+(1-n)^2}(1) = f^{2n^2-2n}(0) = 0. \tag{♡}$$

Assume that there exists some $m \neq 0$ such that $f(m) \neq 0$. Choose such an m for which $|m|$ is minimal possible. Then $|m| > 1$ due to (♠); $f(|m|) \neq 0$ due to (♣); and $f(1 - |m|) \neq 0$ due to (♡) for $n = |m|$. This contradicts to the minimality assumption.

So, $f(n) = 0$ for $n \neq 0$. Finally, $f(0) = f^3(0) = f^4(2) = 2f(2) = 0$. Clearly, the function $f(x) \equiv 0$ satisfies the problem condition, which provides the first of the two answers.

Case 2: All orbits are infinite.

Since the orbits $\mathcal{O}(a)$ and $\mathcal{O}(a-1)$ differ by finitely many terms for all $a \in \mathbb{Z}$, each two orbits $\mathcal{O}(a)$ and $\mathcal{O}(b)$ have infinitely many common terms for arbitrary $a, b \in \mathbb{Z}$.

For a minute, fix any $a, b \in \mathbb{Z}$. We claim that all pairs (n, m) of nonnegative integers such that $f^n(a) = f^m(b)$ have the same difference $n - m$. Arguing indirectly, we have $f^n(a) = f^m(b)$ and $f^p(a) = f^q(b)$ with, say, $n - m > p - q$, then $f^{p+m+k}(b) = f^{p+n+k}(a) = f^{q+n+k}(b)$, for all nonnegative integers k. This means that $f^{\ell+(n-m)-(p-q)}(b) = f^\ell(b)$ for all sufficiently large ℓ, i.e., that the sequence $\big(f^n(b)\big)$ is eventually periodic, so $\mathcal{O}(b)$ is finite, which is impossible.

Now, for every $a, b \in \mathbb{Z}$, denote the common difference $n - m$ defined above by $X(a, b)$. We have $X(a-1, a) = 1$ by (1). Trivially, $X(a, b) + X(b, c) = X(a, c)$, as if $f^n(a) = f^m(b)$ and $f^p(b) = f^q(c)$, then $f^{p+n}(a) = f^{p+m}(b) = f^{q+m}(c)$. These two properties imply that $X(a, b) = b - a$ for all $a, b \in \mathbb{Z}$.

But (1) yields $f^{a^2+1}(f(a-1)) = f^{a^2}(f(a))$, so

$$1 = X\big(f(a-1), f(a)\big) = f(a) - f(a-1) \quad \text{for all } a \in \mathbb{Z}.$$

Recalling that $f(-1) = 0$, we conclude by (two-sided) induction on x that $f(x) = x + 1$ for all $x \in \mathbb{Z}$.

Finally, the obtained function also satisfies the assumption. Indeed, $f^n(x) = x + n$ for all $n \geqslant 0$, so

$$f^{a^2+b^2}(a+b) = a + b + a^2 + b^2 = af(a) + bf(b).$$

Comment. There are many possible variations of the solution above, but it seems that finiteness of orbits seems to be a crucial distinction in all solutions. However, the case distinction could be made in different ways; in particular, there exist some versions of Case 1 which work whenever there is at least *one* finite orbit.

We believe that Case 2 is conceptually harder than Case 1.

A7. Let n and k be positive integers. Prove that for $a_1, \ldots, a_n \in [1, 2^k]$ one has

$$\sum_{i=1}^{n} \frac{a_i}{\sqrt{a_1^2 + \ldots + a_i^2}} \leqslant 4\sqrt{kn}.$$

(Iran)

Solution 1. Partition the set of indices $\{1, 2, \ldots, n\}$ into disjoint subsets $M_1, M_2, \ldots, M_k$ so that $a_\ell \in [2^{j-1}, 2^j]$ for $\ell \in M_j$. Then, if $|M_j| =: p_j$, we have

$$\sum_{\ell \in M_j} \frac{a_\ell}{\sqrt{a_1^2 + \ldots + a_\ell^2}} \leqslant \sum_{i=1}^{p_j} \frac{2^j}{2^{j-1}\sqrt{i}} = 2\sum_{i=1}^{p_j} \frac{1}{\sqrt{i}},$$

where we used that $a_\ell \leqslant 2^j$ and in the denominator every index from M_j contributes at least $(2^{j-1})^2$. Now, using $\sqrt{i} - \sqrt{i-1} = \frac{1}{\sqrt{i}+\sqrt{i-1}} \geqslant \frac{1}{2\sqrt{i}}$, we deduce that

$$\sum_{\ell \in M_j} \frac{a_\ell}{\sqrt{a_1^2 + \ldots + a_\ell^2}} \leqslant 2\sum_{i=1}^{p_j} \frac{1}{\sqrt{i}} \leqslant 2\sum_{i=1}^{p_j} 2(\sqrt{i} - \sqrt{i-1}) = 4\sqrt{p_j}.$$

Therefore, summing over $j = 1, \ldots, k$ and using the QM–AM inequality, we obtain

$$\sum_{\ell=1}^{n} \frac{a_\ell}{\sqrt{a_1^2 + \ldots + a_\ell^2}} \leqslant 4\sum_{j=1}^{k} \sqrt{|M_j|} \leqslant 4\sqrt{k\sum_{j=1}^{k} |M_j|} = 4\sqrt{kn}.$$

Comment. Consider the function $f(a_1, \ldots, a_n) = \sum_{i=1}^{n} \frac{a_i}{\sqrt{a_1^2+\ldots+a_i^2}}$. One can see that rearranging the variables in increasing order can only increase the value of $f(a_1, \ldots, a_n)$. Indeed, if $a_j > a_{j+1}$ for some index j then we have

$$f(a_1, \ldots, a_{j-1}, a_{j+1}, a_j, a_{j+2}, \ldots, a_n) - f(a_1, \ldots, a_n) = \frac{a}{S} + \frac{b}{\sqrt{S^2 - a^2}} - \frac{b}{S} - \frac{a}{\sqrt{S^2 - b^2}}$$

where $a = a_j, b = a_{j+1}$, and $S = \sqrt{a_1^2 + \ldots + a_{j+1}^2}$. The positivity of the last expression above follows from

$$\frac{b}{\sqrt{S^2 - a^2}} - \frac{b}{S} = \frac{a^2 b}{S\sqrt{S^2 - a^2} \cdot (S + \sqrt{S^2 - a^2})} > \frac{ab^2}{S\sqrt{S^2 - b^2} \cdot (S + \sqrt{S^2 - b^2})} = \frac{a}{\sqrt{S^2 - b^2}} - \frac{a}{S}.$$

Comment. If $k < n$, the example $a_m := 2^{k(m-1)/n}$ shows that the problem statement is sharp up to a multiplicative constant. For $k \geqslant n$ the trivial upper bound n becomes sharp up to a multiplicative constant.

Solution 2. Apply induction on n. The base $n \leqslant 16$ is clear: our sum does not exceed $n \leqslant 4\sqrt{nk}$. For the inductive step from $1, \ldots, n-1$ to $n \geqslant 17$ consider two similar cases.

Case 1: $n = 2t$.

Let $x_\ell = \frac{a_\ell}{\sqrt{a_1^2+\ldots+a_\ell^2}}$. We have

$$\exp(-x_{t+1}^2 - \ldots - x_{2t}^2) \geqslant \left(1 - x_{t+1}^2\right) \ldots \left(1 - x_{2t}^2\right) = \frac{a_1^2 + \ldots + a_t^2}{a_1^2 + \ldots + a_{2t}^2} \geqslant \frac{1}{1 + 4^k},$$

where we used that the product is telescopic and then an estimate $a_{t+i} \leqslant 2^k a_i$ for $i = 1, \ldots, t$. Therefore, $x_{t+1}^2 + \ldots + x_{2t}^2 \leqslant \log(4^k + 1) \leqslant 2k$, where log denotes the natural logarithm. This implies $x_{t+1} + \ldots + x_{2t} \leqslant \sqrt{2kt}$. Hence, using the inductive hypothesis for $n = t$ we get

$$\sum_{\ell=1}^{2t} x_\ell \leqslant 4\sqrt{kt} + \sqrt{2kt} \leqslant 4\sqrt{2kt}.$$

Case 2: $n = 2t + 1$.

Analogously, we get $x_{t+2}^2 + \ldots + x_{2t+1}^2 \leqslant \log(4^k + 1) \leqslant 2k$ and

$$\sum_{\ell=1}^{2t+1} x_\ell \leqslant 4\sqrt{k(t+1)} + \sqrt{2kt} \leqslant 4\sqrt{k(2t+1)}.$$

The last inequality is true for all $t \geqslant 8$ since

$$4\sqrt{2t+1} - \sqrt{2t} \geqslant 3\sqrt{2t} = \sqrt{18t} \geqslant \sqrt{16t+16} = 4\sqrt{t+1}.$$

A8. Let $\mathbb{R}^+$ be the set of positive real numbers. Determine all functions $f\colon \mathbb{R}^+ \to \mathbb{R}^+$ such that, for all positive real numbers x and y,

$$f\big(x + f(xy)\big) + y = f(x)f(y) + 1. \tag{$*$}$$

(Ukraine)

Answer: $f(x) = x + 1$.

Solution 1. A straightforward check shows that $f(x) = x + 1$ satisfies $(*)$. We divide the proof of the converse statement into a sequence of steps.

Step 1: f is injective.

Put $x = 1$ in $(*)$ and rearrange the terms to get

$$y = f(1)f(y) + 1 - f\big(1 + f(y)\big).$$

Therefore, if $f(y_1) = f(y_2)$, then $y_1 = y_2$.

Step 2: f is (strictly) monotone increasing.

For any fixed $y \in \mathbb{R}^+$, the function

$$g(x) := f\big(x + f(xy)\big) = f(x)f(y) + 1 - y$$

is injective by Step 1. Therefore, $x_1 + f(x_1y) \neq x_2 + f(x_2y)$ for all $y, x_1, x_2 \in \mathbb{R}^+$ with $x_1 \neq x_2$. Plugging in $z_i = x_iy$, we arrive at

$$\frac{z_1 - z_2}{y} \neq f(z_2) - f(z_1), \quad \text{or} \quad \frac{1}{y} \neq \frac{f(z_2) - f(z_1)}{z_1 - z_2}$$

for all $y, z_1, z_2 \in \mathbb{R}^+$ with $z_1 \neq z_2$. This means that the right-hand side of the rightmost relation is always non-positive, i.e., f is monotone non-decreasing. Since f is injective, it is strictly monotone.

Step 3: There exist constants a and b such that $f(y) = ay + b$ for all $y \in \mathbb{R}^+$.

Since f is monotone and bounded from below by 0, for each $x_0 \geqslant 0$, there exists a right limit $\lim_{x \searrow x_0} f(x) \geqslant 0$. Put $p = \lim_{x \searrow 0} f(x)$ and $q = \lim_{x \searrow p} f(x)$.

Fix an arbitrary y and take the limit of $(*)$ as $x \searrow 0$. We have $f(xy) \searrow p$ and hence $f\big(x + f(xy)\big) \searrow q$; therefore, we obtain

$$q + y = pf(y) + 1, \quad \text{or} \quad f(y) = \frac{q + y - 1}{p}.$$

(Notice that $p \neq 0$, otherwise $q + y = 1$ for all y, which is absurd.) The claim is proved.

Step 4: $f(x) = x + 1$ for all $x \in \mathbb{R}^+$.

Based on the previous step, write $f(x) = ax + b$. Putting this relation into $(*)$ we get

$$a\,(x + axy + b) + b + y = (ax + b)(ay + b) + 1,$$

which can be rewritten as

$$(a - ab)x + (1 - ab)y + ab + b - b^2 - 1 = 0 \qquad \text{for all } x, y \in \mathbb{R}^+.$$

This identity may hold only if all the coefficients are 0, i.e.,

$$a - ab = 1 - ab = ab + b - b^2 - 1 = 0.$$

Hence, $a = b = 1$.

Solution 2. We provide another proof that $f(x) = x + 1$ is the only function satisfying $(*)$.

Put $a = f(1)$. Define the function $\phi\colon \mathbb{R}^+ \to \mathbb{R}$ by

$$\phi(x) = f(x) - x - 1.$$

Then equation $(*)$ reads as

$$\phi(x + f(xy)) = f(x)f(y) - f(xy) - x - y. \tag{1}$$

Since the right-hand side of (1) is symmetric under swapping x and y, we obtain

$$\phi\big(x + f(xy)\big) = \phi\big(y + f(xy)\big).$$

In particular, substituting $(x, y) = (t, 1/t)$ we get

$$\phi(a + t) = \phi\left(a + \frac{1}{t}\right), \qquad t \in \mathbb{R}^+. \tag{2}$$

Notice that the function f is bounded from below by a positive constant. Indeed, for each $y \in \mathbb{R}^+$, the relation $(*)$ yields $f(x)f(y) > y - 1$, hence

$$f(x) > \frac{y-1}{f(y)} \qquad \text{for all } x \in \mathbb{R}^+.$$

If $y > 1$, this provides a desired positive lower bound for $f(x)$.

Now, let $M = \inf\limits_{x \in \mathbb{R}^+} f(x) > 0$. Then, for all $y \in \mathbb{R}^+$,

$$M \geqslant \frac{y-1}{f(y)}, \quad \text{or} \quad f(y) \geqslant \frac{y-1}{M}. \tag{3}$$

Lemma 1. The function $f(x)$ (and hence $\phi(x)$) is bounded on any segment $[p, q]$, where $0 < p < q < +\infty$.

Proof. f is bounded from below by M. It remains to show that f is bounded from above on $[p, q]$. Substituting $y = 1$ into $(*)$, we get

$$f\big(x + f(x)\big) = af(x). \tag{4}$$

Take $z \in [p, q]$ and put $s = f(z)$. By (4), we have

$$f(z + s) = as \quad \text{and} \quad f(z + s + as) = f\big(z + s + f(z + s)\big) = a^2 s.$$

Plugging in $(x, y) = \big(z, 1 + \frac{s}{z}\big)$ to $(*)$ and using (3), we obtain

$$f(z + as) = f\big(z + f(z + s)\big) = sf\left(1 + \frac{s}{z}\right) - \frac{s}{z} \geqslant \frac{s^2}{Mz} - \frac{s}{z}.$$

Now, substituting $(x, y) = \big(z + as, \frac{z}{z+as}\big)$ to $(*)$ and applying the above estimate and the estimate $f(y) \geqslant M$, we obtain

$$\begin{aligned} a^2 s = f(z + s + as) = f\big(z + as + f(z)\big) &= f(z + as)f\left(\frac{z}{z + as}\right) + 1 - \frac{z}{z + as} \\ &\geqslant Mf(z + as) \geqslant \frac{s^2}{z} - \frac{Ms}{z} \geqslant \frac{s^2}{q} - \frac{Ms}{p}. \end{aligned}$$

This yields $s \leqslant q\big(\frac{M}{p} + a^2\big) =: L$, and f is bounded from above by L on $[p, q]$. □

Applying Lemma 1 to the segment $[a, a+1]$, we see that ϕ is bounded on it. By (2) we get that ϕ is also bounded on $[a+1, +\infty)$, and hence on $[a, +\infty)$. Put $C = \max\{a, 3\}$.

Lemma 2. For all $x \geqslant C$, we have $\phi(x) = 0$ (and hence $f(x) = x + 1$).

Proof. Substituting $y = x$ to (1), we obtain

$$\phi\big(x + f(x^2)\big) = f(x)^2 - f(x^2) - 2x,$$

hence,

$$\phi\big(x + f(x^2)\big) + \phi(x^2) = f(x)^2 - (x+1)^2 = \phi(x)\big(f(x) + x + 1\big). \tag{5}$$

Since $f(x) + x + 1 \geqslant C + 1 \geqslant 4$, we obtain that

$$|\phi(x)| \leqslant \frac{1}{4}\left(\big|\phi\big(x + f(x^2)\big)\big| + \big|\phi(x^2)\big|\right). \tag{6}$$

Since $C \geqslant a$, there exists a finite supremum $S = \sup\limits_{x \geqslant C} |\phi(x)|$. For each $x \in [C, +\infty)$, both $x + f(x^2)$ and x^2 are greater than x; hence they also lie in $[C, +\infty)$. Therefore, taking the supremum of the left-hand side of (6) over $x \in [C, +\infty)$, we obtain $S \leqslant S/2$ and hence $S = 0$. Thus, $\phi(x) = 0$ for all $x \geqslant C$. □

It remains to show that $f(y) = y + 1$ when $0 < y < C$. For each y, choose $x > \max\big\{C, \frac{C}{y}\big\}$. Then all three numbers x, xy, and $x + f(xy)$ are greater than C, so $(*)$ reads as

$$(x + xy + 1) + 1 + y = (x+1)f(y) + 1, \quad \text{hence} \quad f(y) = y + 1.$$

Comment 1. It may be useful to rewrite $(*)$ in the form

$$\phi\big(x + f(xy)\big) + \phi(xy) = \phi(x)\phi(y) + x\phi(y) + y\phi(x) + \phi(x) + \phi(y).$$

This general identity easily implies both (1) and (5).

Comment 2. There are other ways to prove that $f(x) \geqslant x + 1$. Once one has proved this, they can use this stronger estimate instead of (3) in the proof of Lemma 1. Nevertheless, this does not make this proof simpler. So proving that $f(x) \geqslant x + 1$ does not seem to be a serious progress towards the solution of the problem. In what follows, we outline one possible proof of this inequality.

First of all, we improve inequality (3) by noticing that, in fact, $f(x)f(y) \geqslant y - 1 + M$, and hence

$$f(y) \geqslant \frac{y-1}{M} + 1. \tag{7}$$

Now we divide the argument into two steps.

Step 1: We show that $M \leqslant 1$.

Suppose that $M > 1$; recall the notation $a = f(1)$. Substituting $y = 1/x$ in $(*)$, we get

$$f(x + a) = f(x)f\left(\frac{1}{x}\right) + 1 - \frac{1}{x} \geqslant Mf(x),$$

provided that $x \geqslant 1$. By a straightforward induction on $\lceil (x-1)/a \rceil$, this yields

$$f(x) \geqslant M^{(x-1)/a}. \tag{8}$$

Now choose an arbitrary $x_0 \in \mathbb{R}^+$ and define a sequence $x_0, x_1, \ldots$ by $x_{n+1} = x_n + f(x_n) \geqslant x_n + M$ for all $n \geqslant 0$; notice that the sequence is unbounded. On the other hand, by (4) we get

$$ax_{n+1} > af(x_n) = f(x_{n+1}) \geqslant M^{(x_{n+1}-1)/a},$$

which cannot hold when x_{n+1} is large enough.

Step 2: We prove that $f(y) \geqslant y+1$ for all $y \in \mathbb{R}^+$.

Arguing indirectly, choose $y \in \mathbb{R}^+$ such that $f(y) < y+1$, and choose μ with $f(y) < \mu < y+1$. Define a sequence $x_0, x_1, \ldots$ by choosing a large $x_0 \geqslant 1$ and setting $x_{n+1} = x_n + f(x_n y) \geqslant x_n + M$ for all $n \geqslant 0$ (this sequence is also unbounded). If x_0 is large enough, then (7) implies that $\big(\mu - f(y)\big) f(x_n) \geqslant 1 - y$ for all n. Therefore,

$$f(x_{n+1}) = f(y) f(x_n) + 1 - y \leqslant \mu f(x_n).$$

On the other hand, since $M \leqslant 1$, inequality (7) implies that $f(z) \geqslant z$, provided that $z \geqslant 1$. Hence, if x_0 is large enough, we have $x_{n+1} \geqslant x_n(1+y)$ for all n. Therefore,

$$x_0(1+y)^n \leqslant x_n \leqslant f(x_n) \leqslant \mu^n f(x_0),$$

which cannot hold when n is large enough.

Combinatorics

C1. Let n be a positive integer. Find the number of permutations $a_1, a_2, \ldots, a_n$ of the sequence $1, 2, \ldots, n$ satisfying

$$a_1 \leqslant 2a_2 \leqslant 3a_3 \leqslant \ldots \leqslant na_n. \qquad (*)$$

(United Kingdom)

Answer: The number of such permutations is F_{n+1}, where F_k is the kth Fibonacci number: $F_1 = F_2 = 1$, $F_{n+1} = F_n + F_{n-1}$.

Solution 1. Denote by P_n the number of permutations that satisfy $(*)$. It is easy to see that $P_1 = 1$ and $P_2 = 2$.

Lemma 1. Let $n \geqslant 3$. If a permutation $a_1, \ldots, a_n$ satisfies $(*)$ then either $a_n = n$, or $a_{n-1} = n$ and $a_n = n - 1$.

Proof. Let k be the index for which $a_k = n$. If $k = n$ then we are done.

If $k = n - 1$ then, by $(*)$, we have $n(n-1) = (n-1)a_{n-1} \leqslant na_n$, so $a_n \geqslant n - 1$. Since $a_n \neq a_{n-1} = n$, the only choice for a_n is $a_n = n - 1$.

Now suppose that $k \leqslant n - 2$. For every $k < i < n$ we have $kn = ka_k \leqslant ia_i < na_i$, so $a_i \geqslant k + 1$. Moreover, $na_n \geqslant (n-1)a_{n-1} \geqslant (n-1)(k+1) = nk + (n-1-k) > nk$, so $a_n \geqslant k + 1$. Now the $n - k + 1$ numbers $a_k, a_{k+1}, \ldots, a_n$ are all greater than k; but there are only $n - k$ such values; this is not possible. □

If $a_n = n$ then $a_1, a_2, \ldots, a_{n-1}$ must be a permutation of the numbers $1, \ldots, n-1$ satisfying $a_1 \leqslant 2a_2 \leqslant \ldots \leqslant (n-1)a_{n-1}$; there are P_{n-1} such permutations. The last inequality in $(*)$, $(n-1)a_{n-1} \leqslant na_n = n^2$, holds true automatically.

If $(a_{n-1}, a_n) = (n, n-1)$, then $a_1, \ldots, a_{n-2}$ must be a permutation of $1, \ldots, n-2$ satisfying $a_1 \leqslant \ldots \leqslant (n-2)a_{n-2}$; there are P_{n-2} such permutations. The last two inequalities in $(*)$ hold true automatically by $(n-2)a_{n-2} \leqslant (n-2)^2 < n(n-1) = (n-1)a_{n-1} = na_n$.

Hence, the sequence $(P_1, P_2, \ldots)$ satisfies the recurrence relation $P_n = P_{n-1} + P_{n-2}$ for $n \geqslant 3$. The first two elements are $P_1 = F_2$ and $P_2 = F_3$, so by a trivial induction we have $P_n = F_{n+1}$.

Solution 2. We claim that all sought permutations are of the following kind. Split $\{1, 2, \ldots, n\}$ into singletons and pairs of adjacent numbers. In each pair, swap the two numbers and keep the singletons unchanged.

Such permutations correspond to tilings of a $1 \times n$ chessboard using dominoes and unit squares; it is well-known that the number of such tilings is the Fibonacci number F_{n+1}.

The claim follows by induction from

Lemma 2. Assume that $a_1, \ldots, a_n$ is a permutation satisfying $(*)$, and k is an integer such that $1 \leqslant k \leqslant n$ and $\{a_1, a_2, \ldots, a_{k-1}\} = \{1, 2, \ldots, k-1\}$. (If $k = 1$, the condition is empty.) Then either $a_k = k$, or $a_k = k + 1$ and $a_{k+1} = k$.

Proof. Choose t with $a_t = k$. Since $k \notin \{a_1, \ldots, a_{k-1}\}$, we have either $t = k$ or $t > k$. If $t = k$ then we are done, so assume $t > k$.

Notice that one of the numbers among the $t - k$ numbers $a_k, a_{k+1}, \ldots, a_{t-1}$ is at least t, because there are only $t - k - 1$ values between k and t. Let i be an index with $k \leqslant i < t$ and $a_i \geqslant t$; then $kt = ta_t \geqslant ia_i \geqslant it \geqslant kt$, so that all the inequalities turn into equalities, hence $i = k$ and $a_k = t$. If $t = k + 1$, we are done.

Suppose that $t > k + 1$. Then the chain of inequalities $kt = ka_k \leqslant \ldots \leqslant ta_t = kt$ should also turn into a chain of equalities. From this point we can find contradictions in several ways; for example by pointing to $a_{t-1} = \frac{kt}{t-1} = k + \frac{k}{t-1}$ which cannot be an integer, or considering

the product of the numbers $(k+1)a_{k+1}, \ldots, (t-1)a_{t-1}$; the numbers $a_{k+1}, \ldots, a_{t-1}$ are distinct and greater than k, so

$$(kt)^{t-k-1} = (k+1)a_{k+1} \cdot (k+2)a_{k+2} \cdot \ldots \cdot (t-1)a_{t-1} \geqslant \big((k+1)(k+2) \cdot \ldots \cdot (t-1)\big)^2.$$

Notice that $(k+i)(t-i) = kt + i(t-k-i) > kt$ for $1 \leqslant i < t-k$. This leads to the contradiction

$$(kt)^{t-k-1} \geqslant \big((k+1)(k+2) \cdot \ldots \cdot (t-1)\big)^2 = \prod_{i=1}^{t-k-1} (k+i)(t-i) > (kt)^{t-k-1}.$$

Therefore, the case $t > k+1$ is not possible. □

C2. In a regular 100-gon, 41 vertices are colored black and the remaining 59 vertices are colored white. Prove that there exist 24 convex quadrilaterals $Q_1, \ldots, Q_{24}$ whose corners are vertices of the 100-gon, so that

- the quadrilaterals $Q_1, \ldots, Q_{24}$ are pairwise disjoint, and
- every quadrilateral Q_i has three corners of one color and one corner of the other color.

(Austria)

Solution. Call a quadrilateral *skew-colored*, if it has three corners of one color and one corner of the other color. We will prove the following

Claim. If the vertices of a convex $(4k+1)$-gon P are colored black and white such that each color is used at least k times, then there exist k pairwise disjoint skew-colored quadrilaterals whose vertices are vertices of P. (One vertex of P remains unused.)

The problem statement follows by removing 3 arbitrary vertices of the 100-gon and applying the Claim to the remaining 97 vertices with $k = 24$.

Proof of the Claim. We prove by induction. For $k = 1$ we have a pentagon with at least one black and at least one white vertex. If the number of black vertices is even then remove a black vertex; otherwise remove a white vertex. In the remaining quadrilateral, there are an odd number of black and an odd number of white vertices, so the quadrilateral is skew-colored.

For the induction step, assume $k \geqslant 2$. Let b and w be the numbers of black and white vertices, respectively; then $b, w \geqslant k$ and $b + w = 4k + 1$. Without loss of generality we may assume $w \geqslant b$, so $k \leqslant b \leqslant 2k$ and $2k + 1 \leqslant w \leqslant 3k + 1$.

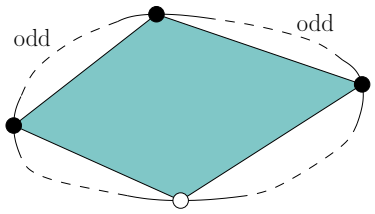
We want to find four consecutive vertices such that three of them are white, the fourth one is black. Denote the vertices by $V_1, V_2, \ldots, V_{4k+1}$ in counterclockwise order, such that V_{4k+1} is black, and consider the following k groups of vertices:

$$(V_1, V_2, V_3, V_4), (V_5, V_6, V_7, V_8), \ldots, (V_{4k-3}, V_{4k-2}, V_{4k-1}, V_{4k})$$

In these groups there are w white and $b - 1$ black vertices. Since $w > b - 1$, there is a group, $(V_i, V_{i+1}, V_{i+2}, V_{i+3})$ that contains more white than black vertices. If three are white and one is black in that group, we are done. Otherwise, if $V_i, V_{i+1}, V_{i+2}, V_{i+3}$ are all white then let V_j be the first black vertex among $V_{i+4}, \ldots, V_{4k+1}$ (recall that V_{4k+1} is black); then V_{j-3}, V_{j-2} and V_{j-1} are white and V_j is black.

Now we have four consecutive vertices $V_i, V_{i+1}, V_{i+2}, V_{i+3}$ that form a skew-colored quadrilateral. The remaining vertices form a convex $(4k-3)$-gon; $w - 3$ of them are white and $b - 1$ are black. Since $b - 1 \geqslant k - 1$ and $w - 3 \geqslant (2k + 1) - 3 > k - 1$, we can apply the Claim with $k - 1$. □

Comment. It is not true that the vertices of the 100-gon can be split into 25 skew-colored quadrilaterals. A possible counter-example is when the vertices $V_1, V_3, V_5, \ldots, V_{81}$ are black and the other vertices, $V_2, V_4, \ldots, V_{80}$ and $V_{82}, V_{83}, \ldots, V_{100}$ are white. For having 25 skew-colored quadrilaterals, there should be 8 containing three black vertices. But such a quadrilateral splits the other 96 vertices into four sets in such a way that at least two sets contain odd numbers of vertices and therefore they cannot be grouped into disjoint quadrilaterals.



C3. Let n be an integer with $n \geqslant 2$. On a slope of a mountain, n^2 checkpoints are marked, numbered from 1 to n^2 from the bottom to the top. Each of two cable car companies, A and B, operates k cable cars numbered from 1 to k; each cable car provides a transfer from some checkpoint to a higher one. For each company, and for any i and j with $1 \leqslant i < j \leqslant k$, the starting point of car j is higher than the starting point of car i; similarly, the finishing point of car j is higher than the finishing point of car i. Say that two checkpoints are *linked* by some company if one can start from the lower checkpoint and reach the higher one by using one or more cars of that company (no movement on foot is allowed).

Determine the smallest k for which one can guarantee that there are two checkpoints that are linked by each of the two companies.

(India)

Answer: $k = n^2 - n + 1$.

Solution. We start with showing that for any $k \leqslant n^2 - n$ there may be no pair of checkpoints linked by both companies. Clearly, it suffices to provide such an example for $k = n^2 - n$.

Let company A connect the pairs of checkpoints of the form $(i, i+1)$, where $n \nmid i$. Then all pairs of checkpoints (i, j) linked by A satisfy $\lceil i/n \rceil = \lceil j/n \rceil$.

Let company B connect the pairs of the form $(i, i+n)$, where $1 \leqslant i \leqslant n^2 - n$. Then pairs of checkpoints (i, j) linked by B satisfy $i \equiv j \pmod{n}$. Clearly, no pair (i, j) satisfies both conditions, so there is no pair linked by both companies.

Now we show that for $k = n^2 - n + 1$ there always exist two required checkpoints. Define an *A-chain* as a sequence of checkpoints $a_1 < a_2 < \ldots < a_t$ such that company A connects a_i with a_{i+1} for all $1 \leqslant i \leqslant t-1$, but there is no A-car transferring from some checkpoint to a_1 and no A-car transferring from a_t to any other checkpoint. Define *B-chains* similarly. Moving forth and back, one easily sees that any checkpoint is included in a unique A-chain (possibly consisting of that single checkpoint), as well as in a unique B-chain. Now, put each checkpoint into a correspondence to the pair of the A-chain and the B-chain it belongs to.

All finishing points of A-cars are distinct, so there are $n^2 - k = n - 1$ checkpoints that are not such finishing points. Each of them is a starting point of a unique A-chain, so the number of A-chains is $n-1$. Similarly, the number of B-chains also equals $n-1$. Hence, there are $(n-1)^2$ pairs consisting of an A- and a B-chain. Therefore, two of the n^2 checkpoints correspond to the same pair, so that they belong to the same A-chain, as well as to the same B-chain. This means that they are linked by both companies, as required.

Comment 1. The condition that the ith car starts and finishes lower than the jth one is used only in the "moving forth and back" argument and in the counting of starting points of the chains. In both cases, the following weaker assumption suffices: No two cars of the same company start at the same checkpoint, and no two such cars finish at the same checkpoint.

Thus, the problem conditions could be weakened in this way,, with no affect on the solution.

Comment 2. If the number of checkpoints were N, then the answer would be $N - \lceil \sqrt{N} \rceil + 1$. The solution above works verbatim for this generalization.

C4. The Fibonacci numbers $F_0, F_1, F_2, \ldots$ are defined inductively by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \geqslant 1$. Given an integer $n \geqslant 2$, determine the smallest size of a set S of integers such that for every $k = 2, 3, \ldots, n$ there exist some $x, y \in S$ such that $x - y = F_k$.

(Croatia)

Answer: $\lceil n/2 \rceil + 1$.

Solution. First we show that if a set $S \subset \mathbb{Z}$ satisfies the conditions then $|S| \geqslant \frac{n}{2} + 1$.

Let $d = \lceil n/2 \rceil$, so $n \leqslant 2d \leqslant n + 1$. In order to prove that $|S| \geqslant d + 1$, construct a graph as follows. Let the vertices of the graph be the elements of S. For each $1 \leqslant k \leqslant d$, choose two elements $x, y \in S$ such that $x - y = F_{2k-1}$, and add the pair (x, y) to the graph as edge. (Note that by the problem's constraints, there must be a pair (x, y) with $x - y = F_{2k-1}$ for every $3 \leqslant 2k - 1 \leqslant 2d - 1 \leqslant n$; moreover, due to $F_1 = F_2$ we have a pair with $x - y = F_1$ as well.) We will say that the *length* of the edge (x, y) is $|x - y|$.

We claim that the graph contains no cycle. For the sake of contradiction, suppose that the graph contains a cycle $(x_1, \ldots, x_\ell)$, and let the longest edge in the cycle be (x_1, x_ℓ) with length F_{2m+1}. The other edges $(x_1, x_2), \ldots, (x_{\ell-1}, x_\ell)$ in the cycle are shorter than F_{2m+1} and distinct, their lengths form a subset of $\{F_1, F_3, \ldots, F_{2m-1}\}$. But this is not possible because

$$\begin{aligned} F_{2m+1} = |x_\ell - x_1| &\leqslant \sum_{i=1}^{\ell-1} |x_{i+1} - x_i| \leqslant F_1 + F_3 + F_5 + \ldots + F_{2m-1} \\ &= F_2 + (F_4 - F_2) + (F_6 - F_4) + \ldots + (F_{2m} - F_{2m-2}) = F_{2m} < F_{2m+1}. \end{aligned}$$

Hence, the graph has d edges and cannot contain a cycle, therefore it must contain at least $d + 1$ vertices, so $|S| \geqslant d + 1$.

Now we show a suitable set with $d + 1$ elements. Let

$$S = \{F_0, F_2, F_4, F_5, \ldots, F_{2d}\}.$$

For $1 \leqslant k \leqslant d$ we have $F_0, F_{2k-2}, F_{2k} \in S$ with differences $F_{2k} - F_{2k-2} = F_{2k-1}$ and $F_{2k} - F_0 = F_{2k}$, so each of $F_1, F_2, \ldots, F_{2d}$ occurs as difference between two elements in S. So this set containing $d + 1$ numbers is suitable.

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C5. Let p be an odd prime, and put $N = \frac{1}{4}(p^3 - p) - 1$. The numbers $1, 2, \ldots, N$ are painted arbitrarily in two colors, red and blue. For any positive integer $n \leqslant N$, denote by $r(n)$ the fraction of integers in $\{1, 2, \ldots, n\}$ that are red.

Prove that there exists a positive integer $a \in \{1, 2, \ldots, p-1\}$ such that $r(n) \neq a/p$ for all $n = 1, 2, \ldots, N$.

(Netherlands)

Solution. Denote by $R(n)$ the number of red numbers in $\{1, 2, \ldots, n\}$, i.e., $R(n) = nr(n)$. Similarly, denote by $B(n)$ and $b(n) = B(n)/n$ the number and proportion of blue numbers in $\{1, 2, \ldots, n\}$, respectively. Notice that $B(n) + R(n) = n$ and $b(n) + r(n) = 1$. Therefore, the statement of the problem does not change after swapping the colors.

Arguing indirectly, for every $a \in \{1, 2, \ldots, p-1\}$ choose some positive integer n_a such that $r(n_a) = a/p$ and, hence, $R(n_a) = an_a/p$. Clearly, $p \mid n_a$, so that $n_a = pm_a$ for some positive integer m_a, and $R(n_a) = am_a$. Without loss of generality, we assume that $m_1 < m_{p-1}$, as otherwise one may swap the colors. Notice that

$$m_a \leqslant \frac{N}{p} < \frac{p^2 - 1}{4} \qquad \text{for all } a = 1, 2, \ldots, p-1. \tag{1}$$

The solution is based on a repeated application of the following simple observation.

Claim. Assume that $m_a < m_b$ for some $a, b \in \{1, 2, \ldots, p-1\}$. Then

$$m_b \geqslant \frac{a}{b} m_a \qquad \text{and} \qquad m_b \geqslant \frac{p-a}{p-b} m_a.$$

Proof. The first inequality follows from $bm_b = R(n_b) \geqslant R(n_a) = am_a$. The second inequality is obtained by swapping colors . □

Let $q = (p-1)/2$. We distinguish two cases.

Case 1: All q numbers $m_1, m_2, \ldots, m_q$ are smaller than m_{p-1}.

Let m_a be the maximal number among $m_1, m_2, \ldots, m_q$; then $m_a \geqslant q \geqslant a$. Applying the Claim, we get

$$m_{p-1} \geqslant \frac{p-a}{p-(p-1)} m_a \geqslant (p-q)q = \frac{p^2-1}{4},$$

which contradicts (1).

Case 2: There exists $k \leqslant q$ such that $m_k > m_{p-1}$.

Choose k to be the smallest index satisfying $m_k > m_{p-1}$; by our assumptions, we have $1 < k \leqslant q < p - 1$.

Let m_a be the maximal number among $m_1, m_2, \ldots, m_{k-1}$; then $a \leqslant k - 1 \leqslant m_a < m_{p-1}$. Applying the Claim, we get

$$\begin{aligned} m_k \geqslant \frac{p-1}{k} m_{p-1} &\geqslant \frac{p-1}{k} \cdot \frac{p-a}{p-(p-1)} m_a \\ &\geqslant \frac{p-1}{k} \cdot (p-k+1)(k-1) \geqslant \frac{k-1}{k} \cdot (p-1)(p-q) \geqslant \frac{1}{2} \cdot \frac{p^2-1}{2} \end{aligned}$$

which contradicts (1) again.

Comment 1. The argument in Case 2, after a slight modification of estimates at the end, applies as soon as there exists $k < \frac{3(p+1)}{4}$ with $a_k < a_{p-1}$. However, this argument does not seem to work if there is no such k.

Comment 2. If p is small enough, then one can color $\{1, 2, \ldots, N+1\}$ so that there exist numbers m_1, m_2, $\ldots$, m_{p-1} satisfying $r(pm_a) = a/p$. For $p = 3, 5, 7$, one can find colorings providing the following sequences:

$$(m_1, m_2) = (1, 2), \qquad (m_1, m_2, m_3, m_4) = (1, 2, 3, 6), \quad \text{and} \quad (m_1, \ldots, , m_6) = (1, 2, 3, 4, 6, 12),$$

respectively.

Thus, for small values of p, the number N in the problem statement cannot be increased. However, a careful analysis of the estimates shows that this number can be slightly increased for $p \geqslant 11$.

C6. $4n$ coins of weights $1, 2, 3, \ldots, 4n$ are given. Each coin is colored in one of n colors and there are four coins of each color. Show that all these coins can be partitioned into two sets with the same total weight, such that each set contains two coins of each color.

(Hungary)

Solution 1. Let us pair the coins with weights summing up to $4n + 1$, resulting in the set S of $2n$ pairs: $\{1, 4n\}$, $\{2, 4n - 1\}$, $\ldots$, $\{2n, 2n + 1\}$. It suffices to partition S into two sets, each consisting of n pairs, such that each set contains two coins of each color.

Introduce a multi-graph G (i.e., a graph with loops and multiple edges allowed) on n vertices, so that each vertex corresponds to a color. For each pair of coins from S, we add an edge between the vertices corresponding to the colors of those coins. Note that each vertex has degree 4. Also, a desired partition of the coins corresponds to a coloring of the edges of G in two colors, say red and blue, so that each vertex has degree 2 with respect to each color (i.e., each vertex has equal red and blue degrees).

To complete the solution, it suffices to provide such a coloring for each component G' of G. Since all degrees of the vertices are even, in G' there exists an Euler circuit C (i.e., a circuit passing through each edge of G' exactly once). Note that the number of edges in C is even (it equals twice the number of vertices in G'). Hence all the edges can be colored red and blue so that any two edges adjacent in C have different colors (one may move along C and color the edges one by one alternating red and blue colors). Thus in G' each vertex has equal red and blue degrees, as desired.

Comment 1. To complete Solution 1, any partition of the edges of G into circuits of even lengths could be used. In the solution above it was done by the reference to the well-known Euler Circuit Lemma: Let G be a connected graph with all its vertices of even degrees. Then there exists a circuit passing through each edge of G exactly once.

Solution 2. As in Solution 1, we will show that it is possible to partition $2n$ pairs $\{1, 4n\}$, $\{2, 4n-1\}$, $\ldots$, $\{2n, 2n+1\}$ into two sets, each consisting of n pairs, such that each set contains two coins of each color.

Introduce a multi-graph (i.e., a graph with multiple edges allowed) Γ whose vertices correspond to coins; thus we have $4n$ vertices of n colors so that there are four vertices of each color. Connect pairs of vertices $\{1, 4n\}$, $\{2, 4n - 1\}$, $\ldots$, $\{2n, 2n + 1\}$ by $2n$ *black* edges.

Further, for each monochromatic quadruple of vertices i, j, k, ℓ we add a pair of *grey* edges forming a matching, e.g., (i, j) and (k, ℓ). In each of n colors of coins we can choose one of three possible matchings; this results in 3^n ways of constructing grey edges. Let us call each of 3^n possible graphs Γ a *cyclic* graph. Note that in a cyclic graph Γ each vertex has both black and grey degrees equal to 1. Hence Γ is a union of disjoint cycles, and in each cycle black and grey edges alternate (in particular, all cycles have even lengths).

It suffices to find a cyclic graph with all its cycle lengths divisible by 4. Indeed, in this case, for each cycle we start from some vertex, move along the cycle and recolor the black edges either to red or to blue, alternating red and blue colors. Now blue and red edges define the required partition, since for each monochromatic quadruple of vertices the grey edges provide a bijection between the endpoints of red and blue edges.

Among all possible cyclic graphs, let us choose graph Γ_0 having the minimal number of components (i.e., cycles). The following claim completes the solution.

Claim. In Γ_0, all cycle lengths are divisible by 4.

Proof. Assuming the contrary, choose a cycle C_1 with an odd number of grey edges. For some color c the cycle C_1 contains exactly one grey edge joining two vertices i, j of color c, while the other edge joining two vertices k, ℓ of color c lies in another cycle C_2. Now delete edges (i, j) and (k, ℓ) and add edges (i, k) and (j, ℓ). By this switch we again obtain a cyclic graph Γ_0' and decrease the number of cycles by 1. This contradicts the choice of Γ_0. □

Comment 2. Use of an auxiliary graph and reduction to a new problem in terms of this graph is one of the crucial steps in both solutions presented. In fact, graph G from Solution 1 could be obtained from any graph Γ from Solution 2 by merging the vertices of the same color.

C7. Consider any rectangular table having finitely many rows and columns, with a real number $a(r,c)$ in the cell in row r and column c. A pair (R,C), where R is a set of rows and C a set of columns, is called a *saddle pair* if the following two conditions are satisfied:

(i) For each row r', there is $r\in R$ such that $a(r,c)\geqslant a(r',c)$ for all $c\in C$;

(ii) For each column c', there is $c\in C$ such that $a(r,c)\leqslant a(r,c')$ for all $r\in R$.

A saddle pair (R,C) is called a *minimal pair* if for each saddle pair (R',C') with $R'\subseteq R$ and $C'\subseteq C$, we have $R'=R$ and $C'=C$.

Prove that any two minimal pairs contain the same number of rows.

(Thailand)

Solution 1. We say that a pair (R',C') of nonempty sets is a *subpair* of a pair (R,C) if $R'\subseteq R$ and $C'\subseteq C$. The subpair is *proper* if at least one of the inclusions is strict.

Let (R_1,C_1) and (R_2,C_2) be two saddle pairs with $|R_1|>|R_2|$. We will find a saddle subpair (R',C') of (R_1,C_1) with $|R'|\leqslant|R_2|$; clearly, this implies the desired statement.

Step 1: We construct maps $\rho\colon R_1\to R_1$ and $\sigma\colon C_1\to C_1$ such that $|\rho(R_1)|\leqslant|R_2|$, and $a\big(\rho(r_1),c_1\big)\geqslant a\big(r_1,\sigma(c_1)\big)$ for all $r_1\in R_1$ and $c_1\in C_1$.

Since (R_1,C_1) is a saddle pair, for each $r_2\in R_2$ there is $r_1\in R_1$ such that $a(r_1,c_1)\geqslant a(r_2,c_1)$ for all $c_1\in C_1$; denote one such an r_1 by $\rho_1(r_2)$. Similarly, we define four functions

$$\begin{array}{llllll}
\rho_1\colon R_2\to R_1 & \text{such that} & a\big(\rho_1(r_2),c_1\big)\geqslant a(r_2,c_1) & \text{for all} & r_2\in R_2, & c_1\in C_1;\\
\rho_2\colon R_1\to R_2 & \text{such that} & a\big(\rho_2(r_1),c_2\big)\geqslant a(r_1,c_2) & \text{for all} & r_1\in R_1, & c_2\in C_2;\\
\sigma_1\colon C_2\to C_1 & \text{such that} & a\big(r_1,\sigma_1(c_2)\big)\leqslant a(r_1,c_2) & \text{for all} & r_1\in R_1, & c_2\in C_2;\\
\sigma_2\colon C_1\to C_2 & \text{such that} & a\big(r_2,\sigma_2(c_1)\big)\leqslant a(r_2,c_1) & \text{for all} & r_2\in R_2, & c_1\in C_1.
\end{array}\tag{1}$$

Set now $\rho=\rho_1\circ\rho_2\colon R_1\to R_1$ and $\sigma=\sigma_1\circ\sigma_2\colon C_1\to C_1$. We have

$$|\rho(R_1)|=|\rho_1(\rho_2(R_1))|\leqslant|\rho_1(R_2)|\leqslant|R_2|.$$

Moreover, for all $r_1\in R_1$ and $c_1\in C_1$, we get

$$\begin{aligned}a\big(\rho(r_1),c_1\big)=a\big(\rho_1(\rho_2(r_1)),c_1\big)&\geqslant a\big(\rho_2(r_1),c_1\big)\geqslant a\big(\rho_2(r_1),\sigma_2(c_1)\big)\\&\geqslant a\big(r_1,\sigma_2(c_1)\big)\geqslant a\big(r_1,\sigma_1(\sigma_2(c_1))\big)=a\big(r_1,\sigma(c_1)\big),\end{aligned}\tag{2}$$

as desired.

Step 2: Given maps ρ and σ, we construct a proper saddle subpair (R',C') of (R_1,C_1).

The properties of ρ and σ yield that

$$a\big(\rho^i(r_1),c_1\big)\geqslant a\big(\rho^{i-1}(r_1),\sigma(c_1)\big)\geqslant\ldots\geqslant a\big(r_1,\sigma^i(c_1)\big),$$

for each positive integer i and all $r_1\in R_1$, $c_1\in C_1$.

Consider the images $R^i=\rho^i(R_1)$ and $C^i=\sigma^i(C_1)$. Clearly, $R_1=R^0\supseteq R^1\supseteq R^2\supseteq\ldots$ and $C_1=C^0\supseteq C^1\supseteq C^2\supseteq\ldots$. Since both chains consist of finite sets, there is an index n such that $R^n=R^{n+1}=\ldots$ and $C^n=C^{n+1}=\ldots$. Then $\rho^n(R^n)=R^{2n}=R^n$, so ρ^n restricted to R^n is a bijection. Similarly, σ^n restricted to C^n is a bijection from C^n to itself. Therefore, there exists a positive integer k such that ρ^{nk} acts identically on R^n, and σ^{nk} acts identically on C^n.

We claim now that (R^n,C^n) is a saddle subpair of (R_1,C_1), with $|R^n|\leqslant|R^1|=|\rho(R_1)|\leqslant|R_2|$, which is what we needed. To check that this is a saddle pair, take any row r'; since (R_1,C_1) is a saddle pair, there exists $r_1\in R_1$ such that $a(r_1,c_1)\geqslant a(r',c_1)$ for all $c_1\in C_1$. Set now $r_*=\rho^{nk}(r_1)\in R^n$. Then, for each $c\in C^n$ we have $c=\sigma^{nk}(c)$ and hence

$$a(r_*,c)=a\big(\rho^{nk}(r_1),c\big)\geqslant a\big(r_1,\sigma^{nk}(c)\big)=a(r_1,c)\geqslant a(r',c),$$

which establishes condition (i). Condition (ii) is checked similarly.

Solution 2. Denote by $\mathcal{R}$ and $\mathcal{C}$ the set of all rows and the set of all columns of the table, respectively. Let $\mathcal{T}$ denote the given table; for a set R of rows and a set C of columns, let $\mathcal{T}[R, C]$ denote the subtable obtained by intersecting rows from R and columns from C.

We say that row r_1 *exceeds* row r_2 *in range of columns* C (where $C \subseteq \mathcal{C}$) and write $r_1 \succeq_C r_2$ or $r_2 \preceq_C r_1$, if $a(r_1, c) \geqslant a(r_2, c)$ for all $c \in C$. We say that a row r_1 is *equal* to a row r_2 *in range of columns* C and write $r_1 \equiv_C r_2$, if $a(r_1, c) = a(r_2, c)$ for all $c \in C$. We introduce similar notions, and use the same notation, for columns. Then conditions (i) and (ii) in the definition of a saddle pair can be written as (i) for each $r' \in \mathcal{R}$ there exists $r \in R$ such that $r \succeq_C r'$; and (ii) for each $c' \in \mathcal{C}$ there exists $c \in C$ such that $c \preceq_R c'$.

Lemma. Suppose that (R, C) is a minimal pair. Remove from the table several rows outside of R and/or several columns outside of C. Then (R, C) remains a minimal pair in the new table.

Proof. Obviously, (R, C) remains a saddle pair. Suppose (R', C') is a proper subpair of (R, C). Since (R, C) is a saddle pair, for each row r^* of the initial table, there is a row $r \in R$ such that $r \succeq_C r^*$. If (R', C') became saddle after deleting rows not in R and/or columns not in C, there would be a row $r' \in R'$ satisfying $r' \succeq_{C'} r$. Therefore, we would obtain that $r' \succeq_{C'} r^*$, which is exactly condition (i) for the pair (R', C') in the initial table; condition (ii) is checked similarly. Thus, (R', C') was saddle in the initial table, which contradicts the hypothesis that (R, C) was minimal. Hence, (R, C) remains minimal after deleting rows and/or columns. □

By the Lemma, it suffices to prove the statement of the problem in the case $\mathcal{R} = R_1 \cup R_2$ and $\mathcal{C} = C_1 \cup C_2$. Further, suppose that there exist rows that belong both to R_1 and R_2. Duplicate every such row, and refer one copy of it to the set R_1, and the other copy to the set R_2. Then (R_1, C_1) and (R_2, C_2) will remain minimal pairs in the new table, with the same numbers of rows and columns, but the sets R_1 and R_2 will become disjoint. Similarly duplicating columns in $C_1 \cap C_2$, we make C_1 and C_2 disjoint. Thus it is sufficient to prove the required statement in the case $R_1 \cap R_2 = \varnothing$ and $C_1 \cap C_2 = \varnothing$.

The rest of the solution is devoted to the proof of the following claim including the statement of the problem.

Claim. Suppose that (R_1, C_1) and (R_2, C_2) are minimal pairs in table $\mathcal{T}$ such that $R_2 = \mathcal{R} \setminus R_1$ and $C_2 = \mathcal{C} \setminus C_1$. Then $|R_1| = |R_2|$, $|C_1| = |C_2|$; moreover, there are four bijections

$$\begin{aligned}
&\rho_1 \colon R_2 \to R_1 \quad \text{such that} \quad \rho_1(r_2) \equiv_{C_1} r_2 \quad \text{for all} \quad r_2 \in R_2;\\
&\rho_2 \colon R_1 \to R_2 \quad \text{such that} \quad \rho_2(r_1) \equiv_{C_2} r_1 \quad \text{for all} \quad r_1 \in R_1;\\
&\sigma_1 \colon C_2 \to C_1 \quad \text{such that} \quad \sigma_1(c_2) \equiv_{R_1} c_2 \quad \text{for all} \quad c_2 \in C_2;\\
&\sigma_2 \colon C_1 \to C_2 \quad \text{such that} \quad \sigma_2(c_1) \equiv_{R_2} c_1 \quad \text{for all} \quad c_1 \in C_1.
\end{aligned} \tag{3}$$

We prove the Claim by induction on $|\mathcal{R}| + |\mathcal{C}|$. In the base case we have $|R_1| = |R_2| = |C_1| = |C_2| = 1$; let $R_i = \{r_i\}$ and $C_i = \{c_i\}$. Since (R_1, C_1) and (R_2, C_2) are saddle pairs, we have $a(r_1, c_1) \geqslant a(r_2, c_1) \geqslant a(r_2, c_2) \geqslant a(r_1, c_2) \geqslant a(r_1, c_1)$, hence, the table consists of four equal numbers, and the statement follows.

To prove the inductive step, introduce the maps ρ_1, ρ_2, σ_1, and σ_2 as in Solution 1, see (1). Suppose first that all four maps are surjective. Then, in fact, we have $|R_1| = |R_2|$, $|C_1| = |C_2|$, and all maps are bijective. Moreover, for all $r_2 \in R_2$ and $c_2 \in C_2$ we have

$$\begin{aligned}
a(r_2, c_2) \leqslant a\big(r_2, \sigma_2^{-1}(c_2)\big) \leqslant a\big(\rho_1(r_2), \sigma_2^{-1}(c_2)\big) \leqslant a\big(\rho_1(r_2), \sigma_1^{-1} \circ \sigma_2^{-1}(c_2)\big)\\
\leqslant a\big(\rho_2 \circ \rho_1(r_2), \sigma_1^{-1} \circ \sigma_2^{-1}(c_2)\big).
\end{aligned} \tag{4}$$

Summing up, we get

$$\sum_{\substack{r_2 \in R_2\\ c_2 \in C_2}} a(r_2, c_2) \leqslant \sum_{\substack{r_2 \in R_2\\ c_2 \in C_2}} a\big(\rho_2 \circ \rho_1(r_2), \sigma_1^{-1} \circ \sigma_2^{-1}(c_2)\big).$$

Since $\rho_1 \circ \rho_2$ and $\sigma_1^{-1} \circ \sigma_2^{-1}$ are permutations of R_2 and C_2, respectively, this inequality is in fact equality. Therefore, all inequalities in (4) turn into equalities, which establishes the inductive step in this case.

It remains to show that all four maps are surjective. For the sake of contradiction, we assume that ρ_1 is not surjective. Now let $R_1' = \rho_1(R_2)$ and $C_1' = \sigma_1(C_2)$, and set $R^* = R_1 \setminus R_1'$ and $C^* = C_1 \setminus C_1'$. By our assumption, $R^* \neq \varnothing$.

Let $\mathcal{Q}$ be the table obtained from $\mathcal{T}$ by removing the rows in R^* and the columns in C^*; in other words, $\mathcal{Q} = \mathcal{T}[R_1' \cup R_2, C_1' \cup C_2]$. By the definition of ρ_1, for each $r_2 \in R_2$ we have $\rho_1(r_2) \succeq_{C_1} r_2$, so a fortiori $\rho_1(r_2) \succeq_{C_1'} r_2$; moreover, $\rho_1(r_2) \in R_1'$. Similarly, $C_1' \ni \sigma_1(c_2) \preceq_{R_1'} c_2$ for each $c_2 \in C_2$. This means that (R_1', C_1') is a saddle pair in $\mathcal{Q}$. Recall that (R_2, C_2) remains a minimal pair in $\mathcal{Q}$, due to the Lemma.

Therefore, $\mathcal{Q}$ admits a minimal pair $(\overline{R}_1, \overline{C}_1)$ such that $\overline{R}_1 \subseteq R_1'$ and $\overline{C}_1 \subseteq C_1'$. For a minute, confine ourselves to the subtable $\overline{\mathcal{Q}} = \mathcal{Q}[\overline{R}_1 \cup R_2, \overline{C}_1 \cup C_2]$. By the Lemma, the pairs $(\overline{R}_1, \overline{C}_1)$ and (R_2, C_2) are also minimal in $\overline{\mathcal{Q}}$. By the inductive hypothesis, we have $|R_2| = |\overline{R}_1| \leqslant |R_1'| = |\rho_1(R_2)| \leqslant |R_2|$, so all these inequalities are in fact equalities. This implies that $\overline{R}_2 = R_2'$ and that ρ_1 is a bijection $R_2 \to R_1'$. Similarly, $\overline{C}_1 = C_1'$, and σ_1 is a bijection $C_2 \to C_1'$. In particular, (R_1', C_1') is a minimal pair in $\mathcal{Q}$.

Now, by inductive hypothesis again, we have $|R_1'| = |R_2|$, $|C_1'| = |C_2|$, and there exist four bijections

$$\begin{aligned}
&\rho_1' \colon R_2 \to R_1' \quad \text{such that} \quad \rho_1'(r_2) \equiv_{C_1'} r_2 \quad \text{for all} \quad r_2 \in R_2;\\
&\rho_2' \colon R_1' \to R_2 \quad \text{such that} \quad \rho_2'(r_1) \equiv_{C_2} r_1 \quad \text{for all} \quad r_1 \in R_1';\\
&\sigma_1' \colon C_2 \to C_1' \quad \text{such that} \quad \sigma_1'(c_2) \equiv_{R_1'} c_2 \quad \text{for all} \quad c_2 \in C_2;\\
&\sigma_2' \colon C_1' \to C_2 \quad \text{such that} \quad \sigma_2'(c_1) \equiv_{R_2} c_1 \quad \text{for all} \quad c_1 \in C_1'.
\end{aligned}$$

Notice here that σ_1 and σ_1' are two bijections $C_2 \to C_1'$ satisfying $\sigma_1'(c_2) \equiv_{R_1'} c_2 \succeq_{R_1} \sigma_1(c_2)$ for all $c_2 \in C_2$. Now, if $\sigma_1'(c_2) \neq \sigma_1(c_2)$ for some $c_2 \in C_2$, then we could remove column $\sigma_1'(c_2)$ from C_1' obtaining another saddle pair $\big(R_1', C_1' \setminus \{\sigma_1'(c_2)\}\big)$ in $\mathcal{Q}$. This is impossible for a minimal pair (R_1', C_1'); hence the maps σ_1 and σ_1' coincide.

Now we are prepared to show that (R_1', C_1') is a saddle pair in $\mathcal{T}$, which yields a desired contradiction (since (R_1, C_1) is not minimal). By symmetry, it suffices to find, for each $r' \in \mathcal{R}$, a row $r_1 \in R_1'$ such that $r_1 \succeq_{C_1'} r'$. If $r' \in R_2$, then we may put $r_1 = \rho_1(r')$; so, in the sequel we assume $r' \in R_1$.

There exists $r_2 \in R_2$ such that $r' \preceq_{C_2} r_2$; set $r_1 = (\rho_2')^{-1}(r_2) \in R_1'$ and recall that $r_1 \equiv_{C_2} r_2 \succeq_{C_2} r'$. Therefore, implementing the bijection $\sigma_1 = \sigma_1'$, for each $c_1 \in C_1'$ we get

$$a(r', c_1) \leqslant a\big(r', \sigma_1^{-1}(c_1)\big) \leqslant a\big(r_1, \sigma_1^{-1}(c_1)\big) = a\big(r_1, \sigma_1' \circ \sigma_1^{-1}(c_1)\big) = a\big(r_1, c_1\big),$$

which shows $r' \preceq_{C_1'} r_1$, as desired. The inductive step is completed.

Comment 1. For two minimal pairs (R_1, C_1) and (R_2, C_2), Solution 2 not only proves the required equalities $|R_1| = |R_2|$ and $|C_1| = |C_2|$, but also shows the existence of bijections (3). In simple words, this means that the four subtables $\mathcal{T}[R_1, C_1]$, $\mathcal{T}[R_1, C_2]$, $\mathcal{T}[R_2, C_1]$, and $\mathcal{T}[R_2, C_2]$ differ only by permuting rows/columns. Notice that the existence of such bijections immediately implies that (R_1, C_2) and (R_2, C_1) are also minimal pairs.

This stronger claim may also be derived directly from the arguments in Solution 1, even without the assumptions $R_1 \cap R_2 = \varnothing$ and $C_1 \cap C_2 = \varnothing$. Indeed, if $|R_1| = |R_2|$ and $|C_1| = |C_2|$, then similar arguments show that $R^n = R_1$, $C^n = C_1$, and for any $r \in R^n$ and $c \in C^n$ we have

$$a(r, c) = a\big(\rho^{nk}(r), c\big) \geqslant a\big(\rho^{nk-1}(r), \sigma(c)\big) \geqslant \ldots \geqslant a\big(r, \sigma^{nk}(c)\big) = a(r, c).$$

This yields that all above inequalities turn into equalities. Moreover, this yields that all inequalities in (2) turn into equalities. Hence ρ_1, ρ_2, σ_1, and σ_2 satisfy (3).

It is perhaps worth mentioning that one cannot necessarily find the maps in (3) so as to satisfy $\rho_1 = \rho_2^{-1}$ and $\sigma_1 = \sigma_2^{-1}$, as shown by the table below.

1	0	0	1
0	1	1	0
1	0	1	0
0	1	0	1

Comment 2. One may use the following, a bit more entertaining formulation of the same problem.

On a specialized market, a finite number of products are being sold, and there are finitely many retailers each selling all the products by some prices. Say that retailer r_1 *dominates* retailer r_2 with respect to a set of products P if r_1's price of each $p \in P$ does not exceed r_2's price of p. Similarly, product p_1 *exceeds* product p_2 with respect to a set of retailers R, if r's price of p_1 is not less than r's price of p_2, for each $r \in R$.

Say that a set R of retailers and a set P of products form a *saddle pair* if for each retailer r' there is $r \in R$ dominating r' with respect to P, and for each product p' there is $p \in P$ exceeding p' with respect to R. A saddle pair (R, P) is called a *minimal pair* if for each saddle pair (R', P') with $R' \subseteq R$ and $P' \subseteq P$, we have $R' = R$ and $P' = P$.

Prove that any two minimal pairs contain the same number of retailers.

C8. Players A and B play a game on a blackboard that initially contains 2020 copies of the number 1. In every round, player A erases two numbers x and y from the blackboard, and then player B writes one of the numbers $x+y$ and $|x-y|$ on the blackboard. The game terminates as soon as, at the end of some round, one of the following holds:

(1) one of the numbers on the blackboard is larger than the sum of all other numbers;

(2) there are only zeros on the blackboard.

Player B must then give as many cookies to player A as there are numbers on the blackboard. Player A wants to get as many cookies as possible, whereas player B wants to give as few as possible. Determine the number of cookies that A receives if both players play optimally.

(Austria)

Answer: 7.

Solution. For a positive integer n, we denote by $S_2(n)$ the sum of digits in its binary representation. We prove that, in fact, if a board initially contains an even number $n > 1$ of ones, then A can guarantee to obtain $S_2(n)$, but not more, cookies. The binary representation of 2020 is $2020 = \overline{11111100100}_2$, so $S_2(2020) = 7$, and the answer follows.

A strategy for A. *At any round, while possible, A chooses two equal nonzero numbers on the board.* Clearly, while A can make such choice, the game does not terminate. On the other hand, A can follow this strategy unless the game has already terminated. Indeed, if A always chooses two equal numbers, then each number appearing on the board is either 0 or a power of 2 with non-negative integer exponent, this can be easily proved using induction on the number of rounds. At the moment when A is unable to follow the strategy all nonzero numbers on the board are distinct powers of 2. If the board contains at least one such power, then the largest of those powers is greater than the sum of the others. Otherwise there are only zeros on the blackboard, in both cases the game terminates.

For every number on the board, define its *range* to be the number of ones it is obtained from. We can prove by induction on the number of rounds that for any nonzero number k written by B its range is k, and for any zero written by B its range is a power of 2. Thus at the end of each round all the ranges are powers of two, and their sum is n. Since $S_2(a+b) \leqslant S_2(a) + S_2(b)$ for any positive integers a and b, the number n cannot be represented as a sum of less than $S_2(n)$ powers of 2. Thus at the end of each round the board contains at least $S_2(n)$ numbers, while A follows the above strategy. So A can guarantee at least $S_2(n)$ cookies for himself.

Comment. There are different proofs of the fact that the presented strategy guarantees at least $S_2(n)$ cookies for A. For instance, one may denote by Σ the sum of numbers on the board, and by z the number of zeros. Then the board contains at least $S_2(\Sigma) + z$ numbers; on the other hand, during the game, the number $S_2(\Sigma) + z$ does not decrease, and its initial value is $S_2(n)$. The claim follows.

A strategy for B. Denote $s = S_2(n)$.

Let $x_1, \ldots, x_k$ be the numbers on the board at some moment of the game after B's turn or at the beginning of the game. Say that a collection of k signs $\varepsilon_1, \ldots, \varepsilon_k \in \{+1, -1\}$ is *balanced* if

$$\sum_{i=1}^{k} \varepsilon_i x_i = 0.$$

We say that a situation on the board is *good* if 2^{s+1} does not divide the number of balanced collections. An appropriate strategy for B can be explained as follows: *Perform a move so that the situation remains good, while it is possible.* We intend to show that in this case B will not lose more than $S_2(n)$ cookies. For this purpose, we prove several lemmas.

For a positive integer k, denote by $\nu_2(k)$ the exponent of the largest power of 2 that divides k. Recall that, by Legendre's formula, $\nu_2(n!) = n - S_2(n)$ for every positive integer n.

Lemma 1. The initial situation is good.

Proof. In the initial configuration, the number of balanced collections is equal to $\binom{n}{n/2}$. We have

$$\nu_2\left(\binom{n}{n/2}\right) = \nu_2(n!) - 2\nu_2\big((n/2)!\big) = \big(n - S_2(n)\big) - 2\left(\frac{n}{2} - S_2(n/2)\right) = S_2(n) = s.$$

Hence 2^{s+1} does not divide the number of balanced collections, as desired. □

Lemma 2. B may play so that after each round the situation remains good.

Proof. Assume that the situation $(x_1, \ldots, x_k)$ before a round is good, and that A erases two numbers, x_p and x_q.

Let N be the number of all balanced collections, N_+ be the number of those having $\varepsilon_p = \varepsilon_q$, and N_- be the number of other balanced collections. Then $N = N_+ + N_-$. Now, if B replaces x_p and x_q by $x_p + x_q$, then the number of balanced collections will become N_+. If B replaces x_p and x_q by $|x_p - x_q|$, then this number will become N_-. Since 2^{s+1} does not divide N, it does not divide one of the summands N_+ and N_-, hence B can reach a good situation after the round. □

Lemma 3. Assume that the game terminates at a good situation. Then the board contains at most s numbers.

Proof. Suppose, one of the numbers is greater than the sum of the other numbers. Then the number of balanced collections is 0 and hence divisible by 2^{s+1}. Therefore, the situation is not good.

Then we have only zeros on the blackboard at the moment when the game terminates. If there are k of them, then the number of balanced collections is 2^k. Since the situation is good, we have $k \leqslant s$. □

By Lemmas 1 and 2, B may act in such way that they keep the situation good. By Lemma 3, when the game terminates, the board contains at most s numbers. This is what we aimed to prove.

Comment 1. If the initial situation had some odd number $n > 1$ of ones on the blackboard, player A would still get $S_2(n)$ cookies, provided that both players act optimally. The proof of this fact is similar to the solution above, after one makes some changes in the definitions. Such changes are listed below.

Say that a collection of k signs $\varepsilon_1, \ldots, \varepsilon_k \in \{+1, -1\}$ is *positive* if

$$\sum_{i=1}^{k} \varepsilon_i x_i > 0.$$

For every index $i = 1, 2, \ldots, k$, we denote by N_i the number of positive collections such that $\varepsilon_i = 1$. Finally, say that a situation on the board is *good* if 2^{s-1} does not divide at least one of the numbers N_i. Now, a strategy for B again consists in preserving the situation good, after each round.

Comment 2. There is an easier strategy for B, allowing, in the game starting with an even number n of ones, to lose no more than $d = \lfloor \log_2(n+2) \rfloor - 1$ cookies. If the binary representation of n contains at most two zeros, then $d = S_2(n)$, and hence the strategy is optimal in that case. We outline this approach below.

First of all, we can assume that A never erases zeros from the blackboard. Indeed, A may skip such moves harmlessly, ignoring the zeros in the further process; this way, A's win will just increase.

We say that a situation on the blackboard is *pretty* if the numbers on the board can be partitioned into two groups with equal sums. Clearly, if the situation before some round is pretty, then B may play so as to preserve this property after the round. The strategy for B is as follows:

• B always chooses a move that leads to a pretty situation.

• If both possible moves of B lead to pretty situations, then B writes the sum of the two numbers erased by A.

Since the situation always remains pretty, the game terminates when all numbers on the board are zeros.

Suppose that, at the end of the game, there are $m \geqslant d+1 = \lfloor \log_2(n+2) \rfloor$ zeros on the board; then $2^m - 1 > n/2$.

Now we analyze the whole process of the play. Let us number the zeros on the board in order of appearance. During the play, each zero had appeared after subtracting two equal numbers. Let $n_1, \ldots, n_m$ be positive integers such that the first zero appeared after subtracting n_1 from n_1, the second zero appeared after subtracting n_2 from n_2, and so on. Since the sum of the numbers on the blackboard never increases, we have $2n_1 + \ldots + 2n_m \leqslant n$, hence

$$n_1 + \ldots + n_m \leqslant n/2 < 2^m - 1.$$

There are 2^m subsets of the set $\{1, 2, \ldots, m\}$. For $I \subseteq \{1, 2, \ldots, m\}$, denote by $f(I)$ the sum $\sum_{i \in I} n_i$. There are less than 2^m possible values for $f(I)$, so there are two distinct subsets I and J with $f(I) = f(J)$. Replacing I and J with $I \setminus J$ and $J \setminus I$, we assume that I and J are disjoint.

Let i_0 be the smallest number in $I \cup J$; without loss of generality, $i_0 \in I$. Consider the round when A had erased two numbers equal to n_{i_0}, and B had put the $i_0{}^{\text{th}}$ zero instead, and the situation *before* that round.

For each nonzero number z which is on the blackboard at this moment, we can keep track of it during the further play until it enters one of the numbers n_i, $i \geqslant i_0$, which then turn into zeros. For every $i = i_0, i_0 + 1, \ldots, m$, we denote by X_i the collection of all numbers on the blackboard that finally enter the first copy of n_i, and by Y_i the collection of those finally entering the second copy of n_i. Thus, each of X_{i_0} and Y_{i_0} consists of a single number. Since A never erases zeros, the $2(m - i_0 + 1)$ defined sets are pairwise disjoint.

Clearly, for either of the collections X_i and Y_i, a signed sum of its elements equals n_i, for a proper choice of the signs. Therefore, for every $i = i_0, i_0 + 1, \ldots, m$ one can endow numbers in $X_i \cup Y_i$ with signs so that their sum becomes any of the numbers $-2n_i$, 0, or $2n_i$. Perform such endowment so as to get $2n_i$ from each collection $X_i \cup Y_i$ with $i \in I$, $-2n_j$ from each collection $X_j \cup Y_j$ with $j \in J$, and 0 from each remaining collection. The obtained signed sum of all numbers on the blackboard equals

$$\sum_{i \in I} 2n_i - \sum_{i \in J} 2n_i = 0,$$

and the numbers in X_{i_0} and Y_{i_0} have the same (positive) sign.

This means that, at this round, B could add up the two numbers n_{i_0} to get a pretty situation. According to the strategy, B should have performed that, instead of subtracting the numbers. This contradiction shows that $m \leqslant d$, as desired.

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Geometry

G1. Let ABC be an isosceles triangle with $BC = CA$, and let D be a point inside side AB such that $AD < DB$. Let P and Q be two points inside sides BC and CA, respectively, such that $\angle DPB = \angle DQA = 90^\circ$. Let the perpendicular bisector of PQ meet line segment CQ at E, and let the circumcircles of triangles ABC and CPQ meet again at point F, different from C.

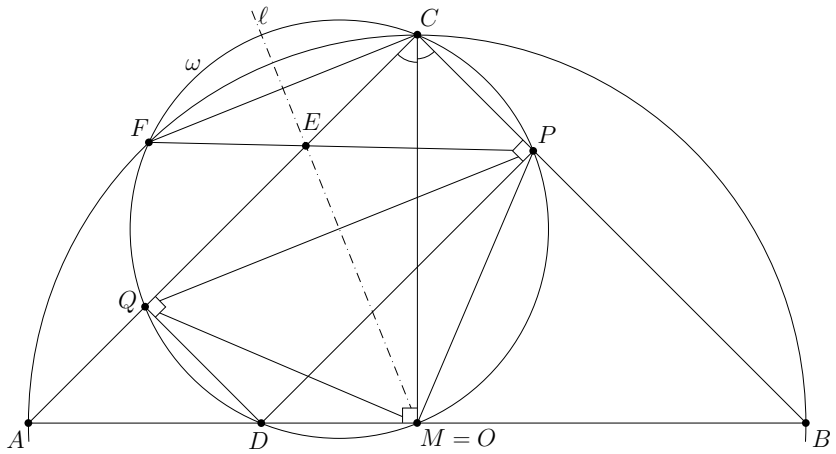
Suppose that P, E, F are collinear. Prove that $\angle ACB = 90^\circ$.

(Luxembourg)

Solution 1. Let ℓ be the perpendicular bisector of PQ, and denote by ω the circle $CFPQ$. By $DP \perp BC$ and $DQ \perp AC$, the circle ω passes through D; moreover, CD is a diameter of ω.

The lines QE and PE are symmetric about ℓ, and ℓ is a symmetry axis of ω as well; it follows that the chords CQ and FP are symmetric about ℓ, hence C and F are symmetric about ℓ. Therefore, the perpendicular bisector of CF coincides with ℓ. Thus ℓ passes through the circumcenter O of ABC.

Let M be the midpoint of AB. Since $CM \perp DM$, M also lies on ω. By $\angle ACM = \angle BCM$, the chords MP and MQ of ω are equal. Then, from $MP = MQ$ it follows that ℓ passes through M.



Finally, both O and M lie on lines ℓ and CM, therefore $O = M$, and $\angle ACB = 90^\circ$ follows.

Solution 2. Like in the first solution, we conclude that points C, P, Q, D, F and the midpoint M of AB lie on one circle ω with diameter CD, and M lies on ℓ, the perpendicular bisector of PQ.

Let BF and CM meet at G and let $\alpha = \angle ABF$. Then, since E lies on ℓ, and the quadrilaterals $FCBA$ and $FCPQ$ are cyclic, we have

$$\angle CQP = \angle FPQ = \angle FCQ = \angle FCA = \angle FBA = \alpha.$$

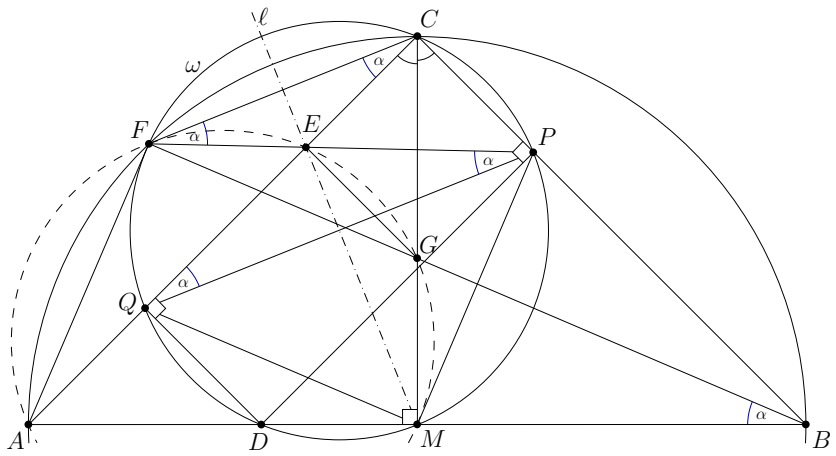
Since points P, E, F are collinear, we have

$$\angle FEM = \angle FEQ + \angle QEM = 2\alpha + (90^\circ - \alpha) = 90^\circ + \alpha.$$

But $\angle FGM = 90^\circ + \alpha$, so $FEGM$ is cyclic. Hence

$$\angle EGC = \angle EFM = \angle PFM = \angle PCM.$$

Thus $GE \parallel BC$. It follows that $\angle FAC = \angle CBF = \angle EGF$, so $FEGA$ is cyclic, too. Hence $\angle ACB = \angle AFB = \angle AFG = 180^\circ - \angle AMG = 90^\circ$, that completes the proof.



Comment 1. The converse statement is true: if $\angle ACB = 90^\circ$ then points P, E and F are collinear. This direction is easier to prove.

Comment 2. The statement of the problem remains true if the projection P of D onto BC lies outside line segment BC. The restriction that P lies inside line segment BC is given to reduce case-sensitivity.

G2. Let $ABCD$ be a convex quadrilateral. Suppose that P is a point in the interior of $ABCD$ such that

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC.$$

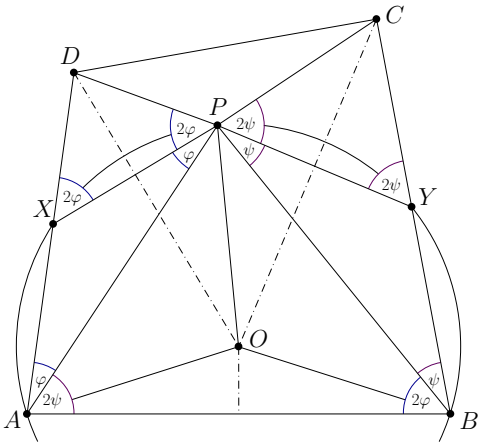
The internal bisectors of angles ADP and PCB meet at a point Q inside the triangle ABP. Prove that $AQ = BQ$.

(Poland)

Solution 1. Let $\varphi = \angle PAD$ and $\psi = \angle CBP$; then we have $\angle PBA = 2\varphi$, $\angle DPA = 3\varphi$, $\angle BAP = 2\psi$ and $\angle BPC = 3\psi$. Let X be the point on segment AD with $\angle XPA = \varphi$. Then

$$\angle PXD = \angle PAX + \angle XPA = 2\varphi = \angle DPA - \angle XPA = \angle DPX.$$

It follows that triangle DPX is isosceles with $DX = DP$ and therefore the internal angle bisector of $\angle ADP$ coincides with the perpendicular bisector of XP. Similarly, if Y is a point on BC such that $\angle BPY = \psi$, then the internal angle bisector of $\angle PCB$ coincides with the perpendicular bisector of PY. Hence, we have to prove that the perpendicular bisectors of XP, PY, and AB are concurrent.



Notice that

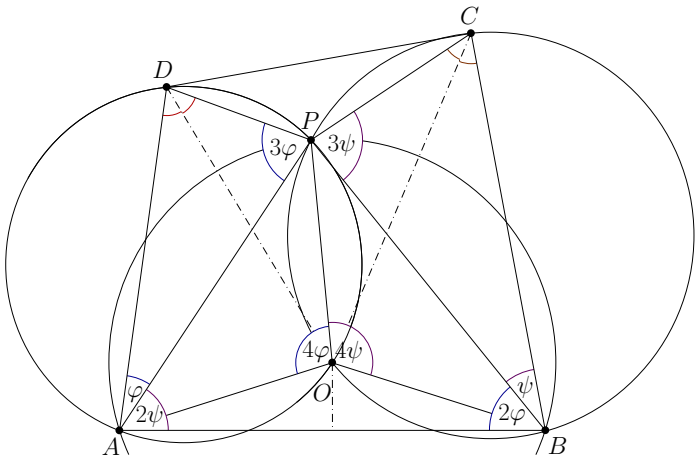
$$\angle AXP = 180^\circ - \angle PXD = 180^\circ - 2\varphi = 180^\circ - \angle PBA.$$

Hence the quadrilateral $AXPB$ is cyclic; in other words, X lies on the circumcircle of triangle APB. Similarly, Y lies on the circumcircle of triangle APB. It follows that the perpendicular bisectors of XP, PY, and AB all pass through the center of circle $(ABYPX)$. This finishes the proof.

Comment. Introduction of points X and Y seems to be the key step in the solution above. Note that the same point X could be introduced in different ways, e.g., as the point on the ray CP beyond P such that $\angle PBX = \varphi$, or as a point where the circle (APB) meets again AB. Different definitions of X could lead to different versions of the further solution.

Solution 2. We define the angles $\varphi = \angle PAD$, $\psi = \angle CBP$ and use $\angle PBA = 2\varphi$, $\angle DPA = 3\varphi$, $\angle BAP = 2\psi$ and $\angle BPC = 3\psi$ again. Let O be the circumcenter of $\triangle APB$.

Notice that $\angle ADP = 180^\circ - \angle PAD - \angle DPA = 180^\circ - 4\varphi$, which, in particular, means that $4\varphi < 180^\circ$. Further, $\angle POA = 2\angle PBA = 4\varphi = 180^\circ - \angle ADP$, therefore the quadrilateral $ADPO$ is cyclic. By $AO = OP$, it follows that $\angle ADO = \angle ODP$. Thus DO is the internal bisector of $\angle ADP$. Similarly, CO is the internal bisector of $\angle PCB$.



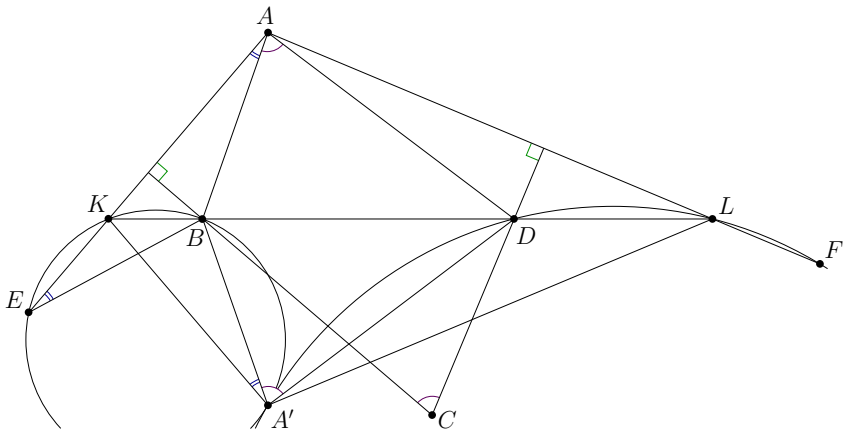
Finally, O lies on the perpendicular bisector of AB as it is the circumcenter of $\triangle APB$. Therefore the three given lines in the problem statement concur at point O.

G3. Let $ABCD$ be a convex quadrilateral with $\angle ABC > 90^\circ$, $\angle CDA > 90^\circ$, and $\angle DAB = \angle BCD$. Denote by E and F the reflections of A in lines BC and CD, respectively. Suppose that the segments AE and AF meet the line BD at K and L, respectively. Prove that the circumcircles of triangles BEK and DFL are tangent to each other.

(Slovakia)

Solution 1. Denote by A' the reflection of A in BD. We will show that that the quadrilaterals $A'BKE$ and $A'DLF$ are cyclic, and their circumcircles are tangent to each other at point A'.

From the symmetry about line BC we have $\angle BEK = \angle BAK$, while from the symmetry in BD we have $\angle BAK = \angle BA'K$. Hence $\angle BEK = \angle BA'K$, which implies that the quadrilateral $A'BKE$ is cyclic. Similarly, the quadrilateral $A'DLF$ is also cyclic.



For showing that circles $A'BKE$ and $A'DLF$ are tangent it suffices to prove that

$$\angle A'KB + \angle A'LD = \angle BA'D.$$

Indeed, by $AK \perp BC$, $AL \perp CD$, and again the symmetry in BD we have

$$\angle A'KB + \angle A'LD = 180^\circ - \angle KA'L = 180^\circ - \angle KAL = \angle BCD = \angle BAD = \angle BA'D,$$

as required.

Comment 1. The key to the solution above is introducing the point A'; then the angle calculations can be done in many different ways.

Solution 2. Note that $\angle KAL = 180^\circ - \angle BCD$, since AK and AL are perpendicular to BC and CD, respectively. Reflect both circles (BEK) and (DFL) in BD. Since $\angle KEB = \angle KAB$ and $\angle DFL = \angle DAL$, the images are the circles (KAB) and (LAD), respectively; so they meet at A. We need to prove that those two reflections are tangent at A.

For this purpose, we observe that

$$\angle AKB + \angle ALD = 180^\circ - \angle KAL = \angle BCD = \angle BAD.$$

Thus, there exists a ray AP inside angle $\angle BAD$ such that $\angle BAP = \angle AKB$ and $\angle DAP = \angle DLA$. Hence the line AP is a common tangent to the circles (KAB) and (LAD), as desired.

Comment 2. The statement of the problem remains true for a more general configuration, e.g., if line BD intersect the extension of segment AE instead of the segment itself, etc. The corresponding restrictions in the statement are given to reduce case sensitivity.

G4. In the plane, there are $n \geqslant 6$ pairwise disjoint disks $D_1, D_2, \ldots, D_n$ with radii $R_1 \geqslant R_2 \geqslant \ldots \geqslant R_n$. For every $i = 1, 2, \ldots, n$, a point P_i is chosen in disk D_i. Let O be an arbitrary point in the plane. Prove that

$$OP_1 + OP_2 + \ldots + OP_n \geqslant R_6 + R_7 + \ldots + R_n.$$

(A disk is assumed to contain its boundary.)

(Iran)

Solution. We will make use of the following lemma.

Lemma. Let $D_1, \ldots, D_6$ be disjoint disks in the plane with radii $R_1, \ldots, R_6$. Let P_i be a point in D_i, and let O be an arbitrary point. Then there exist indices i and j such that $OP_i \geqslant R_j$.

Proof. Let O_i be the center of D_i. Consider six rays OO_1, …, OO_6 (if $O = O_i$, then the ray OO_i may be assumed to have an arbitrary direction). These rays partition the plane into six angles (one of which may be non-convex) whose measures sum up to 360°; hence one of the angles, say $\angle O_iOO_j$, has measure at most 60°. Then O_iO_j cannot be the unique largest side in (possibly degenerate) triangle OO_iO_j, so, without loss of generality, $OO_i \geqslant O_iO_j \geqslant R_i + R_j$. Therefore, $OP_i \geqslant OO_i - R_i \geqslant (R_i + R_j) - R_i = R_j$, as desired. □

Now we prove the required inequality by induction on $n \geqslant 5$. The base case $n = 5$ is trivial. For the inductive step, apply the Lemma to the six largest disks, in order to find indices i and j such that $1 \leqslant i, j \leqslant 6$ and $OP_i \geqslant R_j \geqslant R_6$. Removing D_i from the configuration and applying the inductive hypothesis, we get

$$\sum_{k \neq i} OP_k \geqslant \sum_{\ell \geqslant 7} R_\ell.$$

Adding up this inequality with $OP_i \geqslant R_6$ we establish the inductive step.

Comment 1. It is irrelevant to the problem whether the disks contain their boundaries or not. This condition is included for clarity reasons only. The problem statement remains true, and the solution works verbatim, if the disks are assumed to have disjoint interiors.

Comment 2. There are several variations of the above solution. In particular, while performing the inductive step, one may remove the disk with the largest value of OP_i and apply the inductive hypothesis to the remaining disks (the Lemma should still be applied to the six largest disks).

Comment 3. While proving the Lemma, one may reduce it to a particular case when the disks are congruent, as follows: Choose the smallest radius r of the disks in the Lemma statement, and then replace, for each i, the ith disk with its homothetic copy, using the homothety centered at P_i with ratio r/R_i.

This argument shows that the Lemma is tightly connected to a circle packing problem, see, e.g., https://en.wikipedia.org/wiki/Circle_packing_in_a_circle. The known results on that problem provide versions of the Lemma for different numbers of disks, which lead to different inequalities of the same kind. E.g., for 4 disks the best possible estimate in the Lemma is $OP_i \geqslant (\sqrt{2} - 1)R_j$, while for 13 disks it has the form $OP_i \geqslant \sqrt{5}\, R_j$. Arguing as in the above solution, one obtains the inequalities

$$\sum_{i=1}^{n} OP_i \geqslant (\sqrt{2} - 1) \sum_{j=4}^{n} R_j \quad \text{and} \quad \sum_{i=1}^{n} OP_i \geqslant \sqrt{5} \sum_{j=13}^{n} R_j.$$

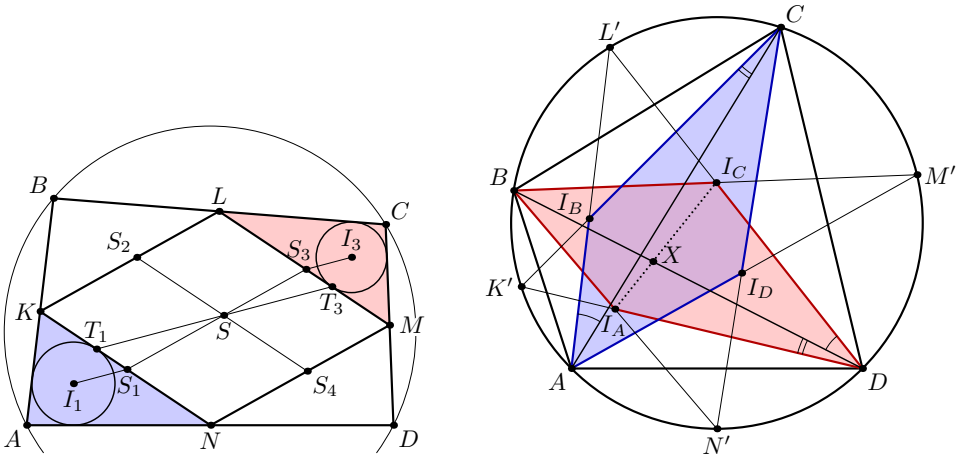
However, there are some harder arguments which allow to improve these inequalities, meaning that the R_j with large indices may be taken with much greater factors.

G5. Let $ABCD$ be a cyclic quadrilateral with no two sides parallel. Let K, L, M, and N be points lying on sides AB, BC, CD, and DA, respectively, such that $KLMN$ is a rhombus with $KL \parallel AC$ and $LM \parallel BD$. Let ω_1, ω_2, ω_3, and ω_4 be the incircles of triangles ANK, BKL, CLM, and DMN, respectively. Prove that the internal common tangents to ω_1 and ω_3 and the internal common tangents to ω_2 and ω_4 are concurrent.

(Poland)

Solution 1. Let I_i be the center of ω_i, and let r_i be its radius for $i = 1, 2, 3, 4$. Denote by T_1 and T_3 the points of tangency of ω_1 and ω_3 with NK and LM, respectively. Suppose that the internal common tangents to ω_1 and ω_3 meet at point S, which is the center of homothety h with negative ratio (namely, with ratio $-\dfrac{r_3}{r_1}$) mapping ω_1 to ω_3. This homothety takes T_1 to T_3 (since the tangents to ω_1 and ω_3 at T_1 to T_3 are parallel), hence S is a point on the segment T_1T_3 with $T_1S : ST_3 = r_1 : r_3$.

Construct segments $S_1S_3 \parallel KL$ and $S_2S_4 \parallel LM$ through S with $S_1 \in NK$, $S_2 \in KL$, $S_3 \in LM$, and $S_4 \in MN$. Note that h takes S_1 to S_3, hence $I_1S_1 \parallel I_3S_3$, and $S_1S : SS_3 = r_1 : r_3$. We will prove that $S_2S : SS_4 = r_2 : r_4$ or, equivalently, $KS_1 : S_1N = r_2 : r_4$. This will yield the problem statement; indeed, applying similar arguments to the intersection point S' of the internal common tangents to ω_2 and ω_4, we see that S' satisfies similar relations, and there is a unique point inside $KLMN$ satisfying them. Therefore, $S' = S$.



Further, denote by I_A, I_B, I_C, I_D and r_A, r_B, r_C, r_D the incenters and inradii of triangles DAB, ABC, BCD, and CDA, respectively. One can shift triangle CLM by $\overrightarrow{LK}$ to glue it with triangle AKN into a quadrilateral $AKC'N$ similar to $ABCD$. In particular, this shows that $r_1 : r_3 = r_A : r_C$; similarly, $r_2 : r_4 = r_B : r_D$. Moreover, the same shift takes S_3 to S_1, and it also takes I_3 to the incenter I_3' of triangle $KC'N$. Since $I_1S_1 \parallel I_3S_3$, the points I_1, S_1, I_3' are collinear. Thus, in order to complete the solution, it suffices to apply the following Lemma to quadrilateral $AKC'N$.

Lemma 1. Let $ABCD$ be a cyclic quadrilateral, and define I_A, I_C, r_B, and r_D as above. Let I_AI_C meet BD at X; then $BX : XD = r_B : r_D$.

Proof. Consider an inversion centered at X; the images under that inversion will be denoted by primes, e.g., A' is the image of A.

By properties of inversion, we have

$$\angle I_C'I_A'D' = \angle XI_A'D' = \angle XDI_A = \angle BDA/2 = \angle BCA/2 = \angle ACI_B.$$

We obtain $\angle I_A'I_C'D' = \angle CAI_B$ likewise; therefore, $\triangle I_C'I_A'D' \sim \triangle ACI_B$. In the same manner, we get $\triangle I_C'I_A'B' \sim \triangle ACI_D$, hence the quadrilaterals $I_C'B'I_A'D'$ and AI_DCI_B are also similar. But the diagonals AC and I_BI_D of quadrilateral AI_DCI_B meet at a point Y such that I_BY :

$YI_D = r_B : r_D$. By similarity, we get $D'X : B'X = r_B : r_D$ and hence $BX : XD = D'X : B'X = r_B : r_D$. □

Comment 1. The solution above shows that the problem statement holds also for any *parallelogram* $KLMN$ whose sides are parallel to the diagonals of $ABCD$, as no property specific for a rhombus has been used. This solution works equally well when two sides of quadrilateral $ABCD$ are parallel.

Comment 2. The problem may be reduced to Lemma 1 by using different tools, e.g., by using mass point geometry, linear motion of K, L, M, and N, etc.

Lemma 1 itself also can be proved in different ways. We present below one alternative proof.

Proof. In the circumcircle of $ABCD$, let K', L'. M', and N' be the midpoints of arcs AB, BC, CD, and DA containing no other vertices of $ABCD$, respectively. Thus, $K' = CI_B \cap DI_A$, etc. In the computations below, we denote by $[P]$ the area of a polygon P. We use similarities $\triangle I_A BK' \sim \triangle I_A DN'$, $\triangle I_B K'L' \sim \triangle I_B AC$, etc., as well as congruences $\triangle I_B K'L' = \triangle BK'L'$ and $\triangle I_D M'N' = \triangle DM'N'$ (e.g., the first congruence holds because $K'L'$ is a common internal bisector of angles $BK'I_B$ and $BL'I_B$).

We have

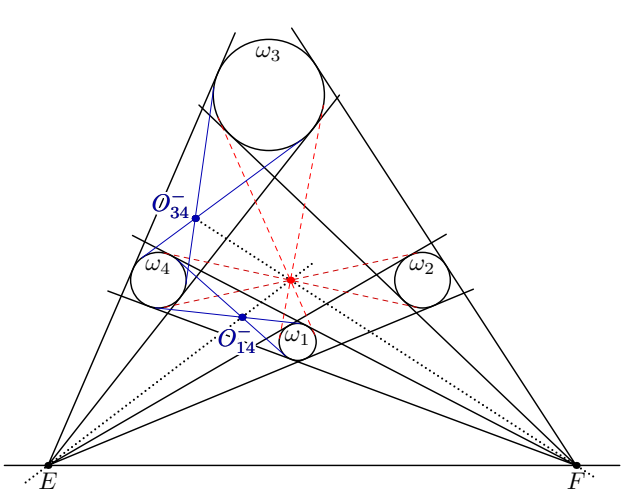
$$\frac{BX}{DX} = \frac{[I_A BI_C]}{[I_A DI_C]} = \frac{BI_A \cdot BI_C \cdot \sin I_A BI_C}{DI_A \cdot DI_C \cdot \sin I_A DI_C} = \frac{BI_A}{DI_A} \cdot \frac{BI_C}{DI_C} \cdot \frac{\sin N'BM'}{\sin K'DL'}$$
$$= \frac{BK'}{DN'} \cdot \frac{BL'}{DM'} \cdot \frac{\sin N'DM'}{\sin K'BL'} = \frac{BK' \cdot BL' \cdot \sin K'BL'}{DN' \cdot DM' \cdot \sin N'DM'} \cdot \frac{\sin^2 N'DM'}{\sin^2 K'BL'}$$
$$= \frac{[K'BL']}{[M'DN']} \cdot \frac{N'M'^2}{K'L'^2} = \frac{[K'I_B L'] \cdot \frac{A'C'^2}{K'L'^2}}{[M'I_D N'] \cdot \frac{A'C'^2}{N'M'^2}} = \frac{[AI_B C]}{[AI_D C]} = \frac{r_B}{r_D},$$

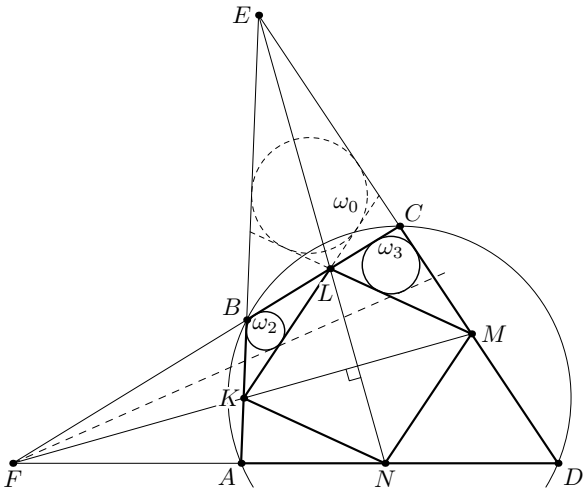
as required. □

Solution 2. This solution is based on the following general Lemma.

Lemma 2. Let E and F be distinct points, and let ω_i, $i = 1, 2, 3, 4$, be circles lying in the same halfplane with respect to EF. For distinct indices $i, j \in \{1, 2, 3, 4\}$, denote by O_{ij}^+ (respectively, O_{ij}^-) the center of homothety with positive (respectively, negative) ratio taking ω_i to ω_j. Suppose that $E = O_{12}^+ = O_{34}^+$ and $F = O_{23}^+ = O_{41}^+$. Then $O_{13}^- = O_{24}^-$.

Proof. Applying Monge's theorem to triples of circles $\omega_1, \omega_2, \omega_4$ and $\omega_1, \omega_3, \omega_4$, we get that both points O_{24}^- and O_{13}^- lie on line EO_{14}^-. Notice that this line is distinct from EF. Similarly we obtain that both points O_{24}^- and O_{13}^- lie on FO_{34}^-. Since the lines EO_{14}^- and FO_{34}^- are distinct, both points coincide with the meeting point of those lines. □





Turning back to the problem, let AB intersect CD at E and let BC intersect DA at F. Assume, without loss of generality, that B lies on segments AE and CF. We will show that the points E and F, and the circles ω_i satisfy the conditions of Lemma 2, so the problem statement follows. In the sequel, we use the notation of $O_{ij}^{\pm}$ from the statement of Lemma 2, applied to circles ω_1, ω_2, ω_3, and ω_4.

Using the relations $\triangle ECA \sim \triangle EBD$, $KN \parallel BD$, and $MN \parallel AC$. we get

$$\frac{AN}{ND} = \frac{AN}{AD} \cdot \frac{AD}{ND} = \frac{KN}{BD} \cdot \frac{AC}{NM} = \frac{AC}{BD} = \frac{AE}{ED}.$$

Therefore, by the angle bisector theorem, point N lies on the internal angle bisector of $\angle AED$. We prove similarly that L also lies on that bisector, and that the points K and M lie on the internal angle bisector of $\angle AFB$.

Since $KLMN$ is a rhombus, points K and M are symmetric in line ELN. Hence, the convex quadrilateral determined by the lines EK, EM, KL, and ML is a kite, and therefore it has an incircle ω_0. Applying Monge's theorem to ω_0, ω_2, and ω_3, we get that O_{23}^{+} lies on KM. On the other hand, O_{23}^{+} lies on BC, as BC is an external common tangent to ω_2 and ω_3. It follows that $F = O_{23}^{+}$. Similarly, $E = O_{12}^{+} = O_{34}^{+}$, and $F = O_{41}^{+}$.

Comment 3. The reduction to Lemma 2 and the proof of Lemma 2 can be performed with the use of different tools, e.g., by means of Menelaus theorem, by projecting harmonic quadruples, by applying Monge's theorem in some other ways, etc.

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G6. Let I and I_A be the incenter and the A-excenter of an acute-angled triangle ABC with $AB < AC$. Let the incircle meet BC at D. The line AD meets BI_A and CI_A at E and F, respectively. Prove that the circumcircles of triangles AID and I_AEF are tangent to each other.

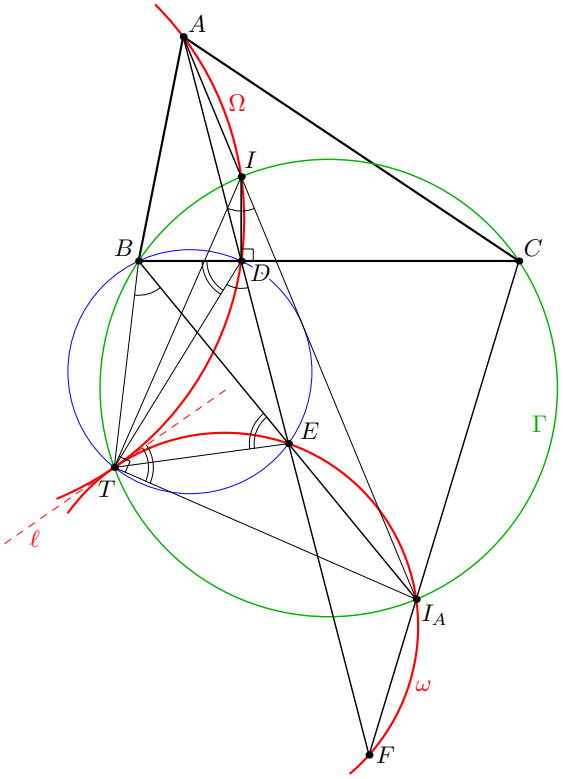
(Slovakia)

Solution 1. Let $\measuredangle(p, q)$ denote the directed angle between lines p and q.

The points B, C, I, and I_A lie on the circle Γ with diameter II_A. Let ω and Ω denote the circles (I_AEF) and (AID), respectively. Let T be the second intersection point of ω and Γ. Then T is the Miquel point of the complete quadrilateral formed by the lines BC, BI_A, CI_A, and DEF, so T also lies on circle (BDE) (as well as on circle (CDF)). We claim that T is a desired tangency point of ω and Ω.

In order to show that T lies on Ω, use cyclic quadrilaterals $BDET$ and BII_AT to write

$$\measuredangle(DT, DA) = \measuredangle(DT, DE) = \measuredangle(BT, BE) = \measuredangle(BT, BI_A) = \measuredangle(IT, II_A) = \measuredangle(IT, IA).$$



To show that ω and Ω are tangent at T, let ℓ be the tangent to ω at T, so that $\measuredangle(TI_A, \ell) = \measuredangle(EI_A, ET)$. Using circles $(BDET)$ and $(BICI_A)$, we get

$$\measuredangle(EI_A, ET) = \measuredangle(EB, ET) = \measuredangle(DB, DT).$$

Therefore,

$$\measuredangle(TI, \ell) = 90^\circ + \measuredangle(TI_A, \ell) = 90^\circ + \measuredangle(DB, DT) = \measuredangle(DI, DT),$$

which shows that ℓ is tangent to Ω at T.

Solution 2. We use the notation of circles Γ, ω, and Ω as in the previous solution.

Let L be the point opposite to I in circle Ω. Then $\angle IAL = \angle IDL = 90^\circ$, which means that L is the foot of the external bisector of $\angle A$ in triangle ABC. Let LI cross Γ again at M.

Let T be the foot of the perpendicular from I onto I_AL. Then T is the second intersection point of Γ and Γ. We will show that T is the desired tangency point.

First, we show that T lies on circle ω. Notice that

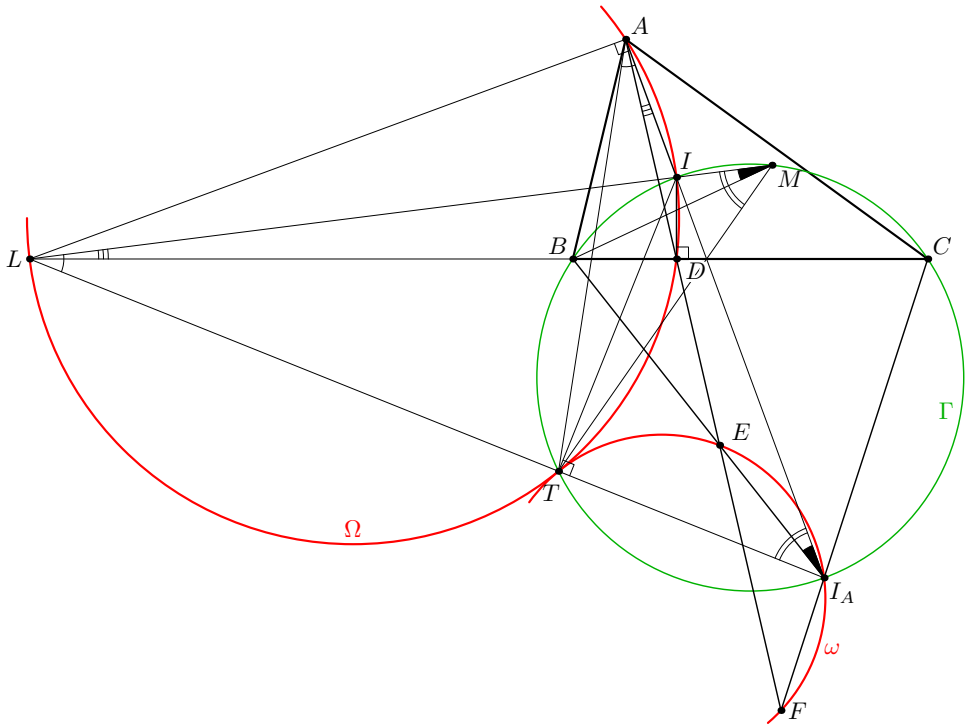
$$\sphericalangle(LT, LM) = \sphericalangle(AT, AI) \quad \text{and} \quad \sphericalangle(MT, ML) = \sphericalangle(MT, MI) = \sphericalangle(I_AT, I_AI),$$

which shows that triangles TML and TI_AA are similar and equioriented. So there exists a rotational homothety τ mapping TML to TI_AA.

Since $\sphericalangle(ML, LD) = \sphericalangle(AI, AD)$, we get $\tau(BC) = AD$. Next, since

$$\sphericalangle(MB, ML) = \sphericalangle(MB, MI) = \sphericalangle(I_AB, I_AI) = \sphericalangle(I_AE, I_AA),$$

we get $\tau(B) = E$. Similarly, $\tau(C) = F$. Since the points M, B, C, and T are concyclic, so are their τ-images, which means that T lies on $\omega = \tau(\Gamma)$.



Finally, since $\tau(L) = A$ and $\tau(B) = E$, triangles ATL and ETB are similar so that

$$\sphericalangle(AT, AL) = \sphericalangle(ET, EB) = \sphericalangle(EI_A, ET).$$

This means that the tangents to Ω and ω at T make the same angle with the line I_ATL, so the circles are indeed tangent at T.

Comment. In both solutions above, a crucial step is a guess that the desired tangency point lies on Γ. There are several ways to recognize this helpful property.

E.g. one may perform some angle chasing to see that the tangents to Ω at L and to ω at I_A are parallel (and the circles lie on different sides of the tangents). This yields that, under the assumption that the circles are tangent externally, the tangency point must lie on I_AL. Since IL is a diameter in Ω, this, in turn, implies that T is the projection of I onto I_AL.

Another way to see the same fact is to perform a homothety centered at A and mapping I to I_A (and D to some point D'). The image Ω' of Ω is tangent to ω at I_A, because $\angle BI_AA + \angle CI_AD' = 180^\circ$. This yields that the tangents to Ω at I and to ω at I_A are parallel.

There are other ways to describe the tangency point. The next solution presents one of them.

Solution 3. We also use the notation of circles ω, and Ω from the previous solutions.

Perform an inversion centered at D. The images of the points will be denoted by primes, e.g., A' is the image of A.

For convenience, we use the notation $\angle BID = \beta$, $\angle CID = \gamma$, and $\alpha = 180^\circ - \beta - \gamma = 90^\circ - \angle BAI$. We start with computing angles appearing after inversion. We get

$$\angle DB'I' = \beta, \quad \angle DC'I' = \gamma, \quad \text{and hence} \quad \angle B'I'C' = \alpha;$$
$$\angle E'I'_AF' = \angle E'I'_AD - \angle F'I'_AD = \angle I_AED - \angle I_AFD = \angle EI_AF = 180^\circ - \alpha.$$

Next, we have

$$\angle A'E'B' = \angle DE'B' = \angle DBE = \beta = 90^\circ - \frac{\angle DBA}{2} = 90^\circ - \frac{\angle E'A'B'}{2},$$

which yields that triangle $A'B'E'$ is isosceles with $A'B' = A'E'$. Similarly, $A'F' = A'C'$.

Finally, we get

$$\angle A'B'I' = \angle I'B'D - \angle A'B'D = \beta - \angle BAD = \beta - (90^\circ - \alpha) + \angle IAD$$
$$= \angle ICD + \angle IAD = \angle C'I'D + \angle A'I'D = \angle C'I'A';$$

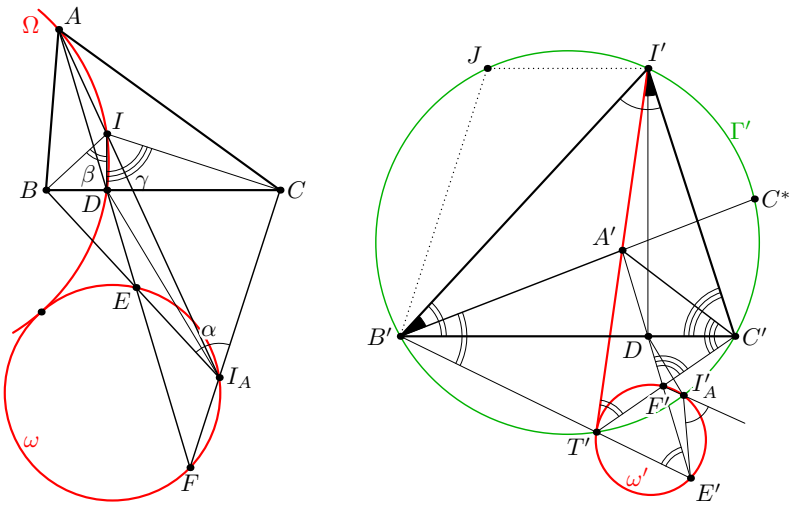
similarly, $\angle A'C'I' = \angle A'I'B'$, so that triangles $A'B'I'$ and $A'I'C'$ are similar. Therefore, $A'I'^2 = A'B' \cdot A'C'$.

Recall that we need to prove the tangency of line $A'I' = \Omega'$ with circle $(E'F'I'_A) = \omega'$. A desired tangency point T' must satisfy $A'T'^2 = A'E' \cdot A'F'$; the relations obtained above yield

$$A'E' \cdot A'F' = A'B' \cdot A'C' = A'I'^2,$$

so that T' should be symmetric to I' with respect to A'.

Thus, let us *define* a point T' as the reflection of I' in A', and show that T' lies on circle Ω'; the equalities above will then imply that $A'T'$ is tangent to Ω', as desired.



The property that triangles $B'A'I'$ and $I'A'C'$ are similar means that quadrilateral $B'I'C'T'$ is harmonic. Indeed, let C^* be the reflection of C' in the perpendicular bisector of $I'T'$; then C^* lies on $B'A'$ by $\angle B'A'I' = \angle A'I'C' = \angle T'I'C^*$, and then C^* lies on circle $(I'B'T')$ since $A'B' \cdot A'C^* = A'B' \cdot A'C' = A'I'^2 = A'I' \cdot A'T'$. Therefore, C' also lies on that circle (and the circle is $(B'I'C') = \Gamma'$). Moreover, $B'C^*$ is a median in triangle $B'I'T'$, so $B'C'$ is its symmedian, which establishes harmonicity.

Now we have $\angle A'B'T' = \angle I'B'C' = \beta = \angle A'B'E'$; which shows that E' lies on $B'T'$. Similarly, F' lies on $C'T'$. Hence, $\angle E'T'F' = \angle B'I'C' = 180^\circ - \angle E'I'_AF'$, which establishes $T' \in \omega'$.

Comment 2. The solution above could be finished without use of harmonicity. E.g., one may notice that both triangles $A'T'F'$ and $A'E'T'$ are similar to triangle $B'I'J$, where J is the point symmetric to I' in the perpendicular bisector of $B'C'$; indeed, we have $\angle T'A'E' = \gamma - \beta = \angle I'B'J'$ and $\frac{B'I'}{B'J'} = \frac{B'I'}{C'I'} = \frac{B'A'}{A'I'} = \frac{A'E'}{A'T'}$. This also allows to compute $\angle E'T'F' = \angle E'T'A' - \angle F'T'A' = \angle I'JB' - \angle JI'B' = \alpha$.

Comment 3. Here we list several properties of the configuration in the problem, which can be derived from the solutions above.

The quadrilateral $IBTC$ (as well as $I'B'T'C'$) is harmonic. Hence, line IT contains the meeting point of tangents to Γ at B and C, i.e., the midpoint N of arc BAC in the circumcircle of $\triangle ABC$.

G7. Let P be a point on the circumcircle of an acute-angled triangle ABC. Let D, E, and F be the reflections of P in the midlines of triangle ABC parallel to BC, CA, and AB, respectively. Denote by ω_A, ω_B, and ω_C the circumcircles of triangles ADP, BEP, and CFP, respectively. Denote by ω the circumcircle of the triangle formed by the perpendicular bisectors of segments AD, BE and CF.

Show that ω_A, ω_B, ω_C, and ω have a common point.

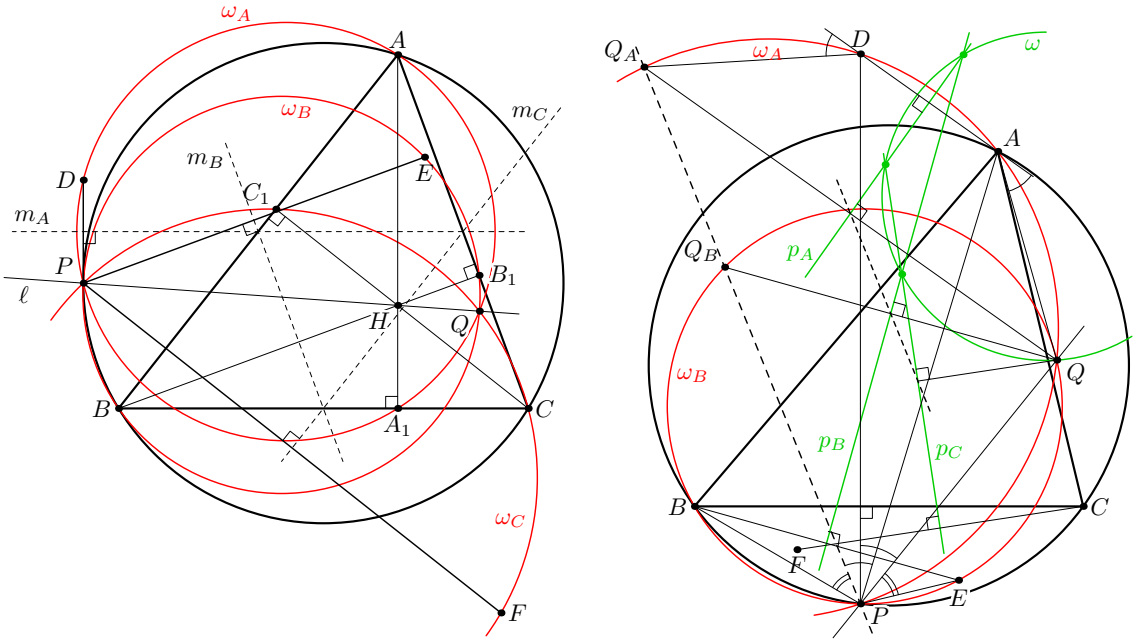
(Denmark)

Solution. Let AA_1, BB_1, and CC_1 be the altitudes in triangle ABC, and let m_A, m_B, and m_C be the midlines parallel to BC, CA, and AB, respectively. We always denote by $\measuredangle(p, q)$ the directed angle from a line p to a line q, taken modulo 180°.

Step 1: Circles ω_A, ω_B, and ω_C share a common point Q different from P.

Notice that m_A is the perpendicular bisector of PD, so ω_A is symmetric with respect to m_A. Since A and A_1 are also symmetric to each other in m_A, the point A_1 lies on ω_A. Similarly, B_1 and C_1 lie on ω_B and ω_C, respectively.

Let H be the orthocenter of $\triangle ABC$. Quadrilaterals ABA_1B_1 and BCB_1C_1 are cyclic, so $AH \cdot HA_1 = BH \cdot HB_1 = CH \cdot HC_1$. This means that H lies on pairwise radical axes of ω_A, ω_B, and ω_C. Point P also lies on those radical axes; hence the three circles have a common radical axis $\ell = PH$, and the second meeting point Q of ℓ with ω_A is the second common point of the three circles. Notice here that H lies inside all three circles, hence $Q \neq P$.



Step 2: Point Q lies on ω.

Let p_A, p_B, and p_C denote the perpendicular bisectors of AD, BE, and CF, respectively; denote by Δ the triangle formed by those perpendicular bisectors. By Simson's theorem, in order to show that Q lies on the circumcircle ω of Δ, it suffices to prove that the projections of Q onto the sidelines p_A, p_B, and p_C are collinear. Alternatively, but equivalently, it suffices to prove that the reflections Q_A, Q_B, and Q_C of Q in those lines, respectively, are collinear. In fact, we will show that four points P, Q_A, Q_B, and Q_C are collinear.

Since p_A is the common perpendicular bisector of AD and QQ_A, the point Q_A lies on ω_A, and, moreover, $\measuredangle(DA, DQ_A) = \measuredangle(AQ, AD)$. Therefore,

$$\measuredangle(PA, PQ_A) = \measuredangle(DA, DQ_A) = \measuredangle(AQ, AD) = \measuredangle(PQ, PD) = \measuredangle(PQ, BC) + 90^\circ.$$

Similarly, we get $\measuredangle(PB, PQ_B) = \measuredangle(PQ, CA) + 90^\circ$. Therefore,

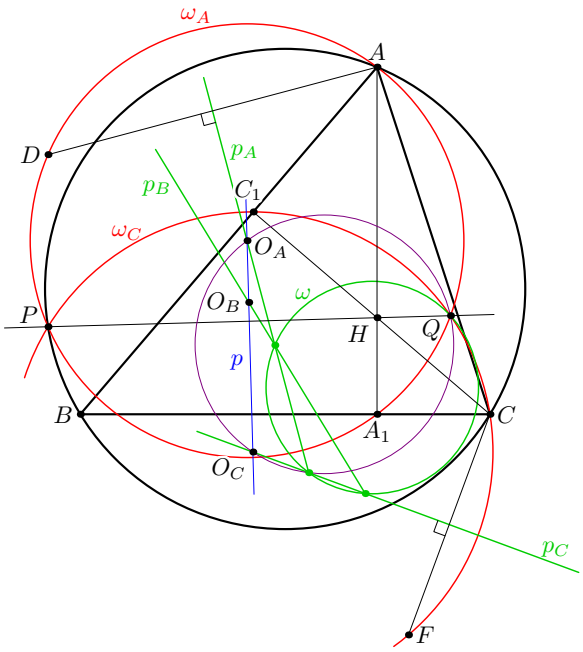
$$\begin{aligned}\measuredangle(PQ_A, PQ_B) &= \measuredangle(PQ_A, PA) + \measuredangle(PA, PB) + \measuredangle(PB, PQ_B)\\ &= \measuredangle(BC, PQ) + 90^\circ + \measuredangle(CA, CB) + \measuredangle(PQ, CA) + 90^\circ = 0,\end{aligned}$$

which shows that P, Q_A, and Q_B are collinear. Similarly, Q_C also lies on PQ_A.

Comment 1. There are several variations of Step 2. In particular, let O_A, O_B, and O_C denote the centers of ω_A, ω_B, and ω_C, respectively; they lie on p_A, p_B, and p_C, respectively. Moreover, all those centers lie on the perpendicular bisector p of PQ. Now one can show that $\measuredangle(QO_A, p_A) = \measuredangle(QO_B, p_B) = \measuredangle(QO_C, p_C)$, and then finish by applying generalized Simson's theorem, Alternatively, but equivalently, those relations show that Q is the Miquel point of the lines p_A, p_B, p_C, and p.

To establish $\measuredangle(QO_A, p_A) = \measuredangle(QO_C, p_C)$, notice that it is equivalent to $\measuredangle(QO_A, QO_C) = \measuredangle(p_A, p_C)$ which may be obtained, e.g., as follows:

$$\begin{aligned}\measuredangle(QO_A, QO_C) &= \measuredangle(QO_A, p) + \measuredangle(p, QO_C) = \measuredangle(AQ, AP) + \measuredangle(CP, CQ)\\ &= \measuredangle(AQ, CQ) + \measuredangle(CP, AP) = \measuredangle(AQ, PQ) + \measuredangle(PQ, CQ) + \measuredangle(CB, AB)\\ &= \measuredangle(AD, AA_1) + \measuredangle(CC_1, CF) + \measuredangle(AA_1, CC_1) = \measuredangle(AD, CF) = \measuredangle(p_A, p_C).\end{aligned}$$



Comment 2. The inversion at H with (negative) power $-AH \cdot HA_1$ maps P to Q, and the circumcircle of $\triangle ABC$ to its Euler circle. Therefore, Q lies on that Euler circle.

G8. Let Γ and I be the circumcircle and the incenter of an acute-angled triangle ABC. Two circles ω_B and ω_C passing through B and C, respectively, are tangent at I. Let ω_B meet the shorter arc AB of Γ and segment AB again at P and M, respectively. Similarly, let ω_C meet the shorter arc AC of Γ and segment AC again at Q and N, respectively. The rays PM and QN meet at X, and the tangents to ω_B and ω_C at B and C, respectively, meet at Y.

Prove that the points A, X, and Y are collinear.

(Netherlands)

Solution 1. Let AI, BI, and CI meet Γ again at D, E, and F, respectively. Let ℓ be the common tangent to ω_B and ω_C at I. We always denote by $\measuredangle(p, q)$ the directed angle from a line p to a line q, taken modulo 180°.

Step 1: We show that Y lies on Γ.

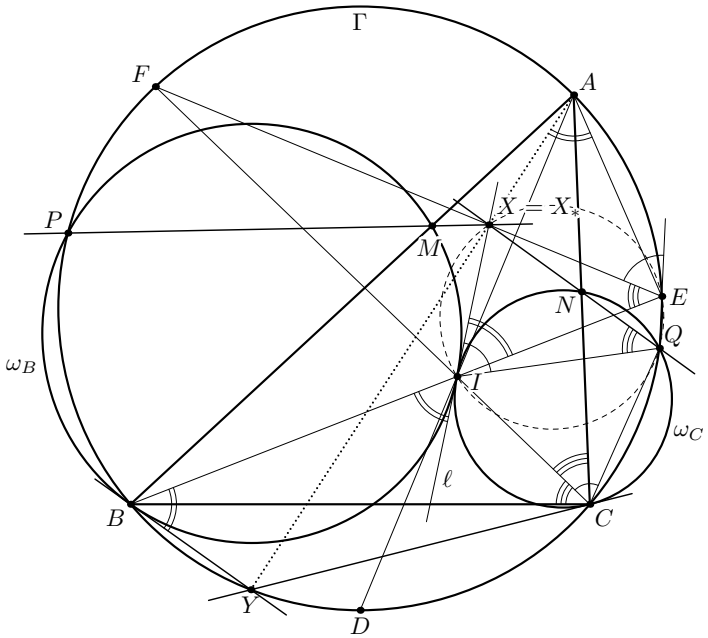
Recall that any chord of a circle makes complementary directed angles with the tangents to the circle at its endpoints. Hence,

$$\measuredangle(BY, BI) + \measuredangle(CI, CY) = \measuredangle(IB, \ell) + \measuredangle(\ell, IC) = \measuredangle(IB, IC).$$

Therefore,

$$\begin{aligned}\measuredangle(BY, BA) + \measuredangle(CA, CY) &= \measuredangle(BI, BA) + \measuredangle(BY, BI) + \measuredangle(CI, CY) + \measuredangle(CA, CI) \\ &= \measuredangle(BC, BI) + \measuredangle(IB, IC) + \measuredangle(CI, CB) = 0,\end{aligned}$$

which yields $Y \in \Gamma$.



Step 2: We show that $X = \ell \cap EF$.

Let $X_* = \ell \cap EF$. To prove our claim, it suffices to show that X_* lies on both PM and QN; this will yield $X_* = X$. Due to symmetry, it suffices to show $X_* \in QN$.

Notice that

$$\measuredangle(IX_*, IQ) = \measuredangle(CI, CQ) = \measuredangle(CF, CQ) = \measuredangle(EF, EQ) = \measuredangle(EX_*, EQ);$$

therefore, the points X_*, I, Q, and E are concyclic (if $Q = E$, then the direction of EQ is supposed to be the direction of a tangent to Γ at Q; in this case, the equality means that the circle (X_*IQ) is tangent to Γ at Q). Then we have

$$\measuredangle(QX_*, QI) = \measuredangle(EX_*, EI) = \measuredangle(EF, EB) = \measuredangle(CA, CF) = \measuredangle(CN, CI) = \measuredangle(QN, QI),$$

which shows that $X_* \in QN$.

Step 3: We finally show that A, X, and Y are collinear.

Recall that I is the orthocenter of triangle DEF, and A is symmetric to I with respect to EF. Therefore,

$$\measuredangle(AX, AE) = \measuredangle(IE, IX) = \measuredangle(BI, \ell) = \measuredangle(BY, BI) = \measuredangle(BY, BE) = \measuredangle(AY, AE),$$

which yields the desired collinearity.

Comment 1. Step 2 in the above solution seems to be crucial. After it has been performed (even without Step 1), there are different ways of finishing the solution.

E.g., one may involve the notion of *isogonal conjugacy*. Let X_1 and Y_1 be isogonal conjugates of X and Y, respectively, with respect to triangle ABC. Since $XA = XI$, triangle AIX is isosceles, and hence the lines AX and XI form equal angles with the internal bisector AI of $\angle BAC$. This means that $AX_1 \parallel XI$, or $AX_1 \parallel \ell$.

On the other hand, the lines BY and ℓ form equal angles with BI, so that $BY_1 \parallel \ell$. Similarly, $CY_1 \parallel \ell$. This means that Y_1 is an ideal point, and $AY_1 \parallel \ell$ as well. Therefore, points A, X_1, and Y_1 are collinear, and hence A, X, and Y are such.

Solution 2. Perform an inversion centered at I; the images of the points are denoted by primes, e.g., A' is the image of A.

On the inverted figure, I and Γ' are the orthocenter and the circumcircle of triangle $A'B'C'$, respectively. The points P' and Q' lie on Γ' such that $B'P' \parallel C'Q'$ (since $B'P' = \omega'_B$ and $C'Q' = \omega'_C$). The points M' and N' are the second intersections of lines $B'P'$ and $C'Q'$ with the circumcircles γ_B and γ_C of triangles $A'IB'$ and $A'IC'$, respectively. Notice here that γ_C is obtained from γ_B by the translation at $\overrightarrow{B'C'}$; the same translation maps line $B'P'$ to $C'Q'$, and hence M' to N'. In other words, $B'M'N'C'$ is a parallelogram, and $P'Q'$ partitions it into two isosceles trapezoids.

Point X' is the second intersection point of circles $(IP'M')$ and $(IQ'N')$ that is — the reflection of I in their line of centers. But the centers lie on the common perpendicular bisector p of $P'M'$ and $Q'N'$, so p is that line of centers. Hence, $IX' \parallel B'P'$, as both lines are perpendicular to p.

Finally, the point Y satisfies $\measuredangle(BY, BI) = \measuredangle(PB, PI)$ and $\measuredangle(CY, CI) = \measuredangle(QC, QI)$, which yields $\measuredangle(Y'B', Y'I) = \measuredangle(B'P', B'I)$ and $\measuredangle(Y'C', Y'I) = \measuredangle(C'Q', C'I)$. Therefore,

$$\measuredangle(Y'B', Y'C') = \measuredangle(B'P', B'I) + \measuredangle(C'I, C'Q') = \measuredangle(C'I, B'I) = \measuredangle(A'B', A'C'),$$

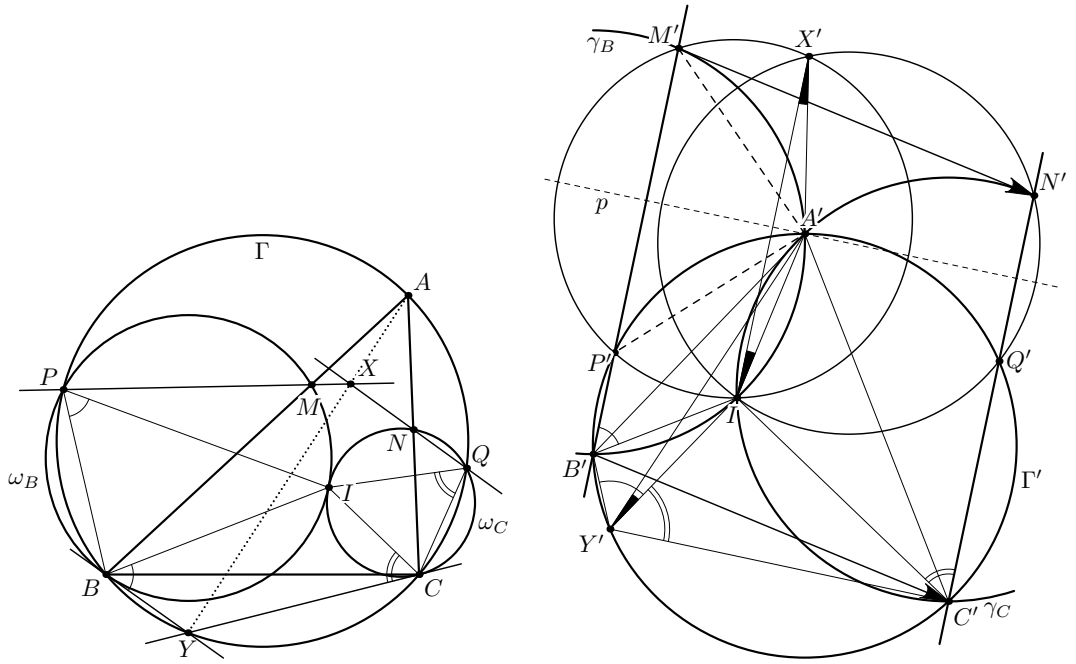
which shows that $Y' \in \Gamma'$.

In congruent circles Γ' and γ_B, the chords $A'P'$ and $A'M'$ subtend the same angle $\angle A'B'P'$; therefore, $A'P' = A'M'$, and hence $A' \in p$. This yields $A'X' = A'I$, and hence $\measuredangle(IA', IX') = \measuredangle(X'I, X'A')$.

Finally, we have

$$\begin{aligned}\measuredangle(Y'I, Y'A') &= \measuredangle(Y'I, Y'B') + \measuredangle(Y'B', Y'A') \\ &= \measuredangle(B'I, B'P') + \measuredangle(IA', IB') = \measuredangle(IA', B'P') = \measuredangle(IA', IX') = \measuredangle(X'I, X'A'),\end{aligned}$$

which yields that the points A', X', Y', and I are concyclic. This means exactly that A, X, and Y are collinear.



Comment 2. An inversion at I may also help in establishing Step 2 in Solution 1. Indeed, relation $A'X' = A'I$ yields $XA = XI$, so that $X \in EF$. On the other hand, $IX' \parallel B'P'$ yields $IX \parallel \ell$, i.e., $X \in \ell$.

G9. Prove that there exists a positive constant c such that the following statement is true:

Assume that n is an integer with $n \geqslant 2$, and let $\mathcal{S}$ be a set of n points in the plane such that the distance between any two distinct points in $\mathcal{S}$ is at least 1. Then there is a line ℓ separating $\mathcal{S}$ such that the distance from any point of $\mathcal{S}$ to ℓ is at least $cn^{-1/3}$.

(A line ℓ *separates* a point set $\mathcal{S}$ if some segment joining two points in $\mathcal{S}$ crosses ℓ.)

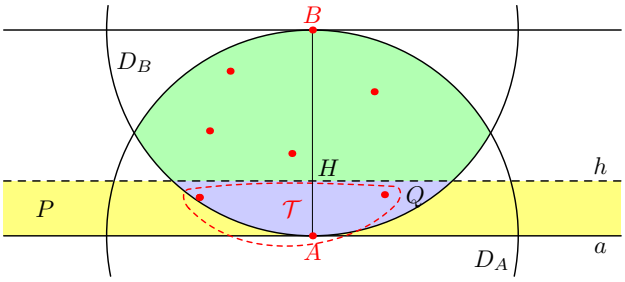
(Taiwan)

Solution. We prove that the desired statement is true with $c = \frac{1}{8}$. Set $\delta = \frac{1}{8}n^{-1/3}$. For any line ℓ and any point X, let X_ℓ denote the projection of X to ℓ; a similar notation applies to sets of points.

Suppose that, for some line ℓ, the set $\mathcal{S}_\ell$ contains two adjacent points X and Y with $XY = 2d$. Then the line perpendicular to ℓ and passing through the midpoint of segment XY separates $\mathcal{S}$, and all points in $\mathcal{S}$ are at least d apart from ℓ. Thus, if $d \geqslant \delta$, then a desired line has been found. For the sake of contradiction, we assume that no such points exist, in any projection.

Choose two points A and B in $\mathcal{S}$ with the maximal distance $M = AB$ (i.e., AB is a *diameter* of $\mathcal{S}$); by the problem condition, $M \geqslant 1$. Denote by ℓ the line AB. The set $\mathcal{S}$ is contained in the intersection of two disks D_A and D_B of radius M centered at A and B, respectively. Hence, the projection $\mathcal{S}_\ell$ is contained in the segment AB. Moreover, the points in $\mathcal{S}_\ell$ divide that segment into at most $n-1$ parts, each of length less than 2δ. Therefore,

$$M < n \cdot 2\delta. \tag{1}$$



Choose a point H on segment AB with $AH = \frac{1}{2}$. Let P be a strip between the lines a and h perpendicular to AB and passing through A and H, respectively; we assume that P contains its boundary, which consists of lines a and h. Set $\mathcal{T} = P \cap \mathcal{S}$ and let $t = |\mathcal{T}|$. By our assumption, segment AH contains at least $\left\lceil \frac{1}{2} : (2\delta) \right\rceil$ points of S_ℓ, which yields

$$t \geqslant \frac{1}{4\delta}. \tag{2}$$

Notice that $\mathcal{T}$ is contained in $Q = P \cap D_B$. The set Q is a circular segment, and its projection Q_a is a line segment of length

$$2\sqrt{M^2 - \left(M - \frac{1}{2}\right)^2} < 2\sqrt{M}.$$

On the other hand, for any two points $X, Y \in \mathcal{T}$, we have $XY \geqslant 1$ and $X_\ell Y_\ell \leqslant \frac{1}{2}$, so $X_a Y_a = \sqrt{XY^2 - X_\ell Y_\ell^2} \geqslant \frac{\sqrt{3}}{2}$. To summarize, t points constituting $\mathcal{T}_a$ lie on the segment of length less than $2\sqrt{M}$, and are at least $\frac{\sqrt{3}}{2}$ apart from each other. This yields $2\sqrt{M} > (t-1)\frac{\sqrt{3}}{2}$, or

$$t < 1 + \frac{4\sqrt{M}}{\sqrt{3}} < 4\sqrt{M}, \tag{3}$$

as $M \geqslant 1$.

Combining the estimates (1), (2), and (3), we finally obtain

$$\frac{1}{4\delta} \leqslant t < 4\sqrt{M} < 4\sqrt{2n\delta}, \quad \text{or} \quad 512n\delta^3 > 1,$$

which does not hold for the chosen value of δ.

Comment 1. As the proposer mentions, the exponent $-1/3$ in the problem statement is optimal. In fact, for any $n \geqslant 2$, there is a configuration $\mathcal{S}$ of n points in the plane such that any two points in $\mathcal{S}$ are at least 1 apart, but every line ℓ separating $\mathcal{S}$ is at most $c'n^{-1/3}\log n$ apart from some point in $\mathcal{S}$; here c' is some absolute constant.

The original proposal suggested to prove the estimate of the form $cn^{-1/2}$. That version admits much easier solutions. E.g., setting $\delta = \frac{1}{16}n^{-1/2}$ and applying (1), we see that $\mathcal{S}$ is contained in a disk D of radius $\frac{1}{8}n^{1/2}$. On the other hand, for each point X of $\mathcal{S}$, let D_X be the disk of radius $\frac{1}{2}$ centered at X; all these disks have disjoint interiors and lie within the disk concentric to D, of radius $\frac{1}{16}n^{1/2} + \frac{1}{2} < \frac{1}{2}n^{1/2}$. Comparing the areas, we get

$$n \cdot \frac{\pi}{4} \leqslant \pi\left(\frac{n^{1/2}}{16} + \frac{1}{2}\right)^2 < \frac{\pi n}{4},$$

which is a contradiction.

The Problem Selection Committee decided to choose a harder version for the Shortlist.

Comment 2. In this comment, we discuss some versions of the solution above, which avoid concentrating on the diameter of $\mathcal{S}$. We start with introducing some terminology suitable for those versions.

Put $\delta = cn^{-1/3}$ for a certain sufficiently small positive constant c. For the sake of contradiction, suppose that, for some set $\mathcal{S}$ satisfying the conditions in the problem statement, there is no separating line which is at least δ apart from each point of $\mathcal{S}$.

Let C be the convex hull of $\mathcal{S}$. A line is separating if and only if it meets C (we assume that a line passing through a point of $\mathcal{S}$ is always separating). Consider a strip between two parallel separating lines a and a' which are, say, $\frac{1}{4}$ apart from each other. Define a *slice* determined by the strip as the intersection of $\mathcal{S}$ with the strip. The *length* of the slice is the diameter of the projection of the slice to a.

In this terminology, the arguments used in the proofs of (2) and (3) show that for any slice $\mathcal{T}$ of length L, we have

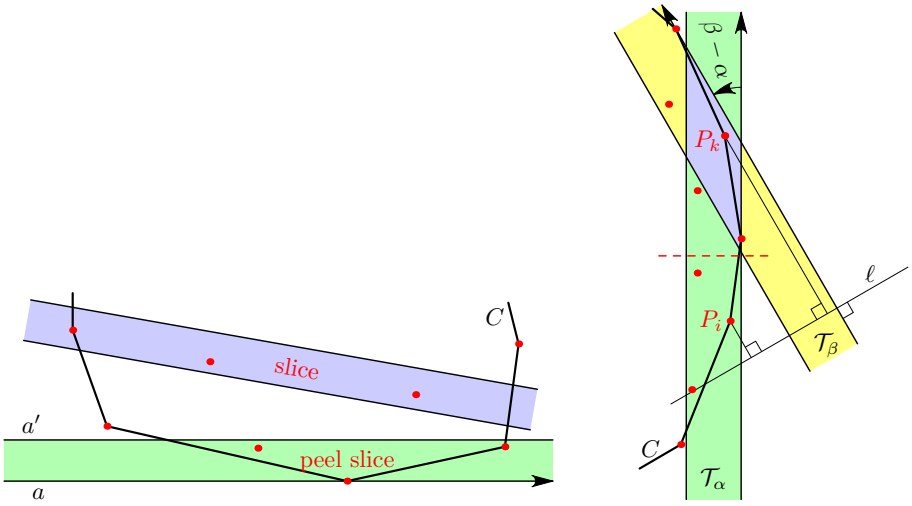
$$\frac{1}{8\delta} \leqslant |\mathcal{T}| \leqslant 1 + \frac{4}{\sqrt{15}}L. \tag{4}$$

The key idea of the solution is to apply these estimates to a *peel slice*, where line a does not cross the interior of C. In the above solution, this idea was applied to one carefully chosen peel slice. Here, we outline some different approach involving many of them. We always assume that n is sufficiently large.

Consider a peel slice determined by lines a and a', where a contains no interior points of C. We orient a so that C lies to the left of a. Line a is called a *supporting line* of the slice, and the obtained direction is the *direction* of the slice; notice that the direction determines uniquely the supporting line and hence the slice. Fix some direction $\mathbf{v}_0$, and for each $\alpha \in [0, 2\pi)$ denote by $\mathcal{T}_\alpha$ the peel slice whose direction is $\mathbf{v}_0$ rotated by α counterclockwise.

When speaking about the slice, we always assume that the figure is rotated so that its direction is vertical from the bottom to the top; then the points in $\mathcal{T}$ get a natural order from the bottom to the top. In particular, we may speak about the *top half* $\mathsf{T}(\mathcal{T})$ consisting of $\lfloor|\mathcal{T}|/2\rfloor$ topmost points in $\mathcal{T}$, and similarly about its *bottom half* $\mathsf{B}(\mathcal{T})$. By (4), each half contains at least 10 points when n is large.

Claim. Consider two angles $\alpha, \beta \in [0, \pi/2]$ with $\beta - \alpha \geqslant 40\delta =: \phi$. Then all common points of $\mathcal{T}_\alpha$ and $\mathcal{T}_\beta$ lie in $\mathsf{T}(\mathcal{T}_\alpha) \cap \mathsf{B}(\mathcal{T}_\beta)$.



Proof. By symmetry, it suffices to show that all those points lie in $\mathsf{T}(\mathcal{T}_\alpha)$. Let a be the supporting line of $\mathcal{T}_\alpha$, and let ℓ be a line perpendicular to the direction of $\mathcal{T}_\beta$. Let $P_1, \ldots, P_k$ list all points in $\mathcal{T}_\alpha$, numbered from the bottom to the top; by (4), we have $k \geqslant \frac{1}{8}\delta^{-1}$.

Introduce the Cartesian coordinates so that the (oriented) line a is the y-axis. Let P_i be any point in $\mathsf{B}(\mathcal{T}_\alpha)$. The difference of ordinates of P_k and P_i is at least $\frac{\sqrt{15}}{4}(k-i) > \frac{1}{3}k$, while their abscissas differ by at most $\frac{1}{4}$. This easily yields that the projections of those points to ℓ are at least

$$\frac{k}{3}\sin\phi - \frac{1}{4} \geqslant \frac{1}{24\delta}\cdot 20\delta - \frac{1}{4} > \frac{1}{4}$$

apart from each other, and P_k is closer to the supporting line of $\mathcal{T}_\beta$ than P_i, so that P_i does not belong to $\mathcal{T}_\beta$. □

Now, put $\alpha_i = 40\delta i$, for $i = 0, 1, \ldots, \left\lfloor \frac{1}{40}\delta^{-1}\cdot\frac{\pi}{2}\right\rfloor$, and consider the slices $\mathcal{T}_{\alpha_i}$. The Claim yields that each point in $\mathcal{S}$ is contained in at most two such slices. Hence, the union $\mathcal{U}$ of those slices contains at least

$$\frac{1}{2}\cdot\frac{1}{8\delta}\cdot\frac{1}{40\delta}\cdot\frac{\pi}{2} = \frac{\lambda}{\delta^2}$$

points (for some constant λ), and each point in $\mathcal{U}$ is at most $\frac{1}{4}$ apart from the boundary of C.

It is not hard now to reach a contradiction with (1). E.g., for each point $X \in \mathcal{U}$, consider a closest point $f(X)$ on the boundary of C. Obviously, $f(X)f(Y) \geqslant XY - \frac{1}{2} \geqslant \frac{1}{2}XY$ for all $X, Y \in \mathcal{U}$. This yields that the perimeter of C is at least $\mu\delta^{-2}$, for some constant μ, and hence the diameter of $\mathcal{S}$ is of the same order.

Alternatively, one may show that the projection of $\mathcal{U}$ to the line at the angle of $\pi/4$ with $\mathbf{v}_0$ has diameter at least $\mu\delta^{-2}$ for some constant μ.

Number Theory

N1. Given a positive integer k, show that there exists a prime p such that one can choose distinct integers $a_1, a_2, \ldots, a_{k+3} \in \{1, 2, \ldots, p-1\}$ such that p divides $a_i a_{i+1} a_{i+2} a_{i+3} - i$ for all $i = 1, 2, \ldots, k$.

(South Africa)

Solution. First we choose distinct positive rational numbers $r_1, \ldots, r_{k+3}$ such that

$$r_i r_{i+1} r_{i+2} r_{i+3} = i \quad \text{for } 1 \leqslant i \leqslant k.$$

Let $r_1 = x$, $r_2 = y$, $r_3 = z$ be some distinct primes greater than k; the remaining terms satisfy $r_4 = \frac{1}{r_1 r_2 r_3}$ and $r_{i+4} = \frac{i+1}{i} r_i$. It follows that if r_i are represented as irreducible fractions, the numerators are divisible by x for $i \equiv 1 \pmod 4$, by y for $i \equiv 2 \pmod 4$, by z for $i \equiv 3 \pmod 4$ and by none for $i \equiv 0 \pmod 4$. Notice that $r_i < r_{i+4}$; thus the sequences $r_1 < r_5 < r_9 < \ldots$, $r_2 < r_6 < r_{10} < \ldots$, $r_3 < r_7 < r_{11} < \ldots$, $r_4 < r_8 < r_{12} < \ldots$ are increasing and have no common terms, that is, all r_i are distinct.

If each r_i is represented by an irreducible fraction $\frac{u_i}{v_i}$, choose a prime p which divides neither v_i, $1 \leqslant i \leqslant k+1$, nor $v_i v_j (r_i - r_j) = v_j u_i - v_i u_j$ for $i < j$, and define a_i by the congruence $a_i v_i \equiv u_i \pmod p$. Since $r_i r_{i+1} r_{i+2} r_{i+3} = i$, we have

$$\begin{aligned} i v_i v_{i+1} v_{i+2} v_{i+3} &= r_i v_i r_{i+1} v_{i+1} r_{i+2} v_{i+2} r_{i+3} v_{i+3} \\ &= u_i u_{i+1} u_{i+2} u_{i+3} \equiv a_i v_i a_{i+1} v_{i+1} a_{i+2} v_{i+2} a_{i+3} v_{i+3} \pmod p \end{aligned}$$

and therefore $a_i a_{i+1} a_{i+2} a_{i+3} \equiv i \pmod p$ for $1 \leqslant i \leqslant k$.

If $a_i \equiv a_j \pmod p$, then $u_i v_j \equiv a_i v_i v_j \equiv u_j v_i \pmod p$, a contradiction.

Comment. One can explicitly express residues $b_i \equiv a_1 a_2 \cdot \ldots \cdot a_i \pmod p$ in terms of b_1, b_2, b_3 and $b_0 = 1$:

$$b_{i+3} = i(i-4)(i-8) \cdot \ldots \cdot (i-4k+4) b_r,$$

where $i + 3 = 4k + r$, $0 \leqslant r < 4$. Then the numbers a_i are found from the congruences $b_{i-1} a_i \equiv b_i \pmod p$, and choosing p so that a_i are not congruent modulo p is done in a way very similar to the above solution.

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N2. For each prime p, there is a kingdom of p-Landia consisting of p islands numbered $1, 2, \ldots, p$. Two distinct islands numbered n and m are connected by a bridge if and only if p divides $(n^2 - m + 1)(m^2 - n + 1)$. The bridges may pass over each other, but cannot cross. Prove that for infinitely many p there are two islands in p-Landia not connected by a chain of bridges.

(Denmark)

Solution 1. We prove that for each prime $p > 3$ dividing a number of the form $x^2 - x + 1$ with integer x there are two unconnected islands in p-Landia.

For brevity's sake, when a bridge connects the islands numbered m and n, we shall speak simply that it connects m and n.

A bridge connects m and n if $n \equiv m^2 + 1 \pmod p$ or $m \equiv n^2 + 1 \pmod p$. If $m^2 + 1 \equiv n \pmod p$, we draw an arrow starting at m on the bridge connecting m and n. Clearly only one arrow starts at m if $m^2 + 1 \not\equiv m \pmod p$, and no arrows otherwise. The total number of bridges does not exceed the total number of arrows.

Suppose $x^2 - x + 1 \equiv 0 \pmod p$. We may assume that $1 \leqslant x \leqslant p$; then there is no arrow starting at x. Since $(1-x)^2 - (1-x) + 1 = x^2 - x + 1$, $(p+1-x)^2 + 1 \equiv (p+1-x) \pmod p$, and there is also no arrow starting at $p + 1 - x$. If $x = p + 1 - x$, that is, $x = \frac{p+1}{2}$, then $4(x^2 - x + 1) = p^2 + 3$ and therefore $x^2 - x + 1$ is not divisible by p. Thus the islands x and $p + 1 - x$ are different, and no arrows start at either of them. It follows that the total number of bridges in p-Landia does not exceed $p - 2$.

Let $1, 2, \ldots, p$ be the vertices of a graph G_p, where an edge connects m and n if and only if there is a bridge between m and n. The number of vertices of G_p is p and the number of edges is less than $p-1$. This means that the graph is not connected, which means that there are two islands not connected by a chain of bridges.

It remains to prove that there are infinitely many primes p dividing x^2-x+1 for some integer x. Let $p_1, p_2, \ldots, p_k$ be any finite set of such primes. The number $(p_1p_2\cdot\ldots\cdot p_k)^2-p_1p_2\cdot\ldots\cdot p_k+1$ is greater than 1 and not divisible by any p_i; therefore it has another prime divisor with the required property.

Solution 2. One can show, by using only arithmetical methods, that for infinitely many p, the kingdom of p-Ladia contains two islands connected to no other island, except for each other.

Let arrows between islands have the same meaning as in the previous solution. Suppose that positive $a < p$ satisfies the congruence $x^2 - x + 1 \equiv 0 \pmod p$. We have seen in the first solution that $b = p+1-a$ satisfies it too, and $b \neq a$ when $p > 3$. It follows that $ab \equiv a(1-a) \equiv 1 \pmod p$. If an arrow goes from t to a, then t must satisfy the congruence $t^2 + 1 \equiv a \equiv a^2 + 1 \pmod p$; the only such $t \neq a$ is $p - a$. Similarly, the only arrow going to b goes from $p - b$. If one of the numbers $p - a$ and $p - b$, say, $p - a$, is not at the end of any arrow, the pair a, $p - a$ is not connected with the rest of the islands. This is true if at least one of the congruences $x^2 + 1 \equiv -a$, $x^2 + 1 \equiv -b$ has no solutions, that is, either $-a - 1$ or $-b - 1$ is a quadratic non-residue modulo p.

Note that $x^2 - x + 1 \equiv x^2 - (a + b)x + ab \equiv (x - a)(x - b) \pmod p$. Substituting $x = -1$ we get $(-1 - a)(-1 - b) \equiv 3 \pmod p$. If 3 is a quadratic non-residue modulo p, so is one of the numbers $-1 - a$ and $-1 - b$.

Thus it is enough to find infinitely many primes $p > 3$ dividing $x^2 - x + 1$ for some integer x and such that 3 is a quadratic non-residue modulo p.

If $x^2 - x + 1 \equiv 0 \pmod p$ then $(2x - 1)^2 \equiv -3 \pmod p$, that is, -3 is a quadratic residue modulo p, so 3 is a quadratic non-residue if and only if -1 is also a non-residue, in other words, $p \equiv -1 \pmod 4$.

Similarly to the first solution, let $p_1, \ldots, p_k$ be primes congruent to -1 modulo 4 and dividing numbers of the form $x^2 - x + 1$. The number $(2p_1 \cdot \ldots \cdot p_k)^2 - 2p_1 \cdot \ldots \cdot p_k + 1$ is

not divisible by any p_i and is congruent to -1 modulo 4, therefore, it has some prime divisor $p \equiv -1 \pmod 4$ which has the required properties.

N3. Let n be an integer with $n \geqslant 2$. Does there exist a sequence $(a_1, \ldots, a_n)$ of positive integers with not all terms being equal such that the arithmetic mean of every two terms is equal to the geometric mean of some (one or more) terms in this sequence?

(Estonia)

Answer: No such sequence exists.

Solution 1. Suppose that $a_1, \ldots, a_n$ satisfy the required properties. Let $d = \gcd(a_1 \ldots, a_n)$. If $d > 1$ then replace the numbers $a_1, \ldots, a_n$ by $\frac{a_1}{d}, \ldots, \frac{a_n}{d}$; all arithmetic and all geometric means will be divided by d, so we obtain another sequence satisfying the condition. Hence, without loss of generality, we can assume that $\gcd(a_1 \ldots, a_n) = 1$.

We show two numbers, a_m and a_k such that their arithmetic mean, $\frac{a_m + a_k}{2}$ is different from the geometric mean of any (nonempty) subsequence of $a_1 \ldots, a_n$. That proves that there cannot exist such a sequence.

Choose the index $m \in \{1, \ldots, n\}$ such that $a_m = \max(a_1, \ldots, a_n)$. Note that $a_m \geqslant 2$, because $a_1, \ldots, a_n$ are not all equal. Let p be a prime divisor of a_m.

Let $k \in \{1, \ldots, n\}$ be an index such that $a_k = \max\{a_i : p \nmid a_i\}$. Due to $\gcd(a_1 \ldots, a_n) = 1$, not all a_i are divisible by p, so such a k exists. Note that $a_m > a_k$ because $a_m \geqslant a_k$, $p \mid a_m$ and $p \nmid a_k$.

Let $b = \frac{a_m + a_k}{2}$; we will show that b cannot be the geometric mean of any subsequence of $a_1, \ldots, a_n$.

Consider the geometric mean, $g = \sqrt[t]{a_{i_1} \cdot \ldots \cdot a_{i_t}}$ of an arbitrary subsequence of $a_1, \ldots, a_n$.

If none of $a_{i_1}, \ldots, a_{i_t}$ is divisible by p, then they are not greater than a_k, so

$$g = \sqrt[t]{a_{i_1} \cdot \ldots \cdot a_{i_t}} \leqslant a_k < \frac{a_m + a_k}{2} = b,$$

and therefore $g \neq b$.

Otherwise, if at least one of $a_{i_1}, \ldots, a_{i_t}$ is divisible by p, then $2g = 2\sqrt[t]{a_{i_1} \cdot \ldots \cdot a_{i_t}}$ is either not an integer or is divisible by p, while $2b = a_m + a_k$ is an integer not divisible by p, so $g \neq b$ again.

Solution 2. Like in the previous solution, we assume that the numbers $a_1, \ldots, a_n$ have no common divisor greater than 1. The arithmetic mean of any two numbers in the sequence is half of an integer; on the other hand, it is a (some integer order) root of an integer. This means each pair's mean is an integer, so all terms in the sequence must be of the same parity; hence they all are odd. Let $d = \min\left\{\gcd(a_i, a_j) \colon a_i \neq a_j\right\}$. By reordering the sequence we can assume that $\gcd(a_1, a_2) = d$, the sum $a_1 + a_2$ is maximal among such pairs, and $a_1 > a_2$.

We will show that $\frac{a_1 + a_2}{2}$ cannot be the geometric mean of any subsequence of $a_1 \ldots, a_n$.

Let $a_1 = xd$ and $a_2 = yd$ where x, y are coprime, and suppose that there exist some $b_1, \ldots, b_t \in \{a_1, \ldots, a_n\}$ whose geometric mean is $\frac{a_1 + a_2}{2}$. Let $d_i = \gcd(a_1, b_i)$ for $i = 1, 2, \ldots, t$ and let $D = d_1 d_2 \cdot \ldots \cdot d_t$. Then

$$D = d_1 d_2 \cdot \ldots \cdot d_t \mid b_1 b_2 \cdot \ldots \cdot b_t = \left(\frac{a_1 + a_2}{2}\right)^t = \left(\frac{x + y}{2}\right)^t d^t.$$

We claim that $D \mid d^t$. Consider an arbitrary prime divisor p of D. Let $\nu_p(x)$ denote the exponent of p in the prime factorization of x. If $p \mid \frac{x+y}{2}$, then $p \nmid x, y$, so p is coprime with x; hence, $\nu_p(d_i) \leqslant \nu_p(a_1) = \nu_p(xd) = \nu_p(d)$ for every $1 \leqslant i \leqslant t$, therefore $\nu_p(D) = \sum_i \nu_p(d_i) \leqslant t\nu_p(d) = \nu_p(d^t)$. Otherwise, if p is coprime to $\frac{x+y}{2}$, we have $\nu_p(D) \leqslant \nu_p(d^t)$ trivially. The claim has been proved.

Notice that $d_i = \gcd(b_i, a_1) \geqslant d$ for $1 \leqslant i \leqslant t$: if $b_i \neq a_1$ then this follows from the definition of d; otherwise we have $b_i = a_1$, so $d_i = a_1 \geqslant d$. Hence, $D = d_1 \cdot \ldots \cdot d_t \geqslant d^t$, and the claim forces $d_1 = \ldots = d_t = d$.

Finally, by $\frac{a_1+a_2}{2} > a_2$ there must be some b_k which is greater than a_2. From $a_1 > a_2 \geqslant d = \gcd(a_1, b_k)$ it follows that $a_1 \neq b_k$. Now the have a pair a_1, b_k such that $\gcd(a_1, b_k) = d$ but $a_1 + b_k > a_1 + a_2$; that contradicts the choice of a_1 and a_2.

Comment. The original problem proposal contained a second question asking if there exists a non-constant sequence $(a_1, \ldots, a_n)$ of positive integers such that the geometric mean of every two terms is equal the arithmetic mean of some terms.

For $n \geqslant 3$ such a sequence is $(4, 1, 1, \ldots, 1)$. The case $n = 2$ can be done by the trivial estimates

$$\min(a_1, a_2) < \sqrt{a_1 a_2} < \frac{a_1 + a_2}{2} < \max(a_1, a_2).$$

The Problem Selection Committee found this variant less interesting and suggests using only the first question.

N4. For any odd prime p and any integer n, let $d_p(n) \in \{0, 1, \ldots, p-1\}$ denote the remainder when n is divided by p. We say that $(a_0, a_1, a_2, \ldots)$ is a *p-sequence*, if a_0 is a positive integer coprime to p, and $a_{n+1} = a_n + d_p(a_n)$ for $n \geqslant 0$.

(a) Do there exist infinitely many primes p for which there exist p-sequences $(a_0, a_1, a_2, \ldots)$ and $(b_0, b_1, b_2, \ldots)$ such that $a_n > b_n$ for infinitely many n, and $b_n > a_n$ for infinitely many n?

(b) Do there exist infinitely many primes p for which there exist p-sequences $(a_0, a_1, a_2, \ldots)$ and $(b_0, b_1, b_2, \ldots)$ such that $a_0 < b_0$, but $a_n > b_n$ for all $n \geqslant 1$?

(United Kingdom)

Answer: Yes, for both parts.

Solution. Fix some odd prime p, and let T be the smallest positive integer such that $p \mid 2^T - 1$; in other words, T is the multiplicative order of 2 modulo p.

Consider any p-sequence $(x_n) = (x_0, x_1, x_2, \ldots)$. Obviously, $x_{n+1} \equiv 2x_n \pmod p$ and therefore $x_n \equiv 2^n x_0 \pmod p$. This yields $x_{n+T} \equiv x_n \pmod p$ and therefore $d(x_{n+T}) = d(x_n)$ for all $n \geqslant 0$. It follows that the sum $d(x_n) + d(x_{n+1}) + \ldots + d(x_{n+T-1})$ does not depend on n and is thus a function of x_0 (and p) only; we shall denote this sum by $S_p(x_0)$, and extend the function $S_p(\cdot)$ to all (not necessarily positive) integers. Therefore, we have $x_{n+kT} = x_n + kS_p(x_0)$ for all positive integers n and k. Clearly, $S_p(x_0) = S_p(2^t x_0)$ for every integer $t \geqslant 0$.

In both parts, we use the notation

$$S_p^+ = S_p(1) = \sum_{i=0}^{T-1} d_p(2^i) \quad \text{and} \quad S_p^- = S_p(-1) = \sum_{i=0}^{T-1} d_p(p - 2^i).$$

(a) Let $q > 3$ be a prime and p a prime divisor of $2^q + 1$ that is greater than 3. We will show that p is suitable for part (a). Notice that $9 \nmid 2^q + 1$, so that such a p exists. Moreover, for any two odd primes $q < r$ we have $\gcd(2^q + 1, 2^r + 1) = 2^{\gcd(q,r)} + 1 = 3$, thus there exist infinitely many such primes p.

For the chosen p, we have $T = 2q$. Since $2^q \equiv -1 \pmod p$, we have $S_p^+ = S_p^-$. Now consider the p-sequences (a_n) and (b_n) with $a_0 = p + 1$ and $b_0 = p - 1$; we claim that these sequences satisfy the required conditions. We have $a_0 > b_0$ and $a_1 = p + 2 < b_1 = 2p - 2$. It follows then that

$$a_{k \cdot 2q} = a_0 + kS_p^+ > b_0 + kS_p^+ = b_{k \cdot 2q} \quad \text{and} \quad a_{k \cdot 2q+1} = a_1 + kS_p^+ < b_1 + kS_p^+ = b_{k \cdot 2q+1}$$

for all $k = 0, 1, \ldots$, as desired.

(b) Let q be an odd prime and p a prime divisor of $2^q - 1$; thus we have $T = q$. We will show that p is suitable for part (b). Notice that the numbers of the form $2^q - 1$ are pairwise coprime (since $\gcd(2^q - 1, 2^r - 1) = 2^{\gcd(q,r)} - 1 = 1$ for any two distinct primes q and r), thus there exist infinitely many such primes p. Notice that $d_p(x) + d_p(p - x) = p$ for all x with $p \nmid x$, so that the sum $S_p^+ + S_p^- = pq$ is odd, which yields $S_p^+ = S_p(1) \neq S_p(-1) = S_p^-$.

Assume that (x_n) and (y_n) are two p-sequences with $S_p(x_0) > S_p(y_0)$ but $x_0 < y_0$. The first condition yields that

$$x_{Mq+r} - y_{Mq+r} = (x_r - y_r) + M\big(S_p(x_0) - S_p(y_0)\big) \geqslant (x_r - y_r) + M$$

for all nonnegative integers M and every $r = 0, 1, \ldots, q-1$. Thus, we have $x_n > y_n$ for every $n \geqslant q + q \cdot \max\{y_r - x_r : r = 0, 1, \ldots, q-1\}$. Now, since $x_0 < y_0$, there exists the largest n_0 with $x_{n_0} < y_{n_0}$. In this case the p-sequences $a_n = x_{n-n_0}$ and $b_n = y_{n-n_0}$ possess the desired property (notice here that $x_n \neq y_n$ for all $n \geqslant 0$, as otherwise we would have $S_p(x_0) = S_p(x_n) = S_p(y_n) = S_p(y_0)$).

It remains to find p-sequences (x_n) and (y_n) satisfying the two conditions. Recall that $S_p^+ \neq S_p^-$. Now, if $S_p^+ > S_p^-$, then we can put $x_0 = 1$ and $y_0 = p - 1$. Otherwise, if $S_p^+ < S_p^-$, then we put $x_0 = p - 1$ and $y_0 = p + 1$.

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N5. Determine all functions f defined on the set of all positive integers and taking non-negative integer values, satisfying the three conditions:

(i) $f(n) \neq 0$ for at least one n;

(ii) $f(xy) = f(x) + f(y)$ for every positive integers x and y;

(iii) there are infinitely many positive integers n such that $f(k) = f(n-k)$ for all $k < n$.

(Croatia)

Answer: The sought functions are those of the form $f(n) = c \cdot \nu_p(n)$, where p is some prime, c is a nonnegative integer, and $\nu_p(n)$ denotes the exponent of p in the prime decomposition of n.

Solution 1. If a number n is a product of primes, $n = p_1 p_2 \cdot \ldots \cdot p_k$, then

$$f(n) = f(p_1) + \ldots + f(p_k),$$

in particular, $f(1) = 0$ (since $f(1) = f(1) + f(1)$).

It is also clear that $f(n) = 0$ implies $f(p) = 0$ for all primes p dividing n.

Let us call positive integer n *good* if $f(k) = f(n-k)$ for $0 < k < n$. If n is good then each its divisor d is also good; indeed, if $n = dm$ then

$$f(k) = f(mk) - f(m) = f(n-mk) - f(m) = f(m(d-k)) - f(m) = f(d-k)$$

for $0 < k < d$. Thus, good numbers are products of good primes.

It follows immediately from (i) that there exists a prime p such that $f(p) \neq 0$; let p be the smallest such prime. Then $f(r) = 0$ for all $r < p$ (since all prime divisors of $r < p$ are less than p). Now every good number $n > p$ must be divisible by p. Indeed, if $n = pk + r$ is a good number, $k > 0$, $0 < r < p$, then $f(p) \leqslant f(pk) = f(n - pk) = f(r) = 0$, a contradiction. Since any divisor of a good number is also good, this means that if a divisor r of a good number is not divisible by p, it is less than p. Thus all good numbers have the form $r \cdot p^k$ with $r < p$. The condition (iii) implies that k can be arbitrarily large, consequently all powers of p are good.

If $q \neq p$ is a prime, $p^{q-1} - 1$ is divisible by q and p^{q-1} is good. Then $f(q) \leqslant f(p^{q-1} - 1) = f(1) = 0$, that is, $f(q) = 0$.

Now we see that $f(n) = \nu_p(n) \cdot c$, where $c = f(p)$. The conditions (i) and (ii) for all such functions with $c \neq 0$ are obvious; the condition (iii) holds for all $n = p^m$, since $\nu_p(p^m - k) = \nu_p(k)$ when $0 < k < p^m$.

Solution 2. We use the notion of a good number from the previous solution. As above, we also denote by $\nu_p(n)$ the exponent of a prime p in the prime decomposition of n.

Say that a positive integer k is *big* if $f(k) > 0$. Let $\mathcal{B}$ be the set of big primes, and let $p_1 < p_2 < \ldots$ list the elements of $\mathcal{B}$ (this set might be either finite or infinite). By the problem conditions, we have

$$f(n) = \sum_i \nu_{p_i}(n) f(p_i); \tag{1}$$

thus, the big numbers are those divisible by at least one big prime.

For a positive integer k, define its *essence* $e(k)$ to be the largest product e of (not necessarily different) big primes such that $e \mid k$. In other words,

$$e(n) = \prod_{p_i \in \mathcal{B}} p_i^{\nu_{p_i}(n)}.$$

This yields that $k/e(k)$ is not big, so $f(k) = f(e(k)) + f(k/e(k)) = f(e(k))$.

Lemma. Assume that n is a good number. Then $e(k) = e(n-k)$ for all $k < n$.

Proof. Arguing indirectly, choose a minimal k for which the claim of the lemma is violated. Clearly, k is big, as otherwise $f(k) = f(n-k) = 0$ and hence $e(k) = e(n-k) = 1$.

There are $t = k/e(k)$ multiples of $e(k)$ in each of the segments $[1, k]$ and $[n-k, n-1]$. On the other hand, there are $t-1$ such multiples on $[1, k-1]$ — and, by minimality of k, on $[n-k+1, n-1]$ as well. This yields that $n-k$ is a multiple of $e(k)$. Therefore,

$$f(e(k)) = f(k) = f(n-k) = f(e(k)) + f\left(\frac{n-k}{e(k)}\right),$$

so the last summand vanishes, hence $\frac{n-k}{e(k)}$ has no big prime divisors, that is, $e(n-k) = e(k)$. This contradicts our choice. □

Back to the problem, assume that $|\mathcal{B}| \geqslant 2$. Take any good number $n > p_1p_2$, and let p_1^α be the largest power of p_1 smaller than n, so that $n \leqslant p_1^{\alpha+1} < p_1^\alpha p_2$. By the lemma, $e(n - p_1^\alpha) = e(p_1^\alpha) = p_1^\alpha$, which yields $p_1^\alpha \mid n$. Similarly, $p_2 \mid n$, so that $n \geqslant p_1^\alpha p_2$. This contradiction shows that $|\mathcal{B}| \leqslant 1$, which by (1) yields that f is listed in the answer.

Solution 3. We have $f(\prod p_i^{\alpha_i}) = \sum \alpha_i f(p_i)$. Note that

$$f(n-1) + f(n-2) + \ldots + \ldots f(n-k) \geqslant f(1) + \ldots + f(k)$$

for all $k = 1, 2, \ldots, n-1$, since the difference LHS−RHS is just $f(\binom{n-1}{k})$. Assume that $f(p) > 0$. If $f(k) = f(n-k)$ for all k, it implies that $\binom{n-1}{k}$ is not divisible by p for all $k = 1, 2, \ldots, n-2$. It is well known that it implies $n = a \cdot p^s$, $a < p$. If there are two primes p, q such that $f(p) > 0, f(q) > 0$, there exist only finitely many n which are equal both to $a \cdot p^s$, $a < p$, and $b \cdot q^t, b < q$. So there exists at most one such p, and therefore $f(n) = C \cdot \nu_p(n)$ for some constant C.

Solution 4. We call a function $f : \mathbb{N} \to \mathbb{N}_0$ satisfying (ii) *additive*. We call a pair (f, n), where f is an additive function and $n \in \mathbb{N}$, *good*, if for all $k < n$ it holds $f(k) = f(n-k)$. For an additive function f and a prime number p the number $\frac{f(p)}{\ln p}$ is denoted by $g(f, p)$.

Let (f, n) be a good pair such that $f(p) > 0$ for at least two primes less than n. Let p_0 be the prime with maximal $g(f, p)$ among all primes $p < n$. Let a_0 be the maximal exponent such that $p_0^{a_0} < n$. Then $f(k) < f(p_0^{a_0})$ for all $k < p_0^{a_0}$. Indeed, if $k = p_1^{a_1} \ldots p_m^{a_m} < p_0^{a_0}$, then

$$\begin{aligned} f(k) &= a_1 f(p_1) + \ldots + a_m f(p_m) = g(f, p_1) a_1 \ln p_1 + \ldots + g(f, p_m) a_m \ln a_m \\ &< g(f, p_0) a_0 \ln p_0 = f(p_0^{a_0}). \end{aligned}$$

Let $n = bp_0^{a_0} + r$, where $0 < r < p_0^{a_0}$. Then $f(r) = f(bp_0^{a_0}) \geqslant f(p_0^{a_0})$. This contradiction shows that $p_0^{a_0} | n$. Then $n = p_0^{\nu_{p_0}(n)} n'$, where $n' \leqslant p_0$.

The functions $f_1(m) := f(p_0)\nu_{p_0}(m)$ and $f_2 := f - f_1$ are additive (obviously $f(m) \geqslant f(p_0^{\nu_{p_0}(m)}) = f_1(m)$, since $p_0^{\nu_{p_0}(m)}$ divides m). For $k < n$, $\nu_p(k) = \nu_p(n-k)$. Hence the pair (f_2, n) is also good. Note that $f_2(p_0) = 0$.

Choose among all primes $p < n$ the prime q_0 with maximal $g(f_2, p)$. As above we can prove that $n = q_0^{\nu_{q_0}(n)} n''$ with $n'' < q_0$. Since $p_0 \neq q_0$, we get a contradiction. Thus $f(n) = f(p) \cdot \nu_p(n)$.

N6. For a positive integer n, let $d(n)$ be the number of positive divisors of n, and let $\varphi(n)$ be the number of positive integers not exceeding n which are coprime to n. Does there exist a constant C such that

$$\frac{\varphi(d(n))}{d(\varphi(n))} \leqslant C$$

for all $n \geqslant 1$?

(Cyprus)

Answer: No, such constant does not exist.

Solution 1. Fix $N > 1$, let $p_1, \ldots, p_k$ be all primes between 1 and N and $p_{k+1}, \ldots, p_{k+s}$ be all primes between $N+1$ and $2N$. Since for $j \leqslant k+s$ all prime divisors of $p_j - 1$ do not exceed N, we have

$$\prod_{j=1}^{k+s}(p_j - 1) = \prod_{i=1}^{k} p_i^{c_i},$$

with some fixed exponents $c_1, \ldots, c_k$. Choose a huge prime number q and consider a number

$$n = (p_1 \cdot \ldots \cdot p_k)^{q-1} \cdot (p_{k+1} \cdot \ldots \cdot p_{k+s}).$$

Then

$$\varphi(d(n)) = \varphi(q^k \cdot 2^s) = q^{k-1}(q-1)2^{s-1}$$

and

$$d(\varphi(n)) = d\left((p_1 \cdot \ldots \cdot p_k)^{q-2} \prod_{i=1}^{k+s}(p_i - 1)\right) = d\left(\prod_{i=1}^{k} p_i^{q-2+c_i}\right) = \prod_{i=1}^{k}(q - 1 + c_i),$$

so

$$\frac{\varphi(d(n))}{d(\varphi(n))} = \frac{q^{k-1}(q-1)2^{s-1}}{\prod_{i=1}^{k}(q-1+c_i)} = 2^{s-1} \cdot \frac{q-1}{q} \cdot \prod_{i=1}^{k} \frac{q}{q-1+c_i},$$

which can be made arbitrarily close to 2^{s-1} by choosing q large enough. It remains to show that s can be arbitrarily large, i.e. that there can be arbitrarily many primes between N and $2N$.

This follows, for instance, from the well-known fact that $\sum \frac{1}{p} = \infty$, where the sum is taken over the set $\mathbb{P}$ of prime numbers. Indeed, if, for some constant C, there were always at most C primes between 2^ℓ and $2^{\ell+1}$, we would have

$$\sum_{p \in \mathbb{P}} \frac{1}{p} = \sum_{\ell=0}^{\infty} \sum_{\substack{p \in \mathbb{P} \\ p \in [2^\ell, 2^{\ell+1})}} \frac{1}{p} \leqslant \sum_{\ell=0}^{\infty} \frac{C}{2^\ell} < \infty,$$

which is a contradiction.

Comment 1. Here we sketch several alternative elementary self-contained ways to perform the last step of the solution above. In particular, they avoid using divergence of $\sum \frac{1}{p}$.

Suppose that for some constant C and for every $k = 1, 2, \ldots$ there exist at most C prime numbers between 2^k and 2^{k+1}. Consider the prime factorization of the factorial $(2^n)! = \prod p^{\alpha_p}$. We have $\alpha_p = \lfloor 2^n/p \rfloor + \lfloor 2^n/p^2 \rfloor + \ldots$. Thus, for $p \in [2^k, 2^{k+1})$, we get $\alpha_p \leqslant 2^n/2^k + 2^n/2^{k+1} + \ldots = 2^{n-k+1}$, therefore $p^{\alpha_p} \leqslant 2^{(k+1)2^{n-k+1}}$. Combining this with the bound $(2m)! \geqslant m(m+1) \cdot \ldots \cdot (2m-1) \geqslant m^m$ for $m = 2^{n-1}$ we get

$$2^{(n-1)\cdot 2^{n-1}} \leqslant (2^n)! \leqslant \prod_{k=1}^{n-1} 2^{C(k+1)2^{n-k+1}},$$

or

$$\sum_{k=1}^{n-1} C(k+1)2^{1-k} \geqslant \frac{n-1}{2}$$

that fails for large n since $C(k+1)2^{1-k} < 1/3$ for all but finitely many k.

In fact, a much stronger inequality can be obtained in an elementary way: Note that the formula for $\nu_p(n!)$ implies that if p^α is the largest power of p dividing $\binom{n}{n/2}$, then $p^\alpha \leqslant n$. By looking at prime factorization of $\binom{n}{n/2}$ we instantaneously infer that

$$\pi(n) \geqslant \log_n \binom{n}{n/2} \geqslant \frac{\log\left(2^n/n\right)}{\log n} \geqslant \frac{n}{2\log n}.$$

This, in particular, implies that for infinitely many n there are at least $\frac{n}{3\log n}$ primes between n and $2n$.

Solution 2. In this solution we will use the Prime Number Theorem which states that

$$\pi(m) = \frac{m}{\log m} \cdot (1 + o(1)),$$

as m tends to infinity. Here and below $\pi(m)$ denotes the number of primes not exceeding m, and log the natural logarithm.

Let $m > 5$ be a large positive integer and let $n := p_1 p_2 \cdot \ldots \cdot p_{\pi(m)}$ be the product of all primes not exceeding m. Then $\varphi(d(n)) = \varphi\left(2^{\pi(m)}\right) = 2^{\pi(m)-1}$. Consider the number

$$\varphi(n) = \prod_{k=1}^{\pi(m)} (p_k - 1) = \prod_{s=1}^{\pi(m/2)} q_s^{\alpha_s},$$

where $q_1, \ldots, q_{\pi(m/2)}$ are primes not exceeding $m/2$. Note that every term $p_k - 1$ contributes at most one prime $q_s > \sqrt{m}$ into the product $\prod_s q_s^{\alpha_s}$, so we have

$$\sum_{s\colon q_s > \sqrt{m}} \alpha_s \leqslant \pi(m) \quad \Longrightarrow \quad \sum_{s\colon q_s > \sqrt{m}} (1 + \alpha_s) \leqslant \pi(m) + \pi(m/2).$$

Hence, applying the AM–GM inequality and the inequality $(A/x)^x \leqslant e^{A/e}$, we obtain

$$\prod_{s\colon q_s > \sqrt{m}} (\alpha_s + 1) \leqslant \left(\frac{\pi(m) + \pi(m/2)}{\ell}\right)^{\ell} \leqslant \exp\left(\frac{\pi(m) + \pi(m/2)}{e}\right),$$

where ℓ is the number of primes in the interval $(\sqrt{m}, m]$.

We then use a trivial bound $\alpha_i \leqslant \log_2\left(\varphi(n)\right) \leqslant \log_2 n < \log_2\left(m^m\right) < m^2$ for each i with $q_i < \sqrt{m}$ to obtain

$$\prod_{s=1}^{\pi(\sqrt{m})} (\alpha_s + 1) \leqslant \left(m^2\right)^{\sqrt{m}} = m^{2\sqrt{m}}.$$

Putting this together we obtain

$$d(\varphi(n)) = \prod_{s=1}^{\pi(m/2)} (\alpha_s + 1) \leqslant \exp\left(2\sqrt{m} \cdot \log m + \frac{\pi(m) + \pi(m/2)}{e}\right).$$

The prime number theorem then implies that

$$\limsup_{m\to\infty} \frac{\log\left(d(\varphi(n))\right)}{m/\log m} \leqslant \limsup_{m\to\infty} \frac{2\sqrt{m} \cdot \log m}{m/\log m} + \limsup_{m\to\infty} \frac{\pi(m) + \pi(m/2)}{e \cdot m/\log m} = \frac{3}{2e}.$$

Whereas, again by prime number theorem, we have

$$\liminf_{m\to\infty} \frac{\log\left(\varphi(d(n))\right)}{m/\log m} = \liminf_{m\to\infty} \frac{\log\left(2^{\pi(m)-1}\right)}{m/\log m} = \log 2.$$

Since $\frac{3}{2e} < \frac{3}{5} < \log 2$, this implies that $\varphi(d(n))/d(\varphi(n))$ can be arbitrarily large.

Comment 2. The original formulation of the problem was asking whether $d(\varphi(n)) \geqslant \varphi(d(n))$ for all but finitely many values of n. The Problem Selection Committee decided that the presented version is better suited for the Shortlist.

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N7. Let $\mathcal{S}$ be a set consisting of $n \geqslant 3$ positive integers, none of which is a sum of two other distinct members of $\mathcal{S}$. Prove that the elements of $\mathcal{S}$ may be ordered as $a_1, a_2, \ldots, a_n$ so that a_i does not divide $a_{i-1} + a_{i+1}$ for all $i = 2, 3, \ldots, n-1$.

(Ukraine)

Common remarks. In all solutions, we call a set $\mathcal{S}$ of positive integers *good* if no its element is a sum of two other distinct members of $\mathcal{S}$. We will use the following simple observation.

Observation A. If a, b, and c are three distinct elements of a good set $\mathcal{S}$ with $b > a, c$, then $b \nmid a + c$. Otherwise, since $b \neq a + c$, we would have $b \leqslant (a + c)/2 < \max\{a, c\}$.

Solution 1. We prove the following stronger statement.

Claim. Let $\mathcal{S}$ be a good set consisting of $n \geqslant 2$ positive integers. Then the elements of $\mathcal{S}$ may be ordered as $a_1, a_2, \ldots, a_n$ so that $a_i \nmid a_{i-1} + a_{i+1}$ and $a_i \nmid a_{i-1} - a_{i+1}$, for all $i = 2, 3, \ldots, n-1$.

Proof. Say that the ordering $a_1, \ldots, a_n$ of $\mathcal{S}$ is *nice* if it satisfies the required property.

We proceed by induction on n. The base case $n = 2$ is trivial, as there are no restrictions on the ordering.

To perform the step of induction, suppose that $n \geqslant 3$. Let $a = \max \mathcal{S}$, and set $\mathcal{T} = \mathcal{S} \setminus \{a\}$. Use the inductive hypothesis to find a nice ordering $b_1, \ldots, b_{n-1}$ of $\mathcal{T}$. We will show that a may be inserted into this sequence so as to reach a nice ordering of $\mathcal{S}$. In other words, we will show that there exists a $j \in \{1, 2, \ldots, n\}$ such that the ordering

$$N_j = (b_1, \ldots, b_{j-1}, a, b_j, b_{j+1}, \ldots, b_{n-1})$$

is nice.

Assume that, for some j, the ordering N_j is not nice, so that some element x in it divides either the sum or the difference of two adjacent ones. This did not happen in the ordering of $\mathcal{T}$, hence $x \in \{b_{j-1}, a, b_j\}$ (if, say, b_{j-1} does not exist, then $x \in \{a, b_j\}$; a similar agreement is applied hereafter). But the case $x = a$ is impossible: a cannot divide $b_{j-1} - b_j$, since $0 < |b_{j-1} - b_j| < a$, while $a \nmid b_{j-1} + b_j$ by Observation A. Therefore $x \in \{b_{j-1}, b_j\}$. In this case, assign the number x to the index j.

Suppose now that none of the N_j is nice. Since there are n possible indices j, and only $n-1$ elements in $\mathcal{T}$, one of those elements (say, b_k) is assigned to two different indices, which then should equal k and $k + 1$. This means that b_k divides the numbers $b_{k-1} + \varepsilon_1 a$ and $a + \varepsilon_2 b_{k+1}$, for some signs $\varepsilon_1, \varepsilon_2 \in \{-1, 1\}$. But then

$$b_{k-1} \equiv -\varepsilon_1 a \equiv \varepsilon_1 \varepsilon_2 b_{k+1} \pmod{b_k},$$

and therefore $b_k \mid b_{k-1} - \varepsilon_1 \varepsilon_2 b_{k+1}$, which means that the ordering of $\mathcal{T}$ was not nice. This contradiction proves the step of induction. □

Solution 2. We again prove a stronger statement.

Claim. Let $\mathcal{S}$ be an arbitrary set of $n \geqslant 3$ positive integers. Then its elements can be ordered as $a_1, \ldots, a_n$ so that, if $a_i \mid a_{i-1} + a_{i+1}$, then $a_i = \max \mathcal{S}$.

The claim easily implies what we need to prove, due to Observation A.

To prove the Claim, introduce the function f which assigns to any two elements $a, b \in \mathcal{S}$ with $a < b$ the unique integer $f(a, b) \in \{1, 2, \ldots, a\}$ such that $a \mid b + f(a, b)$. Hence, if $b \mid a + c$ for some $a, b, c \in \mathcal{S}$ with $a < b < c$, then $a = f(b, c)$. Therefore, the Claim is a consequence of the following combinatorial lemma.

Lemma. Let $\mathcal{S}$ be a set of $n \geqslant 3$ positive integers, and let f be a function which assigns to any $a, b \in \mathcal{S}$ with $a < b$ some integer from the range $\{1, \ldots, a\}$. Then the elements of $\mathcal{S}$ may be ordered as $a_1, a_2, \ldots, a_n$ so as to satisfy the following two conditions simultaneously:

(i) **Unimodality:** There exists a $j \in \{1, 2, \ldots, n\}$ such that $a_1 < a_2 < \ldots < a_j > a_{j+1} > \ldots > a_n$; and

(ii) **f-avoidance:** If $a < b$ are two elements of $\mathcal{S}$, which are adjacent in the ordering, then $f(a, b)$ is not adjacent to a.

Proof. We call an ordering of $\mathcal{S}$ satisfying (i) and (ii) *f-nice*. We agree that $f(x, y) = x$ for $x \geqslant y$; this agreement puts no extra restriction.

We proceed by induction; for the base case $n = 3$, it suffices to put the maximal element in $\mathcal{S}$ onto the middle position.

To perform the step of induction, let $p < q$ be the two minimal elements of $\mathcal{S}$, and set $\mathcal{T} = \mathcal{S} \setminus \{p\}$. Define a function g by assigning to any elements $a < b$ of $\mathcal{T}$ the value

$$g(a, b) = \begin{cases} q, & \text{if } f(a, b) = p; \\ f(a, b), & \text{otherwise.} \end{cases} \tag{1}$$

Notice that $g(a, b) \leqslant a$ for all $a, b \in \mathcal{T}$.

Use the inductive hypothesis to get a g-nice ordering $b_1, b_2, \ldots, b_{n-1}$ of $\mathcal{T}$. By unimodality, either b_1 or b_{n-1} equals q; these cases differ only by reverting the order, so we assume $b_1 = q$.

Notice that, according to (1), the number $f(b_2, b_3)$ differs from both p and q. On the other hand, the number $f(b_{n-1}, b_{n-2})$ differs from at least one of them — say, from r; set $s = p + q - r$, so that $\{r, s\} = \{p, q\}$. Now, order $\mathcal{S}$ as

$$s, b_2, b_3, \ldots, b_{n-1}, r.$$

By the induction hypothesis and the above choice, this ordering is nice. □

Comment. In the original proposal, the numbers in the set were assumed to be odd (which implies that none is a sum of two others); moreover, the proposal requested to arrange in a row all numbers but one.

On the other hand, Solution 2 shows that the condition of $\mathcal{S}$ being good may be relaxed to the condition that the maximal element of $\mathcal{S}$ is not a sum of two other elements in $\mathcal{S}$. On the other hand, the set $\{1, 2, 3\}$ shows that the condition cannot be merely omitted.

The Problem Selection Committee considered several versions of the problem and chose the best version in their opinion for the Shortlist.

IMO 2020

61st International
Mathematical
Olympiad
Saint Petersburg
Russia

ORGANIZERS

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IMO 2019
60TH INTERNATIONAL
MATHEMATICAL OLYMPIAD
July 11th - July 22nd,
Bath, United Kingdom
SHORTLISTED
PROBLEMS
WITH SOLUTIONS





Shortlisted Problems

(with solutions)

60th International Mathematical Olympiad

Bath — UK, 11th–22nd July 2019

Note of Confidentiality

The Shortlist has to be kept strictly confidential until the conclusion of the following International Mathematical Olympiad.

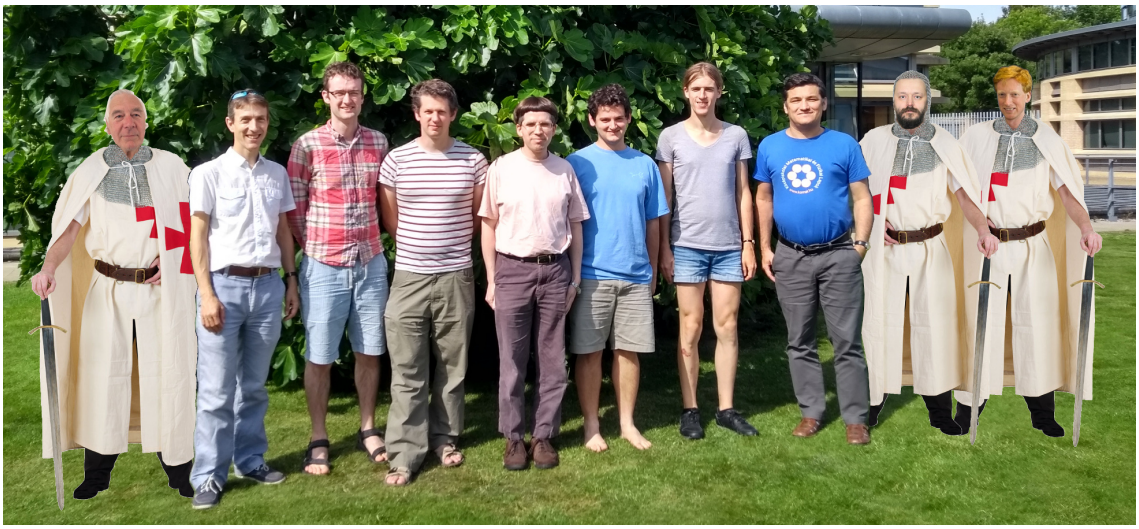
IMO General Regulations §6.6

Contributing Countries

The Organising Committee and the Problem Selection Committee of IMO 2019 thank the following 58 countries for contributing 204 problem proposals:

Albania, Armenia, Australia, Austria, Belarus, Belgium, Brazil, Bulgaria, Canada, China, Croatia, Cuba, Cyprus, Czech Republic, Denmark, Ecuador, El Salvador, Estonia, Finland, France, Georgia, Germany, Greece, Hong Kong, Hungary, India, Indonesia, Iran, Ireland, Israel, Italy, Japan, Kazakhstan, Kosovo, Luxembourg, Mexico, Netherlands, New Zealand, Nicaragua, Nigeria, North Macedonia, Philippines, Poland, Russia, Serbia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Sweden, Switzerland, Taiwan, Tanzania, Thailand, Ukraine, USA, Vietnam.

Problem Selection Committee



Tony Gardiner, Edward Crane, Alexander Betts, James Cranch, Joseph Myers (chair), James Aaronson, Andrew Carlotti, Géza Kós, Ilya I. Bogdanov, Jack Shotton

Problems

Algebra

A1. Let $\mathbb{Z}$ be the set of integers. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that, for all integers a and b,

$$f(2a) + 2f(b) = f(f(a + b)).$$

(South Africa)

A2. Let $u_1, u_2, \ldots, u_{2019}$ be real numbers satisfying

$$u_1 + u_2 + \cdots + u_{2019} = 0 \quad \text{and} \quad u_1^2 + u_2^2 + \cdots + u_{2019}^2 = 1.$$

Let $a = \min(u_1, u_2, \ldots, u_{2019})$ and $b = \max(u_1, u_2, \ldots, u_{2019})$. Prove that

$$ab \leqslant -\frac{1}{2019}.$$

(Germany)

A3. Let $n \geqslant 3$ be a positive integer and let $(a_1, a_2, \ldots, a_n)$ be a strictly increasing sequence of n positive real numbers with sum equal to 2. Let X be a subset of $\{1, 2, \ldots, n\}$ such that the value of

$$\left| 1 - \sum_{i \in X} a_i \right|$$

is minimised. Prove that there exists a strictly increasing sequence of n positive real numbers $(b_1, b_2, \ldots, b_n)$ with sum equal to 2 such that

$$\sum_{i \in X} b_i = 1.$$

(New Zealand)

A4. Let $n \geqslant 2$ be a positive integer and $a_1, a_2, \ldots, a_n$ be real numbers such that

$$a_1 + a_2 + \cdots + a_n = 0.$$

Define the set A by

$$A = \big\{(i, j) \;\big|\; 1 \leqslant i < j \leqslant n, |a_i - a_j| \geqslant 1\big\}.$$

Prove that, if A is not empty, then

$$\sum_{(i,j) \in A} a_i a_j < 0.$$

(China)

A5. Let $x_1, x_2, \ldots, x_n$ be different real numbers. Prove that

$$\sum_{1 \leqslant i \leqslant n} \prod_{j \neq i} \frac{1 - x_i x_j}{x_i - x_j} = \begin{cases} 0, & \text{if } n \text{ is even;} \\ 1, & \text{if } n \text{ is odd.} \end{cases}$$

(Kazakhstan)

A6. A polynomial $P(x, y, z)$ in three variables with real coefficients satisfies the identities

$$P(x, y, z) = P(x, y, xy - z) = P(x, zx - y, z) = P(yz - x, y, z).$$

Prove that there exists a polynomial $F(t)$ in one variable such that

$$P(x, y, z) = F(x^2 + y^2 + z^2 - xyz).$$

(Russia)

A7. Let $\mathbb{Z}$ be the set of integers. We consider functions $f\colon \mathbb{Z} \to \mathbb{Z}$ satisfying

$$f\big(f(x + y) + y\big) = f\big(f(x) + y\big)$$

for all integers x and y. For such a function, we say that an integer v is *f-rare* if the set

$$X_v = \{x \in \mathbb{Z}\colon f(x) = v\}$$

is finite and nonempty.

(a) Prove that there exists such a function f for which there is an f-rare integer.

(b) Prove that no such function f can have more than one f-rare integer.

(Netherlands)

Combinatorics

C1. The infinite sequence a_0, a_1, a_2, ... of (not necessarily different) integers has the following properties: $0 \leqslant a_i \leqslant i$ for all integers $i \geqslant 0$, and

$$\binom{k}{a_0} + \binom{k}{a_1} + \cdots + \binom{k}{a_k} = 2^k$$

for all integers $k \geqslant 0$.

Prove that all integers $N \geqslant 0$ occur in the sequence (that is, for all $N \geqslant 0$, there exists $i \geqslant 0$ with $a_i = N$).

(Netherlands)

C2. You are given a set of n blocks, each weighing at least 1; their total weight is $2n$. Prove that for every real number r with $0 \leqslant r \leqslant 2n - 2$ you can choose a subset of the blocks whose total weight is at least r but at most $r + 2$.

(Thailand)

C3. Let n be a positive integer. Harry has n coins lined up on his desk, each showing heads or tails. He repeatedly does the following operation: if there are k coins showing heads and $k > 0$, then he flips the kth coin over; otherwise he stops the process. (For example, the process starting with THT would be $THT \to HHT \to HTT \to TTT$, which takes three steps.)

Letting C denote the initial configuration (a sequence of n H's and T's), write $\ell(C)$ for the number of steps needed before all coins show T. Show that this number $\ell(C)$ is finite, and determine its average value over all 2^n possible initial configurations C.

(USA)

C4. On a flat plane in Camelot, King Arthur builds a labyrinth $\mathfrak{L}$ consisting of n walls, each of which is an infinite straight line. No two walls are parallel, and no three walls have a common point. Merlin then paints one side of each wall entirely red and the other side entirely blue.

At the intersection of two walls there are four corners: two diagonally opposite corners where a red side and a blue side meet, one corner where two red sides meet, and one corner where two blue sides meet. At each such intersection, there is a two-way door connecting the two diagonally opposite corners at which sides of different colours meet.

After Merlin paints the walls, Morgana then places some knights in the labyrinth. The knights can walk through doors, but cannot walk through walls.

Let $k(\mathfrak{L})$ be the largest number k such that, no matter how Merlin paints the labyrinth $\mathfrak{L}$, Morgana can always place at least k knights such that no two of them can ever meet. For each n, what are all possible values for $k(\mathfrak{L})$, where $\mathfrak{L}$ is a labyrinth with n walls?

(Canada)

C5. On a certain social network, there are 2019 users, some pairs of which are friends, where friendship is a symmetric relation. Initially, there are 1010 people with 1009 friends each and 1009 people with 1010 friends each. However, the friendships are rather unstable, so events of the following kind may happen repeatedly, one at a time:

> Let A, B, and C be people such that A is friends with both B and C, but B and C are not friends; then B and C become friends, but A is no longer friends with them.

Prove that, regardless of the initial friendships, there exists a sequence of such events after which each user is friends with at most one other user.

(Croatia)

C6. Let $n > 1$ be an integer. Suppose we are given $2n$ points in a plane such that no three of them are collinear. The points are to be labelled A_1, A_2, ..., A_{2n} in some order. We then consider the $2n$ angles $\angle A_1A_2A_3$, $\angle A_2A_3A_4$, ..., $\angle A_{2n-2}A_{2n-1}A_{2n}$, $\angle A_{2n-1}A_{2n}A_1$, $\angle A_{2n}A_1A_2$. We measure each angle in the way that gives the smallest positive value (i.e. between 0° and 180°). Prove that there exists an ordering of the given points such that the resulting $2n$ angles can be separated into two groups with the sum of one group of angles equal to the sum of the other group.

(USA)

C7. There are 60 empty boxes B_1, ..., B_{60} in a row on a table and an unlimited supply of pebbles. Given a positive integer n, Alice and Bob play the following game.

In the first round, Alice takes n pebbles and distributes them into the 60 boxes as she wishes. Each subsequent round consists of two steps:

(a) Bob chooses an integer k with $1 \leqslant k \leqslant 59$ and splits the boxes into the two groups B_1, ..., B_k and B_{k+1}, ..., B_{60}.

(b) Alice picks one of these two groups, adds one pebble to each box in that group, and removes one pebble from each box in the other group.

Bob wins if, at the end of any round, some box contains no pebbles. Find the smallest n such that Alice can prevent Bob from winning.

(Czech Republic)

C8. Alice has a map of Wonderland, a country consisting of $n \geqslant 2$ towns. For every pair of towns, there is a narrow road going from one town to the other. One day, all the roads are declared to be "one way" only. Alice has no information on the direction of the roads, but the King of Hearts has offered to help her. She is allowed to ask him a number of questions. For each question in turn, Alice chooses a pair of towns and the King of Hearts tells her the direction of the road connecting those two towns.

Alice wants to know whether there is at least one town in Wonderland with at most one outgoing road. Prove that she can always find out by asking at most $4n$ questions.

Comment. This problem could be posed with an explicit statement about points being awarded for weaker bounds cn for some $c > 4$, in the style of IMO 2014 Problem 6.

(Thailand)

C9. For any two different real numbers x and y, we define $D(x, y)$ to be the unique integer d satisfying $2^d \leqslant |x - y| < 2^{d+1}$. Given a set of reals $\mathcal{F}$, and an element $x \in \mathcal{F}$, we say that the *scales* of x in $\mathcal{F}$ are the values of $D(x, y)$ for $y \in \mathcal{F}$ with $x \neq y$.

Let k be a given positive integer. Suppose that each member x of $\mathcal{F}$ has at most k different scales in $\mathcal{F}$ (note that these scales may depend on x). What is the maximum possible size of $\mathcal{F}$?

(Italy)

Geometry

G1. Let ABC be a triangle. Circle Γ passes through A, meets segments AB and AC again at points D and E respectively, and intersects segment BC at F and G such that F lies between B and G. The tangent to circle BDF at F and the tangent to circle CEG at G meet at point T. Suppose that points A and T are distinct. Prove that line AT is parallel to BC.

(Nigeria)

G2. Let ABC be an acute-angled triangle and let D, E, and F be the feet of altitudes from A, B, and C to sides BC, CA, and AB, respectively. Denote by ω_B and ω_C the incircles of triangles BDF and CDE, and let these circles be tangent to segments DF and DE at M and N, respectively. Let line MN meet circles ω_B and ω_C again at $P \neq M$ and $Q \neq N$, respectively. Prove that $MP = NQ$.

(Vietnam)

G3. In triangle ABC, let A_1 and B_1 be two points on sides BC and AC, and let P and Q be two points on segments AA_1 and BB_1, respectively, so that line PQ is parallel to AB. On ray PB_1, beyond B_1, let P_1 be a point so that $\angle PP_1C = \angle BAC$. Similarly, on ray QA_1, beyond A_1, let Q_1 be a point so that $\angle CQ_1Q = \angle CBA$. Show that points P, Q, P_1, and Q_1 are concyclic.

(Ukraine)

G4. Let P be a point inside triangle ABC. Let AP meet BC at A_1, let BP meet CA at B_1, and let CP meet AB at C_1. Let A_2 be the point such that A_1 is the midpoint of PA_2, let B_2 be the point such that B_1 is the midpoint of PB_2, and let C_2 be the point such that C_1 is the midpoint of PC_2. Prove that points A_2, B_2, and C_2 cannot all lie strictly inside the circumcircle of triangle ABC.

(Australia)

G5. Let $ABCDE$ be a convex pentagon with $CD = DE$ and $\angle EDC \neq 2 \cdot \angle ADB$. Suppose that a point P is located in the interior of the pentagon such that $AP = AE$ and $BP = BC$. Prove that P lies on the diagonal CE if and only if $\text{area}(BCD) + \text{area}(ADE) = \text{area}(ABD) + \text{area}(ABP)$.

(Hungary)

G6. Let I be the incentre of acute-angled triangle ABC. Let the incircle meet BC, CA, and AB at D, E, and F, respectively. Let line EF intersect the circumcircle of the triangle at P and Q, such that F lies between E and P. Prove that $\angle DPA + \angle AQD = \angle QIP$.

(Slovakia)

G7. The incircle ω of acute-angled scalene triangle ABC has centre I and meets sides BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω again at R. Line AR meets ω again at P. The circumcircles of triangles PCE and PBF meet again at $Q \neq P$. Prove that lines DI and PQ meet on the external bisector of angle BAC.

(India)

G8. Let $\mathcal{L}$ be the set of all lines in the plane and let f be a function that assigns to each line $\ell \in \mathcal{L}$ a point $f(\ell)$ on ℓ. Suppose that for any point X, and for any three lines ℓ_1, ℓ_2, ℓ_3 passing through X, the points $f(\ell_1)$, $f(\ell_2)$, $f(\ell_3)$ and X lie on a circle.

Prove that there is a unique point P such that $f(\ell) = P$ for any line ℓ passing through P.

(Australia)

Number Theory

N1. Find all pairs (m, n) of positive integers satisfying the equation

$$(2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}) = m!$$

(El Salvador)

N2. Find all triples (a, b, c) of positive integers such that $a^3 + b^3 + c^3 = (abc)^2$.

(Nigeria)

N3. We say that a set S of integers is *rootiful* if, for any positive integer n and any $a_0, a_1, \ldots, a_n \in S$, all integer roots of the polynomial $a_0 + a_1 x + \cdots + a_n x^n$ are also in S. Find all rootiful sets of integers that contain all numbers of the form $2^a - 2^b$ for positive integers a and b.

(Czech Republic)

N4. Let $\mathbb{Z}_{>0}$ be the set of positive integers. A positive integer constant C is given. Find all functions $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that, for all positive integers a and b satisfying $a + b > C$,

$$a + f(b) \mid a^2 + b\, f(a).$$

(Croatia)

N5. Let a be a positive integer. We say that a positive integer b is *a-good* if $\binom{an}{b} - 1$ is divisible by $an + 1$ for all positive integers n with $an \geqslant b$. Suppose b is a positive integer such that b is a-good, but $b + 2$ is not a-good. Prove that $b + 1$ is prime.

(Netherlands)

N6. Let $H = \left\{ \left\lfloor i\sqrt{2} \right\rfloor : i \in \mathbb{Z}_{>0} \right\} = \{1, 2, 4, 5, 7, \ldots\}$, and let n be a positive integer. Prove that there exists a constant C such that, if $A \subset \{1, 2, \ldots, n\}$ satisfies $|A| \geqslant C\sqrt{n}$, then there exist $a, b \in A$ such that $a - b \in H$. (Here $\mathbb{Z}_{>0}$ is the set of positive integers, and $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z.)

(Brazil)

N7. Prove that there is a constant $c > 0$ and infinitely many positive integers n with the following property: there are infinitely many positive integers that cannot be expressed as the sum of fewer than $cn \log(n)$ pairwise coprime n^{th} powers.

(Canada)

N8. Let a and b be two positive integers. Prove that the integer

$$a^2 + \left\lceil \frac{4a^2}{b} \right\rceil$$

is not a square. (Here $\lceil z \rceil$ denotes the least integer greater than or equal to z.)

(Russia)

Solutions

Algebra

A1. Let $\mathbb{Z}$ be the set of integers. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that, for all integers a and b,

$$f(2a) + 2f(b) = f(f(a+b)). \tag{1}$$

(South Africa)

Answer: The solutions are $f(n) = 0$ and $f(n) = 2n + K$ for any constant $K \in \mathbb{Z}$.

Common remarks. Most solutions to this problem first prove that f must be linear, before determining all linear functions satisfying (1).

Solution 1. Substituting $a = 0, b = n + 1$ gives $f(f(n+1)) = f(0) + 2f(n+1)$. Substituting $a = 1, b = n$ gives $f(f(n+1)) = f(2) + 2f(n)$.

In particular, $f(0) + 2f(n+1) = f(2) + 2f(n)$, and so $f(n+1) - f(n) = \frac{1}{2}(f(2) - f(0))$. Thus $f(n+1) - f(n)$ must be constant. Since f is defined only on $\mathbb{Z}$, this tells us that f must be a linear function; write $f(n) = Mn + K$ for arbitrary constants M and K, and we need only determine which choices of M and K work.

Now, (1) becomes

$$2Ma + K + 2(Mb + K) = M(M(a+b) + K) + K$$

which we may rearrange to form

$$(M-2)\big(M(a+b) + K\big) = 0.$$

Thus, either $M = 2$, or $M(a+b) + K = 0$ for all values of $a + b$. In particular, the only possible solutions are $f(n) = 0$ and $f(n) = 2n + K$ for any constant $K \in \mathbb{Z}$, and these are easily seen to work.

Solution 2. Let $K = f(0)$.

First, put $a = 0$ in (1); this gives

$$f(f(b)) = 2f(b) + K \tag{2}$$

for all $b \in \mathbb{Z}$.

Now put $b = 0$ in (1); this gives

$$f(2a) + 2K = f(f(a)) = 2f(a) + K,$$

where the second equality follows from (2). Consequently,

$$f(2a) = 2f(a) - K \tag{3}$$

for all $a \in \mathbb{Z}$.

Substituting (2) and (3) into (1), we obtain

$$\begin{aligned} f(2a) + 2f(b) &= f(f(a+b)) \\ 2f(a) - K + 2f(b) &= 2f(a+b) + K \\ f(a) + f(b) &= f(a+b) + K. \end{aligned}$$

Thus, if we set $g(n) = f(n) - K$ we see that g satisfies the Cauchy equation $g(a + b) = g(a) + g(b)$. The solution to the Cauchy equation over $\mathbb{Z}$ is well-known; indeed, it may be proven by an easy induction that $g(n) = Mn$ for each $n \in \mathbb{Z}$, where $M = g(1)$ is a constant.

Therefore, $f(n) = Mn + K$, and we may proceed as in Solution 1.

Comment 1. Instead of deriving (3) by substituting $b = 0$ into (1), we could instead have observed that the right hand side of (1) is symmetric in a and b, and thus

$$f(2a) + 2f(b) = f(2b) + 2f(a).$$

Thus, $f(2a) - 2f(a) = f(2b) - 2f(b)$ for any $a, b \in \mathbb{Z}$, and in particular $f(2a) - 2f(a)$ is constant. Setting $a = 0$ shows that this constant is equal to $-K$, and so we obtain (3).

Comment 2. Some solutions initially prove that $f(f(n))$ is linear (sometimes via proving that $f(f(n)) - 3K$ satisfies the Cauchy equation). However, one can immediately prove that f is linear by substituting something of the form $f(f(n)) = M'n + K'$ into (2).

A2. Let $u_1, u_2, \ldots, u_{2019}$ be real numbers satisfying

$$u_1 + u_2 + \cdots + u_{2019} = 0 \quad \text{and} \quad u_1^2 + u_2^2 + \cdots + u_{2019}^2 = 1.$$

Let $a = \min(u_1, u_2, \ldots, u_{2019})$ and $b = \max(u_1, u_2, \ldots, u_{2019})$. Prove that

$$ab \leqslant -\frac{1}{2019}.$$

(Germany)

Solution 1. Notice first that $b > 0$ and $a < 0$. Indeed, since $\sum_{i=1}^{2019} u_i^2 = 1$, the variables u_i cannot be all zero, and, since $\sum_{i=1}^{2019} u_i = 0$, the nonzero elements cannot be all positive or all negative.

Let $P = \{i : u_i > 0\}$ and $N = \{i : u_i \leqslant 0\}$ be the indices of positive and nonpositive elements in the sequence, and let $p = |P|$ and $n = |N|$ be the sizes of these sets; then $p + n = 2019$. By the condition $\sum_{i=1}^{2019} u_i = 0$ we have $0 = \sum_{i=1}^{2019} u_i = \sum_{i\in P} u_i - \sum_{i\in N} |u_i|$, so

$$\sum_{i\in P} u_i = \sum_{i\in N} |u_i|. \tag{1}$$

After this preparation, estimate the sum of squares of the positive and nonpositive elements as follows:

$$\sum_{i\in P} u_i^2 \leqslant \sum_{i\in P} bu_i = b\sum_{i\in P} u_i = b\sum_{i\in N} |u_i| \leqslant b\sum_{i\in N} |a| = -nab; \tag{2}$$

$$\sum_{i\in N} u_i^2 \leqslant \sum_{i\in N} |a|\cdot|u_i| = |a|\sum_{i\in N} |u_i| = |a|\sum_{i\in P} u_i \leqslant |a|\sum_{i\in P} b = -pab. \tag{3}$$

The sum of these estimates is

$$1 = \sum_{i=1}^{2019} u_i^2 = \sum_{i\in P} u_i^2 + \sum_{i\in N} u_i^2 \leqslant -(p+n)ab = -2019ab;$$

that proves $ab \leqslant \frac{-1}{2019}$.

Comment 1. After observing $\sum_{i\in P} u_i^2 \leqslant b\sum_{i\in P} u_i$ and $\sum_{i\in N} u_i^2 \leqslant |a|\sum_{i\in P} |u_i|$, instead of (2, 3) an alternative continuation is

$$|ab| \geqslant \frac{\sum_{i\in P} u_i^2}{\sum_{i\in P} u_i} \cdot \frac{\sum_{i\in N} u_i^2}{\sum_{i\in N} |u_i|} = \frac{\sum_{i\in P} u_i^2}{\left(\sum_{i\in P} u_i\right)^2} \sum_{i\in N} u_i^2 \geqslant \frac{1}{p}\sum_{i\in N} u_i^2$$

(by the AM-QM or the Cauchy–Schwarz inequality) and similarly $|ab| \geqslant \frac{1}{n}\sum_{i\in P} u_i^2$.

Solution 2. As in the previous solution we conclude that $a < 0$ and $b > 0$.

For every index i, the number u_i is a convex combination of a and b, so

$$u_i = x_i a + y_i b \quad \text{with some weights } 0 \leqslant x_i, y_i \leqslant 1, \text{ with } x_i + y_i = 1.$$

Let $X = \sum_{i=1}^{2019} x_i$ and $Y = \sum_{i=1}^{2019} y_i$. From $0 = \sum_{i=1}^{2019} u_i = \sum_{i=1}^{2019} (x_i a + y_i b) = -|a|X + bY$, we get

$$|a|X = bY. \tag{4}$$

From $\sum_{i=1}^{2019}(x_i+y_i)=2019$ we have

$$X+Y=2019. \tag{5}$$

The system of linear equations $(4,5)$ has a unique solution:

$$X=\frac{2019b}{|a|+b},\quad Y=\frac{2019|a|}{|a|+b}.$$

Now apply the following estimate to every u_i^2 in their sum:

$$u_i^2=x_i^2a^2+2x_iy_iab+y_i^2b^2\leqslant x_ia^2+y_ib^2;$$

we obtain that

$$1=\sum_{i=1}^{2019}u_i^2\leqslant\sum_{i=1}^{2019}(x_ia^2+y_ib^2)=Xa^2+Yb^2=\frac{2019b}{|a|+b}|a|^2+\frac{2019|a|}{|a|+b}b^2=2019|a|b=-2019ab.$$

Hence, $ab\leqslant\dfrac{-1}{2019}$.

Comment 2. The idea behind Solution 2 is the following thought. Suppose we fix $a<0$ and $b>0$, fix $\sum u_i=0$ and vary the u_i to achieve the maximum value of $\sum u_i^2$. Considering varying any two of the u_i while preserving their sum: the maximum value of $\sum u_i^2$ is achieved when those two are as far apart as possible, so all but at most one of the u_i are equal to a or b. Considering a weighted version of the problem, we see the maximum (with fractional numbers of u_i having each value) is achieved when $\dfrac{2019b}{|a|+b}$ of them are a and $\dfrac{2019|a|}{|a|+b}$ are b.

In fact, this happens in the solution: the number u_i is replaced by x_i copies of a and y_i copies of b.

A3. Let $n \geqslant 3$ be a positive integer and let $(a_1, a_2, \ldots, a_n)$ be a strictly increasing sequence of n positive real numbers with sum equal to 2. Let X be a subset of $\{1, 2, \ldots, n\}$ such that the value of

$$\left|1 - \sum_{i \in X} a_i\right|$$

is minimised. Prove that there exists a strictly increasing sequence of n positive real numbers $(b_1, b_2, \ldots, b_n)$ with sum equal to 2 such that

$$\sum_{i \in X} b_i = 1.$$

(New Zealand)

Common remarks. In all solutions, we say an index set X is (a_i)-*minimising* if it has the property in the problem for the given sequence (a_i). Write X^c for the complement of X, and $[a, b]$ for the interval of integers k such that $a \leqslant k \leqslant b$. Note that

$$\left|1 - \sum_{i \in X} a_i\right| = \left|1 - \sum_{i \in X^c} a_i\right|,$$

so we may exchange X and X^c where convenient. Let

$$\Delta = \sum_{i \in X^c} a_i - \sum_{i \in X} a_i$$

and note that X is (a_i)-minimising if and only if it minimises $|\Delta|$, and that $\sum_{i \in X} a_i = 1$ if and only if $\Delta = 0$.

In some solutions, a scaling process is used. If we have a strictly increasing sequence of positive real numbers c_i (typically obtained by perturbing the a_i in some way) such that

$$\sum_{i \in X} c_i = \sum_{i \in X^c} c_i,$$

then we may put $b_i = 2c_i / \sum_{j=1}^{n} c_j$. So it suffices to construct such a sequence without needing its sum to be 2.

The solutions below show various possible approaches to the problem. Solutions 1 and 2 perturb a few of the a_i to form the b_i (with scaling in the case of Solution 1, without scaling in the case of Solution 2). Solutions 3 and 4 look at properties of the index set X. Solution 3 then perturbs many of the a_i to form the b_i, together with scaling. Rather than using such perturbations, Solution 4 constructs a sequence (b_i) directly from the set X with the required properties. Solution 4 can be used to give a complete description of sets X that are (a_i)-minimising for some (a_i).

Solution 1. Without loss of generality, assume $\sum_{i \in X} a_i \leqslant 1$, and we may assume strict inequality as otherwise $b_i = a_i$ works. Also, X clearly cannot be empty.

If $n \in X$, add Δ to a_n, producing a sequence of c_i with $\sum_{i \in X} c_i = \sum_{i \in X^c} c_i$, and then scale as described above to make the sum equal to 2. Otherwise, there is some k with $k \in X$ and $k + 1 \in X^c$. Let $\delta = a_{k+1} - a_k$.

- If $\delta > \Delta$, add Δ to a_k and then scale.
- If $\delta < \Delta$, then considering $X \cup \{k + 1\} \setminus \{k\}$ contradicts X being (a_i)-minimising.
- If $\delta = \Delta$, choose any $j \neq k, k + 1$ (possible since $n \geqslant 3$), and any ϵ less than the least of a_1 and all the differences $a_{i+1} - a_i$. If $j \in X$ then add $\Delta - \epsilon$ to a_k and ϵ to a_j, then scale; otherwise, add Δ to a_k and $\epsilon/2$ to a_{k+1}, and subtract $\epsilon/2$ from a_j, then scale.

Solution 2. This is similar to Solution 1, but without scaling. As in that solution, without loss of generality, assume $\sum_{i\in X} a_i < 1$.

Suppose there exists $1 \leqslant j \leqslant n-1$ such that $j \in X$ but $j+1 \in X^c$. Then $a_{j+1} - a_j \geqslant \Delta$, because otherwise considering $X \cup \{j+1\} \setminus \{j\}$ contradicts X being (a_i)-minimising.

If $a_{j+1} - a_j > \Delta$, put

$$b_i = \begin{cases} a_j + \Delta/2, & \text{if } i = j; \\ a_{j+1} - \Delta/2, & \text{if } i = j+1; \\ a_i, & \text{otherwise.} \end{cases}$$

If $a_{j+1} - a_j = \Delta$, choose any ϵ less than the least of $\Delta/2$, a_1 and all the differences $a_{i+1} - a_i$. If $|X| \geqslant 2$, choose $k \in X$ with $k \neq j$, and put

$$b_i = \begin{cases} a_j + \Delta/2 - \epsilon, & \text{if } i = j; \\ a_{j+1} - \Delta/2, & \text{if } i = j+1; \\ a_k + \epsilon, & \text{if } i = k; \\ a_i, & \text{otherwise.} \end{cases}$$

Otherwise, $|X^c| \geqslant 2$, so choose $k \in X^c$ with $k \neq j+1$, and put

$$b_i = \begin{cases} a_j + \Delta/2, & \text{if } i = j; \\ a_{j+1} - \Delta/2 + \epsilon, & \text{if } i = j+1; \\ a_k - \epsilon, & \text{if } i = k; \\ a_i, & \text{otherwise.} \end{cases}$$

If there is no $1 \leqslant j \leqslant n$ such that $j \in X$ but $j+1 \in X^c$, there must be some $1 < k \leqslant n$ such that $X = [k, n]$ (certainly X cannot be empty). We must have $a_1 > \Delta$, as otherwise considering $X \cup \{1\}$ contradicts X being (a_i)-minimising. Now put

$$b_i = \begin{cases} a_1 - \Delta/2, & \text{if } i = 1; \\ a_n + \Delta/2, & \text{if } i = n; \\ a_i, & \text{otherwise.} \end{cases}$$

Solution 3. Without loss of generality, assume $\sum_{i\in X} a_i \leqslant 1$, so $\Delta \geqslant 0$. If $\Delta = 0$ we can take $b_i = a_i$, so now assume that $\Delta > 0$.

Suppose that there is some $k \leqslant n$ such that $|X \cap [k, n]| > |X^c \cap [k, n]|$. If we choose the largest such k then $|X \cap [k, n]| - |X^c \cap [k, n]| = 1$. We can now find the required sequence (b_i) by starting with $c_i = a_i$ for $i < k$ and $c_i = a_i + \Delta$ for $i \geqslant k$, and then scaling as described above.

If no such k exists, we will derive a contradiction. For each $i \in X$ we can choose $i < j_i \leqslant n$ in such a way that $j_i \in X^c$ and all the j_i are different. (For instance, note that necessarily $n \in X^c$ and now just work downwards; each time an $i \in X$ is considered, let j_i be the least element of X^c greater than i and not yet used.) Let Y be the (possibly empty) subset of $[1, n]$ consisting of those elements in X^c that are also not one of the j_i. In any case

$$\Delta = \sum_{i\in X} (a_{j_i} - a_i) + \sum_{j\in Y} a_j$$

where each term in the sums is positive. Since $n \geqslant 3$ the total number of terms above is at least two. Take a least such term and its corresponding index i and consider the set Z which we form from X by removing i and adding j_i (if it is a term of the first type) or just by adding j if it is a term of the second type. The corresponding expression of Δ for Z has the sign of its least term changed, meaning that the sum is still nonnegative but strictly less than Δ, which contradicts X being (a_i)-minimising.

Solution 4. This uses some similar ideas to Solution 3, but describes properties of the index sets X that are sufficient to describe a corresponding sequence (b_i) that is not derived from (a_i).

Note that, for two subsets X, Y of $[1, n]$, the following are equivalent:

- $|X \cap [i, n]| \leqslant |Y \cap [i, n]|$ for all $1 \leqslant i \leqslant n$;
- Y is at least as large as X, and for all $1 \leqslant j \leqslant |Y|$, the jth largest element of Y is at least as big as the jth largest element of X;
- there is an injective function $f : X \to Y$ such that $f(i) \geqslant i$ for all $i \in X$.

If these equivalent conditions are satisfied, we write $X \preceq Y$. We write $X \prec Y$ if $X \preceq Y$ and $X \neq Y$.

Note that if $X \prec Y$, then $\sum_{i\in X} a_i < \sum_{i\in Y} a_i$ (the second description above makes this clear).

We claim first that, if $n \geqslant 3$ and $X \prec X^c$, then there exists Y with $X \prec Y \prec X^c$. Indeed, as $|X| \leqslant |X^c|$, we have $|X^c| \geqslant 2$. Define Y to consist of the largest element of X^c, together with all but the largest element of X; it is clear both that Y is distinct from X and X^c, and that $X \preceq Y \preceq X^c$, which is what we need.

But, in this situation, we have

$$\sum_{i\in X} a_i < \sum_{i\in Y} a_i < \sum_{i\in X^c} a_i \qquad \text{and} \qquad 1 - \sum_{i\in X} a_i = -\left(1 - \sum_{i\in X^c} a_i\right),$$

so $|1 - \sum_{i\in Y} a_i| < |1 - \sum_{i\in X} a_i|$.

Hence if X is (a_i)-minimising, we do not have $X \prec X^c$, and similarly we do not have $X^c \prec X$.

Considering the first description above, this immediately implies the following Claim.

Claim. There exist $1 \leqslant k, \ell \leqslant n$ such that $|X \cap [k, n]| > \frac{n-k+1}{2}$ and $|X \cap [\ell, n]| < \frac{n-\ell+1}{2}$.

We now construct our sequence (b_i) using this claim. Let k and ℓ be the greatest values satisfying the claim, and without loss of generality suppose $k = n$ and $\ell < n$ (otherwise replace X by its complement). As ℓ is maximal, $n - \ell$ is even and $|X \cap [\ell, n]| = \frac{n-\ell}{2}$. For sufficiently small positive ϵ, we take

$$b_i = i\epsilon + \begin{cases} 0, & \text{if } i < \ell; \\ \delta, & \text{if } \ell \leqslant i \leqslant n-1; \\ \gamma, & \text{if } i = n. \end{cases}$$

Let $M = \sum_{i\in X} i$. So we require

$$M\epsilon + \left(\frac{n-\ell}{2} - 1\right)\delta + \gamma = 1$$

and

$$\frac{n(n+1)}{2}\epsilon + (n-\ell)\delta + \gamma = 2.$$

These give

$$\gamma = 2\delta + \left(\frac{n(n+1)}{2} - 2M\right)\epsilon$$

and for sufficiently small positive ϵ, solving for γ and δ gives $0 < \delta < \gamma$ (since $\epsilon = 0$ gives $\delta = 1/(\frac{n-\ell}{2} + 1)$ and $\gamma = 2\delta$), so the sequence is strictly increasing and has positive values.

Comment. This solution also shows that the claim gives a complete description of sets X that are (a_i)-minimising for some (a_i).

Another approach to proving the claim is as follows. We prove the existence of ℓ with the claimed property; the existence of k follows by considering the complement of X.

Suppose, for a contradiction, that for all $1 \leqslant \ell \leqslant n$ we have $|X \cap [\ell, n]| \geqslant \left\lceil \frac{n-\ell+1}{2} \right\rceil$. If we ever have strict inequality, consider the set $Y = \{n, n-2, n-4, \ldots\}$. This set may be obtained from X by possibly removing some elements and reducing the values of others. (To see this, consider the largest $k \in X \setminus Y$, if any; remove it, and replace it by the greatest $j \in X^c$ with $j < k$, if any. Such steps preserve the given inequality, and are possible until we reach the set Y.) So if we had strict inequality, and so $X \neq Y$, we have

$$\sum_{i \in X} a_i > \sum_{i \in Y} a_i > 1,$$

contradicting X being (a_i)-minimising. Otherwise, we always have equality, meaning that $X = Y$. But now consider $Z = Y \cup \{n-1\} \setminus \{n\}$. Since $n \geqslant 3$, we have

$$\sum_{i \in Y} a_i > \sum_{i \in Z} a_i > \sum_{i \in Y^c} a_i = 2 - \sum_{i \in Y} a_i,$$

and so Z contradicts X being (a_i)-minimising.

A4. Let $n \geqslant 2$ be a positive integer and $a_1, a_2, \ldots, a_n$ be real numbers such that

$$a_1 + a_2 + \cdots + a_n = 0.$$

Define the set A by

$$A = \big\{(i,j) \;\big|\; 1 \leqslant i < j \leqslant n, |a_i - a_j| \geqslant 1\big\}.$$

Prove that, if A is not empty, then

$$\sum_{(i,j)\in A} a_i a_j < 0.$$

(China)

Solution 1. Define sets B and C by

$$B = \big\{(i,j) \;\big|\; 1 \leqslant i, j \leqslant n, |a_i - a_j| \geqslant 1\big\},$$
$$C = \big\{(i,j) \;\big|\; 1 \leqslant i, j \leqslant n, |a_i - a_j| < 1\big\}.$$

We have

$$\sum_{(i,j)\in A} a_i a_j = \frac{1}{2} \sum_{(i,j)\in B} a_i a_j$$
$$\sum_{(i,j)\in B} a_i a_j = \sum_{1\leqslant i,j\leqslant n} a_i a_j - \sum_{(i,j)\notin B} a_i a_j = 0 - \sum_{(i,j)\in C} a_i a_j.$$

So it suffices to show that if A (and hence B) are nonempty, then

$$\sum_{(i,j)\in C} a_i a_j > 0.$$

Partition the indices into sets P, Q, R, and S such that

$$P = \big\{i \;\big|\; a_i \leqslant -1\big\} \qquad R = \big\{i \;\big|\; 0 < a_i < 1\big\}$$
$$Q = \big\{i \;\big|\; -1 < a_i \leqslant 0\big\} \qquad S = \big\{i \;\big|\; 1 \leqslant a_i\big\}.$$

Then

$$\sum_{(i,j)\in C} a_i a_j \geqslant \sum_{i\in P\cup S} a_i^2 + \sum_{i,j\in Q\cup R} a_i a_j = \sum_{i\in P\cup S} a_i^2 + \left(\sum_{i\in Q\cup R} a_i\right)^2 \geqslant 0.$$

The first inequality holds because all of the positive terms in the RHS are also in the LHS, and all of the negative terms in the LHS are also in the RHS. The first inequality attains equality only if both sides have the same negative terms, which implies $|a_i - a_j| < 1$ whenever $i, j \in Q \cup R$; the second inequality attains equality only if $P = S = \varnothing$. But then we would have $A = \varnothing$. So A nonempty implies that the inequality holds strictly, as required.

Solution 2. Consider P, Q, R, S as in Solution 1, set

$$p = \sum_{i\in P} a_i, \quad q = \sum_{i\in Q} a_i, \quad r = \sum_{i\in R} a_i, \quad s = \sum_{i\in S} a_i,$$

and let

$$t_+ = \sum_{(i,j)\in A,\ a_i a_j \geqslant 0} a_i a_j, \quad t_- = \sum_{(i,j)\in A,\ a_i a_j \leqslant 0} a_i a_j.$$

We know that $p + q + r + s = 0$, and we need to prove that $t_+ + t_- < 0$.

Notice that $t_+ \leqslant p^2/2 + pq + rs + s^2/2$ (with equality only if $p = s = 0$), and $t_- \leqslant pr + ps + qs$ (with equality only if there do not exist $i \in Q$ and $j \in R$ with $a_j - a_i > 1$). Therefore,

$$t_+ + t_- \leqslant \frac{p^2 + s^2}{2} + pq + rs + pr + ps + qs = \frac{(p+q+r+s)^2}{2} - \frac{(q+r)^2}{2} = -\frac{(q+r)^2}{2} \leqslant 0.$$

If A is not empty and $p = s = 0$, then there must exist $i \in Q, j \in R$ with $|a_i - a_j| > 1$, and hence the earlier equality conditions cannot both occur.

Comment. The RHS of the original inequality cannot be replaced with any constant $c < 0$ (independent of n). Indeed, take

$$a_1 = -\frac{n}{n+2}, a_2 = \cdots = a_{n-1} = \frac{1}{n+2}, a_n = \frac{2}{n+2}.$$

Then $\sum_{(i,j)\in A} a_i a_j = -\frac{2n}{(n+2)^2}$, which converges to zero as $n \to \infty$.

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A5. Let $x_1, x_2, \ldots, x_n$ be different real numbers. Prove that

$$\sum_{1\leqslant i\leqslant n}\prod_{j\neq i}\frac{1-x_ix_j}{x_i-x_j}=\begin{cases}0, & \text{if } n \text{ is even;}\\ 1, & \text{if } n \text{ is odd.}\end{cases}$$

(Kazakhstan)

Common remarks. Let $G(x_1, x_2, \ldots, x_n)$ be the function of the n variables $x_1, x_2, \ldots, x_n$ on the LHS of the required identity.

Solution 1 (Lagrange interpolation). Since both sides of the identity are rational functions, it suffices to prove it when all $x_i \notin \{\pm 1\}$. Define

$$f(t)=\prod_{i=1}^{n}(1-x_it)\,,$$

and note that

$$f(x_i)=(1-x_i^2)\prod_{j\neq i}1-x_ix_j\,.$$

Using the nodes $+1, -1, x_1, \ldots, x_n$, the Lagrange interpolation formula gives us the following expression for f:

$$\sum_{i=1}^{n}f(x_i)\frac{(x-1)(x+1)}{(x_i-1)(x_i+1)}\prod_{j\neq i}\frac{x-x_j}{x_i-x_j}\;+\;f(1)\frac{x+1}{1+1}\prod_{1\leqslant i\leqslant n}\frac{x-x_i}{1-x_i}\;+\;f(-1)\frac{x-1}{-1-1}\prod_{1\leqslant i\leqslant n}\frac{x-x_i}{1-x_i}\,.$$

The coefficient of t^{n+1} in $f(t)$ is 0, since f has degree n. The coefficient of t^{n+1} in the above expression of f is

$$\begin{aligned}0&=\sum_{1\leqslant i\leqslant n}\frac{f(x_i)}{\prod_{j\neq i}(x_i-x_j)\cdot(x_i-1)(x_i+1)}+\frac{f(1)}{\prod_{1\leqslant j\leqslant n}(1-x_j)\cdot(1+1)}+\frac{f(-1)}{\prod_{1\leqslant j\leqslant n}(-1-x_j)\cdot(-1-1)}\\&=-G(x_1,\ldots,x_n)+\frac{1}{2}+\frac{(-1)^{n+1}}{2}.\end{aligned}$$

Comment. The main difficulty is to think of including the two extra nodes ± 1 and evaluating the coefficient t^{n+1} in f when $n+1$ is higher than the degree of f.

It is possible to solve the problem using Lagrange interpolation on the nodes $x_1, \ldots, x_n$, but the definition of the polynomial being interpolated should depend on the parity of n. For n even, consider the polynomial

$$P(x)=\prod_i(1-xx_i)-\prod_i(x-x_i).$$

Lagrange interpolation shows that G is the coefficient of x^{n-1} in the polynomial $P(x)/(1-x^2)$, i.e. 0. For n odd, consider the polynomial

$$P(x)=\prod_i(1-xx_i)-x\prod_i(x-x_i).$$

Now G is the coefficient of x^{n-1} in $P(x)/(1-x^2)$, which is 1.

Solution 2 (using symmetries). Observe that G is symmetric in the variables $x_1, \ldots, x_n$. Define $V = \prod_{i<j}(x_j - x_i)$ and let $F = G \cdot V$, which is a polynomial in $x_1, \ldots, x_n$. Since V is alternating, F is also alternating (meaning that, if we exchange any two variables, then F changes sign). Every alternating polynomial in n variables $x_1, \ldots, x_n$ vanishes when any two variables x_i, x_j $(i \neq j)$ are equal, and is therefore divisible by $x_i - x_j$ for each pair $i \neq j$. Since these linear factors are pairwise coprime, V divides F exactly as a polynomial. Thus G is in fact a symmetric polynomial in $x_1, \ldots, x_n$.

Now observe that if all x_i are nonzero and we set $y_i = 1/x_i$ for $i = 1, \ldots, n$, then we have

$$\frac{1 - y_i y_j}{y_i - y_j} = \frac{1 - x_i x_j}{x_i - x_j},$$

so that

$$G\left(\frac{1}{x_1}, \ldots, \frac{1}{x_n}\right) = G(x_1, \ldots, x_n).$$

By continuity this is an identity of rational functions. Since G is a polynomial, it implies that G is constant. (If G were not constant, we could choose a point $(c_1, \ldots, c_n)$ with all $c_i \neq 0$, such that $G(c_1, \ldots, c_n) \neq G(0, \ldots, 0)$; then $g(x) := G(c_1 x, \ldots, c_n x)$ would be a nonconstant polynomial in the variable x, so $|g(x)| \to \infty$ as $x \to \infty$, hence $\left|G\left(\frac{y}{c_1}, \ldots, \frac{y}{c_n}\right)\right| \to \infty$ as $y \to 0$, which is impossible since G is a polynomial.)

We may identify the constant by substituting $x_i = \zeta^i$, where ζ is a primitive n^{th} root of unity in $\mathbb{C}$. In the ith term in the sum in the original expression we have a factor $1 - \zeta^i \zeta^{n-i} = 0$, unless $i = n$ or $2i = n$. In the case where n is odd, the only exceptional term is $i = n$, which gives the value $\prod_{j \neq n} \frac{1-\zeta^j}{1-\zeta^j} = 1$. When n is even, we also have the term $\prod_{j \neq \frac{n}{2}} \frac{1+\zeta^j}{-1-\zeta^j} = (-1)^{n-1} = -1$, so the sum is 0.

Comment. If we write out an explicit expression for F,

$$F = \sum_{1 \leqslant i \leqslant n} (-1)^{n-i} \prod_{\substack{j<k \\ j,k \neq i}} (x_k - x_j) \prod_{j \neq i} (1 - x_i x_j)$$

then to prove directly that F vanishes when $x_i = x_j$ for some $i \neq j$, but no other pair of variables coincide, we have to check carefully that the two nonzero terms in this sum cancel.

A different and slightly less convenient way to identify the constant is to substitute $x_i = 1 + \epsilon \zeta^i$, and throw away terms that are $O(\epsilon)$ as $\epsilon \to 0$.

Solution 3 (breaking symmetry). Consider G as a rational function in x_n with coefficients that are rational functions in the other variables. We can write

$$G(x_1, \ldots, x_n) = \frac{P(x_n)}{\prod_{j \neq n} (x_n - x_j)}$$

where $P(x_n)$ is a polynomial in x_n whose coefficients are rational functions in the other variables. We then have

$$P(x_n) = \left(\prod_{j \neq n} (1 - x_n x_j)\right) + \sum_{1 \leqslant i \leqslant n-1} (x_i x_n - 1) \left(\prod_{j \neq i,n} (x_n - x_j)\right) \left(\prod_{j \neq i,n} \frac{1 - x_i x_j}{x_i - x_j}\right).$$

For any $k \neq n$, substituting $x_n = x_k$ (which is valid when manipulating the numerator $P(x_n)$

on its own), we have (noting that $x_n - x_j$ vanishes when $j = k$)

$$\begin{aligned} P(x_k) &= \left(\prod_{j\neq n}(1-x_kx_j)\right) + \sum_{1\leqslant i\leqslant n-1}(x_ix_k-1)\left(\prod_{j\neq i,n}(x_k-x_j)\right)\left(\prod_{j\neq i,n}\frac{1-x_ix_j}{x_i-x_j}\right) \\ &= \left(\prod_{j\neq n}(1-x_kx_j)\right) + \left(x_k^2-1\right)\left(\prod_{j\neq k,n}(x_k-x_j)\right)\left(\prod_{j\neq k,n}\frac{1-x_kx_j}{x_k-x_j}\right) \\ &= \left(\prod_{j\neq n}(1-x_kx_j)\right) + \left(x_k^2-1\right)\left(\prod_{j\neq k,n}(1-x_kx_j)\right) \\ &= 0. \end{aligned}$$

Note that P is a polynomial in x_n of degree $n-1$. For any choice of distinct real numbers $x_1, \ldots, x_{n-1}$, P has those real numbers as its roots, and the denominator has the same degree and the same roots. This shows that G is constant in x_n, for any fixed choice of distinct $x_1, \ldots, x_{n-1}$. Now, G is symmetric in all n variables, so it must be also be constant in each of the other variables. G is therefore a constant that depends only on n. The constant may be identified as in the previous solution.

Comment. There is also a solution in which we recognise the expression for F in the comment after Solution 2 as the final column expansion of a certain matrix obtained by modifying the final column of the Vandermonde matrix. The task is then to show that the matrix can be modified by column operations either to make the final column identically zero (in the case where n even) or to recover the Vandermonde matrix (in the case where n odd). The polynomial $P/(1-x^2)$ is helpful for this task, where P is the parity-dependent polynomial defined in the comment after Solution 1.

A6. A polynomial $P(x, y, z)$ in three variables with real coefficients satisfies the identities

$$P(x, y, z) = P(x, y, xy - z) = P(x, zx - y, z) = P(yz - x, y, z). \qquad (*)$$

Prove that there exists a polynomial $F(t)$ in one variable such that

$$P(x, y, z) = F(x^2 + y^2 + z^2 - xyz).$$

(Russia)

Common remarks. The polynomial $x^2 + y^2 + z^2 - xyz$ satisfies the condition $(*)$, so every polynomial of the form $F(x^2+y^2+z^2-xyz)$ does satisfy $(*)$. We will use without comment the fact that two polynomials have the same coefficients if and only if they are equal as functions.

Solution 1. In the first two steps, we deal with any polynomial $P(x, y, z)$ satisfying $P(x, y, z) = P(x, y, xy - z)$. Call such a polynomial *weakly symmetric*, and call a polynomial satisfying the full conditions in the problem *symmetric*.

Step 1. We start with the description of weakly symmetric polynomials. We claim that they are exactly the polynomials in x, y, and $z(xy - z)$. Clearly, all such polynomials are weakly symmetric. For the converse statement, consider $P_1(x, y, z) := P(x, y, z + \frac{1}{2}xy)$, which satisfies $P_1(x, y, z) = P_1(x, y, -z)$ and is therefore a polynomial in x, y, and z^2. This means that P is a polynomial in x, y, and $(z - \frac{1}{2}xy)^2 = -z(xy - z) + \frac{1}{4}x^2y^2$, and therefore a polynomial in x, y, and $z(xy - z)$.

Step 2. Suppose that P is weakly symmetric. Consider the monomials in $P(x, y, z)$ of highest total degree. Our aim is to show that in each such monomial $\mu x^a y^b z^c$ we have $a, b \geqslant c$. Consider the expansion

$$P(x, y, z) = \sum_{i,j,k} \mu_{ijk} x^i y^j \big(z(xy - z)\big)^k. \qquad (1.1)$$

The maximal total degree of a summand in (1.1) is $m = \max_{i,j,k:\, \mu_{ijk} \neq 0}(i + j + 3k)$. Now, for any i, j, k satisfying $i + j + 3k = m$ the summand $\mu_{i,j,k} x^i y^j \big(z(xy - z)\big)^k$ has leading term of the form $\mu x^{i+k} y^{j+k} z^k$. No other nonzero summand in (1.1) may have a term of this form in its expansion, hence this term does not cancel in the whole sum. Therefore, $\deg P = m$, and the leading component of P is exactly

$$\sum_{i+j+3k=m} \mu_{i,j,k} x^{i+k} y^{j+k} z^k,$$

and each summand in this sum satisfies the condition claimed above.

Step 3. We now prove the problem statement by induction on $m = \deg P$. For $m = 0$ the claim is trivial. Consider now a symmetric polynomial P with $\deg P > 0$. By Step 2, each of its monomials $\mu x^a y^b z^c$ of the highest total degree satisfies $a, b \geqslant c$. Applying other weak symmetries, we obtain $a, c \geqslant b$ and $b, c \geqslant a$; therefore, P has a unique leading monomial of the form $\mu(xyz)^c$. The polynomial $P_0(x, y, z) = P(x, y, z) - \mu\big(xyz - x^2 - y^2 - z^2\big)^c$ has smaller total degree. Since P_0 is symmetric, it is representable as a polynomial function of $xyz - x^2 - y^2 - z^2$. Then P is also of this form, completing the inductive step.

Comment. We could alternatively carry out Step 1 by an induction on $n = \deg_z P$, in a manner similar to Step 3. If $n = 0$, the statement holds. Assume that $n > 0$ and check the leading component of P with respect to z:

$$P(x, y, z) = Q_n(x, y) z^n + R(x, y, z)\,,$$

where $\deg_z R < n$. After the change $z \mapsto xy - z$, the leading component becomes $Q_n(x,y)(-z)^n$; on the other hand, it should remain the same. Hence n is even. Now consider the polynomial

$$P_0(x,y,z) = P(x,y,z) - Q_n(x,y) \cdot \big(z(z-xy)\big)^{n/2}.$$

It is also weakly symmetric, and $\deg_z P_0 < n$. By the inductive hypothesis, it has the form $P_0(x,y,z) = S\big(x,y,z(z-xy)\big)$. Hence the polynomial

$$P(x,y,z) = S\big(x,y,z(xy-z)\big) + Q_n(x,y)\big(z(z-xy)\big)^{n/2}$$

also has this form. This completes the inductive step.

Solution 2. We will rely on the well-known identity

$$\cos^2 u + \cos^2 v + \cos^2 w - 2\cos u \cos v \cos w - 1 = 0 \quad \text{whenever } u+v+w = 0. \tag{2.1}$$

Claim 1. The polynomial $P(x,y,z)$ is constant on the surface

$$\mathfrak{S} = \big\{(2\cos u, 2\cos v, 2\cos w) : u+v+w = 0\big\}.$$

Proof. Notice that for $x = 2\cos u$, $y = 2\cos v$, $z = 2\cos w$, the Vieta jumps $x \mapsto yz - x$, $y \mapsto zx - y$, $z \mapsto xy - z$ in $(*)$ replace (u,v,w) by $(v-w, -v, w)$, $(u, w-u, -w)$ and $(-u, v, u-v)$, respectively. For example, for the first type of jump we have

$$yz - x = 4\cos v \cos w - 2\cos u = 2\cos(v+w) + 2\cos(v-w) - 2\cos u = 2\cos(v-w).$$

Define $G(u,v,w) = P(2\cos u, 2\cos v, 2\cos w)$. For $u+v+w = 0$, the jumps give

$$\begin{aligned} G(u,v,w) &= G(v-w, -v, w) = G(w-v, -v, (v-w)-(-v)) = G(-u-2v, -v, 2v-w) \\ &= G(u+2v, v, w-2v)\,. \end{aligned}$$

By induction,

$$G(u,v,w) = G\big(u+2kv, v, w-2kv\big) \quad (k \in \mathbb{Z}). \tag{2.2}$$

Similarly,

$$G(u,v,w) = G\big(u, v-2\ell u, w+2\ell u\big) \quad (\ell \in \mathbb{Z}). \tag{2.3}$$

And, of course, we have

$$G(u,v,w) = G\big(u+2p\pi, v+2q\pi, w-2(p+q)\pi\big) \quad (p,q \in \mathbb{Z}). \tag{2.4}$$

Take two nonzero real numbers u, v such that u, v and π are linearly independent over $\mathbb{Q}$. By combining (2.2–2.4), we can see that G is constant on a dense subset of the plane $u+v+w = 0$. By continuity, G is constant on the entire plane and therefore P is constant on $\mathfrak{S}$. □

Claim 2. The polynomial $T(x,y,z) = x^2 + y^2 + z^2 - xyz - 4$ divides $P(x,y,z) - P(2,2,2)$.

Proof. By dividing P by T with remainders, there exist some polynomials $R(x,y,z)$, $A(y,z)$ and $B(y,z)$ such that

$$P(x,y,z) - P(2,2,2) = T(x,y,z) \cdot R(x,y,z) + A(y,z)x + B(y,z). \tag{2.5}$$

On the surface $\mathfrak{S}$ the LHS of (2.5) is zero by Claim 1 (since $(2,2,2) \in \mathfrak{S}$) and $T = 0$ by (2.1). Hence, $A(y,z)x + B(y,z)$ vanishes on $\mathfrak{S}$.

Notice that for every $y = 2\cos v$ and $z = 2\cos w$ with $\frac{\pi}{3} < v, w < \frac{2\pi}{3}$, there are two distinct values of x such that $(x,y,z) \in \mathfrak{S}$, namely $x_1 = 2\cos(v+w)$ (which is negative), and $x_2 = 2\cos(v-w)$ (which is positive). This can happen only if $A(y,z) = B(y,z) = 0$. Hence, $A(y,z) = B(y,z) = 0$ for $|y| < 1$, $|z| < 1$. The polynomials A and B vanish on an open set, so A and B are both the zero polynomial. □

The quotient $(P(x,y,z)-P(2,2,2))/T(x,y,z)$ is a polynomial of lower degree than P and it also satisfies $(*)$. The problem statement can now be proven by induction on the degree of P.

Comment. In the proof of (2.2) and (2.3) we used two consecutive Vieta jumps; in fact from $(*)$ we used only $P(x,y,xy-z)=P(x,zx-y,z)=P(yz-x,y,z)$.

Solution 3 (using algebraic geometry, just for interest). Let $Q=x^2+y^2+z^2-xyz$ and let $t\in\mathbb{C}$. Checking where $Q-t$, $\frac{\partial Q}{\partial x}$, $\frac{\partial Q}{\partial y}$ and $\frac{\partial Q}{\partial z}$ vanish simultaneously, we find that the surface $Q=t$ is smooth except for the cases $t=0$, when the only singular point is $(0,0,0)$, and $t=4$, when the four points $(\pm2,\pm2,\pm2)$ that satisfy $xyz=8$ are the only singular points. The singular points are the fixed points of the group Γ of polynomial automorphisms of $\mathbb{C}^3$ generated by the three *Vieta involutions*

$$\iota_1:(x,y,z)\mapsto(x,y,xy-z),\quad \iota_2:(x,y,z)\mapsto(x,xz-y,z),\quad \iota_3:(x,y,z)\mapsto(yz-x,y,z)\,.$$

Γ acts on each surface $\mathcal{V}_t: Q-t=0$. If $Q-t$ were reducible then the surface $Q=t$ would contain a curve of singular points. Therefore $Q-t$ is irreducible in $\mathbb{C}[x,y,z]$. (One can also prove algebraically that $Q-t$ is irreducible, for example by checking that its discriminant as a quadratic polynomial in x is not a square in $\mathbb{C}[y,z]$, and likewise for the other two variables.) In the following solution we will only use the algebraic surface $\mathcal{V}_0$.

Let U be the Γ-orbit of $(3,3,3)$. Consider $\iota_3\circ\iota_2$, which leaves z invariant. For each fixed value of z, $\iota_3\circ\iota_2$ acts linearly on (x,y) by the matrix

$$M_z:=\begin{pmatrix} z^2-1 & -z\\ z & -1\end{pmatrix}.$$

The reverse composition $\iota_2\circ\iota_3$ acts by $M_z^{-1}=M_z^{adj}$. Note $\det M_z=1$ and $\mathrm{tr}M_z=z^2-2$. When z does not lie in the real interval $[-2,2]$, the eigenvalues of M_z do not have absolute value 1, so every orbit of the group generated by M_z on $\mathbb{C}^2\setminus\{(0,0)\}$ is unbounded. For example, fixing $z=3$ we find $(3F_{2k+1},3F_{2k-1},3)\in U$ for every $k\in\mathbb{Z}$, where $(F_n)_{n\in\mathbb{Z}}$ is the Fibonacci sequence with $F_0=0$, $F_1=1$.

Now we may start at any point $(3F_{2k+1},3F_{2k-1},3)$ and iteratively apply $\iota_1\circ\iota_2$ to generate another infinite sequence of distinct points of U, Zariski dense in the hyperbola cut out of $\mathcal{V}_0$ by the plane $x-3F_{2k+1}=0$. (The plane $x=a$ cuts out an irreducible conic when $a\notin\{-2,0,2\}$.) Thus the Zariski closure $\overline{U}$ of U contains infinitely many distinct algebraic curves in $\mathcal{V}_0$. Since $\mathcal{V}_0$ is an irreducible surface this implies that $\overline{U}=\mathcal{V}_0$.

For any polynomial P satisfying $(*)$, we have $P-P(3,3,3)=0$ at each point of U. Since $\overline{U}=\mathcal{V}_0$, $P-P(3,3,3)$ vanishes on $\mathcal{V}_0$. Then Hilbert's Nullstellensatz and the irreducibility of Q imply that $P-P(3,3,3)$ is divisible by Q. Now $(P-P(3,3,3))/Q$ is a polynomial also satisfying $(*)$, so we may complete the proof by an induction on the total degree, as in the other solutions.

Comment. We remark that Solution 2 used a trigonometric parametrisation of a real component of $\mathcal{V}_4$; in contrast $\mathcal{V}_0$ is birationally equivalent to the projective space $\mathbb{P}^2$ under the maps

$$(x,y,z)\to(x:y:z),\quad (a:b:c)\to\left(\frac{a^2+b^2+c^2}{bc},\frac{a^2+b^2+c^2}{ac},\frac{a^2+b^2+c^2}{ab}\right).$$

The set U in Solution 3 is contained in $\mathbb{Z}^3$ so it is nowhere dense in $\mathcal{V}_0$ in the classical topology.

Comment (background to the problem). A triple $(a,b,c)\in\mathbb{Z}^3$ is called a *Markov triple* if $a^2+b^2+c^2=3abc$, and an integer that occurs as a coordinate of some Markov triple is called a *Markov number*. (The spelling *Markoff* is also frequent.) Markov triples arose in A. Markov's work in the 1870s on the reduction theory of indefinite binary quadratic forms. For every Markov triple,

$(3a, 3b, 3c)$ lies on $Q = 0$. It is well known that all nonzero Markov triples can be generated from $(1, 1, 1)$ by sequences of Vieta involutions, which are the substitutions described in equation $(*)$ in the problem statement. There has been recent work by number theorists about the properties of Markov numbers (see for example Jean Bourgain, Alex Gamburd and Peter Sarnak, *Markoff triples and strong approximation*, Comptes Rendus Math. **345**, no. 2, 131–135 (2016), arXiv:1505.06411). Each Markov number occurs in infinitely many triples, but a famous old open problem is the unicity conjecture, which asserts that each Markov number occurs in only one Markov triple (up to permutations and sign changes) as the largest coordinate in absolute value in that triple. It is a standard fact in the modern literature on Markov numbers that the Markov triples are Zariski dense in the Markov surface. Proving this is the main work of Solution 3. Algebraic geometry is definitely off-syllabus for the IMO, and one still has to work a bit to prove the Zariski density. On the other hand the approaches of Solutions 1 and 2 are elementary and only use tools expected to be known by IMO contestants. Therefore we do not think that the existence of a solution using algebraic geometry necessarily makes this problem unsuitable for the IMO.

A7. Let $\mathbb{Z}$ be the set of integers. We consider functions $f\colon \mathbb{Z} \to \mathbb{Z}$ satisfying

$$f\big(f(x+y)+y\big) = f\big(f(x)+y\big)$$

for all integers x and y. For such a function, we say that an integer v is *f-rare* if the set

$$X_v = \{x \in \mathbb{Z}\colon f(x) = v\}$$

is finite and nonempty.

(a) Prove that there exists such a function f for which there is an f-rare integer.

(b) Prove that no such function f can have more than one f-rare integer.

(Netherlands)

Solution 1. a) Let f be the function where $f(0) = 0$ and $f(x)$ is the largest power of 2 dividing $2x$ for $x \neq 0$. The integer 0 is evidently f-rare, so it remains to verify the functional equation.

Since $f(2x) = 2f(x)$ for all x, it suffices to verify the functional equation when at least one of x and y is odd (the case $x = y = 0$ being trivial). If y is odd, then we have

$$f(f(x+y)+y) = 2 = f(f(x)+y)$$

since all the values attained by f are even. If, on the other hand, x is odd and y is even, then we already have

$$f(x+y) = 2 = f(x)$$

from which the functional equation follows immediately.

b) An easy inductive argument (substituting $x + ky$ for x) shows that

$$f(f(x+ky)+y) = f(f(x)+y) \tag{$*$}$$

for all integers x, y and k. If v is an f-rare integer and a is the least element of X_v, then by substituting $y = a - f(x)$ in the above, we see that

$$f(x + k\cdot(a - f(x))) - f(x) + a \in X_v$$

for all integers x and k, so that in particular

$$f(x + k\cdot(a - f(x))) \geqslant f(x)$$

for all integers x and k, by assumption on a. This says that on the (possibly degenerate) arithmetic progression through x with common difference $a - f(x)$, the function f attains its minimal value at x.

Repeating the same argument with a replaced by the greatest element b of X_v shows that

$$f(x + k\cdot(b - f(x)) \leqslant f(x)$$

for all integers x and k. Combined with the above inequality, we therefore have

$$f(x + k\cdot(a - f(x))\cdot(b - f(x))) = f(x) \tag{$\dagger$}$$

for all integers x and k.

Thus if $f(x) \neq a, b$, then the set $X_{f(x)}$ contains a nondegenerate arithmetic progression, so is infinite. So the only possible f-rare integers are a and b.

In particular, the f-rare integer v we started with must be one of a or b, so that $f(v) = f(a) = f(b) = v$. This means that there cannot be any other f-rare integers v', as they would on the one hand have to be either a or b, and on the other would have to satisfy $f(v') = v'$. Thus v is the unique f-rare integer.

Comment 1. If f is a solution to the functional equation, then so too is any conjugate of f by a translation, i.e. any function $x \mapsto f(x+n)-n$ for an integer n. Thus in proving part (b), one is free to consider only functions f for which 0 is f-rare, as in the following solution.

Solution 2, part (b) only. Suppose v is f-rare, and let a and b be the least and greatest elements of X_v, respectively. Substituting $x = v$ and $y = a - v$ into the equation shows that

$$f(v) - v + a \in X_v$$

and in particular $f(v) \geqslant v$. Repeating the same argument with $x = v$ and $y = b - v$ shows that $f(v) \leqslant v$, and hence $f(v) = v$.

Suppose now that v' is a second f-rare integer. We may assume that $v = 0$ (see Comment 1). We've seen that $f(v') = v'$; we claim that in fact $f(kv') = v'$ for all positive integers k. This gives a contradiction unless $v' = v = 0$.

This claim is proved by induction on k. Supposing it to be true for k, we substitute $y = kv'$ and $x = 0$ into the functional equation to yield

$$f((k+1)v') = f(f(0) + kv') = f(kv') = v'$$

using that $f(0) = 0$. This completes the induction, and hence the proof.

Comment 2. There are many functions f satisfying the functional equation for which there is an f-rare integer. For instance, one may generalise the construction in part (a) of Solution 1 by taking a sequence $1 = a_0, a_1, a_2, \ldots$ of positive integers with each a_i a proper divisor of a_{i+1} and choosing arbitrary functions $f_i \colon (\mathbb{Z}/a_i\mathbb{Z}) \setminus \{0\} \to a_i\mathbb{Z} \setminus \{0\}$ from the nonzero residue classes modulo a_i to the nonzero multiples of a_i. One then defines a function $f \colon \mathbb{Z} \to \mathbb{Z}$ by

$$f(x) := \begin{cases} f_{i+1}(x \bmod a_{i+1}), & \text{if } a_i \mid x \text{ but } a_{i+1} \nmid x; \\ 0, & \text{if } x = 0. \end{cases}$$

If one writes $v(x)$ for the largest i such that $a_i \mid x$ (with $v(0) = \infty$), then it is easy to verify the functional equation for f separately in the two cases $v(y) > v(x)$ and $v(x) \geqslant v(y)$. Hence this f satisfies the functional equation and 0 is an f-rare integer.

Comment 3. In fact, if v is an f-rare integer for an f satisfying the functional equation, then its fibre $X_v = \{v\}$ must be a singleton. We may assume without loss of generality that $v = 0$. We've already seen in Solution 1 that 0 is either the greatest or least element of X_0; replacing f with the function $x \mapsto -f(-x)$ if necessary, we may assume that 0 is the least element of X_0. We write b for the largest element of X_0, supposing for contradiction that $b > 0$, and write $N = (2b)!$.

It now follows from $(*)$ that we have

$$f(f(Nb) + b) = f(f(0) + b) = f(b) = 0,$$

from which we see that $f(Nb) + b \in X_0 \subseteq [0, b]$. It follows that $f(Nb) \in [-b, 0)$, since by construction $Nb \notin X_v$. Now it follows that $(f(Nb) - 0) \cdot (f(Nb) - b)$ is a divisor of N, so from $(\dagger)$ we see that $f(Nb) = f(0) = 0$. This yields the desired contradiction.

Combinatorics

C1. The infinite sequence a_0, a_1, a_2, ... of (not necessarily different) integers has the following properties: $0 \leqslant a_i \leqslant i$ for all integers $i \geqslant 0$, and

$$\binom{k}{a_0} + \binom{k}{a_1} + \cdots + \binom{k}{a_k} = 2^k$$

for all integers $k \geqslant 0$.

Prove that all integers $N \geqslant 0$ occur in the sequence (that is, for all $N \geqslant 0$, there exists $i \geqslant 0$ with $a_i = N$).

(Netherlands)

Solution. We prove by induction on k that every initial segment of the sequence, $a_0, a_1, \ldots, a_k$, consists of the following elements (counted with multiplicity, and not necessarily in order), for some $\ell \geqslant 0$ with $2\ell \leqslant k+1$:

$$0, 1, \ldots, \ell-1, \quad 0, 1, \ldots, k-\ell.$$

For $k = 0$ we have $a_0 = 0$, which is of this form. Now suppose that for $k = m$ the elements $a_0, a_1, \ldots, a_m$ are $0, 0, 1, 1, 2, 2, \ldots, \ell-1, \ell-1, \ell, \ell+1, \ldots, m-\ell-1, m-\ell$ for some ℓ with $0 \leqslant 2\ell \leqslant m+1$. It is given that

$$\binom{m+1}{a_0} + \binom{m+1}{a_1} + \cdots + \binom{m+1}{a_m} + \binom{m+1}{a_{m+1}} = 2^{m+1},$$

which becomes

$$\left(\binom{m+1}{0} + \binom{m+1}{1} + \cdots + \binom{m+1}{\ell-1}\right) + \left(\binom{m+1}{0} + \binom{m+1}{1} + \cdots + \binom{m+1}{m-\ell}\right) + \binom{m+1}{a_{m+1}} = 2^{m+1},$$

or, using $\binom{m+1}{i} = \binom{m+1}{m+1-i}$, that

$$\left(\binom{m+1}{0} + \binom{m+1}{1} + \cdots + \binom{m+1}{\ell-1}\right) + \left(\binom{m+1}{m+1} + \binom{m+1}{m} + \cdots + \binom{m+1}{\ell+1}\right) + \binom{m+1}{a_{m+1}} = 2^{m+1}.$$

On the other hand, it is well known that

$$\binom{m+1}{0} + \binom{m+1}{1} + \cdots + \binom{m+1}{m+1} = 2^{m+1},$$

and so, by subtracting, we get

$$\binom{m+1}{a_{m+1}} = \binom{m+1}{\ell}.$$

From this, using the fact that the binomial coefficients $\binom{m+1}{i}$ are increasing for $i \leqslant \frac{m+1}{2}$ and decreasing for $i \geqslant \frac{m+1}{2}$, we conclude that either $a_{m+1} = \ell$ or $a_{m+1} = m+1-\ell$. In either case, $a_0, a_1, \ldots, a_{m+1}$ is again of the claimed form, which concludes the induction.

As a result of this description, any integer $N \geqslant 0$ appears as a term of the sequence a_i for some $0 \leqslant i \leqslant 2N$.

C2. You are given a set of n blocks, each weighing at least 1; their total weight is $2n$. Prove that for every real number r with $0 \leqslant r \leqslant 2n-2$ you can choose a subset of the blocks whose total weight is at least r but at most $r+2$.

(Thailand)

Solution 1. We prove the following more general statement by induction on n.

Claim. Suppose that you have n blocks, each of weight at least 1, and of total weight $s \leqslant 2n$. Then for every r with $-2 \leqslant r \leqslant s$, you can choose some of the blocks whose total weight is at least r but at most $r+2$.

Proof. The base case $n = 1$ is trivial. To prove the inductive step, let x be the largest block weight. Clearly, $x \geqslant s/n$, so $s - x \leqslant \frac{n-1}{n}s \leqslant 2(n-1)$. Hence, if we exclude a block of weight x, we can apply the inductive hypothesis to show the claim holds (for this smaller set) for any $-2 \leqslant r \leqslant s - x$. Adding the excluded block to each of those combinations, we see that the claim also holds when $x - 2 \leqslant r \leqslant s$. So if $x - 2 \leqslant s - x$, then we have covered the whole interval $[-2, s]$. But each block weight is at least 1, so we have $x - 2 \leqslant (s - (n-1)) - 2 = s - (2n - (n-1)) \leqslant s - (s - (n-1)) \leqslant s - x$, as desired. □

Comment. Instead of inducting on sets of blocks with total weight $s \leqslant 2n$, we could instead prove the result only for $s = 2n$. We would then need to modify the inductive step to scale up the block weights before applying the induction hypothesis.

Solution 2. Let $x_1, \ldots, x_n$ be the weights of the blocks in weakly increasing order. Consider the set S of sums of the form $\sum_{j \in J} x_j$ for a subset $J \subseteq \{1, 2, \ldots, n\}$. We want to prove that the *mesh* of S – i.e. the largest distance between two adjacent elements – is at most 2.

For $0 \leqslant k \leqslant n$, let S_k denote the set of sums of the form $\sum_{i \in J} x_i$ for a subset $J \subseteq \{1, 2, \ldots, k\}$. We will show by induction on k that the mesh of S_k is at most 2.

The base case $k = 0$ is trivial (as $S_0 = \{0\}$). For $k > 0$ we have

$$S_k = S_{k-1} \cup (x_k + S_{k-1})$$

(where $(x_k + S_{k-1})$ denotes $\{x_k + s : s \in S_{k-1}\}$), so it suffices to prove that $x_k \leqslant \sum_{j<k} x_j + 2$. But if this were not the case, we would have $x_l > \sum_{j<k} x_j + 2 \geqslant k + 1$ for all $l \geqslant k$, and hence

$$2n = \sum_{j=1}^{n} x_j > (n + 1 - k)(k + 1) + k - 1.$$

This rearranges to $n > k(n+1-k)$, which is false for $1 \leqslant k \leqslant n$, giving the desired contradiction.

C3. Let n be a positive integer. Harry has n coins lined up on his desk, each showing heads or tails. He repeatedly does the following operation: if there are k coins showing heads and $k > 0$, then he flips the kth coin over; otherwise he stops the process. (For example, the process starting with THT would be $THT \to HHT \to HTT \to TTT$, which takes three steps.)

Letting C denote the initial configuration (a sequence of n H's and T's), write $\ell(C)$ for the number of steps needed before all coins show T. Show that this number $\ell(C)$ is finite, and determine its average value over all 2^n possible initial configurations C.

(USA)

Answer: The average is $\frac{1}{4}n(n+1)$.

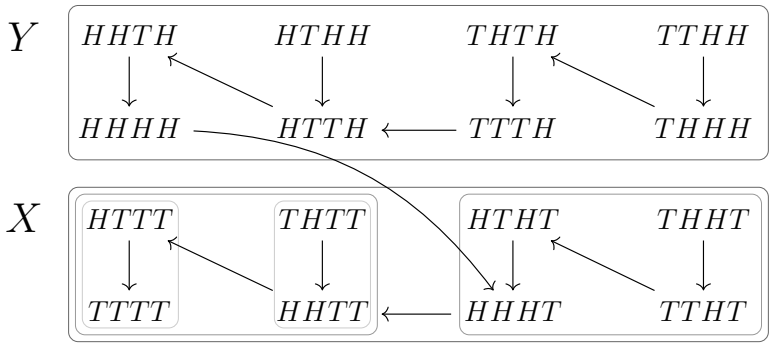
Common remarks. Throughout all these solutions, we let $E(n)$ denote the desired average value.

Solution 1. We represent the problem using a directed graph G_n whose vertices are the length-n strings of H's and T's. The graph features an edge from each string to its successor (except for $TT\cdots TT$, which has no successor). We will also write $\bar{H} = T$ and $\bar{T} = H$.

The graph G_0 consists of a single vertex: the empty string. The main claim is that G_n can be described explicitly in terms of G_{n-1}:

- We take two copies, X and Y, of G_{n-1}.
- In X, we take each string of $n-1$ coins and just append a T to it. In symbols, we replace $s_1 \cdots s_{n-1}$ with $s_1 \cdots s_{n-1}T$.
- In Y, we take each string of $n-1$ coins, flip every coin, reverse the order, and append an H to it. In symbols, we replace $s_1 \cdots s_{n-1}$ with $\bar{s}_{n-1}\bar{s}_{n-2}\cdots\bar{s}_1 H$.
- Finally, we add one new edge from Y to X, namely $HH\cdots HHH \to HH\cdots HHT$.

We depict G_4 below, in a way which indicates this recursive construction:



We prove the claim inductively. Firstly, X is correct as a subgraph of G_n, as the operation on coins is unchanged by an extra T at the end: if $s_1 \cdots s_{n-1}$ is sent to $t_1 \cdots t_{n-1}$, then $s_1 \cdots s_{n-1}T$ is sent to $t_1 \cdots t_{n-1}T$.

Next, Y is also correct as a subgraph of G_n, as if $s_1 \cdots s_{n-1}$ has k occurrences of H, then $\bar{s}_{n-1} \cdots \bar{s}_1 H$ has $(n-1-k)+1 = n-k$ occurrences of H, and thus (provided that $k > 0$), if $s_1 \cdots s_{n-1}$ is sent to $t_1 \cdots t_{n-1}$, then $\bar{s}_{n-1} \cdots \bar{s}_1 H$ is sent to $\bar{t}_{n-1} \cdots \bar{t}_1 H$.

Finally, the one edge from Y to X is correct, as the operation does send $HH\cdots HHH$ to $HH\cdots HHT$.

To finish, note that the sequences in X take an average of $E(n-1)$ steps to terminate, whereas the sequences in Y take an average of $E(n-1)$ steps to reach $HH\cdots H$ and then an additional n steps to terminate. Therefore, we have

$$E(n) = \frac{1}{2}\left(E(n-1) + (E(n-1)+n)\right) = E(n-1) + \frac{n}{2}.$$

We have $E(0) = 0$ from our description of G_0. Thus, by induction, we have $E(n) = \frac{1}{2}(1+\cdots+n) = \frac{1}{4}n(n+1)$, which in particular is finite.

Solution 2. We consider what happens with configurations depending on the coins they start and end with.

- If a configuration starts with H, the last $n-1$ coins follow the given rules, as if they were all the coins, until they are all T, then the first coin is turned over.
- If a configuration ends with T, the last coin will never be turned over, and the first $n-1$ coins follow the given rules, as if they were all the coins.
- If a configuration starts with T and ends with H, the middle $n-2$ coins follow the given rules, as if they were all the coins, until they are all T. After that, there are $2n-1$ more steps: first coins $1, 2, \ldots, n-1$ are turned over in that order, then coins $n, n-1, \ldots, 1$ are turned over in that order.

As this covers all configurations, and the number of steps is clearly finite for 0 or 1 coins, it follows by induction on n that the number of steps is always finite.

We define $E_{AB}(n)$, where A and B are each one of H, T or $*$, to be the average number of steps over configurations of length n restricted to those that start with A, if A is not $*$, and that end with B, if B is not $*$ (so $*$ represents "either H or T"). The above observations tell us that, for $n \geqslant 2$:

- $E_{H*}(n) = E(n-1) + 1$.
- $E_{*T}(n) = E(n-1)$.
- $E_{HT}(n) = E(n-2) + 1$ (by using both the observations for $H*$ and for $*T$).
- $E_{TH}(n) = E(n-2) + 2n - 1$.

Now $E_{H*}(n) = \frac{1}{2}(E_{HH}(n) + E_{HT}(n))$, so $E_{HH}(n) = 2E(n-1) - E(n-2) + 1$. Similarly, $E_{TT}(n) = 2E(n-1) - E(n-2) - 1$. So

$$E(n) = \frac{1}{4}(E_{HT}(n) + E_{HH}(n) + E_{TT}(n) + E_{TH}(n)) = E(n-1) + \frac{n}{2}.$$

We have $E(0) = 0$ and $E(1) = \frac{1}{2}$, so by induction on n we have $E(n) = \frac{1}{4}n(n+1)$.

Solution 3. Let H_i be the number of heads in positions 1 to i inclusive (so H_n is the total number of heads), and let I_i be 1 if the ith coin is a head, 0 otherwise. Consider the function

$$t(i) = I_i + 2(\min\{i, H_n\} - H_i).$$

We claim that $t(i)$ is the total number of times coin i is turned over (which implies that the process terminates). Certainly $t(i) = 0$ when all coins are tails, and $t(i)$ is always a nonnegative integer, so it suffices to show that when the kth coin is turned over (where $k = H_n$), $t(k)$ goes down by 1 and all the other $t(i)$ are unchanged. We show this by splitting into cases:

- If $i < k$, I_i and H_i are unchanged, and $\min\{i, H_n\} = i$ both before and after the coin flip, so $t(i)$ is unchanged.
- If $i > k$, $\min\{i, H_n\} = H_n$ both before and after the coin flip, and both H_n and H_i change by the same amount, so $t(i)$ is unchanged.
- If $i = k$ and the coin is heads, I_i goes down by 1, as do both $\min\{i, H_n\} = H_n$ and H_i; so $t(i)$ goes down by 1.
- If $i = k$ and the coin is tails, I_i goes up by 1, $\min\{i, H_n\} = i$ is unchanged and H_i goes up by 1; so $t(i)$ goes down by 1.

We now need to compute the average value of

$$\sum_{i=1}^{n} t(i) = \sum_{i=1}^{n} I_i + 2\sum_{i=1}^{n} \min\{i, H_n\} - 2\sum_{i=1}^{n} H_i.$$

The average value of the first term is $\frac{1}{2}n$, and that of the third term is $-\frac{1}{2}n(n+1)$. To compute the second term, we sum over choices for the total number of heads, and then over the possible values of i, getting

$$2^{1-n}\sum_{j=0}^{n}\binom{n}{j}\sum_{i=1}^{n}\min\{i,j\} = 2^{1-n}\sum_{j=0}^{n}\binom{n}{j}\left(nj - \binom{j}{2}\right).$$

Now, in terms of trinomial coefficients,

$$\sum_{j=0}^{n} j\binom{n}{j} = \sum_{j=1}^{n}\binom{n}{n-j, j-1, 1} = n\sum_{j=0}^{n-1}\binom{n-1}{j} = 2^{n-1}n$$

and

$$\sum_{j=0}^{n}\binom{j}{2}\binom{n}{j} = \sum_{j=2}^{n}\binom{n}{n-j, j-2, 2} = \binom{n}{2}\sum_{j=0}^{n-2}\binom{n-2}{j} = 2^{n-2}\binom{n}{2}.$$

So the second term above is

$$2^{1-n}\left(2^{n-1}n^2 - 2^{n-2}\binom{n}{2}\right) = n^2 - \frac{n(n-1)}{4},$$

and the required average is

$$E(n) = \frac{1}{2}n + n^2 - \frac{n(n-1)}{4} - \frac{1}{2}n(n+1) = \frac{n(n+1)}{4}.$$

Solution 4. Harry has built a Turing machine to flip the coins for him. The machine is initially positioned at the kth coin, where there are k heads (and the position before the first coin is considered to be the 0th coin). The machine then moves according to the following rules, stopping when it reaches the position before the first coin: if the coin at its current position is H, it flips the coin and moves to the previous coin, while if the coin at its current position is T, it flips the coin and moves to the next position.

Consider the maximal sequences of consecutive moves in the same direction. Suppose the machine has a consecutive moves to the next coin, before a move to the previous coin. After those a moves, the a coins flipped in those moves are all heads, as is the coin the machine is now at, so at least the next $a + 1$ moves will all be moves to the previous coin. Similarly, a consecutive moves to the previous coin are followed by at least $a + 1$ consecutive moves to

the next coin. There cannot be more than n consecutive moves in the same direction, so this proves that the process terminates (with a move from the first coin to the position before the first coin).

Thus we have a (possibly empty) sequence $a_1 < \cdots < a_t \leqslant n$ giving the lengths of maximal sequences of consecutive moves in the same direction, where the final a_t moves must be moves to the previous coin, ending before the first coin. We claim there is a bijection between initial configurations of the coins and such sequences. This gives

$$E(n) = \frac{1}{2}(1 + 2 + \cdots + n) = \frac{n(n+1)}{4}$$

as required, since each i with $1 \leqslant i \leqslant n$ will appear in half of the sequences, and will contribute i to the number of moves when it does.

To see the bijection, consider following the sequence of moves backwards, starting with the machine just before the first coin and all coins showing tails. This certainly determines a unique configuration of coins that could possibly correspond to the given sequence. Furthermore, every coin flipped as part of the a_j consecutive moves is also flipped as part of all subsequent sequences of a_k consecutive moves, for all $k > j$, meaning that, as we follow the moves backwards, each coin is always in the correct state when flipped to result in a move in the required direction. (Alternatively, since there are 2^n possible configurations of coins and 2^n possible such ascending sequences, the fact that the sequence of moves determines at most one configuration of coins, and thus that there is an injection from configurations of coins to such ascending sequences, is sufficient for it to be a bijection, without needing to show that coins are in the right state as we move backwards.)

Solution 5. We explicitly describe what happens with an arbitrary sequence C of n coins. Suppose that C contain k heads at positions $1 \leqslant c_1 < c_2 < \cdots < c_k \leqslant n$.

Let i be the minimal index such that $c_i \geqslant k$. Then the first few steps will consist of turning over the kth, $(k+1)^{\text{th}}$, …, ${c_i}^{\text{th}}$, $(c_i - 1)^{\text{th}}$, $(c_i - 2)^{\text{th}}$, …, k^{th} coins in this order. After that we get a configuration with $k-1$ heads at the same positions as in the initial one, except for c_i. This part of the process takes $2(c_i - k) + 1$ steps.

After that, the process acts similarly; by induction on the number of heads we deduce that the process ends. Moreover, if the c_i disappear in order $c_{i_1}, \ldots, c_{i_k}$, the whole process takes

$$\ell(C) = \sum_{j=1}^{k}\big(2(c_{i_j} - (k+1-j)) + 1\big) = 2\sum_{j=1}^{k} c_j - 2\sum_{j=1}^{k}(k+1-j) + k = 2\sum_{j=1}^{k} c_j - k^2$$

steps.

Now let us find the total value S_k of $\ell(C)$ over all $\binom{n}{k}$ configurations with exactly k heads. To sum up the above expression over those, notice that each number $1 \leqslant i \leqslant n$ appears as c_j exactly $\binom{n-1}{k-1}$ times. Thus

$$\begin{aligned} S_k &= 2\binom{n-1}{k-1}\sum_{i=1}^{n} i - \binom{n}{k}k^2 = 2\frac{(n-1)\cdots(n-k+1)}{(k-1)!}\cdot\frac{n(n+1)}{2} - \frac{n\cdots(n-k+1)}{k!}k^2 \\ &= \frac{n(n-1)\cdots(n-k+1)}{(k-1)!}\big((n+1)-k\big) = n(n-1)\binom{n-2}{k-1} + n\binom{n-1}{k-1}. \end{aligned}$$

Therefore, the total value of $\ell(C)$ over all configurations is

$$\sum_{k=1}^{n} S_k = n(n-1)\sum_{k=1}^{n}\binom{n-2}{k-1} + n\sum_{k=1}^{n}\binom{n-1}{k-1} = n(n-1)2^{n-2} + n2^{n-1} = 2^n\frac{n(n+1)}{4}.$$

Hence the required average is $E(n) = \frac{n(n+1)}{4}$.

C4. On a flat plane in Camelot, King Arthur builds a labyrinth $\mathfrak{L}$ consisting of n walls, each of which is an infinite straight line. No two walls are parallel, and no three walls have a common point. Merlin then paints one side of each wall entirely red and the other side entirely blue.

At the intersection of two walls there are four corners: two diagonally opposite corners where a red side and a blue side meet, one corner where two red sides meet, and one corner where two blue sides meet. At each such intersection, there is a two-way door connecting the two diagonally opposite corners at which sides of different colours meet.

After Merlin paints the walls, Morgana then places some knights in the labyrinth. The knights can walk through doors, but cannot walk through walls.

Let $k(\mathfrak{L})$ be the largest number k such that, no matter how Merlin paints the labyrinth $\mathfrak{L}$, Morgana can always place at least k knights such that no two of them can ever meet. For each n, what are all possible values for $k(\mathfrak{L})$, where $\mathfrak{L}$ is a labyrinth with n walls?

(Canada)

Answer: The only possible value of k is $k = n + 1$, no matter what shape the labyrinth is.

Solution 1. First we show by induction that the n walls divide the plane into $\binom{n+1}{2} + 1$ regions. The claim is true for $n = 0$ as, when there are no walls, the plane forms a single region. When placing the nth wall, it intersects each of the $n-1$ other walls exactly once and hence splits each of n of the regions formed by those other walls into two regions. By the induction hypothesis, this yields $\left(\binom{n}{2} + 1\right) + n = \binom{n+1}{2} + 1$ regions, proving the claim.

Now let G be the graph with vertices given by the $\binom{n+1}{2} + 1$ regions, and with two regions connected by an edge if there is a door between them.

We now show that no matter how Merlin paints the n walls, Morgana can place at least $n + 1$ knights. No matter how the walls are painted, there are exactly $\binom{n}{2}$ intersection points, each of which corresponds to a single edge in G. Consider adding the edges of G sequentially and note that each edge reduces the number of connected components by at most one. Therefore the number of connected components of G is at least $\binom{n+1}{2} + 1 - \binom{n}{2} = n + 1$. If Morgana places a knight in regions corresponding to different connected components of G, then no two knights can ever meet.

Now we give a construction showing that, no matter what shape the labyrinth is, Merlin can colour it such that there are exactly $n + 1$ connected components, allowing Morgana to place at most $n + 1$ knights.

First, we choose a coordinate system on the labyrinth so that none of the walls run due north-south, or due east-west. We then have Merlin paint the west face of each wall red, and the east face of each wall blue. We label the regions according to how many walls the region is on the east side of: the labels are integers between 0 and n.

We claim that, for each i, the regions labelled i are connected by doors. First, we note that for each i with $0 \leqslant i \leqslant n$ there is a unique region labelled i which is unbounded to the north.

Now, consider a knight placed in some region with label i, and ask them to walk north (moving east or west by following the walls on the northern sides of regions, as needed). This knight will never get stuck: each region is convex, and so, if it is bounded to the north, it has a single northernmost vertex with a door northwards to another region with label i.

Eventually it will reach a region which is unbounded to the north, which will be the unique such region with label i. Hence every region with label i is connected to this particular region, and so all regions with label i are connected to each other.

As a result, there are exactly $n + 1$ connected components, and Morgana can place at most $n + 1$ knights.

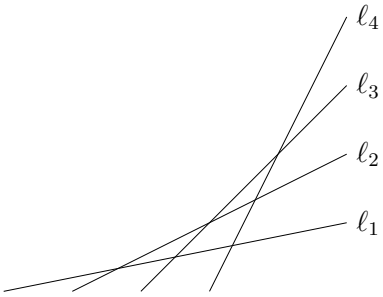
Comment. Variations on this argument exist: some of them capture more information, and some of them capture less information, about the connected components according to this system of numbering.

For example, it can be shown that the unbounded regions are numbered $0, 1, \ldots, n-1, n, n-1, \ldots, 1$ as one cycles around them, that the regions labelled 0 and n are the only regions in their connected components, and that each other connected component forms a single chain running between the two unbounded ones. It is also possible to argue that the regions are acyclic without revealing much about their structure.

Solution 2. We give another description of a strategy for Merlin to paint the walls so that Morgana can place no more than $n + 1$ knights.

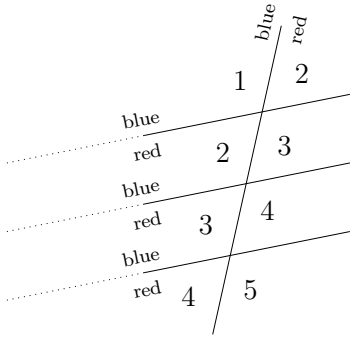
Merlin starts by building a labyrinth of n walls of his own design. He places walls in turn with increasing positive gradients, placing each so far to the right that all intersection points of previously-placed lines lie to the left of it. He paints each in such a way that blue is on the left and red is on the right.

For example, here is a possible sequence of four such lines $\ell_1, \ell_2, \ell_3, \ell_4$:



We say that a region is "on the right" if it has x-coordinate unbounded above (note that if we only have one wall, then both regions are on the right). We claim inductively that, after placing n lines, there are $n + 1$ connected components in the resulting labyrinth, each of which contains exactly one region on the right. This is certainly true after placing 0 lines, as then there is only one region (and hence one connected component) and it is on the right.

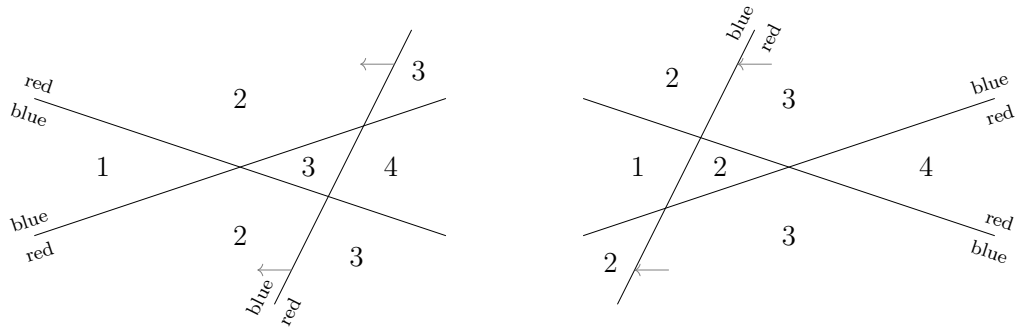
When placing the nth line, it then cuts every one of the $n - 1$ previously placed lines, and since it is to the right of all intersection points, the regions it cuts are exactly the n regions on the right.



The addition of this line leaves all previous connected components with exactly one region on the right, and creates a new connected component containing exactly one region, and that region is also on the right. As a result, by induction, this particular labyrinth will have $n + 1$ connected components.

Having built this labyrinth, Merlin then moves the walls one-by-one (by a sequence of continuous translations and rotations of lines) into the proper position of the given labyrinth, in such a way that no two lines ever become parallel.

The only time the configuration is changed is when one wall is moved through an intersection point of two others:



Note that all moves really do switch between two configurations like this: all sets of three lines have this colour configuration initially, and the rules on rotations mean they are preserved (in particular, we cannot create three lines creating a triangle with three red edges inwards, or three blue edges inwards).

However, as can be seen, such a move preserves the number of connected components, so in the painting this provides for Arthur's actual labyrinth, Morgana can still only place at most $n + 1$ knights.

Comment. While these constructions are superficially distinct, they in fact result in the same colourings for any particular labyrinth. In fact, using the methods of Solution 2, it is possible to show that these are the only colourings that result in exactly $n + 1$ connected components.

C5. On a certain social network, there are 2019 users, some pairs of which are friends, where friendship is a symmetric relation. Initially, there are 1010 people with 1009 friends each and 1009 people with 1010 friends each. However, the friendships are rather unstable, so events of the following kind may happen repeatedly, one at a time:

> Let A, B, and C be people such that A is friends with both B and C, but B and C are not friends; then B and C become friends, but A is no longer friends with them.

Prove that, regardless of the initial friendships, there exists a sequence of such events after which each user is friends with at most one other user.

(Croatia)

Common remarks. The problem has an obvious rephrasing in terms of graph theory. One is given a graph G with 2019 vertices, 1010 of which have degree 1009 and 1009 of which have degree 1010. One is allowed to perform operations on G of the following kind:

> Suppose that vertex A is adjacent to two distinct vertices B and C which are not adjacent to each other. Then one may remove the edges AB and AC from G and add the edge BC into G.

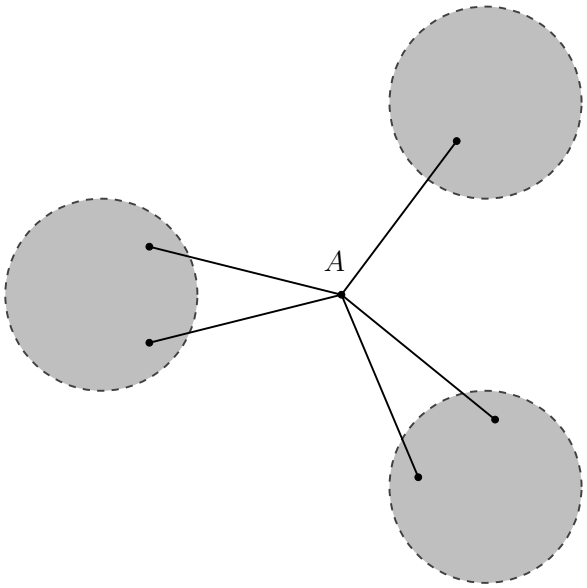
Call such an operation a *refriending*. One wants to prove that, via a sequence of such refriendings, one can reach a graph which is a disjoint union of single edges and vertices.

All of the solutions presented below will use this reformulation.

Solution 1. Note that the given graph is connected, since the total degree of any two vertices is at least 2018 and hence they are either adjacent or have at least one neighbour in common. Hence the given graph satisfies the following condition:

> Every connected component of G with at least three vertices is not complete and has a vertex of odd degree. (1)

We will show that if a graph G satisfies condition (1) and has a vertex of degree at least 2, then there is a refriending on G that preserves condition (1). Since refriendings decrease the total number of edges of G, by using a sequence of such refriendings, we must reach a graph G with maximal degree at most 1, so we are done.



Pick a vertex A of degree at least 2 in a connected component G' of G. Since no component of G with at least three vertices is complete we may assume that not all of the neighbours of A are adjacent to one another. (For example, pick a maximal complete subgraph K of G'. Some vertex A of K has a neighbour outside K, and this neighbour is not adjacent to every vertex of K by maximality.) Removing A from G splits G' into smaller connected components $G_1, \ldots, G_k$ (possibly with $k = 1$), to each of which A is connected by at least one edge. We divide into several cases.

Case 1: $k \geqslant 2$ and A is connected to some G_i by at least two edges.

Choose a vertex B of G_i adjacent to A, and a vertex C in another component G_j adjacent to A. The vertices B and C are not adjacent, and hence removing edges AB and AC and adding in edge BC does not disconnect G'. It is easy to see that this preserves the condition, since the refriending does not change the parity of the degrees of vertices.

Case 2: $k \geqslant 2$ and A is connected to each G_i by exactly one edge.

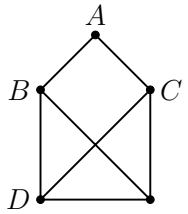
Consider the induced subgraph on any G_i and the vertex A. The vertex A has degree 1 in this subgraph; since the number of odd-degree vertices of a graph is always even, we see that G_i has a vertex of odd degree (in G). Thus if we let B and C be any distinct neighbours of A, then removing edges AB and AC and adding in edge BC preserves the above condition: the refriending creates two new components, and if either of these components has at least three vertices, then it cannot be complete and must contain a vertex of odd degree (since each G_i does).

Case 3: $k = 1$ and A is connected to G_1 by at least three edges.

By assumption, A has two neighbours B and C which are not adjacent to one another. Removing edges AB and AC and adding in edge BC does not disconnect G'. We are then done as in Case 1.

Case 4: $k = 1$ and A is connected to G_1 by exactly two edges.

Let B and C be the two neighbours of A, which are not adjacent. Removing edges AB and AC and adding in edge BC results in two new components: one consisting of a single vertex; and the other containing a vertex of odd degree. We are done unless this second component would be a complete graph on at least 3 vertices. But in this case, G_1 would be a complete graph minus the single edge BC, and hence has at least 4 vertices since G' is not a 4-cycle. If we let D be a third vertex of G_1, then removing edges BA and BD and adding in edge AD does not disconnect G'. We are then done as in Case 1.

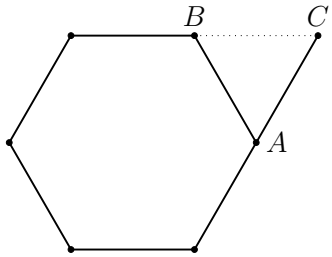


Comment. In fact, condition 1 above *precisely* characterises those graphs which can be reduced to a graph of maximal degree $\leqslant 1$ by a sequence of refriendings.

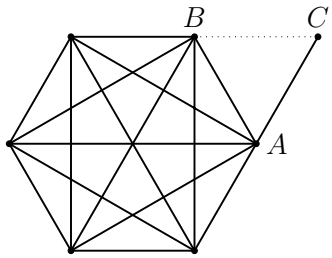
Solution 2. As in the previous solution, note that a refriending preserves the property that a graph has a vertex of odd degree and (trivially) the property that it is not complete; note also that our initial graph is connected. We describe an algorithm to reduce our initial graph to a graph of maximal degree at most 1, proceeding in two steps.

Step 1: There exists a sequence of refriendings reducing the graph to a tree.

Proof. Since the number of edges decreases with each refriending, it suffices to prove the following: as long as the graph contains a cycle, there exists a refriending such that the resulting graph is still connected. We will show that the graph in fact contains a cycle Z and vertices A, B, C such that A and B are adjacent in the cycle Z, C is not in Z, and is adjacent to A but not B. Removing edges AB and AC and adding in edge BC keeps the graph connected, so we are done.



To find this cycle Z and vertices A, B, C, we pursue one of two strategies. If the graph contains a triangle, we consider a largest complete subgraph K, which thus contains at least three vertices. Since the graph itself is not complete, there is a vertex C not in K connected to a vertex A of K. By maximality of K, there is a vertex B of K not connected to C, and hence we are done by choosing a cycle Z in K through the edge AB.



If the graph is triangle-free, we consider instead a smallest cycle Z. This cycle cannot be Hamiltonian (i.e. it cannot pass through every vertex of the graph), since otherwise by minimality the graph would then have no other edges, and hence would have even degree at every vertex. We may thus choose a vertex C not in Z adjacent to a vertex A of Z. Since the graph is triangle-free, it is not adjacent to any neighbour B of A in Z, and we are done. □

Step 2: Any tree may be reduced to a disjoint union of single edges and vertices by a sequence of refriendings.

Proof. The refriending preserves the property of being acyclic. Hence, after applying a sequence of refriendings, we arrive at an acyclic graph in which it is impossible to perform any further refriendings. The maximal degree of any such graph is 1: if it had a vertex A with two neighbours B, C, then B and C would necessarily be nonadjacent since the graph is cycle-free, and so a refriending would be possible. Thus we reach a graph with maximal degree at most 1 as desired. □

C6. Let $n > 1$ be an integer. Suppose we are given $2n$ points in a plane such that no three of them are collinear. The points are to be labelled A_1, A_2, …, A_{2n} in some order. We then consider the $2n$ angles $\angle A_1A_2A_3$, $\angle A_2A_3A_4$, …, $\angle A_{2n-2}A_{2n-1}A_{2n}$, $\angle A_{2n-1}A_{2n}A_1$, $\angle A_{2n}A_1A_2$. We measure each angle in the way that gives the smallest positive value (i.e. between 0° and 180°). Prove that there exists an ordering of the given points such that the resulting $2n$ angles can be separated into two groups with the sum of one group of angles equal to the sum of the other group.

(USA)

Comment. The first three solutions all use the same construction involving a line separating the points into groups of n points each, but give different proofs that this construction works. Although Solution 1 is very short, the Problem Selection Committee does not believe any of the solutions is easy to find and thus rates this as a problem of medium difficulty.

Solution 1. Let ℓ be a line separating the points into two groups (L and R) with n points in each. Label the points A_1, A_2, …, A_{2n} so that $L = \{A_1, A_3, \ldots, A_{2n-1}\}$. We claim that this labelling works.

Take a line $s = A_{2n}A_1$.

(a) Rotate s around A_1 until it passes through A_2; the rotation is performed in a direction such that s is never parallel to ℓ.

(b) Then rotate the new s around A_2 until it passes through A_3 in a similar manner.

(c) Perform $2n-2$ more such steps, after which s returns to its initial position.

The total (directed) rotation angle Θ of s is clearly a multiple of 180°. On the other hand, s was never parallel to ℓ, which is possible only if $\Theta = 0$. Now it remains to partition all the $2n$ angles into those where s is rotated anticlockwise, and the others.

Solution 2. When tracing a cyclic path through the A_i in order, with straight line segments between consecutive points, let θ_i be the exterior angle at A_i, with a sign convention that it is positive if the path turns left and negative if the path turns right. Then $\sum_{i=1}^{2n} \theta_i = 360k^\circ$ for some integer k. Let $\phi_i = \angle A_{i-1}A_iA_{i+1}$ (indices mod $2n$), defined as in the problem; thus $\phi_i = 180^\circ - |\theta_i|$.

Let L be the set of i for which the path turns left at A_i and let R be the set for which it turns right. Then $S = \sum_{i\in L} \phi_i - \sum_{i\in R} \phi_i = (180(|L| - |R|) - 360k)^\circ$, which is a multiple of 360° since the number of points is even. We will show that the points can be labelled such that $S = 0$, in which case L and R satisfy the required condition of the problem.

Note that the value of S is defined for a slightly larger class of configurations: it is OK for two points to coincide, as long as they are not consecutive, and OK for three points to be collinear, as long as A_i, A_{i+1} and A_{i+2} do not appear on a line *in that order*. In what follows it will be convenient, although not strictly necessary, to consider such configurations.

Consider how S changes if a single one of the A_i is moved along some straight-line path (not passing through any A_j and not lying on any line A_jA_k, but possibly crossing such lines). Because S is a multiple of 360°, and the angles change continuously, S can only change when a point moves between R and L. Furthermore, if $\phi_j = 0$ when A_j moves between R and L, S is unchanged; it only changes if $\phi_j = 180^\circ$ when A_j moves between those sets.

For any starting choice of points, we will now construct a new configuration, with labels such that $S = 0$, that can be perturbed into the original one without any ϕ_i passing through 180°, so that $S = 0$ for the original configuration with those labels as well.

Take some line such that there are n points on each side of that line. The new configuration has n copies of a single point on each side of the line, and a path that alternates between

sides of the line; all angles are 0, so this configuration has $S = 0$. Perturbing the points into their original positions, while keeping each point on its side of the line, no angle ϕ_i can pass through 180°, because no straight line can go from one side of the line to the other and back. So the perturbation process leaves $S = 0$.

Comment. More complicated variants of this solution are also possible; for example, a path defined using four quadrants of the plane rather than just two half-planes.

Solution 3. First, let ℓ be a line in the plane such that there are n points on one side and the other n points on the other side. For convenience, assume ℓ is horizontal (otherwise, we can rotate the plane). Then we can use the terms "above", "below", "left" and "right" in the usual way. We denote the n points above the line in an arbitrary order as $P_1, P_2, \ldots, P_n$, and the n points below the line as $Q_1, Q_2, \ldots, Q_n$.

If we connect P_i and Q_j with a line segment, the line segment will intersect with the line ℓ. Denote the intersection as I_{ij}. If P_i is connected to Q_j and Q_k, where $j < k$, then I_{ij} and I_{ik} are two different points, because P_i, Q_j and Q_k are not collinear.

Now we define a "sign" for each angle $\angle Q_jP_iQ_k$. Assume $j < k$. We specify that the sign is positive for the following two cases:

- if i is odd and I_{ij} is to the left of I_{ik},
- if i is even and I_{ij} is to the right of I_{ik}.

Otherwise the sign of the angle is negative. If $j > k$, then the sign of $\angle Q_jP_iQ_k$ is taken to be the same as for $\angle Q_kP_iQ_j$.

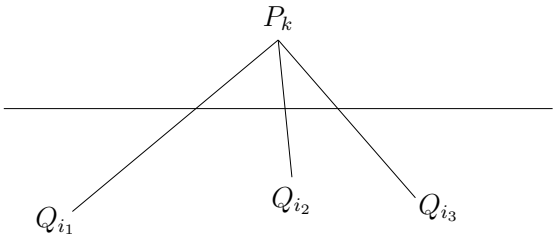
Similarly, we can define the sign of $\angle P_jQ_iP_k$ with $j < k$ (or equivalently $\angle P_kQ_iP_j$). For example, it is positive when i is odd and I_{ji} is to the left of I_{ki}.

Henceforth, whenever we use the notation $\angle Q_jP_iQ_k$ or $\angle P_jQ_iP_k$ for a numerical quantity, it is understood to denote either the (geometric) measure of the angle or the negative of this measure, depending on the sign as specified above.

We now have the following important fact for signed angle measures:

$$\angle Q_{i_1}P_kQ_{i_3} = \angle Q_{i_1}P_kQ_{i_2} + \angle Q_{i_2}P_kQ_{i_3} \tag{1}$$

for all points P_k, Q_{i_1}, Q_{i_2} and Q_{i_3} with $i_1 < i_2 < i_3$. The following figure shows a "natural" arrangement of the points. Equation (1) still holds for any other arrangement, as can be easily verified.



Similarly, we have

$$\angle P_{i_1}Q_kP_{i_3} = \angle P_{i_1}Q_kP_{i_2} + \angle P_{i_2}Q_kP_{i_3}, \tag{2}$$

for all points Q_k, P_{i_1}, P_{i_2} and P_{i_3}, with $i_1 < i_2 < i_3$.

We are now ready to specify the desired ordering $A_1, \ldots, A_{2n}$ of the points:

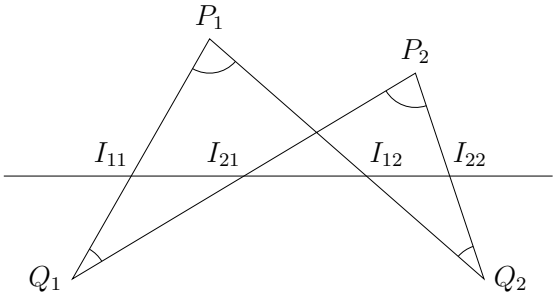
- if $i \leqslant n$ is odd, put $A_i = P_i$ and $A_{2n+1-i} = Q_i$;
- if $i \leqslant n$ is even, put $A_i = Q_i$ and $A_{2n+1-i} = P_i$.

For example, for $n = 3$ this ordering is $P_1, Q_2, P_3, Q_3, P_2, Q_1$. This sequence alternates between P's and Q's, so the above conventions specify a sign for each of the angles $A_{i-1}A_iA_{i+1}$. We claim that the sum of these $2n$ signed angles equals 0. If we can show this, it would complete the proof.

We prove the claim by induction. For brevity, we use the notation $\angle P_i$ to denote whichever of the $2n$ angles has its vertex at P_i, and $\angle Q_i$ similarly.

First let $n = 2$. If the four points can be arranged to form a convex quadrilateral, then the four line segments P_1Q_1, P_1Q_2, P_2Q_1 and P_2Q_2 constitute a self-intersecting quadrilateral. We use several figures to illustrate the possible cases.

The following figure is one possible arrangement of the points.



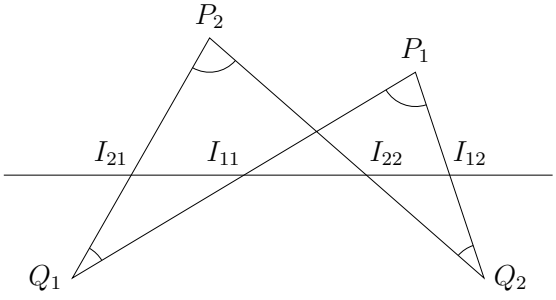
Then $\angle P_1$ and $\angle Q_1$ are positive, $\angle P_2$ and $\angle Q_2$ are negative, and we have

$$|\angle P_1| + |\angle Q_1| = |\angle P_2| + |\angle Q_2|.$$

With signed measures, we have

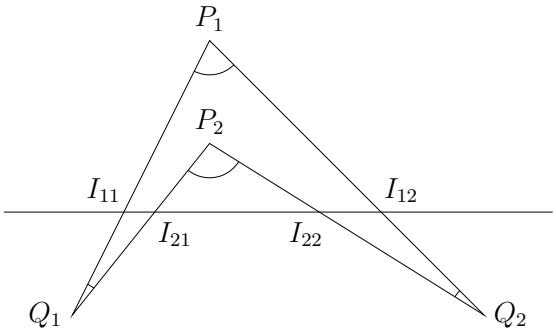
$$\angle P_1 + \angle Q_1 + \angle P_2 + \angle Q_2 = 0. \tag{3}$$

If we switch the labels of P_1 and P_2, we have the following picture:



Switching labels P_1 and P_2 has the effect of flipping the sign of all four angles (as well as swapping the magnitudes on the relabelled points); that is, the new values of $(\angle P_1, \angle P_2, \angle Q_1, \angle Q_2)$ equal the old values of $(-\angle P_2, -\angle P_1, -\angle Q_1, -\angle Q_2)$. Consequently, equation (3) still holds. Similarly, when switching the labels of Q_1 and Q_2, or both the P's and the Q's, equation (3) still holds.

The remaining subcase of $n = 2$ is that one point lies inside the triangle formed by the other three. We have the following picture.



We have

$$|\angle P_1| + |\angle Q_1| + |\angle Q_2| = |\angle P_2|.$$

and equation (3) holds.

Again, switching the labels for P's or the Q's will not affect the validity of equation (3). Also, if the point lying inside the triangle of the other three is one of the Q's rather than the P's, the result still holds, since our sign convention is preserved when we relabel Q's as P's and vice-versa and reflect across ℓ.

We have completed the proof of the claim for $n = 2$.

Assume the claim holds for $n = k$, and we wish to prove it for $n = k + 1$. Suppose we are given our $2(k + 1)$ points. First ignore P_{k+1} and Q_{k+1}, and form $2k$ angles from P_1, ..., P_k, Q_1, ..., Q_k as in the $n = k$ case. By the induction hypothesis we have

$$\sum_{i=1}^{k} (\angle P_i + \angle Q_i) = 0.$$

When we add in the two points P_{k+1} and Q_{k+1}, this changes our angles as follows:

- the angle at P_k changes from $\angle Q_{k-1}P_kQ_k$ to $\angle Q_{k-1}P_kQ_{k+1}$;
- the angle at Q_k changes from $\angle P_{k-1}Q_kP_k$ to $\angle P_{k-1}Q_kP_{k+1}$;
- two new angles $\angle Q_kP_{k+1}Q_{k+1}$ and $\angle P_kQ_{k+1}P_{k+1}$ are added.

We need to prove the changes have no impact on the total sum. In other words, we need to prove

$$(\angle Q_{k-1}P_kQ_{k+1} - \angle Q_{k-1}P_kQ_k) + (\angle P_{k-1}Q_kP_{k+1} - \angle P_{k-1}Q_kP_k) + (\angle P_{k+1} + \angle Q_{k+1}) = 0. \quad (4)$$

In fact, from equations (1) and (2), we have

$$\angle Q_{k-1}P_kQ_{k+1} - \angle Q_{k-1}P_kQ_k = \angle Q_kP_kQ_{k+1},$$

and

$$\angle P_{k-1}Q_kP_{k+1} - \angle P_{k-1}Q_kP_k = \angle P_kQ_kP_{k+1}.$$

Therefore, the left hand side of equation (4) becomes $\angle Q_kP_kQ_{k+1}+\angle P_kQ_kP_{k+1}+\angle Q_kP_{k+1}Q_{k+1}+\angle P_kQ_{k+1}P_{k+1}$, which equals 0, simply by applying the $n = 2$ case of the claim. This completes the induction.

Solution 4. We shall think instead of the problem as asking us to assign a weight ± 1 to each angle, such that the weighted sum of all the angles is zero.

Given an ordering $A_1, \ldots, A_{2n}$ of the points, we shall assign weights according to the following recipe: walk in order from point to point, and assign the left turns $+1$ and the right turns -1. This is the same weighting as in Solution 3, and as in that solution, the weighted sum is a multiple of 360°.

We now aim to show the following:

Lemma. Transposing any two consecutive points in the ordering changes the weighted sum by $\pm 360^\circ$ or 0.

Knowing that, we can conclude quickly: if the ordering $A_1, \ldots, A_{2n}$ has weighted angle sum $360k^\circ$, then the ordering $A_{2n}, \ldots, A_1$ has weighted angle sum $-360k^\circ$ (since the angles are the same, but left turns and right turns are exchanged). We can reverse the ordering of $A_1, \ldots, A_{2n}$ by a sequence of transpositions of consecutive points, and in doing so the weighted angle sum must become zero somewhere along the way.

We now prove that lemma:

Proof. Transposing two points amounts to taking a section $A_kA_{k+1}A_{k+2}A_{k+3}$ as depicted, reversing the central line segment $A_{k+1}A_{k+2}$, and replacing its two neighbours with the dotted lines.

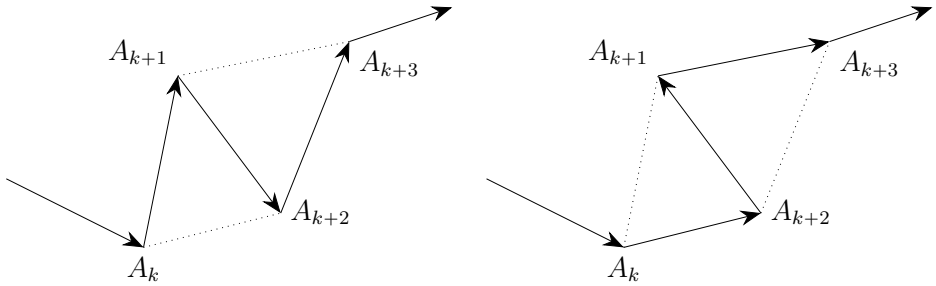


Figure 1: Transposing two consecutive vertices: before (left) and afterwards (right)

In each triangle, we alter the sum by $\pm 180^\circ$. Indeed, using (anticlockwise) directed angles modulo 360°, we either add or subtract all three angles of each triangle.

Hence both triangles together alter the sum by $\pm 180 \pm 180^\circ$, which is $\pm 360^\circ$ or 0. □

C7. There are 60 empty boxes $B_1, \ldots, B_{60}$ in a row on a table and an unlimited supply of pebbles. Given a positive integer n, Alice and Bob play the following game.

In the first round, Alice takes n pebbles and distributes them into the 60 boxes as she wishes. Each subsequent round consists of two steps:

(a) Bob chooses an integer k with $1 \leqslant k \leqslant 59$ and splits the boxes into the two groups $B_1, \ldots, B_k$ and $B_{k+1}, \ldots, B_{60}$.

(b) Alice picks one of these two groups, adds one pebble to each box in that group, and removes one pebble from each box in the other group.

Bob wins if, at the end of any round, some box contains no pebbles. Find the smallest n such that Alice can prevent Bob from winning.

(Czech Republic)

Answer: $n = 960$. In general, if there are $N > 1$ boxes, the answer is $n = \left\lfloor \frac{N}{2} + 1 \right\rfloor \left\lceil \frac{N}{2} + 1 \right\rceil - 1$.

Common remarks. We present solutions for the general case of $N > 1$ boxes, and write $M = \left\lfloor \frac{N}{2} + 1 \right\rfloor \left\lceil \frac{N}{2} + 1 \right\rceil - 1$ for the claimed answer. For $1 \leqslant k < N$, say that Bob makes a *k-move* if he splits the boxes into a left group $\{B_1, \ldots, B_k\}$ and a right group $\{B_{k+1}, \ldots, B_N\}$. Say that one configuration *dominates* another if it has at least as many pebbles in each box, and say that it *strictly dominates* the other configuration if it also has more pebbles in at least one box. (Thus, if Bob wins in some configuration, he also wins in every configuration that it dominates.)

It is often convenient to consider 'V-shaped' configurations; for $1 \leqslant i \leqslant N$, let V_i be the configuration where B_j contains $1 + |j - i|$ pebbles (i.e. where the ith box has a single pebble and the numbers increase by one in both directions, so the first box has i pebbles and the last box has $N + 1 - i$ pebbles). Note that V_i contains $\frac{1}{2}i(i+1) + \frac{1}{2}(N+1-i)(N+2-i) - 1$ pebbles. If $i = \left\lceil \frac{N}{2} \right\rceil$, this number equals M.

Solutions split naturally into a strategy for Alice (starting with M pebbles and showing she can prevent Bob from winning) and a strategy for Bob (showing he can win for any starting configuration with at most $M - 1$ pebbles). The following observation is also useful to simplify the analysis of strategies for Bob.

Observation A. Consider two consecutive rounds. Suppose that in the first round Bob made a k-move and Alice picked the left group, and then in the second round Bob makes an ℓ-move, with $\ell > k$. We may then assume, without loss of generality, that Alice again picks the left group.

Proof. Suppose Alice picks the right group in the second round. Then the combined effect of the two rounds is that each of the boxes $B_{k+1}, \ldots, B_\ell$ lost two pebbles (and the other boxes are unchanged). Hence this configuration is strictly dominated by that before the first round, and it suffices to consider only Alice's other response. □

Solution 1 (Alice). Alice initially distributes pebbles according to $V_{\left\lceil \frac{N}{2} \right\rceil}$. Suppose the current configuration of pebbles dominates V_i. If Bob makes a k-move with $k \geqslant i$ then Alice picks the left group, which results in a configuration that dominates V_{i+1}. Likewise, if Bob makes a k-move with $k < i$ then Alice picks the right group, which results in a configuration that dominates V_{i-1}. Since none of $V_1, \ldots, V_N$ contains an empty box, Alice can prevent Bob from ever winning.

Solution 1 (Bob). The key idea in this solution is the following claim.

Claim. If there exist a positive integer k such that there are at least $2k$ boxes that have at most k pebbles each then Bob can force a win.

Proof. We ignore the other boxes. First, Bob makes a k-move (splits the $2k$ boxes into two groups of k boxes each). Without loss of generality, Alice picks the left group. Then Bob makes a $(k+1)$-move, ..., a $(2k-1)$-move. By Observation A, we may suppose Alice always picks the left group. After Bob's $(2k-1)$-move, the rightmost box becomes empty and Bob wins. □

Now, we claim that if $n < M$ then either there already exists an empty box, or there exist a positive integer k and $2k$ boxes with at most k pebbles each (and thus Bob can force a win). Otherwise, assume each box contains at least 1 pebble, and for each $1 \leqslant k \leqslant \left\lfloor \frac{N}{2} \right\rfloor$, at least $N - (2k-1) = N + 1 - 2k$ boxes contain at least $k+1$ pebbles. Summing, there are at least as many pebbles in total as in $V_{\left\lceil \frac{N}{2} \right\rceil}$; that is, at least M pebbles, as desired.

Solution 2 (Alice). Let $K = \left\lfloor \frac{N}{2} + 1 \right\rfloor$. Alice starts with the boxes in the configuration V_K. For each of Bob's $N-1$ possible choices, consider the subset of rounds in which he makes that choice. In that subset of rounds, Alice alternates between picking the left group and picking the right group; the first time Bob makes that choice, Alice picks the group containing the Kth box. Thus, at any time during the game, the number of pebbles in each box depends only on which choices Bob has made an odd number of times. This means that the number of pebbles in a box could decrease by at most the number of choices for which Alice would have started by removing a pebble from the group containing that box. These numbers are, for each box,

$$\left\lfloor \tfrac{N}{2} \right\rfloor, \left\lfloor \tfrac{N}{2} - 1 \right\rfloor, \ldots, 1, 0, 1, \ldots, \left\lceil \tfrac{N}{2} - 1 \right\rceil.$$

These are pointwise less than the numbers of pebbles the boxes started with, meaning that no box ever becomes empty with this strategy.

Solution 2 (Bob). Let $K = \left\lfloor \frac{N}{2} + 1 \right\rfloor$. For Bob's strategy, we consider a configuration X with at most $M-1$ pebbles, and we make use of Observation A. Consider two configurations with M pebbles: V_K and V_{N+1-K} (if n is odd, they are the same configuration; if n is even, one is the reverse of the other). The configuration X has fewer pebbles than V_K in at least one box, and fewer pebbles than V_{N+1-K} in at least one box.

Suppose first that, with respect to one of those configurations (without loss of generality V_K), X has fewer pebbles in one of the boxes in the half where they have 1, 2, ..., $\left\lceil \frac{N}{2} \right\rceil$ pebbles (the right half in V_K if N is even; if N is odd, we can take it to be the right half, without loss of generality, as the configuration is symmetric). Note that the number cannot be fewer in the box with 1 pebble in V_K, because then it would have 0 pebbles. Bob then does a K-move. If Alice picks the right group, the total number of pebbles goes down and we restart Bob's strategy with a smaller number of pebbles. If Alice picks the left group, Bob follows with a $(K+1)$-move, a $(K+2)$-move, and so on; by Observation A we may assume Alice always picks the left group. But whichever box in the right half had fewer pebbles in X than in V_K ends up with 0 pebbles at some point in this sequence of moves.

Otherwise, N is even, and for both of those configurations, there are fewer pebbles in X only on the 2, 3, ..., $\frac{N}{2} + 1$ side. That is, the numbers of pebbles in X are at least

$$\tfrac{N}{2}, \tfrac{N}{2} - 1, \ldots, 1, 1, \ldots, \tfrac{N}{2} \qquad (C)$$

with equality occurring at least once on each side. Bob does an $\frac{N}{2}$-move. Whichever group Alice chooses, the total number of pebbles is unchanged, and the side from which pebbles are removed now has a box with fewer pebbles than in (C), so the previous case of Bob's strategy can now be applied.

Solution 3 (Bob). For any configuration C, define $L(C)$ to be the greatest integer such that, for all $0 \leqslant i \leqslant N-1$, the box B_{i+1} contains at least $L(C)-i$ pebbles. Similarly, define $R(C)$ to be greatest integer such that, for all $0 \leqslant i \leqslant N-1$, the box B_{N-i} contains at least $R(C)-i$ pebbles. (Thus, C dominates the 'left half' of $V_{L(C)}$ and the 'right half' of $V_{N+1-R(C)}$.) Then C dominates a 'V-shaped' configuration if and only if $L(C)+R(C) \geqslant N+1$. Note that if C dominates a V-shaped configuration, it has at least M pebbles.

Now suppose that there are fewer than M pebbles, so we have $L(C)+R(C) \leqslant N$. Then Bob makes an $L(C)$-move (or more generally any move with at least $L(C)$ boxes on the left and $R(C)$ boxes on the right). Let C' be the new configuration, and suppose that no box becomes empty (otherwise Bob has won). If Alice picks the left group, we have $L(C') = L(C)+1$ and $R(C') = R(C)-1$. Otherwise, we have $L(C') = L(C)-1$ and $R(C') = R(C)+1$. In either case, we have $L(C')+R(C') \leqslant N$.

Bob then repeats this strategy, until one of the boxes becomes empty. Since the condition in Observation A holds, we may assume that Alice picks a group on the same side each time. Then one of L and R is strictly decreasing; without loss of generality assume that L strictly decreases. At some point we reach $L = 1$. If B_2 is still nonempty, then B_1 must contain a single pebble. Bob makes a 1-move, and by Observation A, Alice must (eventually) pick the right group, making this box empty.

C8. Alice has a map of Wonderland, a country consisting of $n \geqslant 2$ towns. For every pair of towns, there is a narrow road going from one town to the other. One day, all the roads are declared to be "one way" only. Alice has no information on the direction of the roads, but the King of Hearts has offered to help her. She is allowed to ask him a number of questions. For each question in turn, Alice chooses a pair of towns and the King of Hearts tells her the direction of the road connecting those two towns.

Alice wants to know whether there is at least one town in Wonderland with at most one outgoing road. Prove that she can always find out by asking at most $4n$ questions.

Comment. This problem could be posed with an explicit statement about points being awarded for weaker bounds cn for some $c > 4$, in the style of IMO 2014 Problem 6.

(Thailand)

Solution. We will show Alice needs to ask at most $4n - 7$ questions. Her strategy has the following phases. In what follows, S is the set of towns that Alice, so far, does not know to have more than one outgoing road (so initially $|S| = n$).

Phase 1. Alice chooses any two towns, say A and B. Without loss of generality, suppose that the King of Hearts' answer is that the road goes from A to B.

At the end of this phase, Alice has asked 1 question.

Phase 2. During this phase there is a single (variable) town T that is known to have at least one incoming road but not yet known to have any outgoing roads. Initially, T is B. Alice does the following $n - 2$ times: she picks a town X she has not asked about before, and asks the direction of the road between T and X. If it is from X to T, T is unchanged; if it is from T to X, X becomes the new choice of town T, as the previous T is now known to have an outgoing road.

At the end of this phase, Alice has asked a total of $n - 1$ questions. The final town T is not yet known to have any outgoing roads, while every other town has exactly one outgoing road known. The undirected graph of roads whose directions are known is a tree.

Phase 3. During this phase, Alice asks about the directions of all roads between T and another town she has not previously asked about, stopping if she finds two outgoing roads from T. This phase involves at most $n - 2$ questions. If she does not find two outgoing roads from T, she has answered her original question with at most $2n - 3 \leqslant 4n - 7$ questions, so in what follows we suppose that she does find two outgoing roads, asking a total of k questions in this phase, where $2 \leqslant k \leqslant n - 2$ (and thus $n \geqslant 4$ for what follows).

For every question where the road goes towards T, the town at the other end is removed from S (as it already had one outgoing road known), while the last question resulted in T being removed from S. So at the end of this phase, $|S| = n - k + 1$, while a total of $n + k - 1$ questions have been asked. Furthermore, the undirected graph of roads within S whose directions are known contains no cycles (as T is no longer a member of S, all questions asked in this phase involved T and the graph was a tree before this phase started). Every town in S has exactly one outgoing road known (not necessarily to another town in S).

Phase 4. During this phase, Alice repeatedly picks any pair of towns in S for which she does not know the direction of the road between them. Because every town in S has exactly one outgoing road known, this always results in the removal of one of those two towns from S. Because there are no cycles in the graph of roads of known direction within S, this can continue until there are at most 2 towns left in S.

If it ends with t towns left, $n - k + 1 - t$ questions were asked in this phase, so a total of $2n - t$ questions have been asked.

Phase 5. During this phase, Alice asks about all the roads from the remaining towns in S that she has not previously asked about. She has definitely already asked about any road between those towns (if $t = 2$). She must also have asked in one of the first two phases about

at least one other road involving one of those towns (as those phases resulted in a tree with $n > 2$ vertices). So she asks at most $t(n-t)-1$ questions in this phase.

At the end of this phase, Alice knows whether any town has at most one outgoing road. If $t = 1$, at most $3n - 3 \leqslant 4n - 7$ questions were needed in total, while if $t = 2$, at most $4n - 7$ questions were needed in total.

Comment 1. The version of this problem originally submitted asked only for an upper bound of $5n$, which is much simpler to prove. The Problem Selection Committee preferred a version with an asymptotically optimal constant. In the following comment, we will show that the constant is optimal.

Comment 2. We will show that Alice *cannot* always find out by asking at most $4n - 3(\log_2 n) - 15$ questions, if $n \geqslant 8$.

To show this, we suppose the King of Hearts is choosing the directions as he goes along, only picking the direction of a road when Alice asks about it for the first time. We provide a strategy for the King of Hearts that ensures that, after the given number of questions, the map is still consistent both with the existence of a town with at most one outgoing road, and with the nonexistence of such a town. His strategy has the following phases. When describing how the King of Hearts' answer to a question is determined below, we always assume he is being asked about a road for the first time (otherwise, he just repeats his previous answer for that road). This strategy is described throughout in graph-theoretic terms (vertices and edges rather than towns and roads).

Phase 1. In this phase, we consider the undirected graph formed by edges whose directions are known. The phase terminates when there are exactly 8 connected components whose undirected graphs are trees. The following invariant is maintained: in a component with k vertices whose undirected graph is a tree, every vertex has at most $\lfloor \log_2 k \rfloor$ edges into it.

- If the King of Hearts is asked about an edge between two vertices in the same component, or about an edge between two components at least one of which is not a tree, he chooses any direction for that edge arbitrarily.
- If he is asked about an edge between a vertex in component A that has a vertices and is a tree and a vertex in component B that has b vertices and is a tree, suppose without loss of generality that $a \geqslant b$. He then chooses the edge to go from A to B. In this case, the new number of edges into any vertex is at most $\max\{\lfloor \log_2 a \rfloor, \lfloor \log_2 b \rfloor + 1\} \leqslant \lfloor \log_2 (a+b) \rfloor$.

In all cases, the invariant is preserved, and the number of tree components either remains unchanged or goes down by 1. Assuming Alice does not repeat questions, the process must eventually terminate with 8 tree components, and at least $n - 8$ questions having been asked.

Note that each tree component contains at least one vertex with no outgoing edges. Colour one such vertex in each tree component red.

Phase 2. Let V_1, V_2 and V_3 be the three of the red vertices whose components are smallest (so their components together have at most $\left\lfloor \frac{3}{8}n \right\rfloor$ vertices, with each component having at most $\left\lfloor \frac{3}{8}n - 2 \right\rfloor$ vertices). Let sets C_1, C_2, ... be the connected components after removing the V_j. By construction, there are no edges with known direction between C_i and C_j for $i \neq j$, and there are at least five such components.

If at any point during this phase, the King of Hearts is asked about an edge within one of the C_i, he chooses an arbitrary direction. If he is asked about an edge between C_i and C_j for $i \neq j$, he answers so that all edges go from C_i to C_{i+1} and C_{i+2}, with indices taken modulo the number of components, and chooses arbitrarily for other pairs. This ensures that all vertices other than the V_j will have more than one outgoing edge.

For edges involving one of the V_j he answers as follows, so as to remain consistent for as long as possible with both possibilities for whether one of those vertices has at most one outgoing edge. Note that as they were red vertices, they have no outgoing edges at the start of this phase. For edges between two of the V_j, he answers that the edges go from V_1 to V_2, from V_2 to V_3 and from V_3 to V_1. For edges between V_j and some other vertex, he always answers that the edge goes into V_j, except for the last such edge for which he is asked the question for any given V_j, for which he answers that the

edge goes out of V_j. Thus, as long as at least one of the V_j has not had the question answered for all the vertices that are not among the V_j, his answers are still compatible both with all vertices having more than one outgoing edge, and with that V_j having only one outgoing edge.

At the start of this phase, each of the V_j has at most $\left\lfloor \log_2 \left\lfloor \frac{3}{8}n - 2 \right\rfloor \right\rfloor < (\log_2 n) - 1$ incoming edges. Thus, Alice cannot determine whether some vertex has only one outgoing edge within $3(n - 3 - ((\log_2 n) - 1)) - 1$ questions in this phase; that is, $4n - 3(\log_2 n) - 15$ questions total.

Comment 3. We can also improve the upper bound slightly, to $4n - 2(\log_2 n) + 1$. (We do not know where the precise minimum number of questions lies between $4n - 3(\log_2 n) + O(1)$ and $4n - 2(\log_2 n) + O(1)$.) Suppose $n \geqslant 5$ (otherwise no questions are required at all).

To do this, we replace Phases 1 and 2 of the given solution with a different strategy that also results in a spanning tree where one vertex V is not known to have any outgoing edges, and all other vertices have exactly one outgoing edge known, but where there is more control over the numbers of incoming edges. In Phases 3 and 4 we then take more care about the order in which pairs of towns are chosen, to ensure that each of the remaining towns has already had a question asked about at least $\log_2 n + O(1)$ edges.

Define trees T_m with 2^m vertices, exactly one of which (the *root*) has no outgoing edges and the rest of which have exactly one outgoing edge, as follows: T_0 is a single vertex, while T_m is constructed by joining the roots of two copies of T_{m-1} with an edge in either direction. If $n = 2^m$ we can readily ask $n - 1$ questions, resulting in a tree T_m for the edges with known direction: first ask about 2^{m-1} disjoint pairs of vertices, then about 2^{m-2} disjoint pairs of the roots of the resulting T_1 trees, and so on. For the general case, where n is not a power of 2, after k stages of this process we have $\lfloor n/2^k \rfloor$ trees, each of which is like T_k but may have some extra vertices (but, however, a unique root). If there are an even number of trees, then ask about pairs of their roots. If there are an odd number (greater than 1) of trees, when a single T_k is left over, ask about its root together with that of one of the T_{k+1} trees.

Say $m = \lfloor \log_2 n \rfloor$. The result of that process is a single T_m tree, possibly with some extra vertices but still a unique root V. That root has at least m incoming edges, and we may list vertices V_0, …, V_{m-1} with edges to V, such that, for all $0 \leqslant i < m$, vertex V_i itself has at least i incoming edges.

Now divide the vertices other than V into two parts: A has all vertices at an odd distance from V and B has all the vertices at an even distance from B. Both A and B are nonempty; A contains the V_i, while B contains a sequence of vertices with at least 0, 1, …, $m - 2$ incoming edges respectively, similar to the V_i. There are no edges with known direction within A or within B.

In Phase 3, then ask about edges between V and other vertices: first those in B, in order of increasing number of incoming edges to the other vertex, then those in A, again in order of increasing number of incoming edges, which involves asking at most $n - 1 - m$ questions in this phase. If two outgoing edges are not found from V, at most $2n - 2 - m \leqslant 4n - 2(\log_2 n) + 1$ questions needed to be asked in total, so we suppose that two outgoing edges were found, with k questions asked in this phase, where $2 \leqslant k \leqslant n - 1 - m$. The state of S is as described in the solution above, with the additional property that, since S must still contain all vertices with edges to V, it contains the vertices V_i described above.

In Phase 4, consider the vertices left in B, in increasing order of number of edges incoming to a vertex. If s is the least number of incoming edges to such a vertex, then, for any $s \leqslant t \leqslant m - 2$, there are at least $m - t - 2$ vertices with more than t incoming edges. Repeatedly asking about the pair of vertices left in B with the least numbers of incoming edges results in a single vertex left over (if any were in B at all at the start of this phase) with at least $m - 2$ incoming edges. Doing the same with A (which must be nonempty) leaves a vertex with at least $m - 1$ incoming edges.

Thus if only A is nonempty we ask at most $n - m$ questions in Phase 5, so in total at most $3n - m - 1$ questions, while if both are nonempty we ask at most $2n - 2m + 1$ questions in Phase 5, so in total at most $4n - 2m - 1 < 4n - 2(\log_2 n) + 1$ questions.

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C9. For any two different real numbers x and y, we define $D(x,y)$ to be the unique integer d satisfying $2^d \leqslant |x-y| < 2^{d+1}$. Given a set of reals $\mathcal{F}$, and an element $x \in \mathcal{F}$, we say that the *scales* of x in $\mathcal{F}$ are the values of $D(x,y)$ for $y \in \mathcal{F}$ with $x \neq y$.

Let k be a given positive integer. Suppose that each member x of $\mathcal{F}$ has at most k different scales in $\mathcal{F}$ (note that these scales may depend on x). What is the maximum possible size of $\mathcal{F}$?

(Italy)

Answer: The maximum possible size of $\mathcal{F}$ is 2^k.

Common remarks. For convenience, we extend the use of the word *scale*: we say that the scale between two reals x and y is $D(x,y)$.

Solution. We first construct a set $\mathcal{F}$ with 2^k members, each member having at most k different scales in $\mathcal{F}$. Take $\mathcal{F} = \{0, 1, 2, \ldots, 2^k - 1\}$. The scale between any two members of $\mathcal{F}$ is in the set $\{0, 1, \ldots, k-1\}$.

We now show that 2^k is an upper bound on the size of $\mathcal{F}$. For every finite set $\mathcal{S}$ of real numbers, and every real x, let $r_{\mathcal{S}}(x)$ denote the number of different scales of x in $\mathcal{S}$. That is, $r_{\mathcal{S}}(x) = |\{D(x,y) \colon x \neq y \in \mathcal{S}\}|$. Thus, for every element x of the set $\mathcal{F}$ in the problem statement, we have $r_{\mathcal{F}}(x) \leqslant k$. The condition $|\mathcal{F}| \leqslant 2^k$ is an immediate consequence of the following lemma.

Lemma. Let $\mathcal{S}$ be a finite set of real numbers, and define

$$w(\mathcal{S}) = \sum_{x \in \mathcal{S}} 2^{-r_{\mathcal{S}}(x)}.$$

Then $w(\mathcal{S}) \leqslant 1$.

Proof. Induction on $n = |\mathcal{S}|$. If $\mathcal{S} = \{x\}$, then $r_{\mathcal{S}}(x) = 0$, so $w(\mathcal{S}) = 1$.

Assume now $n \geqslant 2$, and let $x_1 < \cdots < x_n$ list the members of $\mathcal{S}$. Let d be the minimal scale between two distinct elements of $\mathcal{S}$; then there exist neighbours x_t and x_{t+1} with $D(x_t, x_{t+1}) = d$. Notice that for any two indices i and j with $j - i > 1$ we have $D(x_i, x_j) > d$, since

$$|x_i - x_j| = |x_{i+1} - x_i| + |x_j - x_{i+1}| \geqslant 2^d + 2^d = 2^{d+1}.$$

Now choose the minimal $i \leqslant t$ and the maximal $j \geqslant t+1$ such that $D(x_i, x_{i+1}) = D(x_{i+1}, x_{i+2}) = \cdots = D(x_{j-1}, x_j) = d$.

Let E be the set of all the x_s with even indices $i \leqslant s \leqslant j$, O be the set of those with odd indices $i \leqslant s \leqslant j$, and R be the rest of the elements (so that $\mathcal{S}$ is the disjoint union of E, O and R). Set $\mathcal{S}_O = R \cup O$ and $\mathcal{S}_E = R \cup E$; we have $|\mathcal{S}_O| < |\mathcal{S}|$ and $|\mathcal{S}_E| < |\mathcal{S}|$, so $w(\mathcal{S}_O), w(\mathcal{S}_E) \leqslant 1$ by the inductive hypothesis.

Clearly, $r_{\mathcal{S}_O}(x) \leqslant r_{\mathcal{S}}(x)$ and $r_{\mathcal{S}_E}(x) \leqslant r_{\mathcal{S}}(x)$ for any $x \in R$, and thus

$$\begin{aligned} \sum_{x \in R} 2^{-r_{\mathcal{S}}(x)} &= \frac{1}{2} \sum_{x \in R} (2^{-r_{\mathcal{S}}(x)} + 2^{-r_{\mathcal{S}}(x)}) \\ &\leqslant \frac{1}{2} \sum_{x \in R} (2^{-r_{\mathcal{S}_O}(x)} + 2^{-r_{\mathcal{S}_E}(x)}). \end{aligned}$$

On the other hand, for every $x \in O$, there is no $y \in \mathcal{S}_O$ such that $D_{\mathcal{S}_O}(x,y) = d$ (as all candidates from $\mathcal{S}$ were in E). Hence, we have $r_{\mathcal{S}_O}(x) \leqslant r_{\mathcal{S}}(x) - 1$, and thus

$$\sum_{x \in O} 2^{-r_{\mathcal{S}}(x)} \leqslant \frac{1}{2} \sum_{x \in O} 2^{-r_{\mathcal{S}_O}(x)}.$$

Similarly, for every $x \in E$, we have

$$\sum_{x \in E} 2^{-r_{\mathcal{S}}(x)} \leqslant \frac{1}{2} \sum_{x \in E} 2^{-r_{\mathcal{S}_E}(x)}.$$

We can then combine these to give

$$\begin{aligned}
w(S) &= \sum_{x \in R} 2^{-r_{\mathcal{S}}(x)} + \sum_{x \in O} 2^{-r_{\mathcal{S}}(x)} + \sum_{x \in E} 2^{-r_{\mathcal{S}}(x)} \\
&\leqslant \frac{1}{2} \sum_{x \in R} (2^{-r_{\mathcal{S}_O}(x)} + 2^{-r_{\mathcal{S}_E}(x)}) + \frac{1}{2} \sum_{x \in O} 2^{-r_{\mathcal{S}_O}(x)} + \frac{1}{2} \sum_{x \in E} 2^{-r_{\mathcal{S}_E}(x)} \\
&= \frac{1}{2} \left(\sum_{x \in \mathcal{S}_O} 2^{-r_{\mathcal{S}_O}(x)} + \sum_{x \in \mathcal{S}_E} 2^{-r_{\mathcal{S}_E}(x)} \right) \qquad \text{(since } \mathcal{S}_O = O \cup R \text{ and } \mathcal{S}_E = E \cup R\text{)} \\
&= \frac{1}{2} \left(w(\mathcal{S}_O) + w(\mathcal{S}_E) \right)) \qquad \text{(by definition of } w(\cdot)\text{)} \\
&\leqslant 1 \qquad \text{(by the inductive hypothesis)}
\end{aligned}$$

which completes the induction. □

Comment 1. The sets O and E above are not the only ones we could have chosen. Indeed, we could instead have used the following definitions:

Let d be the maximal scale between two distinct elements of $\mathcal{S}$; that is, $d = D(x_1, x_n)$. Let $O = \{x \in \mathcal{S} : D(x, x_n) = d\}$ (a 'left' part of the set) and let $E = \{x \in \mathcal{S} : D(x_1, x) = d\}$ (a 'right' part of the set). Note that these two sets are disjoint, and nonempty (since they contain x_1 and x_n respectively). The rest of the proof is then the same as in Solution 1.

Comment 2. Another possible set $\mathcal{F}$ containing 2^k members could arise from considering a binary tree of height k, allocating a real number to each leaf, and trying to make the scale between the values of two leaves dependent only on the (graph) distance between them. The following construction makes this more precise.

We build up sets $\mathcal{F}_k$ recursively. Let $\mathcal{F}_0 = \{0\}$, and then let $\mathcal{F}_{k+1} = \mathcal{F}_k \cup \{x + 3 \cdot 4^k : x \in \mathcal{F}_k\}$ (i.e. each half of $\mathcal{F}_{k+1}$ is a copy of F_k). We have that $\mathcal{F}_k$ is contained in the interval $[0, 4^{k+1})$, and so it follows by induction on k that every member of F_{k+1} has k different scales in its own half of F_{k+1} (by the inductive hypothesis), and only the single scale $2k + 1$ in the other half of F_{k+1}.

Both of the constructions presented here have the property that every member of $\mathcal{F}$ has exactly k different scales in $\mathcal{F}$. Indeed, it can be seen that this must hold (up to a slight perturbation) for any such maximal set. Suppose there were some element x with only $k-1$ different scales in $\mathcal{F}$ (and every other element had at most k different scales). Then we take some positive real ϵ, and construct a new set $\mathcal{F}' = \{y : y \in \mathcal{F}, y \leqslant x\} \cup \{y + \epsilon : y \in \mathcal{F}, y \geqslant x\}$. We have $|\mathcal{F}'| = |\mathcal{F}| + 1$, and if ϵ is sufficiently small then $\mathcal{F}'$ will also satisfy the property that no member has more than k different scales in $\mathcal{F}'$.

This observation might be used to motivate the idea of weighting members of an arbitrary set $\mathcal{S}$ of reals according to how many different scales they have in $\mathcal{S}$.

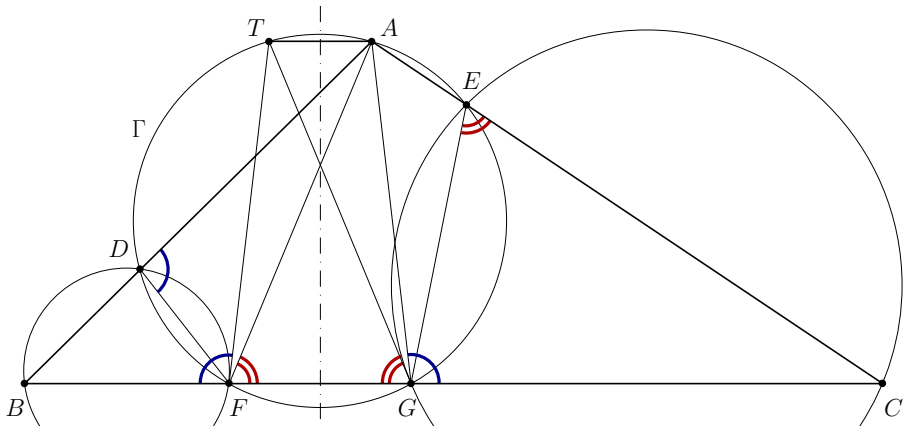
Geometry

G1. Let ABC be a triangle. Circle Γ passes through A, meets segments AB and AC again at points D and E respectively, and intersects segment BC at F and G such that F lies between B and G. The tangent to circle BDF at F and the tangent to circle CEG at G meet at point T. Suppose that points A and T are distinct. Prove that line AT is parallel to BC.

(Nigeria)

Solution. Notice that $\angle TFB = \angle FDA$ because FT is tangent to circle BDF, and moreover $\angle FDA = \angle CGA$ because quadrilateral $ADFG$ is cyclic. Similarly, $\angle TGB = \angle GEC$ because GT is tangent to circle CEG, and $\angle GEC = \angle CFA$. Hence,

$$\angle TFB = \angle CGA \quad \text{and} \quad \angle TGB = \angle CFA. \tag{1}$$



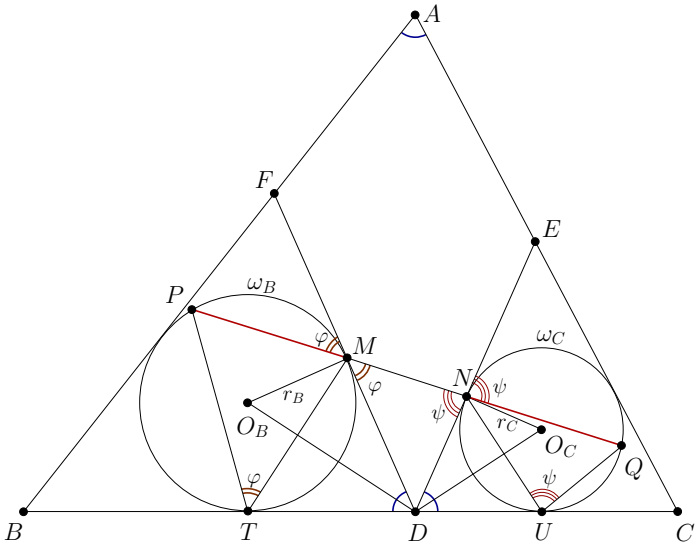
Triangles FGA and GFT have a common side FG, and by (1) their angles at F, G are the same. So, these triangles are congruent. So, their altitudes starting from A and T, respectively, are equal and hence AT is parallel to line $BFGC$.

Comment. Alternatively, we can prove first that T lies on Γ. For example, this can be done by showing that $\angle AFT = \angle AGT$ using (1). Then the statement follows as $\angle TAF = \angle TGF = \angle GFA$.

G2. Let ABC be an acute-angled triangle and let D, E, and F be the feet of altitudes from A, B, and C to sides BC, CA, and AB, respectively. Denote by ω_B and ω_C the incircles of triangles BDF and CDE, and let these circles be tangent to segments DF and DE at M and N, respectively. Let line MN meet circles ω_B and ω_C again at $P \neq M$ and $Q \neq N$, respectively. Prove that $MP = NQ$.

(Vietnam)

Solution. Denote the centres of ω_B and ω_C by O_B and O_C, let their radii be r_B and r_C, and let BC be tangent to the two circles at T and U, respectively.



From the cyclic quadrilaterals $AFDC$ and $ABDE$ we have

$$\angle MDO_B = \frac{1}{2}\angle FDB = \frac{1}{2}\angle BAC = \frac{1}{2}\angle CDE = \angle O_C DN,$$

so the right-angled triangles DMO_B and DNO_C are similar. The ratio of similarity between the two triangles is

$$\frac{DN}{DM} = \frac{O_C N}{O_B M} = \frac{r_C}{r_B}.$$

Let $\varphi = \angle DMN$ and $\psi = \angle MND$. The lines FM and EN are tangent to ω_B and ω_C, respectively, so

$$\angle MTP = \angle FMP = \angle DMN = \varphi \quad \text{and} \quad \angle QUN = \angle QNE = \angle MND = \psi.$$

(It is possible that P or Q coincides with T or U, or lie inside triangles DMT or DUN, respectively. To reduce case-sensitivity, we may use directed angles or simply ignore angles MTP and QUN.)

In the circles ω_B and ω_C the lengths of chords MP and NQ are

$$MP = 2r_B \cdot \sin\angle MTP = 2r_B \cdot \sin\varphi \quad \text{and} \quad NQ = 2r_C \cdot \sin\angle QUN = 2r_C \cdot \sin\psi.$$

By applying the sine rule to triangle DNM we get

$$\frac{DN}{DM} = \frac{\sin\angle DMN}{\sin\angle MND} = \frac{\sin\varphi}{\sin\psi}.$$

Finally, putting the above observations together, we get

$$\frac{MP}{NQ} = \frac{2r_B \sin\varphi}{2r_C \sin\psi} = \frac{r_B}{r_C} \cdot \frac{\sin\varphi}{\sin\psi} = \frac{DM}{DN} \cdot \frac{\sin\varphi}{\sin\psi} = \frac{\sin\psi}{\sin\varphi} \cdot \frac{\sin\varphi}{\sin\psi} = 1,$$

so $MP = NQ$ as required.

G3. In triangle ABC, let A_1 and B_1 be two points on sides BC and AC, and let P and Q be two points on segments AA_1 and BB_1, respectively, so that line PQ is parallel to AB. On ray PB_1, beyond B_1, let P_1 be a point so that $\angle PP_1C = \angle BAC$. Similarly, on ray QA_1, beyond A_1, let Q_1 be a point so that $\angle CQ_1Q = \angle CBA$. Show that points P, Q, P_1, and Q_1 are concyclic.

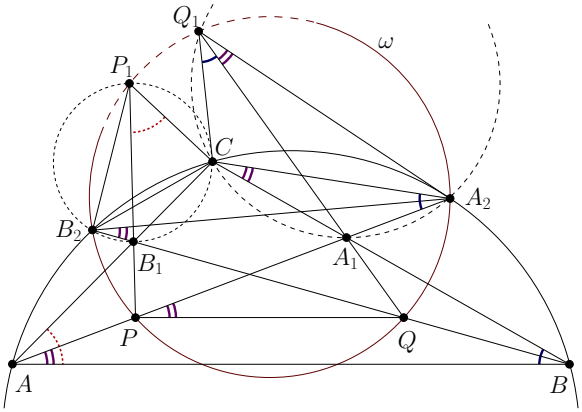
(Ukraine)

Solution 1. Throughout the solution we use oriented angles.

Let rays AA_1 and BB_1 intersect the circumcircle of $\triangle ACB$ at A_2 and B_2, respectively. By

$$\angle QPA_2 = \angle BAA_2 = \angle BB_2A_2 = \angle QB_2A_2,$$

points P, Q, A_2, B_2 are concyclic; denote the circle passing through these points by ω. We shall prove that P_1 and Q_1 also lie on ω.



By

$$\angle CA_2A_1 = \angle CA_2A = \angle CBA = \angle CQ_1Q = \angle CQ_1A_1,$$

points C, Q_1, A_2, A_1 are also concyclic. From that we get

$$\angle QQ_1A_2 = \angle A_1Q_1A_2 = \angle A_1CA_2 = \angle BCA_2 = \angle BAA_2 = \angle QPA_2,$$

so Q_1 lies on ω.

It follows similarly that P_1 lies on ω.

Solution 2. First consider the case when lines PP_1 and QQ_1 intersect each other at some point R.

Let line PQ meet the sides AC and BC at E and F, respectively. Then

$$\angle PP_1C = \angle BAC = \angle PEC,$$

so points C, E, P, P_1 lie on a circle; denote that circle by ω_P. It follows analogously that points C, F, Q, Q_1 lie on another circle; denote it by ω_Q.

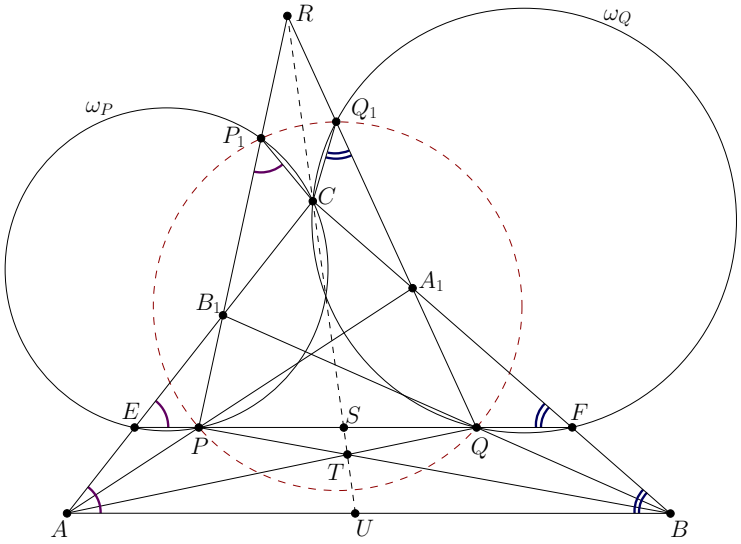
Let AQ and BP intersect at T. Applying Pappus' theorem to the lines AA_1P and BB_1Q provides that points $C = AB_1 \cap BA_1$, $R = A_1Q \cap B_1P$ and $T = AQ \cap BP$ are collinear.

Let line RCT meet PQ and AB at S and U, respectively. From $AB \parallel PQ$ we obtain

$$\frac{SP}{SQ} = \frac{UB}{UA} = \frac{SF}{SE},$$

so

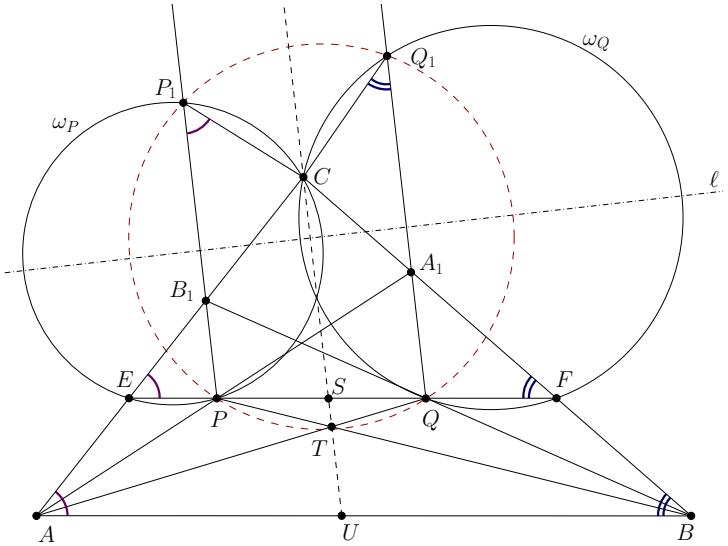
$$SP \cdot SE = SQ \cdot SF.$$



So, point S has equal powers with respect to ω_P and ω_Q, hence line RCS is their radical axis; then R also has equal powers to the circles, so $RP \cdot RP_1 = RQ \cdot RQ_1$, proving that points P, P_1, Q, Q_1 are indeed concyclic.

Now consider the case when PP_1 and QQ_1 are parallel. Like in the previous case, let AQ and BP intersect at T. Applying Pappus' theorem again to the lines AA_1P and BB_1Q, in this limit case it shows that line CT is parallel to PP_1 and QQ_1.

Let line CT meet PQ and AB at S and U, as before. The same calculation as in the previous case shows that $SP \cdot SE = SQ \cdot SF$, so S lies on the radical axis between ω_P and ω_Q.



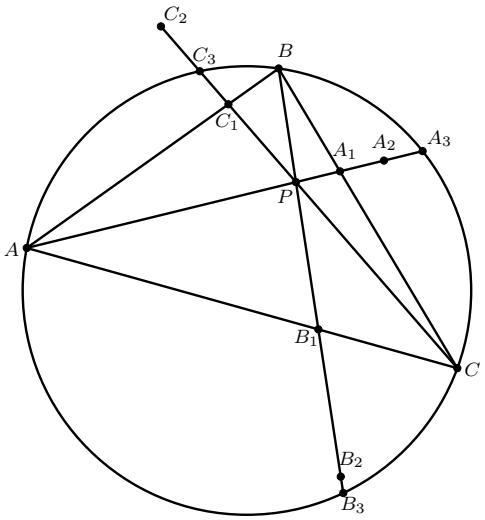
Line CST, that is the radical axis between ω_P and ω_Q, is perpendicular to the line ℓ of centres of ω_P and ω_Q. Hence, the chords PP_1 and QQ_1 are perpendicular to ℓ. So the quadrilateral PP_1Q_1Q is an isosceles trapezium with symmetry axis ℓ, and hence is cyclic.

Comment. There are several ways of solving the problem involving Pappus' theorem. For example, one may consider the points $K = PB_1 \cap BC$ and $L = QA_1 \cap AC$. Applying Pappus' theorem to the lines AA_1P and QB_1B we get that K, L, and $PQ \cap AB$ are collinear, i.e. that $KL \parallel AB$. Therefore, cyclicity of P, Q, P_1, and Q_1 is equivalent to that of K, L, P_1, and Q_1. The latter is easy after noticing that C also lies on that circle. Indeed, e.g. $\angle(LK, LC) = \angle(AB, AC) = \angle(P_1K, P_1C)$ shows that K lies on circle KLC.

This approach also has some possible degeneracy, as the points K and L may happen to be ideal.

G4. Let P be a point inside triangle ABC. Let AP meet BC at A_1, let BP meet CA at B_1, and let CP meet AB at C_1. Let A_2 be the point such that A_1 is the midpoint of PA_2, let B_2 be the point such that B_1 is the midpoint of PB_2, and let C_2 be the point such that C_1 is the midpoint of PC_2. Prove that points A_2, B_2, and C_2 cannot all lie strictly inside the circumcircle of triangle ABC.

(Australia)



Solution 1. Since

$$\angle APB + \angle BPC + \angle CPA = 2\pi = (\pi - \angle ACB) + (\pi - \angle BAC) + (\pi - \angle CBA),$$

at least one of the following inequalities holds:

$$\angle APB \geqslant \pi - \angle ACB, \quad \angle BPC \geqslant \pi - \angle BAC, \quad \angle CPA \geqslant \pi - \angle CBA\,.$$

Without loss of generality, we assume that $\angle BPC \geqslant \pi - \angle BAC$. We have $\angle BPC > \angle BAC$ because P is inside $\triangle ABC$. So $\angle BPC \geqslant \max(\angle BAC, \pi - \angle BAC)$ and hence

$$\sin\angle BPC \leqslant \sin\angle BAC\,. \tag{$*$}$$

Let the rays AP, BP, and CP cross the circumcircle Ω again at A_3, B_3, and C_3, respectively. We will prove that at least one of the ratios $\frac{PB_1}{B_1B_3}$ and $\frac{PC_1}{C_1C_3}$ is at least 1, which yields that one of the points B_2 and C_2 does not lie strictly inside Ω.

Because A, B, C, B_3 lie on a circle, the triangles CB_1B_3 and BB_1A are similar, so

$$\frac{CB_1}{B_1B_3} = \frac{BB_1}{B_1A}\,.$$

Applying the sine rule we obtain

$$\frac{PB_1}{B_1B_3} = \frac{PB_1}{CB_1}\cdot\frac{CB_1}{B_1B_3} = \frac{PB_1}{CB_1}\cdot\frac{BB_1}{B_1A} = \frac{\sin\angle ACP}{\sin\angle BPC}\cdot\frac{\sin\angle BAC}{\sin\angle PBA}\,.$$

Similarly,

$$\frac{PC_1}{C_1C_3} = \frac{\sin\angle PBA}{\sin\angle BPC}\cdot\frac{\sin\angle BAC}{\sin\angle ACP}\,.$$

Multiplying these two equations we get

$$\frac{PB_1}{B_1B_3}\cdot\frac{PC_1}{C_1C_3} = \frac{\sin^2\angle BAC}{\sin^2\angle BPC} \geqslant 1$$

using $(*)$, which yields the desired conclusion.

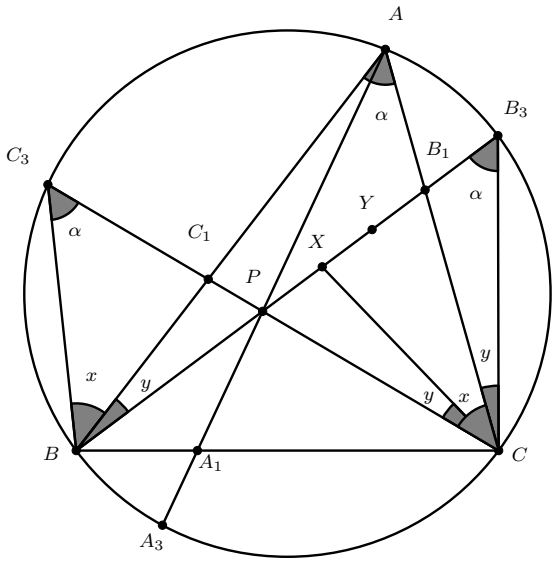
Comment. It also cannot happen that all three points A_2, B_2, and C_2 lie strictly *outside* Ω. The same proof works almost literally, starting by assuming without loss of generality that $\angle BPC \leqslant \pi - \angle BAC$ and using $\angle BPC > \angle BAC$ to deduce that $\sin\angle BPC \geqslant \sin\angle BAC$. It is possible for A_2, B_2, and C_2 all to lie on the circumcircle; from the above solution we may derive that this happens if and only if P is the orthocentre of the triangle ABC, (which lies strictly inside ABC if and only if ABC is acute).

Solution 2. Define points A_3, B_3, and C_3 as in Solution 1. Assume for the sake of contradiction that A_2, B_2, and C_2 all lie strictly inside circle ABC. It follows that $PA_1 < A_1A_3$, $PB_1 < B_1B_3$, and $PC_1 < C_1C_3$.

Observe that $\triangle PBC_3 \sim \triangle PCB_3$. Let X be the point on side PB_3 that corresponds to point C_1 on side PC_3 under this similarity. In other words, X lies on segment PB_3 and satisfies $PX : XB_3 = PC_1 : C_1C_3$. It follows that

$$\angle XCP = \angle PBC_1 = \angle B_3BA = \angle B_3CB_1\,.$$

Hence lines CX and CB_1 are isogonal conjugates in $\triangle PCB_3$.



Let Y be the foot of the bisector of $\angle B_3CP$ in $\triangle PCB_3$. Since $PC_1 < C_1C_3$, we have $PX < XB_3$. Also, we have $PY < YB_3$ because $PB_1 < B_1B_3$ and Y lies between X and B_1. By the angle bisector theorem in $\triangle PCB_3$, we have $PY : YB_3 = PC : CB_3$. So $PC < CB_3$ and it follows that $\angle PB_3C < \angle CPB_3$. Now since $\angle PB_3C = \angle BB_3C = \angle BAC$, we have

$$\angle BAC < \angle CPB_3\,.$$

Similarly, we have

$$\angle CBA < \angle APC_3 \quad \text{and} \quad \angle ACB < \angle BPA_3 = \angle B_3PA\,.$$

Adding these three inequalities yields $\pi < \pi$, and this contradiction concludes the proof.

Solution 3. Choose coordinates such that the circumcentre of $\triangle ABC$ is at the origin and the circumradius is 1. Then we may think of A, B, and C as vectors in $\mathbb{R}^2$ such that

$$|A|^2 = |B|^2 = |C|^2 = 1\,.$$

P may be represented as a convex combination $\alpha A+\beta B+\gamma C$ where $\alpha, \beta, \gamma > 0$ and $\alpha+\beta+\gamma = 1$. Then

$$A_1 = \frac{\beta B + \gamma C}{\beta + \gamma} = \frac{1}{1-\alpha}P - \frac{\alpha}{1-\alpha}A,$$

so

$$A_2 = 2A_1 - P = \frac{1+\alpha}{1-\alpha}P - \frac{2\alpha}{1-\alpha}A\,.$$

Hence

$$|A_2|^2 = \left(\frac{1+\alpha}{1-\alpha}\right)^2 |P|^2 + \left(\frac{2\alpha}{1-\alpha}\right)^2 |A|^2 - \frac{4\alpha(1+\alpha)}{(1-\alpha)^2}\, A\cdot P\,.$$

Using $|A|^2 = 1$ we obtain

$$\frac{(1-\alpha)^2}{2(1+\alpha)}|A_2|^2 = \frac{1+\alpha}{2}|P|^2 + \frac{2\alpha^2}{1+\alpha} - 2\alpha A\cdot P. \tag{1}$$

Likewise

$$\frac{(1-\beta)^2}{2(1+\beta)}|B_2|^2 = \frac{1+\beta}{2}|P|^2 + \frac{2\beta^2}{1+\beta} - 2\beta B\cdot P \tag{2}$$

and

$$\frac{(1-\gamma)^2}{2(1+\gamma)}|C_2|^2 = \frac{1+\gamma}{2}|P|^2 + \frac{2\gamma^2}{1+\gamma} - 2\gamma C\cdot P. \tag{3}$$

Summing (1), (2) and (3) we obtain on the LHS the positive linear combination

$$\text{LHS} = \frac{(1-\alpha)^2}{2(1+\alpha)}|A_2|^2 + \frac{(1-\beta)^2}{2(1+\beta)}|B_2|^2 + \frac{(1-\gamma)^2}{2(1+\gamma)}|C_2|^2$$

and on the RHS the quantity

$$\left(\frac{1+\alpha}{2} + \frac{1+\beta}{2} + \frac{1+\gamma}{2}\right)|P|^2 + \left(\frac{2\alpha^2}{1+\alpha} + \frac{2\beta^2}{1+\beta} + \frac{2\gamma^2}{1+\gamma}\right) - 2(\alpha A\cdot P + \beta B\cdot P + \gamma C\cdot P)\,.$$

The first term is $2|P|^2$ and the last term is $-2P\cdot P$, so

$$\begin{aligned}
\text{RHS} &= \left(\frac{2\alpha^2}{1+\alpha} + \frac{2\beta^2}{1+\beta} + \frac{2\gamma^2}{1+\gamma}\right)\\
&= \frac{3\alpha-1}{2} + \frac{(1-\alpha)^2}{2(1+\alpha)} + \frac{3\beta-1}{2} + \frac{(1-\beta)^2}{2(1+\beta)} + \frac{3\gamma-1}{2} + \frac{(1-\gamma)^2}{2(1+\gamma)}\\
&= \frac{(1-\alpha)^2}{2(1+\alpha)} + \frac{(1-\beta)^2}{2(1+\beta)} + \frac{(1-\gamma)^2}{2(1+\gamma)}\,.
\end{aligned}$$

Here we used the fact that

$$\frac{3\alpha-1}{2} + \frac{3\beta-1}{2} + \frac{3\gamma-1}{2} = 0\,.$$

We have shown that a linear combination of $|A_1|^2$, $|B_1|^2$, and $|C_1|^2$ with positive coefficients is equal to the sum of the coefficients. Therefore at least one of $|A_1|^2$, $|B_1|^2$, and $|C_1|^2$ must be at least 1, as required.

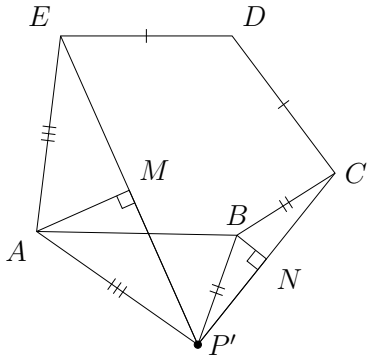
Comment. This proof also works when P is any point for which $\alpha, \beta, \gamma > -1$, $\alpha+\beta+\gamma = 1$, and $\alpha, \beta, \gamma \neq 1$. (In any cases where $\alpha = 1$ or $\beta = 1$ or $\gamma = 1$, some points in the construction are not defined.)

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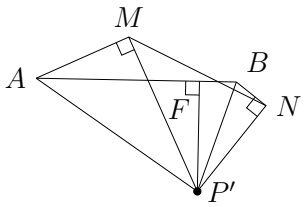
G5. Let $ABCDE$ be a convex pentagon with $CD = DE$ and $\angle EDC \neq 2 \cdot \angle ADB$. Suppose that a point P is located in the interior of the pentagon such that $AP = AE$ and $BP = BC$. Prove that P lies on the diagonal CE if and only if $\text{area}(BCD) + \text{area}(ADE) = \text{area}(ABD) + \text{area}(ABP)$.

(Hungary)

Solution 1. Let P' be the reflection of P across line AB, and let M and N be the midpoints of $P'E$ and $P'C$ respectively. Convexity ensures that P' is distinct from both E and C, and hence from both M and N. We claim that both the area condition and the collinearity condition in the problem are equivalent to the condition that the (possibly degenerate) right-angled triangles $AP'M$ and $BP'N$ are directly similar (equivalently, $AP'E$ and $BP'C$ are directly similar).



For the equivalence with the collinearity condition, let F denote the foot of the perpendicular from P' to AB, so that F is the midpoint of PP'. We have that P lies on CE if and only if F lies on MN, which occurs if and only if we have the equality $\angle AFM = \angle BFN$ of signed angles modulo π. By concyclicity of $AP'FM$ and $BFP'N$, this is equivalent to $\angle AP'M = \angle BP'N$, which occurs if and only if $AP'M$ and $BP'N$ are directly similar.



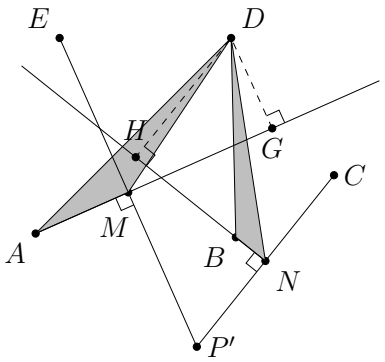
For the other equivalence with the area condition, we have the equality of signed areas $\text{area}(ABD) + \text{area}(ABP) = \text{area}(AP'BD) = \text{area}(AP'D) + \text{area}(BDP')$. Using the identity $\text{area}(ADE) - \text{area}(AP'D) = \text{area}(ADE) + \text{area}(ADP') = 2\,\text{area}(ADM)$, and similarly for B, we find that the area condition is equivalent to the equality

$$\text{area}(DAM) = \text{area}(DBN).$$

Now note that A and B lie on the perpendicular bisectors of $P'E$ and $P'C$, respectively. If we write G and H for the feet of the perpendiculars from D to these perpendicular bisectors respectively, then this area condition can be rewritten as

$$MA \cdot GD = NB \cdot HD.$$

(In this condition, we interpret all lengths as signed lengths according to suitable conventions: for instance, we orient $P'E$ from P' to E, orient the parallel line DH in the same direction, and orient the perpendicular bisector of $P'E$ at an angle $\pi/2$ clockwise from the oriented segment $P'E$ – we adopt the analogous conventions at B.)



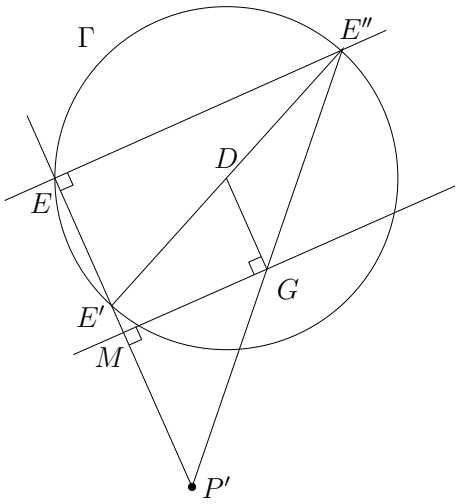
To relate the signed lengths GD and HD to the triangles $AP'M$ and $BP'N$, we use the following calculation.

Claim. Let Γ denote the circle centred on D with both E and C on the circumference, and h the power of P' with respect to Γ. Then we have the equality

$$GD \cdot P'M = HD \cdot P'N = \frac{1}{4}h \neq 0.$$

Proof. Firstly, we have $h \neq 0$, since otherwise P' would lie on Γ, and hence the internal angle bisectors of $\angle EDP'$ and $\angle P'DC$ would pass through A and B respectively. This would violate the angle inequality $\angle EDC \neq 2 \cdot \angle ADB$ given in the question.

Next, let E' denote the second point of intersection of $P'E$ with Γ, and let E'' denote the point on Γ diametrically opposite E', so that $E''E$ is perpendicular to $P'E$. The point G lies on the perpendicular bisectors of the sides $P'E$ and EE'' of the right-angled triangle $P'EE''$; it follows that G is the midpoint of $P'E''$. Since D is the midpoint of $E'E''$, we have that $GD = \frac{1}{2}P'E'$. Since $P'M = \frac{1}{2}P'E$, we have $GD \cdot P'M = \frac{1}{4}P'E' \cdot P'E = \frac{1}{4}h$. The other equality $HD \cdot P'N$ follows by exactly the same argument.



□

From this claim, we see that the area condition is equivalent to the equality

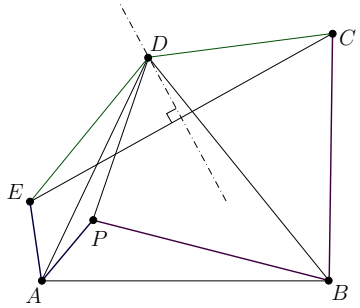
$$(MA : P'M) = (NB : P'N)$$

of ratios of signed lengths, which is equivalent to direct similarity of $AP'M$ and $BP'N$, as desired.

Solution 2. Along the perpendicular bisector of CE, define the linear function

$$f(X) = \text{area}(BCX) + \text{area}(AXE) - \text{area}(ABX) - \text{area}(ABP),$$

where, from now on, we always use signed areas. Thus, we want to show that C, P, E are collinear if and only if $f(D) = 0$.



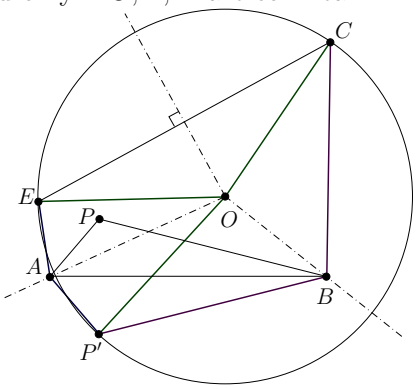
Let P' be the reflection of P across line AB. The point P' does not lie on the line CE. To see this, we let A'' and B'' be the points obtained from A and B by dilating with scale factor 2 about P', so that P is the orthogonal projection of P' onto $A''B''$. Since A lies on the perpendicular bisector of $P'E$, the triangle $A''EP'$ is right-angled at E (and $B''CP'$ similarly). If P' were to lie on CE, then the lines $A''E$ and $B''C$ would be perpendicular to CE and A'' and B'' would lie on the opposite side of CE to D. It follows that the line $A''B''$ does not meet triangle CDE, and hence point P does not lie inside CDE. But then P must lie inside $ABCE$, and it is clear that such a point cannot reflect to a point P' on CE.

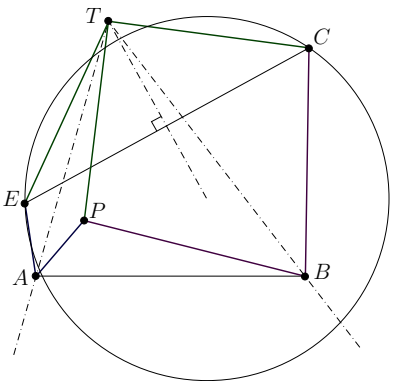
We thus let O be the centre of the circle CEP'. The lines AO and BO are the perpendicular bisectors of EP' and CP', respectively, so

$$\begin{aligned}\text{area}(BCO) + \text{area}(AOE) &= \text{area}(OP'B) + \text{area}(P'OA) = \text{area}(P'BOA) \\ &= \text{area}(ABO) + \text{area}(BAP') = \text{area}(ABO) + \text{area}(ABP),\end{aligned}$$

and hence $f(O) = 0$.

Notice that if point O coincides with D then points A, B lie in angle domain CDE and $\angle EOC = 2 \cdot \angle AOB$, which is not allowed. So, O and D must be distinct. Since f is linear and vanishes at O, it follows that $f(D) = 0$ if and only if f is constant zero – we want to show this occurs if and only if C, P, E are collinear.





In the one direction, suppose firstly that C, P, E are not collinear, and let T be the centre of the circle CEP. The same calculation as above provides

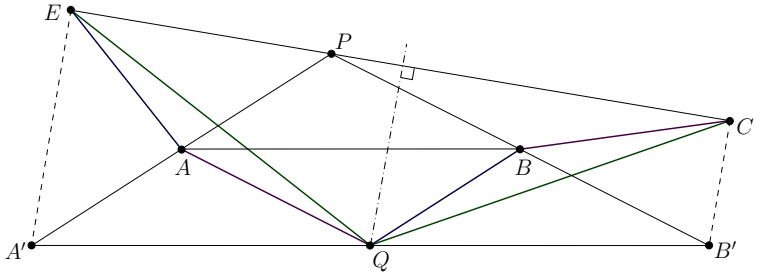
$$\text{area}(BCT) + \text{area}(ATE) = \text{area}(PBTA) = \text{area}(ABT) - \text{area}(ABP)$$

so

$$f(T) = -2\,\text{area}(ABP) \neq 0.$$

Hence, the linear function f is nonconstant with its zero is at O, so that $f(D) \neq 0$.

In the other direction, suppose that the points C, P, E are collinear. We will show that f is constant zero by finding a second point (other than O) at which it vanishes.



Let Q be the reflection of P across the midpoint of AB, so $PAQB$ is a parallelogram. It is easy to see that Q is on the perpendicular bisector of CE; for instance if A' and B' are the points produced from A and B by dilating about P with scale factor 2, then the projection of Q to CE is the midpoint of the projections of A' and B', which are E and C respectively. The triangles BCQ and AQE are indirectly congruent, so

$$f(Q) = \big(\text{area}(BCQ) + \text{area}(AQE)\big) - \big(\text{area}(ABQ) - \text{area}(BAP)\big) = 0 - 0 = 0.$$

The points O and Q are distinct. To see this, consider the circle ω centred on Q with P' on the circumference; since triangle $PP'Q$ is right-angled at P', it follows that P lies outside ω. On the other hand, P lies between C and E on the line CPE. It follows that C and E cannot both lie on ω, so that ω is not the circle CEP' and $Q \neq O$.

Since O and Q are distinct zeroes of the linear function f, we have $f(D) = 0$ as desired.

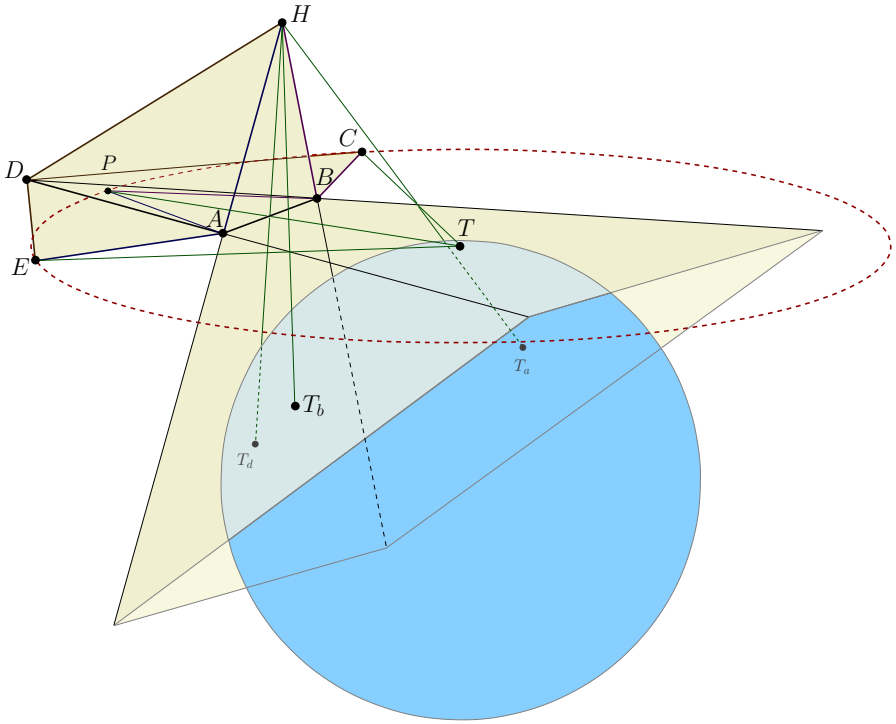
Comment 1. The condition $\angle EDC \neq 2{\cdot}\angle ADB$ cannot be omitted. If D is the centre of circle CEP', then the condition on triangle areas is satisfied automatically, without having P on line CE.

Comment 2. The "only if" part of this problem is easier than the "if" part. For example, in the second part of Solution 2, the triangles EAQ and QBC are indirectly congruent, so the sum of their areas is 0, and $DCQE$ is a kite. Now one can easily see that $\angle(AQ, DE) = \angle(CD, CB)$ and $\angle(BQ, DC) = \angle(ED, EA)$, whence $\text{area}(BCD) = \text{area}(AQD) + \text{area}(EQA)$ and $\text{area}(ADE) = \text{area}(BDQ) + \text{area}(BQC)$, which yields the result.

Comment 3. The origin of the problem is the following observation. Let $ABDH$ be a tetrahedron and consider the sphere $\mathcal{S}$ that is tangent to the four face planes, internally to planes ADH and BDH and externally to ABD and ABH (or vice versa). It is known that the sphere $\mathcal{S}$ exists if and only if $\text{area}(ADH) + \text{area}(BDH) \neq \text{area}(ABH) + \text{area}(ABD)$; this relation comes from the usual formula for the volume of the tetrahedron.

Let T, T_a, T_b, T_d be the points of tangency between the sphere and the four planes, as shown in the picture. Rotate the triangle ABH inward, the triangles BDH and ADH outward, into the triangles ABP, BDC and ADE, respectively, in the plane ABD. Notice that the points T_d, T_a, T_b are rotated to T, so we have $HT_a = HT_b = HT_d = PT = CT = ET$. Therefore, the point T is the centre of the circle CEP. Hence, if the sphere exists then C, E, P cannot be collinear.

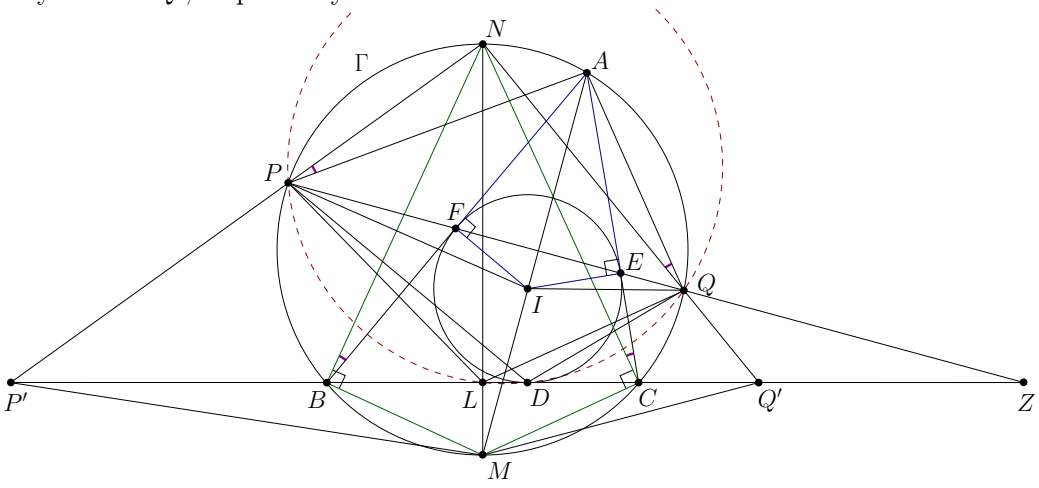
If the condition $\angle EDC \neq 2 \cdot \angle ADB$ is replaced by the constraint that the angles $\angle EDA$, $\angle ADB$ and $\angle BDC$ satisfy the triangle inequality, it enables reconstructing the argument with the tetrahedron and the tangent sphere.



G6. Let I be the incentre of acute-angled triangle ABC. Let the incircle meet BC, CA, and AB at D, E, and F, respectively. Let line EF intersect the circumcircle of the triangle at P and Q, such that F lies between E and P. Prove that $\angle DPA + \angle AQD = \angle QIP$.

(Slovakia)

Solution 1. Let N and M be the midpoints of the arcs $\overset{\frown}{BC}$ of the circumcircle, containing and opposite vertex A, respectively. By $\angle FAE = \angle BAC = \angle BNC$, the right-angled kites $AFIE$ and $NBMC$ are similar. Consider the spiral similarity φ (dilation in case of $AB = AC$) that moves $AFIE$ to $NBMC$. The directed angle in which φ changes directions is $\angle(AF, NB)$, same as $\angle(AP, NP)$ and $\angle(AQ, NQ)$; so lines AP and AQ are mapped to lines NP and NQ, respectively. Line EF is mapped to BC; we can see that the intersection points $P = EF \cap AP$ and $Q = EF \cap AQ$ are mapped to points $BC \cap NP$ and $BC \cap NQ$, respectively. Denote these points by P' and Q', respectively.



Let L be the midpoint of BC. We claim that points P, Q, D, L are concyclic (if $D = L$ then line BC is tangent to circle PQD). Let PQ and BC meet at Z. By applying Menelaus' theorem to triangle ABC and line EFZ, we have

$$\frac{BD}{DC} = \frac{BF}{FA} \cdot \frac{AE}{EC} = -\frac{BZ}{ZC},$$

so the pairs B, C and D, Z are harmonic. It is well-known that this implies $ZB \cdot ZC = ZD \cdot ZL$. (The inversion with pole Z that swaps B and C sends Z to infinity and D to the midpoint of BC, because the cross-ratio is preserved.) Hence, $ZD \cdot ZL = ZB \cdot ZC = ZP \cdot ZQ$ by the power of Z with respect to the circumcircle; this proves our claim.

By $\angle MPP' = \angle MQQ' = \angle MLP' = \angle MLQ' = 90^\circ$, the quadrilaterals $MLPP'$ and $MLQQ'$ are cyclic. Then the problem statement follows by

$$\begin{aligned}\angle DPA + \angle AQD &= 360^\circ - \angle PAQ - \angle QDP = 360^\circ - \angle PNQ - \angle QLP \\ &= \angle LPN + \angle NQL = \angle P'ML + \angle LMQ' = \angle P'MQ' = \angle PIQ.\end{aligned}$$

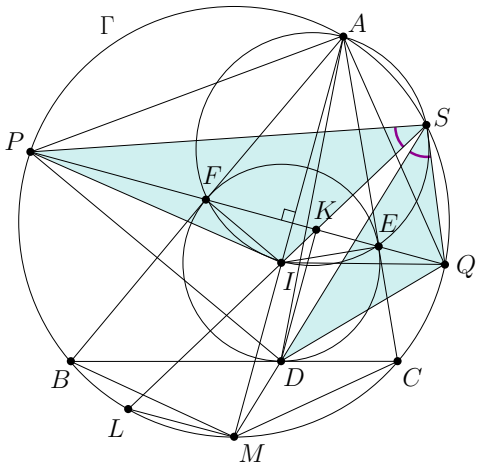
Solution 2. Define the point M and the same spiral similarity φ as in the previous solution. (The point N is not necessary.) It is well-known that the centre of the spiral similarity that maps F, E to B, C is the Miquel point of the lines FE, BC, BF and CE; that is, the second intersection of circles ABC and AEF. Denote that point by S.

By $\varphi(F) = B$ and $\varphi(E) = C$ the triangles SBF and SCE are similar, so we have

$$\frac{SB}{SC} = \frac{BF}{CE} = \frac{BD}{CD}.$$

By the converse of the angle bisector theorem, that indicates that line SD bisects $\angle BSC$ and hence passes through M.

Let K be the intersection point of lines EF and SI. Notice that φ sends points S, F, E, I to S, B, C, M, so $\varphi(K) = \varphi(FE \cap SI) = BC \cap SM = D$. By $\varphi(I) = M$, we have $KD \parallel IM$.



We claim that triangles SPI and SDQ are similar, and so are triangles SPD and SIQ. Let ray SI meet the circumcircle again at L. Note that the segment EF is perpendicular to the angle bisector AM. Then by $\angle AML = \angle ASL = \angle ASI = 90^\circ$, we have $ML \parallel PQ$. Hence, $\widehat{PL} = \widehat{MQ}$ and therefore $\angle PSL = \angle MSQ = \angle DSQ$. By $\angle QPS = \angle QMS$, the triangles SPK and SMQ are similar. Finally,

$$\frac{SP}{SI} = \frac{SP}{SK} \cdot \frac{SK}{SI} = \frac{SM}{SQ} \cdot \frac{SD}{SM} = \frac{SD}{SQ}$$

shows that triangles SPI and SDQ are similar. The second part of the claim can be proved analogously.

Now the problem statement can be proved by

$$\angle DPA + \angle AQD = \angle DPS + \angle SQD = \angle QIS + \angle SIP = \angle QIP.$$

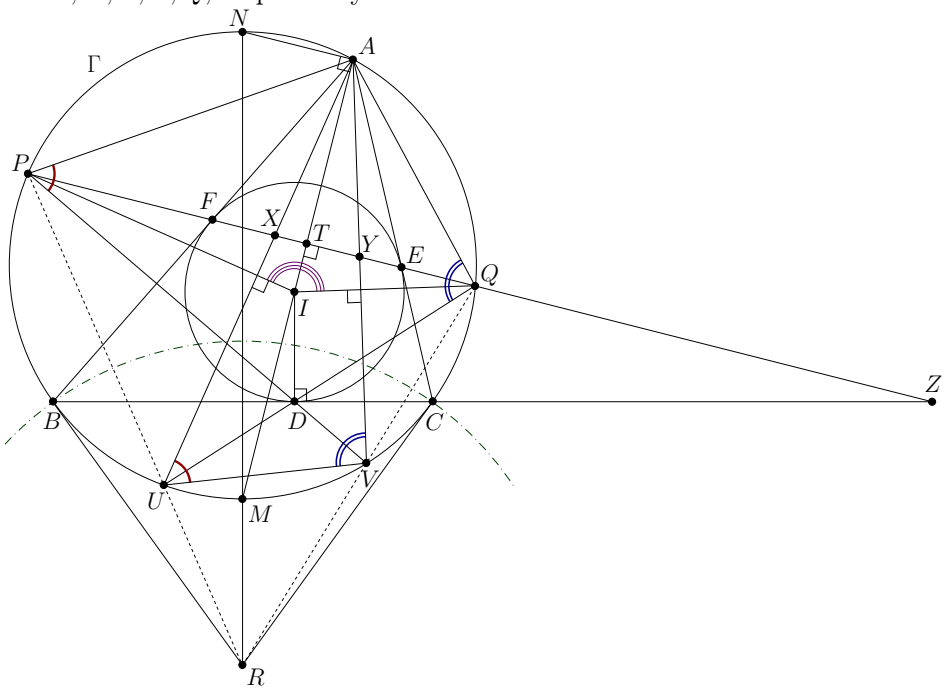
Solution 3. Denote the circumcircle of triangle ABC by Γ, and let rays PD and QD meet Γ again at V and U, respectively. We will show that $AU \perp IP$ and $AV \perp IQ$. Then the problem statement will follow as

$$\angle DPA + \angle AQD = \angle VUA + \angle AVU = 180^\circ - \angle UAV = \angle QIP.$$

Let M be the midpoint of arc $\overset{\frown}{BUVC}$ and let N be the midpoint of arc $\overset{\frown}{CAB}$; the lines AIM and AN being the internal and external bisectors of angle BAC, respectively, are perpendicular. Let the tangents drawn to Γ at B and C meet at R; let line PQ meet AU, AI, AV and BC at X, T, Y and Z, respectively.

As in Solution 1, we observe that the pairs B, C and D, Z are harmonic. Projecting these points from Q onto the circumcircle, we can see that B, C and U, P are also harmonic. Analogously, the pair V, Q is harmonic with B, C. Consider the inversion about the circle with centre R, passing through B and C. Points B and C are fixed points, so this inversion exchanges every point of Γ by its harmonic pair with respect to B, C. In particular, the inversion maps points B, C, N, U, V to points B, C, M, P, Q, respectively.

Combine the inversion with projecting Γ from A to line PQ; the points B, C, M, P, Q are projected to F, E, T, P, Q, respectively.



The combination of these two transformations is projective map from the lines AB, AC, AN, AU, AV to IF, IE, IT, IP, IQ, respectively. On the other hand, we have $AB \perp IF$, $AC \perp IE$ and $AN \perp AT$, so the corresponding lines in these two pencils are perpendicular. This proves $AU \perp IP$ and $AV \perp IQ$, and hence completes the solution.

G7. The incircle ω of acute-angled scalene triangle ABC has centre I and meets sides BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω again at R. Line AR meets ω again at P. The circumcircles of triangles PCE and PBF meet again at $Q \neq P$. Prove that lines DI and PQ meet on the external bisector of angle BAC.

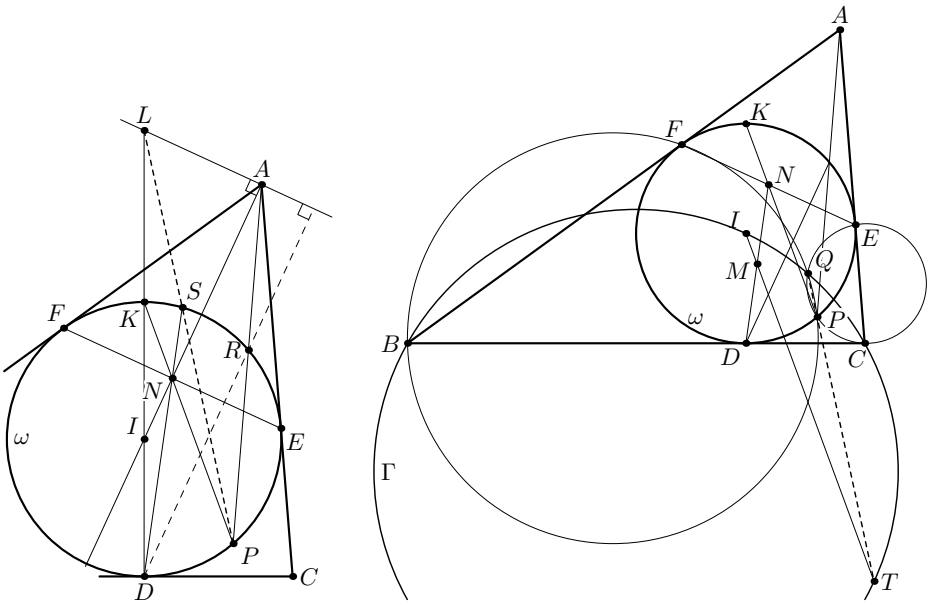
(India)

Common remarks. Throughout the solution, $\angle(a, b)$ denotes the directed angle between lines a and b, measured modulo π.

Solution 1.

Step 1. The external bisector of $\angle BAC$ is the line through A perpendicular to IA. Let DI meet this line at L and let DI meet ω at K. Let N be the midpoint of EF, which lies on IA and is the pole of line AL with respect to ω. Since $AN \cdot AI = AE^2 = AR \cdot AP$, the points R, N, I, and P are concyclic. As $IR = IP$, the line NI is the external bisector of $\angle PNR$, so PN meets ω again at the point symmetric to R with respect to AN – i.e. at K.

Let DN cross ω again at S. Opposite sides of any quadrilateral inscribed in the circle ω meet on the polar line of the intersection of the diagonals with respect to ω. Since L lies on the polar line AL of N with respect to ω, the line PS must pass through L. Thus it suffices to prove that the points S, Q, and P are collinear.

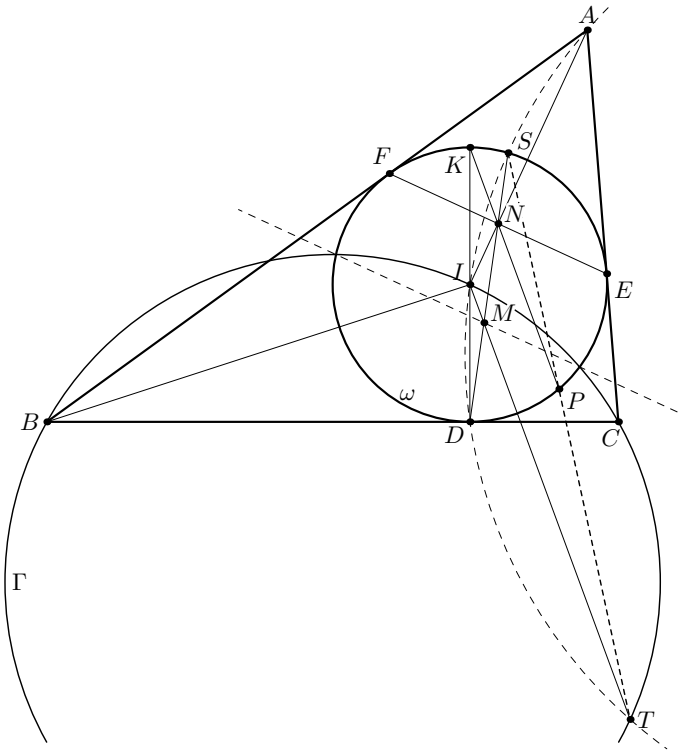


Step 2. Let Γ be the circumcircle of $\triangle BIC$. Notice that

$$\begin{aligned}\angle(BQ, QC) &= \angle(BQ, QP) + \angle(PQ, QC) = \angle(BF, FP) + \angle(PE, EC) \\ &= \angle(EF, EP) + \angle(FP, FE) = \angle(FP, EP) = \angle(DF, DE) = \angle(BI, IC),\end{aligned}$$

so Q lies on Γ. Let QP meet Γ again at T. It will now suffice to prove that S, P, and T are collinear. Notice that $\angle(BI, IT) = \angle(BQ, QT) = \angle(BF, FP) = \angle(FK, KP)$. Note $FD \perp FK$ and $FD \perp BI$ so $FK \parallel BI$ and hence IT is parallel to the line KNP. Since $DI = IK$, the line IT crosses DN at its midpoint M.

Step 3. Let F' and E' be the midpoints of DE and DF, respectively. Since $DE' \cdot E'F = DE'^2 = BE' \cdot E'I$, the point E' lies on the radical axis of ω and Γ; the same holds for F'. Therefore, this radical axis is $E'F'$, and it passes through M. Thus $IM \cdot MT = DM \cdot MS$, so S, I, D, and T are concyclic. This shows $\angle(DS, ST) = \angle(DI, IT) = \angle(DK, KP) = \angle(DS, SP)$, whence the points S, P, and T are collinear, as desired.



Comment. Here is a longer alternative proof in step 1 that P, S, and L are collinear, using a circular inversion instead of the fact that opposite sides of a quadrilateral inscribed in a circle ω meet on the polar line with respect to ω of the intersection of the diagonals. Let G be the foot of the altitude from N to the line $DIKL$. Observe that N, G, K, S are concyclic (opposite right angles) so

$$\angle DIP = 2\angle DKP = \angle GKN + \angle DSP = \angle GSN + \angle NSP = \angle GSP\,,$$

hence I, G, S, P are concyclic. We have $IG \cdot IL = IN \cdot IA = r^2$ since $\triangle IGN \sim \triangle IAL$. Inverting the circle $IGSP$ in circle ω, points P and S are fixed and G is taken to L so we find that P, S, and L are collinear.

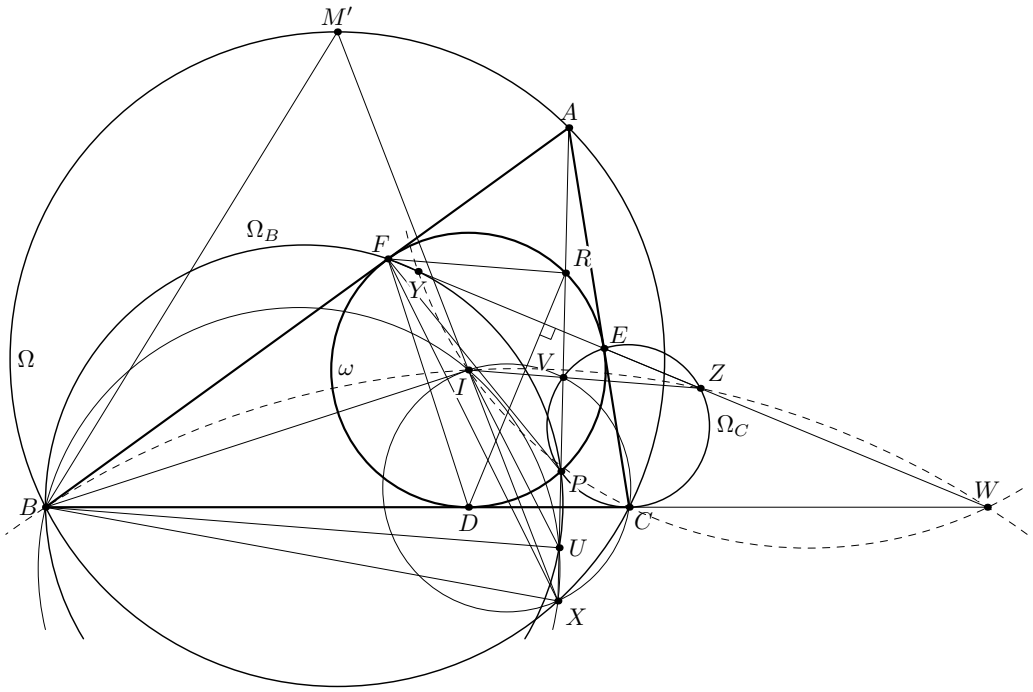
Solution 2. We start as in Solution 1. Namely, we introduce the same points K, L, N, and S, and show that the triples (P, N, K) and (P, S, L) are collinear. We conclude that K and R are symmetric in AI, and reduce the problem statement to showing that P, Q, and S are collinear.

Step 1. Let AR meet the circumcircle Ω of ABC again at X. The lines AR and AK are isogonal in the angle BAC; it is well known that in this case X is the tangency point of Ω with the A-mixtilinear circle. It is also well known that for this point X, the line XI crosses Ω again at the midpoint M' of arc BAC.

Step 2. Denote the circles BFP and CEP by Ω_B and Ω_C, respectively. Let Ω_B cross AR and EF again at U and Y, respectively. We have

$$\angle(UB, BF) = \angle(UP, PF) = \angle(RP, PF) = \angle(RF, FA),$$

so $UB \parallel RF$.



Next, we show that the points B, I, U, and X are concyclic. Since

$$\angle(UB, UX) = \angle(RF, RX) = \angle(AF, AR) + \angle(FR, FA) = \angle(M'B, M'X) + \angle(DR, DF),$$

it suffices to prove $\angle(IB, IX) = \angle(M'B, M'X) + \angle(DR, DF)$, or $\angle(IB, M'B) = \angle(DR, DF)$. But both angles equal $\angle(CI, CB)$, as desired. (This is where we used the fact that M' is the midpoint of arc BAC of Ω.)

It follows now from circles $BUIX$ and $BPUFY$ that

$$\angle(IU, UB) = \angle(IX, BX) = \angle(M'X, BX) = \frac{\pi - \angle A}{2} = \angle(EF, AF) = \angle(YF, BF) = \angle(YU, BU),$$

so the points Y, U, and I are collinear.

Let EF meet BC at W. We have

$$\angle(IY, YW) = \angle(UY, FY) = \angle(UB, FB) = \angle(RF, AF) = \angle(CI, CW),$$

so the points W, Y, I, and C are concyclic.

Similarly, if V and Z are the second meeting points of Ω_C with AR and EF, we get that the 4-tuples (C, V, I, X) and (B, I, Z, W) are both concyclic.

Step 3. Let $Q' = CY \cap BZ$. We will show that $Q' = Q$.

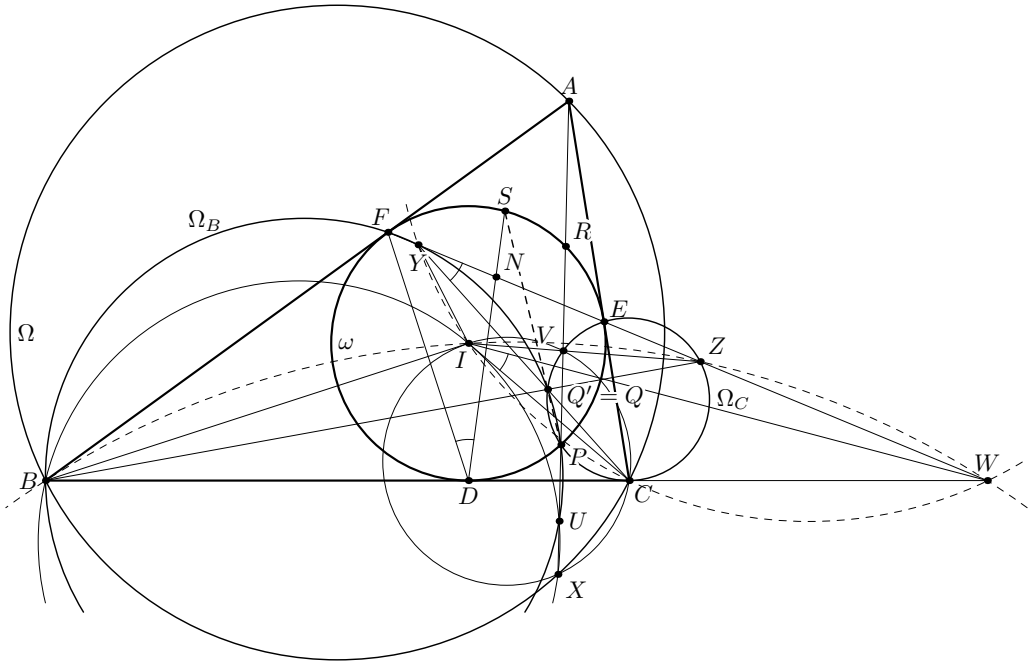
First of all, we have

$$\begin{aligned}\angle(Q'Y, Q'B) = \angle(CY, ZB) &= \angle(CY, ZY) + \angle(ZY, BZ) \\ &= \angle(CI, IW) + \angle(IW, IB) = \angle(CI, IB) = \frac{\pi - \angle A}{2} = \angle(FY, FB),\end{aligned}$$

so $Q' \in \Omega_B$. Similarly, $Q' \in \Omega_C$. Thus $Q' \in \Omega_B \cap \Omega_C = \{P, Q\}$ and it remains to prove that $Q' \neq P$. If we had $Q' = P$, we would have $\angle(PY, PZ) = \angle(Q'Y, Q'Z) = \angle(IC, IB)$. This would imply

$$\angle(PY, YF) + \angle(EZ, ZP) = \angle(PY, PZ) = \angle(IC, IB) = \angle(PE, PF),$$

so circles Ω_B and Ω_C would be tangent at P. That is excluded in the problem conditions, so $Q' = Q$.



Step 4. Now we are ready to show that P, Q, and S are collinear.

Notice that A and D are the poles of EW and DW with respect to ω, so W is the pole of AD. Hence, $WI \perp AD$. Since $CI \perp DE$, this yields $\angle(IC, WI) = \angle(DE, DA)$. On the other hand, DA is a symmedian in $\triangle DEF$, so $\angle(DE, DA) = \angle(DN, DF) = \angle(DS, DF)$. Therefore,

$$\begin{aligned}\angle(PS, PF) = \angle(DS, DF) = \angle(DE, DA) &= \angle(IC, IW) \\ &= \angle(YC, YW) = \angle(YQ, YF) = \angle(PQ, PF),\end{aligned}$$

which yields the desired collinearity.

G8. Let $\mathcal{L}$ be the set of all lines in the plane and let f be a function that assigns to each line $\ell \in \mathcal{L}$ a point $f(\ell)$ on ℓ. Suppose that for any point X, and for any three lines ℓ_1, ℓ_2, ℓ_3 passing through X, the points $f(\ell_1)$, $f(\ell_2)$, $f(\ell_3)$ and X lie on a circle.

Prove that there is a unique point P such that $f(\ell) = P$ for any line ℓ passing through P.

(Australia)

Common remarks. The condition on f is equivalent to the following: There is some function g that assigns to each point X a circle $g(X)$ passing through X such that for any line ℓ passing through X, the point $f(\ell)$ lies on $g(X)$. (The function g may not be uniquely defined for all points, if some points X have at most one value of $f(\ell)$ other than X; for such points, an arbitrary choice is made.)

If there were two points P and Q with the given property, $f(PQ)$ would have to be both P and Q, so there is at most one such point, and it will suffice to show that such a point exists.

Solution 1. We provide a complete characterisation of the functions satisfying the given condition.

Write $\angle(\ell_1, \ell_2)$ for the *directed* angle modulo 180° between the lines ℓ_1 and ℓ_2. Given a point P and an angle $\alpha \in (0, 180^\circ)$, for each line ℓ, let ℓ' be the line through P satisfying $\angle(\ell', \ell) = \alpha$, and let $h_{P,\alpha}(\ell)$ be the intersection point of ℓ and ℓ'. We will prove that there is some pair (P, α) such that f and $h_{P,\alpha}$ are the same function. Then P is the unique point in the problem statement.

Given an angle α and a point P, let a line ℓ be called (P, α)-*good* if $f(\ell) = h_{P,\alpha}(\ell)$. Let a point $X \neq P$ be called (P, α)-*good* if the circle $g(X)$ passes through P and some point $Y \neq P, X$ on $g(X)$ satisfies $\angle(PY, YX) = \alpha$. It follows from this definition that if X is (P, α)-good then every point $Y \neq P, X$ of $g(X)$ satisfies this angle condition, so $h_{P,\alpha}(XY) = Y$ for every $Y \in g(X)$. Equivalently, $f(\ell) \in \{X, h_{P,\alpha}(\ell)\}$ for each line ℓ passing through X. This shows the following lemma.

Lemma 1. If X is (P, α)-good and ℓ is a line passing through X then either $f(\ell) = X$ or ℓ is (P, α)-good.

Lemma 2. If X and Y are different (P, α)-good points, then line XY is (P, α)-good.

Proof. If XY is not (P, α)-good then by the previous Lemma, $f(XY) = X$ and similarly $f(XY) = Y$, but clearly this is impossible as $X \neq Y$. □

Lemma 3. If ℓ_1 and ℓ_2 are different (P, α)-good lines which intersect at $X \neq P$, then either $f(\ell_1) = X$ or $f(\ell_2) = X$ or X is (P, α)-good.

Proof. If $f(\ell_1), f(\ell_2) \neq X$, then $g(X)$ is the circumcircle of X, $f(\ell_1)$ and $f(\ell_2)$. Since ℓ_1 and ℓ_2 are (P, α)-good lines, the angles

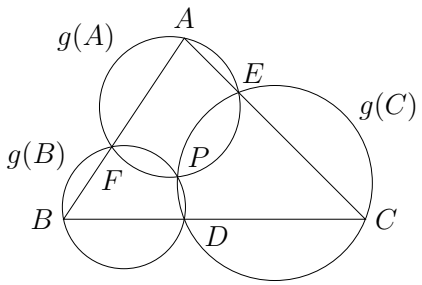
$$\angle(Pf(\ell_1), f(\ell_1)X) = \angle(Pf(\ell_2), f(\ell_2)X) = \alpha,$$

so P lies on $g(X)$. Hence, X is (P, α)-good. □

Lemma 4. If ℓ_1, ℓ_2 and ℓ_3 are different (P, α)-good lines which intersect at $X \neq P$, then X is (P, α)-good.

Proof. This follows from the previous Lemma since at most one of the three lines ℓ_i can satisfy $f(\ell_i) = X$ as the three lines are all (P, α)-good. □

Lemma 5. If ABC is a triangle such that A, B, C, $f(AB)$, $f(AC)$ and $f(BC)$ are all different points, then there is some point P and some angle α such that A, B and C are (P, α)-good points and AB, BC and CA are (P, α)-good lines.



Proof. Let D, E, F denote the points $f(BC)$, $f(AC)$, $f(AB)$, respectively. Then $g(A)$, $g(B)$ and $g(C)$ are the circumcircles of AEF, BDF and CDE, respectively. Let $P \neq F$ be the second intersection of circles $g(A)$ and $g(B)$ (or, if these circles are tangent at F, then $P = F$). By Miquel's theorem (or an easy angle chase), $g(C)$ also passes through P. Then by the cyclic quadrilaterals, the directed angles

$$\angle(PD, DC) = \angle(PF, FB) = \angle(PE, EA) = \alpha,$$

for some angle α. Hence, lines AB, BC and CA are all (P, α)-good, so by Lemma 3, A, B and C are (P, α)-good. (In the case where $P = D$, the line PD in the equation above denotes the line which is tangent to $g(B)$ at $P = D$. Similar definitions are used for PE and PF in the cases where $P = E$ or $P = F$.) □

Consider the set Ω of all points (x, y) with integer coordinates $1 \leqslant x, y \leqslant 1000$, and consider the set L_Ω of all horizontal, vertical and diagonal lines passing through at least one point in Ω. A simple counting argument shows that there are 5998 lines in L_Ω. For each line ℓ in L_Ω we colour the point $f(\ell)$ red. Then there are at most 5998 red points. Now we partition the points in Ω into 10000 ten by ten squares. Since there are at most 5998 red points, at least one of these squares Ω_{10} contains no red points. Let (m, n) be the bottom left point in Ω_{10}. Then the triangle with vertices (m, n), $(m + 1, n)$ and $(m, n + 1)$ satisfies the condition of Lemma 5, so these three vertices are all (P, α)-good for some point P and angle α, as are the lines joining them. From this point on, we will simply call a point or line *good* if it is (P, α)-good for this particular pair (P, α). Now by Lemma 1, the line $x = m + 1$ is good, as is the line $y = n + 1$. Then Lemma 3 implies that $(m+1, n+1)$ is good. By applying these two lemmas repeatedly, we can prove that the line $x+y = m+n+2$ is good, then the points $(m, n+2)$ and $(m+2, n)$ then the lines $x = m+2$ and $y = n+2$, then the points $(m+2, n+1)$, $(m+1, n+2)$ and $(m+2, n+2)$ and so on until we have prove that all points in Ω_{10} are good.

Now we will use this to prove that every point $S \neq P$ is good. Since $g(S)$ is a circle, it passes through at most two points of Ω_{10} on any vertical line, so at most 20 points in total. Moreover, any line ℓ through S intersects at most 10 points in Ω_{10}. Hence, there are at least eight lines ℓ through S which contain a point Q in Ω_{10} which is not on $g(S)$. Since Q is not on $g(S)$, the point $f(\ell) \neq Q$. Hence, by Lemma 1, the line ℓ is good. Hence, at least eight good lines pass through S, so by Lemma 4, the point S is good. Hence, every point $S \neq P$ is good, so by Lemma 2, every line is good. In particular, every line ℓ passing through P is good, and therefore satisfies $f(\ell) = P$, as required.

Solution 2. Note that for any distinct points X, Y, the circles $g(X)$ and $g(Y)$ meet on XY at the point $f(XY) \in g(X) \cap g(Y) \cap (XY)$. We write $s(X, Y)$ for the second intersection point of circles $g(X)$ and $g(Y)$.

Lemma 1. Suppose that X, Y and Z are not collinear, and that $f(XY) \notin \{X, Y\}$ and similarly for YZ and ZX. Then $s(X, Y) = s(Y, Z) = s(Z, X)$.

Proof. The circles $g(X)$, $g(Y)$ and $g(Z)$ through the vertices of triangle XYZ meet pairwise on the corresponding edges (produced). By Miquel's theorem, the second points of intersection of any two of the circles coincide. (See the diagram for Lemma 5 of Solution 1.) □

Now pick any line ℓ and any six different points $Y_1, \ldots, Y_6$ on $\ell \setminus \{f(\ell)\}$. Pick a point X not on ℓ or any of the circles $g(Y_i)$. Reordering the indices if necessary, we may suppose that $Y_1, \ldots, Y_4$ do not lie on $g(X)$, so that $f(XY_i) \notin \{X, Y_i\}$ for $1 \leqslant i \leqslant 4$. By applying the above lemma to triangles XY_iY_j for $1 \leqslant i < j \leqslant 4$, we find that the points $s(Y_i, Y_j)$ and $s(X, Y_i)$ are all equal, to point O say. Note that either O does not lie on ℓ, or $O = f(\ell)$, since $O \in g(Y_i)$.

Now consider an arbitrary point X' not on ℓ or any of the circles $g(Y_i)$ for $1 \leqslant i \leqslant 4$. As above, we see that there are two indices $1 \leqslant i < j \leqslant 4$ such that Y_i and Y_j do not lie on $g(X')$. By applying the above lemma to triangle $X'Y_iY_j$ we see that $s(X', Y_i) = O$, and in particular $g(X')$ passes through O.

We will now show that $f(\ell') = O$ for all lines ℓ' through O. By the above note, we may assume that $\ell' \neq \ell$. Consider a variable point $X' \in \ell' \setminus \{O\}$ not on ℓ or any of the circles $g(Y_i)$ for $1 \leqslant i \leqslant 4$. We know that $f(\ell') \in g(X') \cap \ell' = \{X', O\}$. Since X' was suitably arbitrary, we have $f(\ell') = O$ as desired.

Solution 3. Notice that, for any two different points X and Y, the point $f(XY)$ lies on both $g(X)$ and $g(Y)$, so any two such circles meet in at least one point. We refer to two circles as *cutting* only in the case where they cross, and so meet at exactly two points, thus excluding the cases where they are tangent or are the same circle.

Lemma 1. Suppose there is a point P such that all circles $g(X)$ pass through P. Then P has the given property.

Proof. Consider some line ℓ passing through P, and suppose that $f(\ell) \neq P$. Consider some $X \in \ell$ with $X \neq P$ and $X \neq f(\ell)$. Then $g(X)$ passes through all of P, $f(\ell)$ and X, but those three points are collinear, a contradiction. □

Lemma 2. Suppose that, for all $\epsilon > 0$, there is a point P_ϵ with $g(P_\epsilon)$ of radius at most ϵ. Then there is a point P with the given property.

Proof. Consider a sequence $\epsilon_i = 2^{-i}$ and corresponding points P_{ϵ_i}. Because the two circles $g(P_{\epsilon_i})$ and $g(P_{\epsilon_j})$ meet, the distance between P_{ϵ_i} and P_{ϵ_j} is at most $2^{1-i} + 2^{1-j}$. As $\sum_i \epsilon_i$ converges, these points converge to some point P. For all $\epsilon > 0$, the point P has distance at most 2ϵ from P_ϵ, and all circles $g(X)$ pass through a point with distance at most 2ϵ from P_ϵ, so distance at most 4ϵ from P. A circle that passes distance at most 4ϵ from P for all $\epsilon > 0$ must pass through P, so by Lemma 1 the point P has the given property. □

Lemma 3. Suppose no two of the circles $g(X)$ cut. Then there is a point P with the given property.

Proof. Consider a circle $g(X)$ with centre Y. The circle $g(Y)$ must meet $g(X)$ without cutting it, so has half the radius of $g(X)$. Repeating this argument, there are circles with arbitrarily small radius and the result follows by Lemma 2. □

Lemma 4. Suppose there are six different points A, B_1, B_2, B_3, B_4, B_5 such that no three are collinear, no four are concyclic, and all the circles $g(B_i)$ cut pairwise at A. Then there is a point P with the given property.

Proof. Consider some line ℓ through A that does not pass through any of the B_i and is not tangent to any of the $g(B_i)$. Fix some direction along that line, and let X_ϵ be the point on ℓ that has distance ϵ from A in that direction. In what follows we consider only those ϵ for which X_ϵ does not lie on any $g(B_i)$ (this restriction excludes only finitely many possible values of ϵ).

Consider the circle $g(X_\epsilon)$. Because no four of the B_i are concyclic, at most three of them lie on this circle, so at least two of them do not. There must be some sequence of $\epsilon \to 0$ such that it is the same two of the B_i for all ϵ in that sequence, so now restrict attention to that sequence, and suppose without loss of generality that B_1 and B_2 do not lie on $g(X_\epsilon)$ for any ϵ in that sequence.

Then $f(X_\epsilon B_1)$ is not B_1, so must be the other point of intersection of $X_\epsilon B_1$ with $g(B_1)$, and the same applies with B_2. Now consider the three points X_ϵ, $f(X_\epsilon B_1)$ and $f(X_\epsilon B_2)$. As $\epsilon \to 0$, the angle at X_ϵ tends to $\angle B_1AB_2$ or $180^\circ - \angle B_1AB_2$, which is not 0 or 180° because no three of the points were collinear. All three distances between those points are bounded above by constant multiples of ϵ (in fact, if the triangle is scaled by a factor of $1/\epsilon$, it tends to a fixed triangle). Thus the circumradius of those three points, which is the radius of $g(X_\epsilon)$, is also bounded above by a constant multiple of ϵ, and so the result follows by Lemma 2. □

Lemma 5. Suppose there are two points A and B such that $g(A)$ and $g(B)$ cut. Then there is a point P with the given property.

Proof. Suppose that $g(A)$ and $g(B)$ cut at C and D. One of those points, without loss of generality C, must be $f(AB)$, and so lie on the line AB. We now consider two cases, according to whether D also lies on that line.

Case 1: D does not lie on that line.

In this case, consider a sequence of X_ϵ at distance ϵ from D, tending to D along some line that is not a tangent to either circle, but perturbed slightly (by at most ϵ^2) to ensure that no three of the points A, B and X_ϵ are collinear and no four are concyclic.

Consider the points $f(X_\epsilon A)$ and $f(X_\epsilon B)$, and the circles $g(X_\epsilon)$ on which they lie. The point $f(X_\epsilon A)$ might be either A or the other intersection of $X_\epsilon A$ with the circle $g(A)$, and the same applies for B. If, for some sequence of $\epsilon \to 0$, both those points are the other point of intersection, the same argument as in the proof of Lemma 4 applies to find arbitrarily small circles. Otherwise, we have either infinitely many of those circles passing through A, or infinitely many passing through B; without loss of generality, suppose infinitely many through A.

We now show we can find five points B_i satisfying the conditions of Lemma 4 (together with A). Let B_1 be any of the X_ϵ for which $g(X_\epsilon)$ passes through A. Then repeat the following four times, for $2 \leqslant i \leqslant 5$.

Consider some line $\ell = X_\epsilon A$ (different from those considered for previous i) that is not tangent to any of the $g(B_j)$ for $j < i$, and is such that $f(\ell) = A$, so $g(Y)$ passes through A for all Y on that line. If there are arbitrarily small circles $g(Y)$ we are done by Lemma 2, so the radii of such circles must be bounded below. But as $Y \to A$, along any line not tangent to $g(B_j)$, the radius of a circle through Y and tangent to $g(B_j)$ at A tends to 0. So there must be some Y such that $g(Y)$ cuts $g(B_j)$ at A rather than being tangent to it there, for all of the previous B_j, and we may also pick it such that no three of the B_i and A are collinear and no four are concyclic. Let B_i be this Y. Now the result follows by Lemma 4.

Case 2: D does lie on that line.

In this case, we follow a similar argument, but the sequence of X_ϵ needs to be slightly different. C and D both lie on the line AB, so one must be A and the other must be B. Consider a sequence of X_ϵ tending to B. Rather than tending to B along a straight line (with small perturbations), let the sequence be such that all the points are inside the two circles, with the angle between $X_\epsilon B$ and the tangent to $g(B)$ at B tending to 0.

Again consider the points $f(X_\epsilon A)$ and $f(X_\epsilon B)$. If, for some sequence of $\epsilon \to 0$, both those points are the other point of intersection with the respective circles, we see that the angle at X_ϵ tends to the angle between AB and the tangent to $g(B)$ at B, which is not 0 or 180°, while the distances tend to 0 (although possibly slower than any multiple of ϵ), so we have arbitrarily small circumradii and the result follows by Lemma 2. Otherwise, we have either infinitely many of the circles $g(X_\epsilon)$ passing through A, or infinitely many passing through B, and the same argument as in the previous case enables us to reduce to Lemma 4. □

Lemmas 3 and 5 together cover all cases, and so the required result is proved.

Comment. From the property that all circles $g(X)$ pass through the same point P, it is possible to deduce that the function f has the form given in Solution 1. For any line ℓ not passing through P we may define a corresponding angle $\alpha(\ell)$, which we must show is the same for all such lines. For any point $X \neq P$, with at least one line ℓ through X and not through P, such that $f(\ell) \neq X$, this angle must be equal for all such lines through X (by (directed) angles in the same segment of $g(X)$).

Now consider all horizontal and all vertical lines not through P. For any pair consisting of a horizontal line ℓ_1 and a vertical line ℓ_2, we have $\alpha(\ell_1) = \alpha(\ell_2)$ unless $f(\ell_1)$ or $f(\ell_2)$ is the point of intersection of those lines. Consider the bipartite graph whose vertices are those lines and where an edge joins a horizontal and a vertical line with the same value of α. Considering a subgraph induced by n horizontal and n vertical lines, it must have at least $n^2 - 2n$ edges, so some horizontal line has edges to at least $n - 2$ of the vertical lines. Thus, in the original graph, all but at most two of the vertical lines have the same value of α, and likewise all but at most two of the horizontal lines have the same value of α, and, restricting attention to suitable subsets of those lines, we see that this value must be the same for the vertical lines and for the horizontal lines.

But now we can extend this to *all* vertical and horizontal lines not through P (and thus to lines in other directions as well, since the only requirement for 'vertical' and 'horizontal' above is that they are any two nonparallel directions). Consider any horizontal line ℓ_1 not passing through P, and we wish to show that $\alpha(\ell_1)$ has the same value α it has for all but at most two lines not through P in any direction. Indeed, we can deduce this by considering the intersection with any but at most five of the vertical lines: the only ones to exclude are the one passing through P, the one passing through $f(\ell_1)$, at most two such that $\alpha(\ell) \neq \alpha$, and the one passing through $h_{P,\alpha}(\ell_1)$ (defined as in Solution 1). So all lines ℓ not passing through P have the same value of $\alpha(\ell)$.

Solution 4. For any point X, denote by $t(X)$ the line tangent to $g(X)$ at X; notice that $f(t(X)) = X$, so f is surjective.

Step 1: We find a point P for which there are at least two different lines p_1 and p_2 such that $f(p_i) = P$.

Choose any point X. If X does not have this property, take any $Y \in g(X) \setminus \{X\}$; then $f(XY) = Y$. If Y does not have the property, $t(Y) = XY$, and the circles $g(X)$ and $g(Y)$ meet again at some point Z. Then $f(XZ) = Z = f(YZ)$, so Z has the required property.

We will show that P is the desired point. From now on, we fix two different lines p_1 and p_2 with $f(p_1) = f(p_2) = P$. Assume for contradiction that $f(\ell) = Q \neq P$ for some line ℓ through P. We fix ℓ, and note that $Q \in g(P)$.

Step 2: We prove that $P \in g(Q)$.

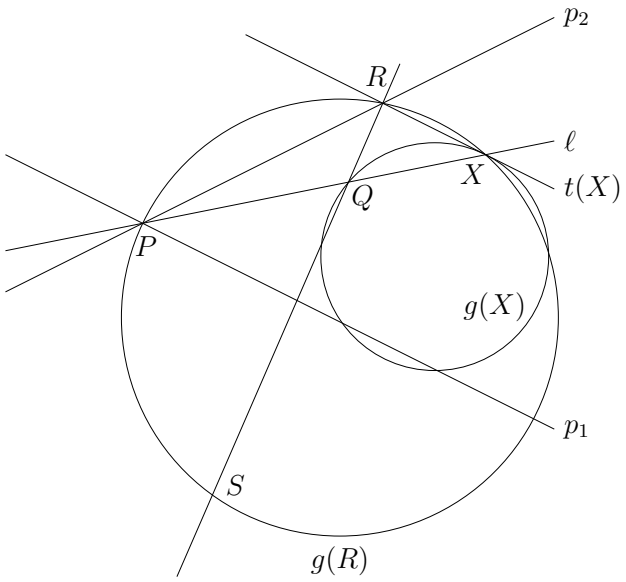
Take an arbitrary point $X \in \ell \setminus \{P, Q\}$. Two cases are possible for the position of $t(X)$ in relation to the p_i; we will show that each case (and subcase) occurs for only finitely many positions of X, yielding a contradiction.

Case 2.1: $t(X)$ is parallel to one of the p_i; say, to p_1.

Let $t(X)$ cross p_2 at R. Then $g(R)$ is the circle (PRX), as $f(RP) = P$ and $f(RX) = X$. Let RQ cross $g(R)$ again at S. Then $f(RQ) \in \{R, S\} \cap g(Q)$, so $g(Q)$ contains one of the points R and S.

If $R \in g(Q)$, then R is one of finitely many points in the intersection $g(Q) \cap p_2$, and each of them corresponds to a unique position of X, since RX is parallel to p_1.

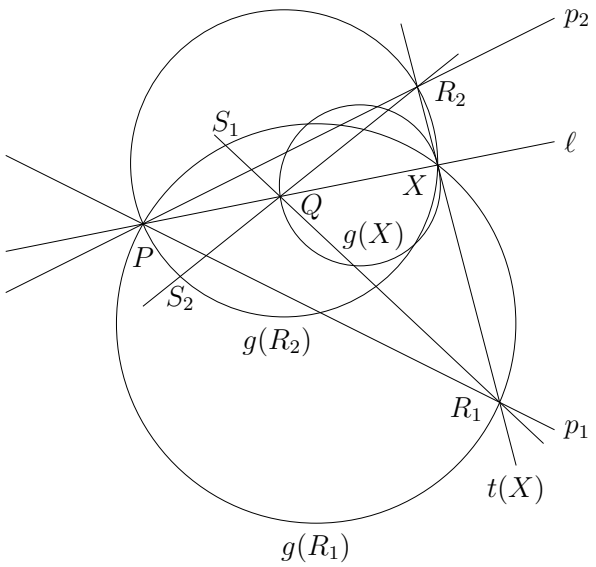
If $S \in g(Q)$, then $\angle(QS, SP) = \angle(RS, SP) = \angle(RX, XP) = \angle(p_1, \ell)$, so $\angle(QS, SP)$ is constant for all such points X, and all points S obtained in such a way lie on one circle γ passing through P and Q. Since $g(Q)$ does not contain P, it is different from γ, so there are only finitely many points S. Each of them uniquely determines R and thus X.



So, Case 2.1 can occur for only finitely many points X.

Case 2.2: $t(X)$ crosses p_1 and p_2 at R_1 and R_2, respectively.

Clearly, $R_1 \neq R_2$, as $t(X)$ is the tangent to $g(X)$ at X, and $g(X)$ meets ℓ only at X and Q. Notice that $g(R_i)$ is the circle (PXR_i). Let R_iQ meet $g(R_i)$ again at S_i; then $S_i \neq Q$, as $g(R_i)$ meets ℓ only at P and X. Then $f(R_iQ) \in \{R_i, S_i\}$, and we distinguish several subcases.



Subcase 2.2.1: $f(R_1Q) = S_1$, $f(R_2Q) = S_2$; so $S_1, S_2 \in g(Q)$.

In this case we have $0 = \angle(R_1X, XP) + \angle(XP, R_2X) = \angle(R_1S_1, S_1P) + \angle(S_2P, S_2R_2) = \angle(QS_1, S_1P) + \angle(S_2P, S_2Q)$, which shows $P \in g(Q)$.

Subcase 2.2.2: $f(R_1Q) = R_1$, $f(R_2Q) = R_2$; so $R_1, R_2 \in g(Q)$.

This can happen for at most four positions of X – namely, at the intersections of ℓ with a line of the form K_1K_2, where $K_i \in g(Q) \cap p_i$.

Subcase 2.2.3: $f(R_1Q) = S_1$, $f(R_2Q) = R_2$ *(the case* $f(R_1Q) = R_1$, $f(R_2Q) = S_2$ *is similar).*

In this case, there are at most two possible positions for R_2 – namely, the meeting points of $g(Q)$ with p_2. Consider one of them. Let X vary on ℓ. Then R_1 is the projection of X to p_1 via R_2, S_1 is the projection of R_1 to $g(Q)$ via Q. Finally, $\angle(QS_1, S_1X) = \angle(R_1S_1, S_1X) = \angle(R_1P, PX) = \angle(p_1, \ell) \neq 0$, so X is obtained by a fixed projective transform $g(Q) \to \ell$ from S_1. So, if there were three points X satisfying the conditions of this subcase, the composition of the three projective transforms would be the identity. But, if we apply it to $X = Q$, we successively get some point R_1', then R_2, and then some point different from Q, a contradiction.

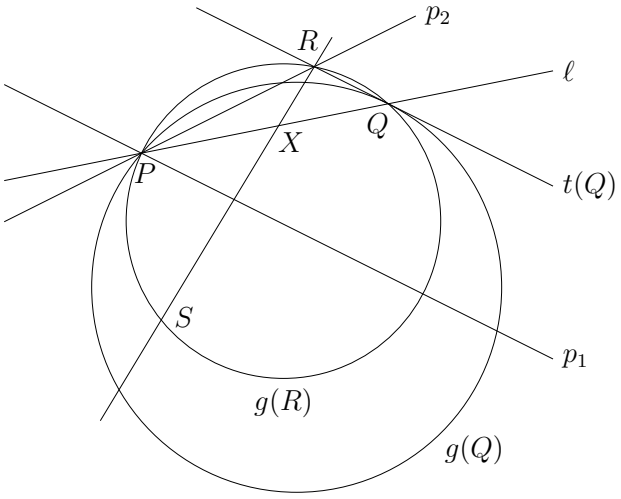
Thus Case 2.2 also occurs for only finitely many points X, as desired.

Step 3: We show that $f(PQ) = P$, *as desired.*

The argument is similar to that in Step 2, with the roles of Q and X swapped. Again, we show that there are only finitely many possible positions for a point $X \in \ell \setminus \{P, Q\}$, which is absurd.

Case 3.1: $t(Q)$ *is parallel to one of the* p_i*; say, to* p_1.

Let $t(Q)$ cross p_2 at R; then $g(R)$ is the circle (PRQ). Let RX cross $g(R)$ again at S. Then $f(RX) \in \{R, S\} \cap g(X)$, so $g(X)$ contains one of the points R and S.



Subcase 3.1.1: $S = f(RX) \in g(X)$.

We have $\angle(t(X), QX) = \angle(SX, SQ) = \angle(SR, SQ) = \angle(PR, PQ) = \angle(p_2, \ell)$. Hence $t(X) \parallel p_2$. Now we recall Case 2.1: we let $t(X)$ cross p_1 at R', so $g(R') = (PR'X)$, and let $R'Q$ meet $g(R')$ again at S'; notice that $S' \neq Q$. Excluding one position of X, we may assume that $R' \notin g(Q)$, so $R' \neq f(R'Q)$. Therefore, $S' = f(R'Q) \in g(Q)$. But then, as in Case 2.1, we get $\angle(t(Q), PQ) = \angle(QS', S'P) = \angle(R'X, XP) = \angle(p_2, \ell)$. This means that $t(Q)$ is parallel to p_2, which is impossible.

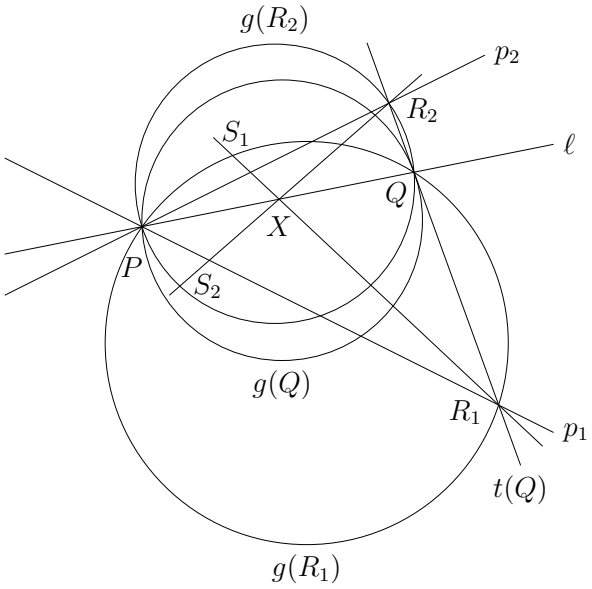
Subcase 3.1.2: $R = f(RX) \in g(X)$.

In this case, we have $\angle(t(X), \ell) = \angle(RX, RQ) = \angle(RX, p_1)$. Again, let $R' = t(X) \cap p_1$; this point exists for all but at most one position of X. Then $g(R') = (R'XP)$; let $R'Q$ meet $g(R')$ again at S'. Due to $\angle(R'X, XR) = \angle(QX, QR) = \angle(\ell, p_1)$, R' determines X in at most two ways, so for all but finitely many positions of X we have $R' \notin g(Q)$. Therefore, for those positions we have $S' = f(R'Q) \in g(Q)$. But then $\angle(RX, p_1) = \angle(R'X, XP) = \angle(R'S', S'P) = \angle(QS', S'P) = \angle(t(Q), QP)$ is fixed, so this case can hold only for one specific position of X as well.

Thus, in Case 3.1, there are only finitely many possible positions of X, yielding a contradiction.

Case 3.2: $t(Q)$ crosses p_1 and p_2 at R_1 and R_2, respectively.

By Step 2, $R_1 \neq R_2$. Notice that $g(R_i)$ is the circle (PQR_i). Let R_iX meet $g(R_i)$ at S_i; then $S_i \neq X$. Then $f(R_iX) \in \{R_i, S_i\}$, and we distinguish several subcases.



Subcase 3.2.1: $f(R_1X) = S_1$ and $f(R_2X) = S_2$, so $S_1, S_2 \in g(X)$.

As in Subcase 2.2.1, we have $0 = \angle(R_1Q, QP) + \angle(QP, R_2Q) = \angle(XS_1, S_1P) + \angle(S_2P, S_2X)$, which shows $P \in g(X)$. But $X, Q \in g(X)$ as well, so $g(X)$ meets ℓ at three distinct points, which is absurd.

Subcase 3.2.2: $f(R_1X) = R_1$, $f(R_2X) = R_2$, so $R_1, R_2 \in g(X)$.

Now three distinct collinear points R_1, R_2, and Q belong to $g(X)$, which is impossible.

Subcase 3.2.3: $f(R_1X) = S_1$, $f(R_2X) = R_2$ (the case $f(R_1X) = R_1$, $f(R_2X) = S_2$ is similar).

We have $\angle(XR_2, R_2Q) = \angle(XS_1, S_1Q) = \angle(R_1S_1, S_1Q) = \angle(R_1P, PQ) = \angle(p_1, \ell)$, so this case can occur for a unique position of X.

Thus, in Case 3.2, there is only a unique position of X, again yielding the required contradiction.

Number Theory

N1. Find all pairs (m, n) of positive integers satisfying the equation

$$(2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}) = m! \tag{1}$$

(El Salvador)

Answer: The only such pairs are $(1, 1)$ and $(3, 2)$.

Common remarks. In all solutions, for any prime p and positive integer N, we will denote by $v_p(N)$ the exponent of the largest power of p that divides N. The left-hand side of (1) will be denoted by L_n; that is, $L_n = (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1})$.

Solution 1. We will get an upper bound on n from the speed at which $v_2(L_n)$ grows.

From

$$L_n = (2^n - 1)(2^n - 2) \cdots (2^n - 2^{n-1}) = 2^{1+2+\cdots+(n-1)}(2^n - 1)(2^{n-1} - 1) \cdots (2^1 - 1)$$

we read

$$v_2(L_n) = 1 + 2 + \cdots + (n - 1) = \frac{n(n-1)}{2}.$$

On the other hand, $v_2(m!)$ is expressed by the Legendre formula as

$$v_2(m!) = \sum_{i=1}^{\infty} \left\lfloor \frac{m}{2^i} \right\rfloor.$$

As usual, by omitting the floor functions,

$$v_2(m!) < \sum_{i=1}^{\infty} \frac{m}{2^i} = m.$$

Thus, $L_n = m!$ implies the inequality

$$\frac{n(n-1)}{2} < m. \tag{2}$$

In order to obtain an opposite estimate, observe that

$$L_n = (2^n - 1)(2^n - 2) \cdots (2^n - 2^{n-1}) < (2^n)^n = 2^{n^2}.$$

We claim that

$$2^{n^2} < \left(\frac{n(n-1)}{2}\right)! \quad \text{for } n \geqslant 6. \tag{3}$$

For $n = 6$ the estimate (3) is true because $2^{6^2} < 6.9 \cdot 10^{10}$ and $\left(\frac{n(n-1)}{2}\right)! = 15! > 1.3 \cdot 10^{12}$.

For $n \geqslant 7$ we prove (3) by the following inequalities:

$$\left(\frac{n(n-1)}{2}\right)! = 15! \cdot 16 \cdot 17 \cdots \frac{n(n-1)}{2} > 2^{36} \cdot 16^{\frac{n(n-1)}{2} - 15}$$
$$= 2^{2n(n-1)-24} = 2^{n^2} \cdot 2^{n(n-2)-24} > 2^{n^2}.$$

Putting together (2) and (3), for $n \geqslant 6$ we get a contradiction, since

$$L_n < 2^{n^2} < \left(\frac{n(n-1)}{2}\right)! < m! = L_n.$$

Hence $n \geqslant 6$ is not possible.

Checking manually the cases $n \leqslant 5$ we find

$$L_1 = 1 = 1!, \qquad L_2 = 6 = 3!, \qquad 5! < L_3 = 168 < 6!,$$
$$7! < L_4 = 20\,160 < 8! \qquad \text{and} \qquad 10! < L_5 = 9\,999\,360 < 11!.$$

So, there are two solutions:

$$(m, n) \in \big\{(1, 1), (3, 2)\big\}.$$

Solution 2. Like in the previous solution, the cases $n = 1, 2, 3, 4$ are checked manually. We will exclude $n \geqslant 5$ by considering the exponents of 3 and 31 in (1).

For odd primes p and distinct integers a, b, coprime to p, with $p \mid a - b$, the Lifting The Exponent lemma asserts that

$$v_p(a^k - b^k) = v_p(a - b) + v_p(k).$$

Notice that 3 divides $2^k - 1$ if only if k is even; moreover, by the Lifting The Exponent lemma we have

$$v_3(2^{2k} - 1) = v_3(4^k - 1) = 1 + v_3(k) = v_3(3k).$$

Hence,

$$v_3(L_n) = \sum_{2k \leqslant n} v_3(4^k - 1) = \sum_{k \leqslant \lfloor \frac{n}{2} \rfloor} v_3(3k).$$

Notice that the last expression is precisely the exponent of 3 in the prime factorisation of $\big(3\lfloor \frac{n}{2} \rfloor\big)!$. Therefore

$$v_3(m!) = v_3(L_n) = v_3\left(\left(3\left\lfloor \frac{n}{2} \right\rfloor\right)!\right)$$
$$3\left\lfloor \frac{n}{2} \right\rfloor \leqslant m \leqslant 3\left\lfloor \frac{n}{2} \right\rfloor + 2. \tag{4}$$

Suppose that $n \geqslant 5$. Note that every fifth factor in L_n is divisible by $31 = 2^5 - 1$, and hence we have $v_{31}(L_n) \geqslant \lfloor \frac{n}{5} \rfloor$. Then

$$\frac{n}{10} \leqslant \left\lfloor \frac{n}{5} \right\rfloor \leqslant v_{31}(L_n) = v_{31}(m!) = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{31^k} \right\rfloor < \sum_{k=1}^{\infty} \frac{m}{31^k} = \frac{m}{30}. \tag{5}$$

By combining (4) and (5),

$$3n < m \leqslant \frac{3n}{2} + 2$$

so $n < \frac{4}{3}$ which is inconsistent with the inequality $n \geqslant 5$.

Comment 1. There are many combinations of the ideas above; for example combining (2) and (4) also provides $n < 5$. Obviously, considering the exponents of any two primes in (1), or considering one prime and the magnitude orders lead to an upper bound on n and m.

Comment 2. This problem has a connection to group theory. Indeed, the left-hand side is the order of the group $GL_n(\mathbb{F}_2)$ of invertible n-by-n matrices with entries modulo 2, while the right-hand side is the order of the symmetric group S_m on m elements. The result thus shows that the only possible isomorphisms between these groups are $GL_1(\mathbb{F}_2) \cong S_1$ and $GL_2(\mathbb{F}_2) \cong S_3$, and there are in fact isomorphisms in both cases. In general, $GL_n(\mathbb{F}_2)$ is a simple group for $n \geqslant 3$, as it is isomorphic to $PSL_n(\mathbb{F}_2)$.

There is also a near-solution of interest: the left-hand side for $n = 4$ is *half* of the right-hand side when $m = 8$; this turns out to correspond to an isomorphism $GL_4(\mathbb{F}_2) \cong A_8$ with the alternating group on eight elements.

However, while this indicates that the problem is a useful one, knowing group theory is of no use in solving it!

N2. Find all triples (a, b, c) of positive integers such that $a^3 + b^3 + c^3 = (abc)^2$.

(Nigeria)

Answer: The solutions are $(1, 2, 3)$ and its permutations.

Common remarks. Note that the equation is symmetric. In all solutions, we will assume without loss of generality that $a \geqslant b \geqslant c$, and prove that the only solution is $(a, b, c) = (3, 2, 1)$.

The first two solutions all start by proving that $c = 1$.

Solution 1. We will start by proving that $c = 1$. Note that

$$3a^3 \geqslant a^3 + b^3 + c^3 > a^3 .$$

So $3a^3 \geqslant (abc)^2 > a^3$ and hence $3a \geqslant b^2c^2 > a$. Now $b^3 + c^3 = a^2(b^2c^2 - a) \geqslant a^2$, and so

$$18b^3 \geqslant 9(b^3 + c^3) \geqslant 9a^2 \geqslant b^4c^4 \geqslant b^3c^5 ,$$

so $18 \geqslant c^5$ which yields $c = 1$.

Now, note that we must have $a > b$, as otherwise we would have $2b^3 + 1 = b^4$ which has no positive integer solutions. So

$$a^3 - b^3 \geqslant (b + 1)^3 - b^3 > 1$$

and

$$2a^3 > 1 + a^3 + b^3 > a^3 ,$$

which implies $2a^3 > a^2b^2 > a^3$ and so $2a > b^2 > a$. Therefore

$$4(1 + b^3) = 4a^2(b^2 - a) \geqslant 4a^2 > b^4 ,$$

so $4 > b^3(b - 4)$; that is, $b \leqslant 4$.

Now, for each possible value of b with $2 \leqslant b \leqslant 4$ we obtain a cubic equation for a with constant coefficients. These are as follows:

$$\begin{aligned} b = 2 &: \quad a^3 - 4a^2 + 9 = 0 \\ b = 3 &: \quad a^3 - 9a^2 + 28 = 0 \\ b = 4 &: \quad a^3 - 16a^2 + 65 = 0. \end{aligned}$$

The only case with an integer solution for a with $b \leqslant a$ is $b = 2$, leading to $(a, b, c) = (3, 2, 1)$.

Comment 1.1. Instead of writing down each cubic equation explicitly, we could have just observed that $a^2 \mid b^3 + 1$, and for each choice of b checked each square factor of $b^3 + 1$ for a^2.

We could also have observed that, with $c = 1$, the relation $18b^3 \geqslant b^4c^4$ becomes $b \leqslant 18$, and we can simply check all possibilities for b (instead of working to prove that $b \leqslant 4$). This check becomes easier after using the factorisation $b^3 + 1 = (b + 1)(b^2 - b + 1)$ and observing that no prime besides 3 can divide both of the factors.

Comment 1.2. Another approach to finish the problem after establishing that $c \leqslant 1$ is to set $k = b^2c^2 - a$, which is clearly an integer and must be positive as it is equal to $(b^3 + c^3)/a^2$. Then we divide into cases based on whether $k = 1$ or $k \geqslant 2$; in the first case, we have $b^3 + 1 = a^2 = (b^2 - 1)^2$ whose only positive root is $b = 2$, and in the second case we have $b^2 \leqslant 3a$, and so

$$b^4 \leqslant (3a)^2 \leqslant \frac{9}{2}(ka^2) = \frac{9}{2}(b^3 + 1),$$

which implies that $b \leqslant 4$.

Solution 2. Again, we will start by proving that $c = 1$. Suppose otherwise that $c \geqslant 2$. We have $a^3 + b^3 + c^3 \leqslant 3a^3$, so $b^2c^2 \leqslant 3a$. Since $c \geqslant 2$, this tells us that $b \leqslant \sqrt{3a/4}$. As the right-hand side of the original equation is a multiple of a^2, we have $a^2 \leqslant 2b^3 \leqslant 2(3a/4)^{3/2}$. In other words, $a \leqslant \frac{27}{16} < 2$, which contradicts the assertion that $a \geqslant c \geqslant 2$. So there are no solutions in this case, and so we must have $c = 1$.

Now, the original equation becomes $a^3 + b^3 + 1 = a^2b^2$. Observe that $a \geqslant 2$, since otherwise $a = b = 1$ as $a \geqslant b$.

The right-hand side is a multiple of a^2, so the left-hand side must be as well. Thus, $b^3 + 1 \geqslant a^2$. Since $a \geqslant b$, we also have

$$b^2 = a + \frac{b^3 + 1}{a^2} \leqslant 2a + \frac{1}{a^2}$$

and so $b^2 \leqslant 2a$ since b^2 is an integer. Thus $(2a)^{3/2} + 1 \geqslant b^3 + 1 \geqslant a^2$, from which we deduce $a \leqslant 8$.

Now, for each possible value of a with $2 \leqslant a \leqslant 8$ we obtain a cubic equation for b with constant coefficients. These are as follows:

$$\begin{aligned}
a = 2 &: \quad b^3 - 4b^2 + 9 = 0 \\
a = 3 &: \quad b^3 - 9b^2 + 28 = 0 \\
a = 4 &: \quad b^3 - 16b^2 + 65 = 0 \\
a = 5 &: \quad b^3 - 25b^2 + 126 = 0 \\
a = 6 &: \quad b^3 - 36b^2 + 217 = 0 \\
a = 7 &: \quad b^3 - 49b^2 + 344 = 0 \\
a = 8 &: \quad b^3 - 64b^2 + 513 = 0.
\end{aligned}$$

The only case with an integer solution for b with $a \geqslant b$ is $a = 3$, leading to $(a, b, c) = (3, 2, 1)$.

Comment 2.1. As in Solution 1, instead of writing down each cubic equation explicitly, we could have just observed that $b^2 \mid a^3 + 1$, and for each choice of a checked each square factor of $a^3 + 1$ for b^2.

Comment 2.2. This solution does not require initially proving that $c = 1$, in which case the bound would become $a \leqslant 108$. The resulting cases could, in principle, be checked by a particularly industrious student.

Solution 3. Set $k = (b^3 + c^3)/a^2 \leqslant 2a$, and rewrite the original equation as $a + k = (bc)^2$. Since b^3 and c^3 are positive integers, we have $(bc)^3 \geqslant b^3 + c^3 - 1 = ka^2 - 1$, so

$$a + k \geqslant (ka^2 - 1)^{2/3}.$$

As in Comment 1.2, k is a positive integer; for each value of $k \geqslant 1$, this gives us a polynomial inequality satisfied by a:

$$k^2a^4 - a^3 - 5ka^2 - 3k^2a - (k^3 - 1) \leqslant 0.$$

We now prove that $a \leqslant 3$. Indeed,

$$0 \geqslant \frac{k^2a^4 - a^3 - 5ka^2 - 3k^2a - (k^3 - 1)}{k^2} \geqslant a^4 - a^3 - 5a^2 - 3a - k \geqslant a^4 - a^3 - 5a^2 - 5a,$$

which fails when $a \geqslant 4$.

This leaves ten triples with $3 \geqslant a \geqslant b \geqslant c \geqslant 1$, which may be checked manually to give $(a, b, c) = (3, 2, 1)$.

Solution 4. Again, observe that $b^3 + c^3 = a^2(b^2c^2 - a)$, so $b \leqslant a \leqslant b^2c^2 - 1$.

We consider the function $f(x) = x^2(b^2c^2 - x)$. It can be seen that that on the interval $[0, b^2c^2 - 1]$ the function f is increasing if $x < \frac{2}{3}b^2c^2$ and decreasing if $x > \frac{2}{3}b^2c^2$. Consequently, it must be the case that

$$b^3 + c^3 = f(a) \geqslant \min\Big(f(b), f(b^2c^2 - 1)\Big).$$

First, suppose that $b^3 + c^3 \geqslant f(b^2c^2 - 1)$. This may be written $b^3 + c^3 \geqslant (b^2c^2 - 1)^2$, and so

$$2b^3 \geqslant b^3 + c^3 \geqslant (b^2c^2 - 1)^2 > b^4c^4 - 2b^2c^2 \geqslant b^4c^4 - 2b^3c^4.$$

Thus, $(b - 2)c^4 < 2$, and the only solutions to this inequality have $(b, c) = (2, 2)$ or $b \leqslant 3$ and $c = 1$. It is easy to verify that the only case giving a solution for $a \geqslant b$ is $(a, b, c) = (3, 2, 1)$.

Otherwise, suppose that $b^3 + c^3 = f(a) \geqslant f(b)$. Then, we have

$$2b^3 \geqslant b^3 + c^3 = a^2(b^2c^2 - a) \geqslant b^2(b^2c^2 - b).$$

Consequently $bc^2 \leqslant 3$, with strict inequality in the case that $b \neq c$. Hence $c = 1$ and $b \leqslant 2$. Both of these cases have been considered already, so we are done.

Comment 4.1. Instead of considering which of $f(b)$ and $f(b^2c^2 - 1)$ is less than $f(a)$, we may also proceed by explicitly dividing into cases based on whether $a \geqslant \frac{2}{3}b^2c^2$ or $a < \frac{2}{3}b^2c^2$. The first case may now be dealt with as follows. We have $b^3c^3 + 1 \geqslant b^3 + c^3$ as b^3 and c^3 are positive integers, so we have

$$b^3c^3 + 1 \geqslant b^3 + c^3 \geqslant a^2 \geqslant \frac{4}{9}b^4c^4.$$

This implies $bc \leqslant 2$, and hence $c = 1$ and $b \leqslant 2$.

N3. We say that a set S of integers is *rootiful* if, for any positive integer n and any $a_0, a_1, \ldots, a_n \in S$, all integer roots of the polynomial $a_0 + a_1x + \cdots + a_nx^n$ are also in S. Find all rootiful sets of integers that contain all numbers of the form $2^a - 2^b$ for positive integers a and b.

(Czech Republic)

Answer: The set $\mathbb{Z}$ of all integers is the only such rootiful set.

Solution 1. The set $\mathbb{Z}$ of all integers is clearly rootiful. We shall prove that any rootiful set S containing all the numbers of the form $2^a - 2^b$ for $a, b \in \mathbb{Z}_{>0}$ must be all of $\mathbb{Z}$.

First, note that $0 = 2^1 - 2^1 \in S$ and $2 = 2^2 - 2^1 \in S$. Now, $-1 \in S$, since it is a root of $2x + 2$, and $1 \in S$, since it is a root of $2x^2 - x - 1$. Also, if $n \in S$ then $-n$ is a root of $x + n$, so it suffices to prove that all positive integers must be in S.

Now, we claim that any positive integer n has a multiple in S. Indeed, suppose that $n = 2^\alpha \cdot t$ for $\alpha \in \mathbb{Z}_{\geqslant 0}$ and t odd. Then $t \mid 2^{\phi(t)} - 1$, so $n \mid 2^{\alpha+\phi(t)+1} - 2^{\alpha+1}$. Moreover, $2^{\alpha+\phi(t)+1} - 2^{\alpha+1} \in S$, and so S contains a multiple of every positive integer n.

We will now prove by induction that all positive integers are in S. Suppose that $0, 1, \ldots, n-1 \in S$; furthermore, let N be a multiple of n in S. Consider the base-n expansion of N, say $N = a_kn^k + a_{k-1}n^{k-1} + \cdots + a_1n + a_0$. Since $0 \leqslant a_i < n$ for each a_i, we have that all the a_i are in S. Furthermore, $a_0 = 0$ since N is a multiple of n. Therefore, $a_kn^k + a_{k-1}n^{k-1} + \cdots + a_1n - N = 0$, so n is a root of a polynomial with coefficients in S. This tells us that $n \in S$, completing the induction.

Solution 2. As in the previous solution, we can prove that $0, 1$ and -1 must all be in any rootiful set S containing all numbers of the form $2^a - 2^b$ for $a, b \in \mathbb{Z}_{>0}$.

We show that, in fact, every integer k with $|k| > 2$ can be expressed as a root of a polynomial whose coefficients are of the form $2^a - 2^b$. Observe that it suffices to consider the case where k is positive, as if k is a root of $a_nx^n + \cdots + a_1x + a_0 = 0$, then $-k$ is a root of $(-1)^n a_nx^n + \cdots - a_1x + a_0 = 0$.

Note that

$$(2^{a_n} - 2^{b_n})k^n + \cdots + (2^{a_0} - 2^{b_0}) = 0$$

is equivalent to

$$2^{a_n}k^n + \cdots + 2^{a_0} = 2^{b_n}k^n + \cdots + 2^{b_0}.$$

Hence our aim is to show that two numbers of the form $2^{a_n}k^n + \cdots + 2^{a_0}$ are equal, for a fixed value of n. We consider such polynomials where every term $2^{a_i}k^i$ is at most $2k^n$; in other words, where $2 \leqslant 2^{a_i} \leqslant 2k^{n-i}$, or, equivalently, $1 \leqslant a_i \leqslant 1 + (n-i)\log_2 k$. Therefore, there must be $1 + \lfloor (n-i)\log_2 k \rfloor$ possible choices for a_i satisfying these constraints.

The number of possible polynomials is then

$$\prod_{i=0}^{n} (1 + \lfloor (n-i)\log_2 k \rfloor) \geqslant \prod_{i=0}^{n-1} (n-i)\log_2 k = n!(\log_2 k)^n$$

where the inequality holds as $1 + \lfloor x \rfloor \geqslant x$.

As there are $(n+1)$ such terms in the polynomial, each at most $2k^n$, such a polynomial must have value at most $2k^n(n+1)$. However, for large n, we have $n!(\log_2 k)^n > 2k^n(n+1)$. Therefore there are more polynomials than possible values, so some two must be equal, as required.

N4. Let $\mathbb{Z}_{>0}$ be the set of positive integers. A positive integer constant C is given. Find all functions $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that, for all positive integers a and b satisfying $a + b > C$,

$$a + f(b) \mid a^2 + b\, f(a). \tag{$*$}$$

(Croatia)

Answer: The functions satisfying $(*)$ are exactly the functions $f(a) = ka$ for some constant $k \in \mathbb{Z}_{>0}$ (irrespective of the value of C).

Common remarks. It is easy to verify that the functions $f(a) = ka$ satisfy $(*)$. Thus, in the proofs below, we will only focus on the converse implication: that condition $(*)$ implies that $f = ka$.

A common minor part of these solutions is the derivation of some relatively easy bounds on the function f. An upper bound is easily obtained by setting $a = 1$ in $(*)$, giving the inequality

$$f(b) \leqslant b \cdot f(1)$$

for all sufficiently large b. The corresponding lower bound is only marginally more difficult to obtain: substituting $b = 1$ in the original equation shows that

$$a + f(1) \mid (a^2 + f(a)) - (a - f(1)) \cdot (a + f(1)) = f(1)^2 + f(a)$$

for all sufficiently large a. It follows from this that one has the lower bound

$$f(a) \geqslant a + f(1) \cdot (1 - f(1)),$$

again for all sufficiently large a.

Each of the following proofs makes use of at least one of these bounds.

Solution 1. First, we show that $b \mid f(b)^2$ for all b. To do this, we choose a large positive integer n so that $nb - f(b) \geqslant C$. Setting $a = nb - f(b)$ in $(*)$ then shows that

$$nb \mid (nb - f(b))^2 + bf(nb - f(b))$$

so that $b \mid f(b)^2$ as claimed.

Now in particular we have that $p \mid f(p)$ for every prime p. If we write $f(p) = k(p) \cdot p$, then the bound $f(p) \leqslant f(1) \cdot p$ (valid for p sufficiently large) shows that some value k of $k(p)$ must be attained for infinitely many p. We will show that $f(a) = ka$ for all positive integers a. To do this, we substitute $b = p$ in $(*)$, where p is any sufficiently large prime for which $k(p) = k$, obtaining

$$a + kp \mid (a^2 + pf(a)) - a(a + kp) = pf(a) - pka.$$

For suitably large p we have $\gcd(a + kp, p) = 1$, and hence we have

$$a + kp \mid f(a) - ka.$$

But the only way this can hold for arbitrarily large p is if $f(a) - ka = 0$. This concludes the proof.

Comment. There are other ways to obtain the divisibility $p \mid f(p)$ for primes p, which is all that is needed in this proof. For instance, if $f(p)$ were not divisible by p then the arithmetic progression $p^2 + bf(p)$ would attain prime values for infinitely many b by Dirichlet's Theorem: hence, for these pairs p, b, we would have $p + f(b) = p^2 + bf(p)$. Substituting $a \mapsto b$ and $b \mapsto p$ in $(*)$ then shows that $(f(p)^2 - p^2)(p - 1)$ is divisible by $b + f(p)$ and hence vanishes, which is impossible since $p \nmid f(p)$ by assumption.

Solution 2. First, we substitute $b = 1$ in (∗) and rearrange to find that

$$\frac{f(a) + f(1)^2}{a + f(1)} = f(1) - a + \frac{a^2 + f(a)}{a + f(1)}$$

is a positive integer for sufficiently large a. Since $f(a) \leqslant af(1)$, for all sufficiently large a, it follows that $\frac{f(a)+f(1)^2}{a+f(1)} \leqslant f(1)$ also and hence there is a positive integer k such that $\frac{f(a)+f(1)^2}{a+f(1)} = k$ for infinitely many values of a. In other words,

$$f(a) = ka + f(1) \cdot (k - f(1))$$

for infinitely many a.

Fixing an arbitrary choice of a in (∗), we have that

$$\frac{a^2 + bf(a)}{a + kb + f(1) \cdot (k - f(1))}$$

is an integer for infinitely many b (the same b as above, maybe with finitely many exceptions). On the other hand, for b taken sufficiently large, this quantity becomes arbitrarily close to $\frac{f(a)}{k}$; this is only possible if $\frac{f(a)}{k}$ is an integer and

$$\frac{a^2 + bf(a)}{a + kb + f(1) \cdot (k - f(1))} = \frac{f(a)}{k}$$

for infinitely many b. This rearranges to

$$\frac{f(a)}{k} \cdot \big(a + f(1) \cdot (k - f(1))\big) = a^2. \tag{∗∗}$$

Hence a^2 is divisible by $a + f(1) \cdot (k - f(1))$, and hence so is $f(1)^2(k - f(1))^2$. The only way this can occur for all a is if $k = f(1)$, in which case (∗∗) provides that $f(a) = ka$ for all a, as desired.

Solution 3. Fix any two distinct positive integers a and b. From (∗) it follows that the two integers

$$(a^2 + cf(a)) \cdot (b + f(c)) \text{ and } (b^2 + cf(b)) \cdot (a + f(c))$$

are both multiples of $(a + f(c)) \cdot (b + f(c))$ for all sufficiently large c. Taking an appropriate linear combination to eliminate the $cf(c)$ term, we find after expanding out that the integer

$$\big[a^2 f(b) - b^2 f(a)\big] \cdot f(c) + \big[(b - a)f(a)f(b)\big] \cdot c + \big[ab(af(b) - bf(a))\big] \tag{†}$$

is also a multiple of $(a + f(c)) \cdot (b + f(c))$.

But as c varies, (†) is bounded above by a positive multiple of c while $(a + f(c)) \cdot (b + f(c))$ is bounded below by a positive multiple of c^2. The only way that such a divisibility can hold is if in fact

$$\big[a^2 f(b) - b^2 f(a)\big] \cdot f(c) + \big[(b - a)f(a)f(b)\big] \cdot c + \big[ab(af(b) - bf(a))\big] = 0 \tag{††}$$

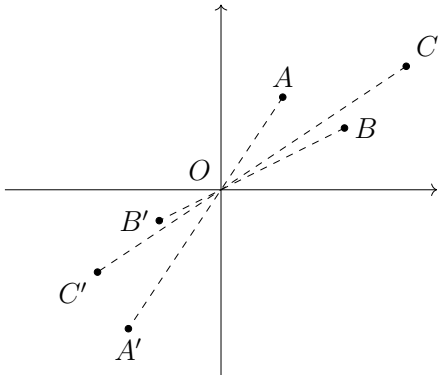
for sufficiently large c. Since the coefficient of c in this linear relation is nonzero, it follows that there are constants k, ℓ such that $f(c) = kc + \ell$ for all sufficiently large c; the constants k and ℓ are necessarily integers.

The value of ℓ satisfies

$$\big[a^2 f(b) - b^2 f(a)\big] \cdot \ell + \big[ab(af(b) - bf(a))\big] = 0 \tag{†††}$$

and hence $b \mid \ell a^2 f(b)$ for all a and b. Taking b sufficiently large so that $f(b) = kb + \ell$, we thus have that $b \mid \ell^2 a^2$ for all sufficiently large b; this implies that $\ell = 0$. From (†††) it then follows that $\frac{f(a)}{a} = \frac{f(b)}{b}$ for all $a \neq b$, so that there is a constant k such that $f(a) = ka$ for all a (k is equal to the constant defined earlier).

Solution 4. Let Γ denote the set of all points $(a, f(a))$, so that Γ is an infinite subset of the upper-right quadrant of the plane. For a point $A = (a, f(a))$ in Γ, we define a point $A' = (-f(a), -f(a)^2/a)$ in the lower-left quadrant of the plane, and let Γ' denote the set of all such points A'.



Claim. For any point $A \in \Gamma$, the set Γ is contained in finitely many lines through the point A'.

Proof. Let $A = (a, f(a))$. The functional equation (with a and b interchanged) can be rewritten as $b + f(a) \mid af(b) - bf(a)$, so that all but finitely many points in Γ are contained in one of the lines with equation

$$ay - f(a)x = m(x + f(a))$$

for m an integer. Geometrically, these are the lines through $A' = (-f(a), -f(a)^2/a)$ with gradient $\frac{f(a)+m}{a}$. Since Γ is contained, with finitely many exceptions, in the region $0 \leqslant y \leqslant f(1) \cdot x$ and the point A' lies strictly in the lower-left quadrant of the plane, there are only finitely many values of m for which this line meets Γ. This concludes the proof of the claim. □

Now consider any distinct points $A, B \in \Gamma$. It is clear that A' and B' are distinct. A line through A' and a line through B' only meet in more than one point if these two lines are equal to the line $A'B'$. It then follows from the above claim that the line $A'B'$ must contain all but finitely many points of Γ. If C is another point of Γ, then the line $A'C'$ also passes through all but finitely many points of Γ, which is only possible if $A'C' = A'B'$.

We have thus seen that there is a line ℓ passing through all points of Γ' and through all but finitely many points of Γ. We claim that this line passes through the origin O and passes through every point of Γ. To see this, note that by construction A, O, A' are collinear for every point $A \in \Gamma$. Since $\ell = AA'$ for all but finitely many points $A \in \Gamma$, it thus follows that $O \in \ell$. Thus any $A \in \Gamma$ lies on the line $\ell = A'O$.

Since Γ is contained in a line through O, it follows that there is a real constant k (the gradient of ℓ) such that $f(a) = ka$ for all a. The number k is, of course, a positive integer.

Comment. Without the $a + b > C$ condition, this problem is approachable by much more naive methods. For instance, using the given divisibility for $a, b \in \{1, 2, 3\}$ one can prove by a somewhat tedious case-check that $f(2) = 2f(1)$ and $f(3) = 3f(1)$; this then forms the basis of an induction establishing that $f(n) = nf(1)$ for all n.

N5. Let a be a positive integer. We say that a positive integer b is *a-good* if $\binom{an}{b} - 1$ is divisible by $an + 1$ for all positive integers n with $an \geqslant b$. Suppose b is a positive integer such that b is a-good, but $b + 2$ is not a-good. Prove that $b + 1$ is prime.

(Netherlands)

Solution 1. For p a prime and n a nonzero integer, we write $v_p(n)$ for the *p-adic valuation of n*: the largest integer t such that $p^t \mid n$.

We first show that b is a-good if and only if b is even, and $p \mid a$ for all primes $p \leqslant b$.

To start with, the condition that $an + 1 \mid \binom{an}{b} - 1$ can be rewritten as saying that

$$\frac{an(an-1)\cdots(an-b+1)}{b!} \equiv 1 \pmod{an+1}. \tag{1}$$

Suppose, on the one hand, there is a prime $p \leqslant b$ with $p \nmid a$. Take $t = v_p(b!)$. Then there exist positive integers c such that $ac \equiv 1 \pmod{p^{t+1}}$. If we take c big enough, and then take $n = (p-1)c$, then $an = a(p-1)c \equiv p-1 \pmod{p^{t+1}}$ and $an \geqslant b$. Since $p \leqslant b$, one of the terms of the numerator $an(an-1)\cdots(an-b+1)$ is $an - p + 1$, which is divisible by p^{t+1}. Hence the p-adic valuation of the numerator is at least $t+1$, but that of the denominator is exactly t. This means that $p \mid \binom{an}{b}$, so $p \nmid \binom{an}{b} - 1$. As $p \mid an + 1$, we get that $an + 1 \nmid \binom{an}{b}$, so b is not a-good.

On the other hand, if for all primes $p \leqslant b$ we have $p \mid a$, then every factor of $b!$ is coprime to $an + 1$, and hence invertible modulo $an + 1$: hence $b!$ is also invertible modulo $an + 1$. Then equation (1) reduces to:

$$an(an-1)\cdots(an-b+1) \equiv b! \pmod{an+1}.$$

However, we can rewrite the left-hand side as follows:

$$an(an-1)\cdots(an-b+1) \equiv (-1)(-2)\cdots(-b) \equiv (-1)^b b! \pmod{an+1}.$$

Provided that $an > 1$, if b is even we deduce $(-1)^b b! \equiv b!$ as needed. On the other hand, if b is odd, and we take $an + 1 > 2(b!)$, then we will not have $(-1)^b b! \equiv b!$, so b is not a-good. This completes the claim.

To conclude from here, suppose that b is a-good, but $b + 2$ is not. Then b is even, and $p \mid a$ for all primes $p \leqslant b$, but there is a prime $q \leqslant b + 2$ for which $q \nmid a$: so $q = b + 1$ or $q = b + 2$. We cannot have $q = b + 2$, as that is even too, so we have $q = b + 1$: in other words, $b + 1$ is prime.

Solution 2. We show only half of the claim of the previous solution: we show that if b is a-good, then $p \mid a$ for all primes $p \leqslant b$. We do this with Lucas' theorem.

Suppose that we have $p \leqslant b$ with $p \nmid a$. Then consider the expansion of b in base p; there will be some digit (not the final digit) which is nonzero, because $p \leqslant b$. Suppose it is the p^t digit for $t \geqslant 1$.

Now, as n varies over the integers, $an + 1$ runs over all residue classes modulo p^{t+1}; in particular, there is a choice of n (with $an > b$) such that the p^0 digit of an is $p - 1$ (so $p \mid an + 1$) and the p^t digit of an is 0. Consequently, $p \mid an + 1$ but $p \mid \binom{an}{b}$ (by Lucas' theorem) so $p \nmid \binom{an}{b} - 1$. Thus b is not a-good.

Now we show directly that if b is a-good but $b+2$ fails to be so, then there must be a prime dividing $an + 1$ for some n, which also divides $(b + 1)(b + 2)$. Indeed, the ratio between $\binom{an}{b+2}$ and $\binom{an}{b}$ is $(b + 1)(b + 2)/(an - b)(an - b - 1)$. We know that there must be a choice of $an + 1$ such that the former binomial coefficient is 1 modulo $an + 1$ but the latter is not, which means that the given ratio must not be 1 mod $an+1$. If $b+1$ and $b+2$ are both coprime to $an+1$ then

the ratio *is* 1, so that must not be the case. In particular, as any prime less than b divides a, it must be the case that either $b+1$ or $b+2$ is prime.

However, we can observe that b must be even by insisting that $an+1$ is prime (which is possible by Dirichlet's theorem) and hence $\binom{an}{b} \equiv (-1)^b = 1$. Thus $b+2$ cannot be prime, so $b+1$ must be prime.

N6. Let $H = \{\lfloor i\sqrt{2}\rfloor : i \in \mathbb{Z}_{>0}\} = \{1, 2, 4, 5, 7, \ldots\}$, and let n be a positive integer. Prove that there exists a constant C such that, if $A \subset \{1, 2, \ldots, n\}$ satisfies $|A| \geqslant C\sqrt{n}$, then there exist $a, b \in A$ such that $a - b \in H$. (Here $\mathbb{Z}_{>0}$ is the set of positive integers, and $\lfloor z\rfloor$ denotes the greatest integer less than or equal to z.)

(Brazil)

Common remarks. In all solutions, we will assume that A is a set such that $\{a-b : a, b \in A\}$ is disjoint from H, and prove that $|A| < C\sqrt{n}$.

Solution 1. First, observe that if n is a positive integer, then $n \in H$ exactly when

$$\left\{\frac{n}{\sqrt{2}}\right\} > 1 - \frac{1}{\sqrt{2}}. \tag{1}$$

To see why, observe that $n \in H$ if and only if $0 < i\sqrt{2} - n < 1$ for some $i \in \mathbb{Z}_{>0}$. In other words, $0 < i - n/\sqrt{2} < 1/\sqrt{2}$, which is equivalent to (1).

Now, write $A = \{a_1 < a_2 < \cdots < a_k\}$, where $k = |A|$. Observe that the set of differences is not altered by shifting A, so we may assume that $A \subseteq \{0, 1, \ldots, n-1\}$ with $a_1 = 0$.

From (1), we learn that $\{a_i/\sqrt{2}\} < 1 - 1/\sqrt{2}$ for each $i > 1$ since $a_i - a_1 \notin H$. Furthermore, we must have $\{a_i/\sqrt{2}\} < \{a_j/\sqrt{2}\}$ whenever $i < j$; otherwise, we would have

$$-\left(1 - \frac{1}{\sqrt{2}}\right) < \left\{\frac{a_j}{\sqrt{2}}\right\} - \left\{\frac{a_i}{\sqrt{2}}\right\} < 0.$$

Since $\{(a_j - a_i)/\sqrt{2}\} = \{a_j/\sqrt{2}\} - \{a_i/\sqrt{2}\} + 1$, this implies that $\{(a_j - a_i)/\sqrt{2}\} > 1/\sqrt{2} > 1 - 1/\sqrt{2}$, contradicting (1).

Now, we have a sequence $0 = a_1 < a_2 < \cdots < a_k < n$, with

$$0 = \left\{\frac{a_1}{\sqrt{2}}\right\} < \left\{\frac{a_2}{\sqrt{2}}\right\} < \cdots < \left\{\frac{a_k}{\sqrt{2}}\right\} < 1 - \frac{1}{\sqrt{2}}.$$

We use the following fact: for any $d \in \mathbb{Z}$, we have

$$\left\{\frac{d}{\sqrt{2}}\right\} > \frac{1}{2d\sqrt{2}}. \tag{2}$$

To see why this is the case, let $h = \lfloor d/\sqrt{2}\rfloor$, so $\{d/\sqrt{2}\} = d/\sqrt{2} - h$. Then

$$\left\{\frac{d}{\sqrt{2}}\right\}\left(\frac{d}{\sqrt{2}} + h\right) = \frac{d^2 - 2h^2}{2} \geqslant \frac{1}{2},$$

since the numerator is a positive integer. Because $d/\sqrt{2} + h < 2d/\sqrt{2}$, inequality (2) follows.

Let $d_i = a_{i+1} - a_i$, for $1 \leqslant i < k$. Then $\{a_{i+1}/\sqrt{2}\} - \{a_i/\sqrt{2}\} = \{d_i/\sqrt{2}\}$, and we have

$$1 - \frac{1}{\sqrt{2}} > \sum_i \left\{\frac{d_i}{\sqrt{2}}\right\} > \frac{1}{2\sqrt{2}} \sum_i \frac{1}{d_i} \geqslant \frac{(k-1)^2}{2\sqrt{2}} \frac{1}{\sum_i d_i} > \frac{(k-1)^2}{2\sqrt{2}} \cdot \frac{1}{n}. \tag{3}$$

Here, the first inequality holds because $\{a_k/\sqrt{2}\} < 1 - 1/\sqrt{2}$, the second follows from (2), the third follows from an easy application of the AM–HM inequality (or Cauchy–Schwarz), and the fourth follows from the fact that $\sum_i d_i = a_k < n$.

Rearranging this, we obtain

$$\sqrt{2\sqrt{2} - 2} \cdot \sqrt{n} > k - 1,$$

which provides the required bound on k.

Solution 2. Let $\alpha = 2 + \sqrt{2}$, so $(1/\alpha) + (1/\sqrt{2}) = 1$. Thus, $J = \{\lfloor i\alpha \rfloor : i \in \mathbb{Z}_{>0}\}$ is the complementary Beatty sequence to H (in other words, H and J are disjoint with $H \cup J = \mathbb{Z}_{>0}$). Write $A = \{a_1 < a_2 < \cdots < a_k\}$. Suppose that A has no differences in H, so all its differences are in J and we can set $a_i - a_1 = \lfloor \alpha b_i \rfloor$ for $b_i \in \mathbb{Z}_{>0}$.

For any $j > i$, we have $a_j - a_i = \lfloor \alpha b_j \rfloor - \lfloor \alpha b_i \rfloor$. Because $a_j - a_i \in J$, we also have $a_j - a_i = \lfloor \alpha t \rfloor$ for some positive integer t. Thus, $\lfloor \alpha t \rfloor = \lfloor \alpha b_j \rfloor - \lfloor \alpha b_i \rfloor$. The right hand side must equal either $\lfloor \alpha(b_j - b_i) \rfloor$ or $\lfloor \alpha(b_j - b_i) \rfloor - 1$, the latter of which is not a member of J as $\alpha > 2$. Therefore, $t = b_j - b_i$ and so we have $\lfloor \alpha b_j \rfloor - \lfloor \alpha b_i \rfloor = \lfloor \alpha(b_j - b_i) \rfloor$.

For $1 \leqslant i < k$ we now put $d_i = b_{i+1} - b_i$, and we have

$$\left\lfloor \alpha \sum_i d_i \right\rfloor = \lfloor \alpha b_k \rfloor = \sum_i \lfloor \alpha d_i \rfloor ;$$

that is, $\sum_i \{\alpha d_i\} < 1$. We also have

$$1 + \left\lfloor \alpha \sum_i d_i \right\rfloor = 1 + a_k - a_1 \leqslant a_k \leqslant n$$

so $\sum_i d_i \leqslant n/\alpha$.

With the above inequalities, an argument similar to (3) (which uses the fact that $\{\alpha d\} = \{d\sqrt{2}\} > 1/(2d\sqrt{2})$ for positive integers d) proves that $1 > \big((k-1)^2/(2\sqrt{2})\big)(\alpha/n)$, which again rearranges to give

$$\sqrt{2\sqrt{2} - 2} \cdot \sqrt{n} > k - 1.$$

Comment. The use of Beatty sequences in Solution 2 is essentially a way to bypass (1). Both Solutions 1 and 2 use the fact that $\sqrt{2} < 2$; the statement in the question would still be true if $\sqrt{2}$ did not have this property (for instance, if it were replaced with α), but any argument along the lines of Solutions 1 or 2 would be more complicated.

Solution 3. Again, define $J = \mathbb{Z}_{>0} \setminus H$, so all differences between elements of A are in J. We start by making the following observation. Suppose we have a set $B \subseteq \{1, 2, \ldots, n\}$ such that all of the differences between elements of B are in H. Then $|A| \cdot |B| \leqslant 2n$.

To see why, observe that any two sums of the form $a + b$ with $a \in A, b \in B$ are different; otherwise, we would have $a_1 + b_1 = a_2 + b_2$, and so $|a_1 - a_2| = |b_2 - b_1|$. However, then the left hand side is in J whereas the right hand side is in H. Thus, $\{a + b : a \in A, b \in B\}$ is a set of size $|A| \cdot |B|$ all of whose elements are no greater than $2n$, yielding the claimed inequality.

With this in mind, it suffices to construct a set B, all of whose differences are in H and whose size is at least $C'\sqrt{n}$ for some constant $C' > 0$.

To do so, we will use well-known facts about the negative Pell equation $X^2 - 2Y^2 = -1$; in particular, that there are infinitely many solutions and the values of X are given by the recurrence $X_1 = 1, X_2 = 7$ and $X_m = 6X_{m-1} - X_{m-2}$. Therefore, we may choose X to be a solution with $\sqrt{n}/6 < X \leqslant \sqrt{n}$.

Now, we claim that we may choose $B = \{X, 2X, \ldots, \lfloor (1/3)\sqrt{n} \rfloor X\}$. Indeed, we have

$$\left(\frac{X}{\sqrt{2}} - Y\right)\left(\frac{X}{\sqrt{2}} + Y\right) = \frac{-1}{2}$$

and so

$$0 > \left(\frac{X}{\sqrt{2}} - Y\right) \geqslant \frac{-3}{\sqrt{2n}},$$

from which it follows that $\{X/\sqrt{2}\} > 1 - (3/\sqrt{2n})$. Combined with (1), this shows that all differences between elements of B are in H.

Comment. Some of the ideas behind Solution 3 may be used to prove that the constant $C = \sqrt{2\sqrt{2}-2}$ (from Solutions 1 and 2) is optimal, in the sense that there are arbitrarily large values of n and sets $A_n \subseteq \{1, 2, \ldots, n\}$ of size roughly $C\sqrt{n}$, all of whose differences are contained in J.

To see why, choose X to come from a sufficiently large solution to the Pell equation $X^2 - 2Y^2 = 1$, so $\{X/\sqrt{2}\} \approx 1/(2X\sqrt{2})$. In particular, $\{X\}, \{2X\}, \ldots, \{\lfloor 2X\sqrt{2}(1-1/\sqrt{2})\rfloor X\}$ are all less than $1 - 1/\sqrt{2}$. Thus, by (1) any positive integer of the form iX for $1 \leqslant i \leqslant \lfloor 2X\sqrt{2}(1-1/\sqrt{2})\rfloor$ lies in J.

Set $n \approx 2X^2\sqrt{2}(1-1/\sqrt{2})$. We now have a set $A = \{iX : i \leqslant \lfloor 2X\sqrt{2}(1-1/\sqrt{2})\rfloor\}$ containing roughly $2X\sqrt{2}(1-1/\sqrt{2})$ elements less than or equal to n such that all of the differences lie in J, and we can see that $|A| \approx C\sqrt{n}$ with $C = \sqrt{2\sqrt{2}-2}$.

Solution 4. As in Solution 3, we will provide a construction of a large set $B \subseteq \{1, 2, \ldots, n\}$, all of whose differences are in H.

Choose Y to be a solution to the Pell-like equation $X^2 - 2Y^2 = \pm 1$; such solutions are given by the recurrence $Y_1 = 1, Y_2 = 2$ and $Y_m = 2Y_{m-1} + Y_{m-2}$, and so we can choose Y such that $n/(3\sqrt{2}) < Y \leqslant n/\sqrt{2}$. Furthermore, it is known that for such a Y and for $1 \leqslant x < Y$,

$$\{x\sqrt{2}\} + \{(Y-x)\sqrt{2}\} = \{Y/\sqrt{2}\} \tag{4}$$

if $X^2 - 2Y^2 = 1$, and

$$\{x\sqrt{2}\} + \{(Y-x)\sqrt{2}\} = 1 + \{Y/\sqrt{2}\} \tag{5}$$

if $X^2 - 2Y^2 = -1$. Indeed, this is a statement of the fact that X/Y is a best rational approximation to $\sqrt{2}$, from below in the first case and from above in the second.

Now, consider the sequence $\{\sqrt{2}\}, \{2\sqrt{2}\}, \ldots, \{(Y-1)\sqrt{2}\}$. The Erdős–Szekeres theorem tells us that this sequence has a monotone subsequence with at least $\sqrt{Y-2} + 1 > \sqrt{Y}$ elements; if that subsequence is decreasing, we may reflect (using (4) or (5)) to ensure that it is increasing. Call the subsequence $\{y_1\sqrt{2}\}, \{y_2\sqrt{2}\}, \ldots, \{y_t\sqrt{2}\}$ for $t > \sqrt{Y}$.

Now, set $B = \{\lfloor y_i\sqrt{2}\rfloor : 1 \leqslant i \leqslant t\}$. We have $\lfloor y_j\sqrt{2}\rfloor - \lfloor y_i\sqrt{2}\rfloor = \lfloor (y_j - y_i)\sqrt{2}\rfloor$ for $i < j$ (because the corresponding inequality for the fractional parts holds by the ordering assumption on the $\{y_i\sqrt{2}\}$), which means that all differences between elements of B are indeed in H. Since $|B| > \sqrt{Y} > \sqrt{n}/\sqrt{3\sqrt{2}}$, this is the required set.

Comment. Any solution to this problem will need to use the fact that $\sqrt{2}$ cannot be approximated well by rationals, either directly or implicitly (for example, by using facts about solutions to Pell-like equations). If $\sqrt{2}$ were replaced by a value of θ with very good rational approximations (from below), then an argument along the lines of Solution 3 would give long arithmetic progressions in $\{\lfloor i\theta\rfloor : 0 \leqslant i < n\}$ (with initial term 0) for certain values of n.

N7. Prove that there is a constant $c > 0$ and infinitely many positive integers n with the following property: there are infinitely many positive integers that cannot be expressed as the sum of fewer than $cn\log(n)$ pairwise coprime n^{th} powers.

(Canada)

Solution 1. Suppose, for an integer n, that we can find another integer N satisfying the following property:

$$n \text{ is divisible by } \varphi(p^e) \text{ for every prime power } p^e \text{ exactly dividing } N. \tag{†}$$

This property ensures that all n^{th} powers are congruent to 0 or 1 modulo each such prime power p^e, and hence that any sum of m pairwise coprime n^{th} powers is congruent to m or $m-1$ modulo p^e, since at most one of the nth powers is divisible by p. Thus, if k denotes the number of distinct prime factors of N, we find by the Chinese Remainder Theorem at most $2^k m$ residue classes modulo N which are sums of at most m pairwise coprime n^{th} powers. In particular, if $N > 2^k m$ then there are infinitely many positive integers not expressible as a sum of at most m pairwise coprime n^{th} powers.

It thus suffices to prove that there are arbitrarily large pairs (n, N) of integers satisfying (†) such that

$$N > c \cdot 2^k n \log(n)$$

for some positive constant c.

We construct such pairs as follows. Fix a positive integer t and choose (distinct) prime numbers $p \mid 2^{2^{t-1}} + 1$ and $q \mid 2^{2^t} + 1$; we set $N = pq$. It is well-known that $2^t \mid p - 1$ and $2^{t+1} \mid q - 1$, hence

$$n = \frac{(p-1)(q-1)}{2^t}$$

is an integer and the pair (n, N) satisfies (†).

Estimating the size of N and n is now straightforward. We have

$$\log_2(n) \leqslant 2^{t-1} + 2^t - t < 2^{t+1} < 2 \cdot \frac{N}{n},$$

which rearranges to

$$N > \frac{1}{8} \cdot 2^2 n \log_2(n)$$

and so we are done if we choose $c < \frac{1}{8\log(2)} \approx 0.18$.

Comment 1. The trick in the above solution was to find prime numbers p and q congruent to 1 modulo some $d = 2^t$ and which are not too large. An alternative way to do this is via Linnik's Theorem, which says that there are absolute constants b and $L > 1$ such that for any coprime integers a and d, there is a prime congruent to a modulo d and of size $\leqslant bd^L$. If we choose some d not divisible by 3 and choose two distinct primes $p, q \leqslant b \cdot (3d)^L$ congruent to 1 modulo d (and, say, distinct modulo 3), then we obtain a pair (n, N) satisfying (†) with $N = pq$ and $n = \frac{(p-1)(q-1)}{d}$. A straightforward computation shows that

$$N > Cn^{1+\frac{1}{2L-1}}$$

for some constant C, which is in particular larger than any $c \cdot 2^2 n \log(n)$ for p large. Thus, the statement of the problem is true for any constant c. More strongly, the statement of the problem is still true when $cn\log(n)$ is replaced by $n^{1+\delta}$ for a sufficiently small $\delta > 0$.

Solution 2, obtaining better bounds. As in the preceding solution, we seek arbitrarily large pairs of integers n and N satisfying (†) such that $N > c2^k n \log(n)$.

This time, to construct such pairs, we fix an integer $x \geqslant 4$, set N to be the lowest common multiple of $1, 2, \ldots, 2x$, and set n to be twice the lowest common multiple of $1, 2, \ldots, x$. The pair (n, N) does indeed satisfy the condition, since if p^e is a prime power divisor of N then $\frac{\varphi(p^e)}{2} \leqslant x$ is a factor of $\frac{n}{2} = \mathrm{lcm}_{r \leqslant x}(r)$.

Now $2N/n$ is the product of all primes having a power lying in the interval $(x, 2x]$, and hence $2N/n > x^{\pi(2x)-\pi(x)}$. Thus for sufficiently large x we have

$$\log\left(\frac{2N}{2^{\pi(2x)}n}\right) > (\pi(2x) - \pi(x))\log(x) - \log(2)\pi(2x) \sim x,$$

using the Prime Number Theorem $\pi(t) \sim t/\log(t)$.

On the other hand, n is a product of at most $\pi(x)$ prime powers less than or equal to x, and so we have the upper bound

$$\log(n) \leqslant \pi(x)\log(x) \sim x,$$

again by the Prime Number Theorem. Combined with the above inequality, we find that for any $\epsilon > 0$, the inequality

$$\log\left(\frac{N}{2^{\pi(2x)}n}\right) > (1-\epsilon)\log(n)$$

holds for sufficiently large x. Rearranging this shows that

$$N > 2^{\pi(2x)}n^{2-\epsilon} > 2^{\pi(2x)}n\log(n)$$

for all sufficiently large x and we are done.

Comment 2. The stronger bound $N > 2^{\pi(2x)}n^{2-\epsilon}$ obtained in the above proof of course shows that infinitely many positive integers cannot be written as a sum of at most $n^{2-\epsilon}$ pairwise coprime n^{th} powers.

By refining the method in Solution 2, these bounds can be improved further to show that infinitely many positive integers cannot be written as a sum of at most n^α pairwise coprime n^{th} powers for any positive $\alpha > 0$. To do this, one fixes a positive integer d, sets N equal to the product of the primes at most dx which are congruent to 1 modulo d, and $n = d\,\mathrm{lcm}_{r \leqslant x}(r)$. It follows as in Solution 2 that (n, N) satisfies (†).

Now the Prime Number Theorem in arithmetic progressions provides the estimates $\log(N) \sim \frac{d}{\varphi(d)}x$, $\log(n) \sim x$ and $\pi(dx) \sim \frac{dx}{\log(x)}$ for any fixed d. Combining these provides a bound

$$N > 2^{\pi(dx)}n^{d/\varphi(d)-\epsilon}$$

for any positive ϵ, valid for x sufficiently large. Since the ratio $\frac{d}{\varphi(d)}$ can be made arbitrarily large by a judicious choice of d, we obtain the n^α bound claimed.

Comment 3. While big results from analytic number theory such as the Prime Number Theorem or Linnik's Theorem certainly can be used in this problem, they do not seem to substantially simplify matters: all known solutions involve first reducing to condition (†), and even then analytic results do not make it clear how to proceed. For this reason, we regard this problem as suitable for the IMO.

Rather than simplifying the problem, what nonelementary results from analytic number theory allow one to achieve is a strengthening of the main bound, typically replacing the $n\log(n)$ growth with a power $n^{1+\delta}$. However, we believe that such stronger bounds are unlikely to be found by students in the exam.

The strongest bound we know how to achieve using purely elementary methods is a bound of the form $N > 2^k n \log(n)^M$ for any positive integer M. This is achieved by a variant of the argument in Solution 1, choosing primes $p_0, \ldots, p_M$ with $p_i \mid 2^{2^{t+i-1}} + 1$ and setting $N = \prod_i p_i$ and $n = 2^{-tM}\prod_i (p_i - 1)$.

N8. Let a and b be two positive integers. Prove that the integer

$$a^2 + \left\lceil \frac{4a^2}{b} \right\rceil$$

is not a square. (Here $\lceil z \rceil$ denotes the least integer greater than or equal to z.)

(Russia)

Solution 1. Arguing indirectly, assume that

$$a^2 + \left\lceil \frac{4a^2}{b} \right\rceil = (a+k)^2, \quad \text{or} \quad \left\lceil \frac{(2a)^2}{b} \right\rceil = (2a+k)k.$$

Clearly, $k \geqslant 1$. In other words, the equation

$$\left\lceil \frac{c^2}{b} \right\rceil = (c+k)k \tag{1}$$

has a positive integer solution (c, k), with an even value of c.

Choose a positive integer solution of (1) with minimal possible value of k, without regard to the parity of c. From

$$\frac{c^2}{b} > \left\lceil \frac{c^2}{b} \right\rceil - 1 = ck + k^2 - 1 \geqslant ck$$

and

$$\frac{(c-k)(c+k)}{b} < \frac{c^2}{b} \leqslant \left\lceil \frac{c^2}{b} \right\rceil = (c+k)k$$

it can be seen that $c > bk > c - k$, so

$$c = kb + r \quad \text{with some } 0 < r < k.$$

By substituting this in (1) we get

$$\left\lceil \frac{c^2}{b} \right\rceil = \left\lceil \frac{(bk+r)^2}{b} \right\rceil = k^2 b + 2kr + \left\lceil \frac{r^2}{b} \right\rceil$$

and

$$(c+k)k = (kb + r + k)k = k^2 b + 2kr + k(k-r),$$

so

$$\left\lceil \frac{r^2}{b} \right\rceil = k(k-r). \tag{2}$$

Notice that relation (2) provides another positive integer solution of (1), namely $c' = r$ and $k' = k - r$, with $c' > 0$ and $0 < k' < k$. That contradicts the minimality of k, and hence finishes the solution.

Solution 2. Suppose that

$$a^2 + \left\lceil \frac{4a^2}{b} \right\rceil = c^2$$

with some positive integer $c > a$, so

$$\begin{aligned} c^2 - 1 < a^2 + \frac{4a^2}{b} \leqslant c^2, \\ 0 \leqslant c^2 b - a^2(b+4) < b. \end{aligned} \tag{3}$$

Let $d = c^2 b - a^2(b+4)$, $x = c + a$ and $y = c - a$; then we have $c = \dfrac{x+y}{2}$ and $a = \dfrac{x-y}{2}$, and (3) can be re-written as follows:

$$\begin{aligned} \left(\frac{x+y}{2}\right)^2 b - \left(\frac{x-y}{2}\right)^2 (b+4) = d, \\ x^2 - (b+2)xy + y^2 + d = 0; \qquad 0 \leqslant d < b. \end{aligned} \tag{4}$$

So, by the indirect assumption, the equation (4) has some positive integer solution (x, y).

Fix b and d, and take a pair (x, y) of positive integers, satisfying (4), such that $x + y$ is minimal. By the symmetry in (4) we may assume that $x \geqslant y \geqslant 1$.

Now we perform a usual "Vieta jump". Consider (4) as a quadratic equation in variable x, and let z be its second root. By the Vieta formulas,

$$x + z = (b+2)y, \quad \text{and} \quad zx = y^2 + d,$$

so

$$z = (b+2)y - x = \frac{y^2 + d}{x}.$$

The first formula shows that z is an integer, and by the second formula z is positive. Hence (z, y) is another positive integer solution of (4). From

$$\begin{aligned} (x-1)(z-1) &= xz - (x+z) + 1 = (y^2 + d) - (b+2)y + 1 \\ &< (y^2 + b) - (b+2)y + 1 = (y-1)^2 - b(y-1) \leqslant (y-1)^2 \leqslant (x-1)^2 \end{aligned}$$

we can see that $z < x$ and therefore $z + y < x + y$. But this contradicts the minimality of $x + y$ among the positive integer solutions of (4).

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